UNIVERSITY OF CALIFORNIA

Los Angeles

Prevocalization:

A Study in the Internal Structure of Consonants

A dissertation submitted in partial satisfaction of the

requirements for the degree Doctor of Philosophy

in Indo-European Studies

by

Natalie Operstein

2007

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ABSTRACT OF THE DISSERTATION

Prevocalization:

A Study in the Internal Structure of Consonants

by

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Doctor of Philosophy in Indo-European Studies

University of California, Los Angeles, 2007

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This work is a detailed cross-linguistic and theoretical study of prevocalization, or the development of a vocalic prearticulation by consonants. The proposed analysis of prevocalization is couched in terms of articulatory phonology (Browman and Goldstein 1986 et seq.), while also incorporating relevant insights of other frameworks, most notably dependency phonology (Anderson and Ewen 1987). It is argued that prevocalization consists in the breaking up of a consonant into a sequence of its internal components, ushering in a vocalic element not present in the original sound chain. It is suggested that the specification of the consonant's vocalic element is present in its underlying gestural score. This makes all consonant-sized units underlyingly bigestural,

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in that they include both a consonantal and a vocalic gesture. The vocalic gesture is identical to the secondary articulation in the case of secondarily modified consonants, and coincides with the position of the tongue body (and, secondarily, lips) in the case of plain consonants. The suggested model of intrasegmental consonant structure makes strong predictions relevant to various areas of phonological theory, diachronic linguistics, and articulatory evolution. The detailed study of prevocalization provided in this work also opens up new vistas for experimental and theoretical research in the areas of syllabic consonants, postvelar consonants, consonant gemination, retroflexion, and the synchronic and diachronic mechanism of such processes as umlaut, vocalization, vowel intrusion, metathesis, and compensatory lengthening, among others.

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1. Prevocalization in synchrony and diachrony

1.1 Introduction

1.1.1 Defining prevocalization

Prevocalization is a relatively common but understudied phonological process which consists in consonants developing vocalic prearticulation, both synchronically and in diachrony. It may be illustrated with the following forms from Apinajé, a Northern Jê language of Central Brazil, in which syllable-final /ɲ/ is productively prevocalized after mid nonnasal vowels (Cunha 2005:55, 77), cf.:[1]

/mbjeɲ/	[mbzeʲɲ]	'husband'
/rɔɲ/	[rɔʲɲ]	'macaúba (palm sp.)'

Depending on the language, the vocoids developing as a result of synchronic prevocalization (which will be referred to here as prevocalization onglides or, following Wetzels 1993, prevowels) may be realized as a range of phonetic variants between a vowel and a glide of varying aperture, tenseness, and duration. For example, the prevowel that develops between labialized consonants and preceding unrounded vowels in Seri is described by Marlett et al. (2005) as "a very short round transitional vowel". Prevowels of Maxakalí are interpreted by Clements (1991:8) as extra-short unstressed vowels. The quality of the high front prevowel in Entraunes Provençal is described by Blinkenberg as being half-way between a [i] and a [j] (1939:40). In general, however, prevowels are problematic for an adequate synchronic description: resulting from a low-

[1]Cunha (2005) states that the vowels in question have to be nonfront, but her example /mbjeɲ/ 'husband', transcribed [mbzeʲɲ] clearly overrides this restriction.

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level (morpho-)phonological process, they are usually not registered by the linguistic consciousness of the speakers and in most cases cannot be reproduced at will, especially in an experimental setting.[2] Their precise phonetic implementation may also be subject to prosodic conditioning and syllable position of the triggering consonant; for example, in Ring Irish the onglide to a final consonant closing a stressed syllable is said to be more salient than the onglide to the same consonant in a nonfinal position (Mhac An Fhailigh 1980:45ff).

While synchronically productive in a number of genetically and areally unrelated languages, prevocalization is perhaps best known from the historical studies in Romance and Indo-European, where it has, in some cases, played a significant role in shaping the phonologies of the individual languages. In addition, scholars working in the two areas have frequently drawn on one another's domains of inquiry, which has served to strengthen their understanding of the underlying unity of the processes involved. For example, when it was suggested for the first time that the initial products of *i*-umlaut in Germanic had had diphthongal shapes, the Germanic forms were compared with developments of the type Latin *gloria, ratio* > French *gloire* 'glory', *raison* 'reason' (Heyse 1838). The two language areas are also explicitly compared by Ascoli (1886-88, 1887) and Wechssler (1900).

The historical effects of prevocalization in Romance and Indo-European language groups may be traced through written records indicative of its workings at the time of

[2] The failure of native speakers to perceive prevocalization onglides is noted by many specialists. For example, Turculeț (1975) comments on this fact with reference to prepalatalization onglides in Moldova Romanian, and McMahon and Faulkes (1994-95) with reference to the schwa onglide to [r] after nonlow vowels in Scots.

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writing. Another source of information is the actual traces of prevocalization in the form of diphthongization and/or modified quality of the vowels preceding the triggering consonants. Thus, prevocalization has served as an important source of falling diphthongs in many varieties of Romance. Such diphthongs may subsequently have been monophthongized, while typically modifying the quality of the original vowels; nevertheless, prevocalization-induced diphthongs can also be directly observed in a number of Romance varieties, which include the forms from Ventimiglia Ligurian cited below. The diphthongal offglide in the Ligurian forms has been generated by the fricative, which resulted from palatalization and weakening of earlier intervocalic /k/ (Azaretti 1977:20f, 30ff), cf.:

Latin				Ventimiglia Ligurian
pace(m)	>	*/paʒe/	>	[pajʒe] 'peace'
dece(m)	>	*/deʒe/	>	[dejʒe] 'ten'

In older Indo-European languages with early written documentation, prevocalization has to be inferred mostly from spelling and other circumstantial evidence rather than direct attestation. As examples of prevocalization in these languages may be cited the Avestan forms *paitiš* 'lord' and *dāuru* 'wood', in which the first <i> and <u>, respectively, are nonetymological or, using the traditional term, "epenthetic". Investigation of prevocalization in these languages poses a special problem as for all such "vowel epentheses"[3] a number of alternative interpretations of the epenthetic

[3] In addition to "epenthesis" (*Epenthese*), other traditional labels include "vowel infixing" (*Vokalinfigierung*), "anticipation" (*Vorklang*), "assimilation", and "vowel infection" (cf. Curtius

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vowels have been proposed, some of which are at variance with the prevocalization view of their origin. Among the alternative interpretations is the view that, in those cases where the original consonant was followed by a glide, the epenthetic offglides may have resulted from simple consonant-glide metatheses. Other interpretations include viewing the epenthetic vowels as secondary modifications of the consonants, or quality changes in the preceding vowels. To cite two well-known examples, Kiparsky (1967) argued that the offglides in the Greek diphthongs /aj, oj/ before the etymological **rj*, **nj/*mj* are the product of metathesis of these consonant groups, while Daunt (1939) argued that, instead of spelling modified vowel shapes that resulted from breaking[4] the Old English digraphs <io>, <eo>, and <ea> indicated velarization of the following consonants. It will be suggested in this work, however, that the metathesis analysis of the Greek facts is not the best one on systemic grounds: as argued by Hock (2003), at the historical stage which attests the development **{a, o}Rj* > *{a, o}jR*, Proto-Greek was undergoing a process of an across-the-board depalatalization of palatalized articulations, which makes prevocalization a better analysis as it often behaves as a depalatalization strategy cross-linguistically. As to the interpretation of the "epenthetic vowels" as markers of secondary features, the combined evidence assembled in this work makes it clear that analyzing them in terms of consonant prevocalization does not preclude the possibility of their simultaneous functioning as orthographic indicators of secondary consonant modification or vowel diphthongization, given that

1866:608ff; Wechssler 1900:493; Vendryes 1906-08:393; Pedersen 1909:336; Kalygin and Korolev 1989:130).

[4] More on Old English breaking in 1.5.2.

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prevocalization is frequently triggered by the former and itself triggers the latter. Finally, despite the existence of alternative views at all periods of linguistic study of Indo-European, much early work in this area, including Sievers (1881) and Brugmann (1897) indicates basic understanding of the fact that "vowel epentheses" represent a process which is distinct from both metathesis and diphthongization. This, for example, is clear from the following discussion of vowel epentheses by Brugmann (1897:33):

> "Die Epenthese setzt Palatalisierung oder Labialisierung eines Consonanten durch nachfolgenden palatalen oder labialen Vocal voraus. Sie entsteht dadurch, dass die dem Consonanten incorporierten Elemente der i- und u-Stellung zeitlich vorgreifen und sich mit vorausgehendem Vocal zu einem Diphthongen vereinigen. Der Gleitlaut, wie er z. B. in av. *a^{i}rišta-*, *a^{u}ruša-*, air. *a^{i}le*, *ne^{u}rt* vorliegt . . . entwickelt sich bei der Epenthese zum Stellungslaut."

1.1.2 Goals and organization of the dissertation

The goal of this dissertation is to provide a comprehensive study of the synchronic and diachronic aspects of prevocalization, elucidate the nature of this process by means of the gestural framework of Browman and Goldstein (1986 et seq.), and propose a model of the internal structure of consonants consistent with the present findings.

Previous work on prevocalization is extremely limited in scope and, in addition, is scattered over a vast range of specialist domains which include synchronic phonological descriptions of an impressive array of languages and dialects, historical phonologies of the better-studied languages of Eurasia, and brief discussions in recent phonology works whose interest in prevocalization is at best tangential. Given the state of previous attention this topic has received, the present work is in many respects a pioneering

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enterprise, and its structure is necessarily a compromise between the stated goal and the availability of, or depth of coverage of aspects of prevocalization in, earlier studies.

The remainder of this chapter is organized as follows. Sections 1.2 through 1.4 provide a detailed introduction to the descriptive facts of prevocalization. Section 1.2 addresses typical factors that condition prevocalization, focusing primarily on /S/-prevocalization in Brazilian Portuguese, with additional examples from other languages. Section 1.3 provides phonetic and phonological evidence for prevocalization being distinct from vowel diphthongization. Section 1.4 addresses differences in the behavior of plain and secondarily modified consonants under prevocalization. Section 1.5 provides an overview of treatments of prevocalization suggested in the phonological literature. Subsection 1.5.1 focuses on the analysis of prevocalization as partial vocalization of the consonants' primary stricture. The bulk of work using this approach was carried out in order to account for prevocalization in Maxakalí, which is given central attention in that subsection. Subsection 1.5.2 focuses on the analysis of prevocalization in terms of vowel diphthongization; this work was carried out based on the facts of English diachronic phonology, which are reviewed there in detail.

Chapter 2 proposes an analysis of prevocalization within the articulatory phonology framework of Browman and Goldstein (1986 et seq.), and puts forward a new model of consonant structure by redefining the existing notions of V- and C-gesture. It is argued that an adequate analysis of prevocalization should be able to accommodate the synchronically subsegmental character of prevocalization vocoids, and the temporal dimension of the synchronic prevocalization events. These aspects of prevocalization

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cannot be addressed by static approaches operating in terms of segment- or feature-sized units, but are straightforwardly accounted for in the gestural framework, which is singularly suited to account for subsegmental dynamics and incorporates a temporal dimension in its representation of speech events. After providing a brief overview of the articulatory phonology, this chapter first applies the gestural framework to prevocalization of secondarily modified consonants, and then extends this analysis to plain consonants by suggesting a modified definition of intrasegmental consonantal and vocalic gestures (C- and V-gestures). The gestural approach to prevocalization is exemplified with an analysis of Maxakalí, followed by a detailed discussion of V-gestures in plain coronals and labials.

Chapters 3 and 4 examine prevocalization in the individual languages and language families. The discussion is organized around the nature of the prevocalization vocoids (roughly, [i]-like, [ɯ]-like, [u]-like, and [ɑ]-like, in that order), and the nature of the triggering consonants. Chapter 3 examines the [i]-like prevocalization vocoids generated by palatalized consonants, palatals, and alveolars, and Chapter 4 examines the vocoids of other qualities generated by secondarily modified consonants (velarized, labialized, or pharyngealized), as well as dentals, labials, velars, uvulars, and pharyngeals. In addition to providing a wealth of factual data to support the analysis of prevocalization suggested in Chapter 2 and to serve as a frame of reference for future studies, this portion of the work also provides an opportunity for discussing a number of issues specific to prevocalization of the individual classes of consonants. These include, but are not limited to, the phonological interpretation of vowel diphthongization before

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nasals (3.4.3), the mechanism of umlaut in Germanic (3.2.4, 4.2.4), the linguistic status of retroflexion (4.4), and the existence of a pharyngeal glide matching glides at other vocalic places of articulation (4.3).

Chapter 5 seeks to provide further evidence for the correctness of the proposed analysis of prevocalization and the model of consonant structure. It approaches this goal from two angles. First, it examines two additional processes capable of revealing the internal composition of consonants: prevocalization and the life cycle / synchronic functioning of syllabic consonants. Both processes are complementary with respect to prevocalization in the sense of affecting consonants in complementary syllable positions, onsets and nuclei, respectively. A brief overview of these phenomena produces results comparable to those of prevocalization, which points to the correctness of the proposed analysis and the underlying structural model. The proposed analysis is then applied to two processes that are frequently found in complementary distribution with prevocalization, compensatory lengthening and vowel epenthesis/intrusion. It is argued that, at least when complementary with prevocalization, compensatory lengthening has the same mechanism as the former, and that vowel intrusion differs from it primarily in the direction of movement of the C-gesture of the triggering consonant.

Chapter 6 contains a brief summary of the main findings, and suggests a place for prevocalization in the recent research on articulatory and phonological evolution.

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1.2 Conditioning factors

Available descriptions of prevocalization show that from a synchronic point of view this is a highly variable and unstable process sensitive to a whole range of conditioning factors; these will be best examined by focusing on the manifestations of this process in the individual languages. One of the best known cases of synchronic prevocalization is the phonetic realization of the coda archiphoneme /S/ in Brazilian Portuguese, which may be used to illustrate many of the factors that affect prevocalization in general. These factors include, but are not limited to, the syllable position of the triggering consonant, articulatory/acoustic distance from the preceding vowel, lexical and phrasal stress, lexical conditioning, morphological conditioning, and sociolinguistic variation.

Brazilian Portuguese /S/, orthographically <s> or <z>, surfaces as a voiced or voiceless alveolar or palatoalveolar fricative [s] ~ [z] (for example, in São Paulo) or [ʃ] ~ [ʒ] (for example, in Rio de Janeiro), its voicing being determined by that of the following segment and its place of articulation by regional and/or stylistic factors (Giangola 1997, 2001:152ff; Barbosa and Albano 2004:229). In some dialects, /S/ is subject to prevocalization after stressed vowels when in absolute coda; its prevocalized allophones are realized as [ʲs] ~ [ʲz] or [ʲʃ] ~ [ʲʒ]. Vowel nasalization does not affect prevocalization, cf. *maçãs* [ma'sɐ̃ʲs ~ ma'sɐ̃ʲʃ] 'apples'; however, it is said to be blocked by a preceding /i/. Words derivationally related to the prevocalized forms as a rule do not attest prevocalization, possibly due to the shift of stress and/or a change in the syllable position of the sibilant (Giangola 2001:88, 152ff), cf.:

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arroz [aˈxoʲs ~ aˈxoʲʃ] 'rice'	*arrozal* [axoˈzaw] 'rice paddy'
luz [ˈluʲs ~ ˈluʲʃ] 'light'	*luzes* [ˈluzis ~ ˈluziʃ] 'lights'

In the dialect of Rio de Janeiro, prevocalization of /S/ is reported to occur most frequently in stressed monosyllables, especially when lexical and sentence stress coincide. For the dialect of São Paulo, Albano (1999) reports the results of a study which indicate that under prosodic conditions that favor lengthening, prevocalization may alternate with compensatory lengthening of the vowel preceding the /S/; for both major dialects, such conditions include location under sentence stress or before a pause.

/S/-prevocalization may also be said to be sensitive to morphological conditioning as, in some dialects, /S/ representing the plural marker *-s* fails to prevocalize. Thus, Wetzels and Sluyters (1995) note that in the dialect of São Paulo, [paʲs] and [pɛʲs] are not permissible as the phonetic realizations of *pá-s* 'shovels' and *pé-s* 'feet', respectively (140 fn. 2). No such restriction is reported for the dialect of Rio de Janeiro, where *pais* /pajS/ 'parents' is said to be (near-)homophonous with both *paz* /paS/ 'peace' and *pá-s* /paS/ 'shovels'.

Another feature of /S/-prevocalization that makes this a typical representative of this process is the fact that it tends to affect high-frequency vocabulary of everyday lexical stock (Reinhardt 1970); this tendency of prevocalization was noted already by Wechssler (1900) and is frequently commented upon by later researchers. While in principle under favorable prosodic conditions prevocalization should be able to affect any eligible lexical item in the language, it is with the high-frequency vocabulary where

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it is reported to be established most firmly. In the dialects of Rio de Janeiro and São Paulo, prevocalization is reported to be almost categorical in the monosyllabic words *mas* 'but', *três* 'three', *faz* 's/he does or makes', and *mês* 'month', as well as the nonmonosyllables *rapaz* 'boy, fellow' and *mesmo* 'even, really'. All of the monosyllabics in this list are high-frequency items, while *rapaz* and *mesmo* are reported to be often used in single word exclamations, which creates a context for their prepausal lengthening (Albano 1999, 2001).

It is possible to argue that at least some of the lexical items affected by prevocalization now include diphthongs underlyingly; at least three unrelated considerations support this analysis. One consideration follows from the complete homophony of these lexical items with the corresponding words with underlying diphthongs. This homophony leads to hypercorrections of the type *mas* 'but' for *mais* 'more' in the writing of semi-literate speakers reported in Albano (1999). In this, the homophonous lexical pair *mas/mais* seems to contrast with pairs like *paz* 'peace' / *pais* 'parents', where the original lexical distinction is said to be "not entirely effaced" (Albano 1999:1787): this may be interpreted as the absence of an underlying glide in *paz* 'peace'. Another consideration follows from the fact that in certain words derivationally related to the prevocalized forms the prevowel is preserved despite the stress shift onto certain derivationally transparent suffixes, and/or shift of the /S/ to an intervocalic position; as mentioned above, such forms as a rule do not attest prevocalization, cf. *arrozal* [axoˈzaw] 'rice paddy' versus *arroz* [aˈxoˑs ~ aˈxoˑʃ] 'rice'.

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Nevertheless, the prevowel may be preserved after the addition of the diminutive suffix *-inho*, cf. *arrozinho* [axoʲˈziɲu] (Giangola 2001:89). The final consideration follows from the workings of a productive phonological rule in the Caipira dialect[5] which creates open syllables by adding a [i] after word-final /s, r, l/. Some of the forms eligible for both prevocalization and the vowel addition rules do not contain the prevowel in the outcome (e.g., *luz* 'light'), while others do (e.g., *rapaz* 'boy, fellow') (cf. Bortoni-Ricardo 1985:56). As may be seen below, *rapaz* behaves in this respect parallel to words with underlying diphthongs like *mais* /majS/ 'more', and it is possible to argue that the diphthong has become underlyingly present in *rapaz* as well.

Spelling	Underlying form	Surface form	Gloss
luz	/luS/	[ˈluzi]	'light'
mais	/majS/	[ˈmajzi]	'more'
rapaz	/rapajS/	[raˈpajzi]	'boy, fellow'

/S/-prevocalization in Brazilian Portuguese illustrates several of the factors that influence prevocalization in general, namely, the fact that it tends to involve consonants in weak syllable positions, is lexically selective in affecting primarily high-frequency words, and is sensitive to lexical and phrasal stress, morphological conditioning, articulatory distance between the prevocalization onglide and the preceding vowel, dialect and register; in addition, Albano (1999) reports that /S/-prevocalization interacts with age and sex. There are, however, additional factors, not illustrated by the above data, that may influence prevocalization, and examples of prevocalization in other

[5] The rural dialect of the state of São Paulo.

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languages where some of the above conditioning factors are seen more clearly. These are briefly discussed below.

Cross-linguistically, the vowel most frequently affected by prevocalization onglides is [a]; in some languages, this is the only vowel after which prevocalization is noted in the literature (e.g., Crowley 1983:318). When other vowels are affected, the glide is usually most salient after a [a] (cf. Broch 1899). Prepalatalization onglides are less prominent after front vowels and the least prominent after a [i]. For example, the onglide triggered by palatalized velars in Irish English is more salient after the more retracted allophone of the preceding /æ/ (Harris 1985), and that triggered by the palatal nasal in Djabugay is more prominent after a /a, u/ than after a /i/ (Patz 1991:253f), cf.:

burrany	[burra^{j}ɲ]	'fly-PAST'
guniny	[guni^{j}ɲ]	'cut-PAST'
dunyu	[du^{j}ɲu]	'husband'

Syllable position of a consonant may affect its functioning as a synchronic trigger of prevocalization, which in turn may lead to differing diachronic outcomes. For example, in Quiaviní Zapotec there is prepalatalization before the sequence /nj/ in coda, but not when the cluster moves to an onset position after the addition of a vowel-initial clitic (Pam Munro, p. c.), cf.:

telebisyoony	[-o'o^{j}nj]	'television'
x-telebisyoony=a Possessive-television=1s	[-o'onja]	'my television'

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In some languages, preconsonantal coda is a stronger conditioning environment than absolute coda. For example, in Majorcan Catalan [ɲ] is prevocalized when preconsonantal but not when coda-final, cf. /aɲs/ [aʲns] 'years' versus /aɲ/ [aɲ] 'year' (Mascaró 1985). In San Dionicio Ocotepec Zapotec (Mexico), the cluster /nj/ is prevocalized only when there is a stop-initial clitic following the word (Broadwell 2005), cf. *mààny=gà'* [maʲnga] 'that bird', literally 'bird-that'. Differences in syllable position of the consonant may lead to a differing treatment of the consonant or consonant cluster in diachrony. Thus, French /ɲ/ does not prevocalize in onsets (cf. *montanea* > *montagne* 'mountain'), but does so in word-final codas (*cuneu* > *coin* 'wedge', *longe* > *loin* 'far') (Straka 1954:79).[6] Likewise, the Latin group *-ntj-* did not prevocalize intervocalically producing French *-ns-*, but did so word-finally producing *-jn(z)*, cf. *cantionem* > *chanson* 'song', *nuntiare* > *noncier* 'announce' versus *nuntium* > Old French *noinz* 'messenger' and place-names like *Sanctus Pontius* > *Saint-Point* and *Degontiu* > *Digoin* (Straka 1954:77ff).

It has been noted above that /S/-prevocalization is tied to environments that condition lengthening, which included lexical and phrasal stress (especially when the two coincide), and location before a pause. Descriptions of prevocalization in other languages note additional lengthening environments, including slow and/or careful enunciation or emphasis. Thus, Straka (1949) notes that in an especially relaxed pronunciation of some Czechs a transitional glide is heard between a /a/ and a following

[6] According to Straka (1954), *baigner*, *témoigner*, *poignée* and similar forms are analogically refashioned after *bain*, *témoin*, *poing*, where the palatal glide is expected (80 fn. 3).

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/s/ (38 fn. 3). Relaxed pronunciation of vowels and slow speech tempo in North-Western Moldova are mentioned as conditioners of prevocalization by Turculeț (1975:14): the relaxed pronunciation of both stressed and unstressed vowels characteristic of that dialect area is responsible for the occurrence of prevocalization in both stressed and unstressed syllables. A connection between emphatic pronunciation and prevocalization has been noted for Irish English (Ó Baoill 1997:78):

> "This palatal quality [of /k/ and /g/, NO] is even more striking at the end of a word or syllable. In words such as *back*, *track*, *thanks*, etc. there is a strong off-glide between the vocalic nucleus and the palatal velar consonant, especially when these words receive heavy stress or undue emphasis. The combinations [a^{I}] and [æ^{I}] are often interpreted as diphthongs by speakers of other varieties of English such as Australian or Southern United States English. Indeed, the realizations [a^{I}] and [æ^{I}] are interpreted as identical to the underlying diphthong in such words as *dime*, *fine*, *side*, etc. in such varieties."

Lastly, the number of syllables per word or a higher syntactic unit may also play a role in conditioning prevocalization. This may be connected with the general phonological tendency of monosyllables to have longer nuclei than polysyllabic words. For example, Howren (1958) notes that the occurrence of [ʌ^{j}] in Louisville, Kentucky is determined by the number of syllables per word or a tightly bound syntactic unit; his examples include such contrastive pronunciations as *lunch* and *not much*, both of which have [ʌ^{j}], versus *lunchbox* and *don't touch it*, both with a [ʌ] (1958:81).

Other factors frequently affecting prevocalization, which are not illustrated by the /S/-prevocalization in Brazilian Portuguese, include length, height, and nasalization of the vowel preceding the triggering consonant. Thus, as mentioned in 1.1.1, prevocalization does not take place after nasalized vowels in Apinajé, while in some

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languages of Australia it is reported to occur only after long vowels, and in Afar only after high vowels. Another characteristic feature of prevocalization is that, rather than replacing the existing pronunciations of the affected forms it mostly creates alternative pronunciations of the relevant lexical items. This is very clear, e.g., from the synchronic descriptions of such languages as present-day European Portuguese, in which the type and scope of prevocalization differ from those of the Brazilian variety. More specifically, in European Portuguese both intervocalic and caudal palatoalveolars /ʎ, ɲ/ and alveolopalatals /ʃ, ʒ/[7] can be prevocalized, provided the preceding vowel is stressed and nonfront. This gives rise to alternative pronunciations some of which are illustrated below (Barbosa 1965:64; Freitas and Rodrigues 2003:63ff).

malha [ˈmaʎə] ~ [ˈmaʲʎə] 'mesh'
unha [ˈuɲə] ~ [ˈuʲɲə] '(finger)nail'
caixa [ˈkaʃə] ~ [ˈkaʲʃə] 'box'
texto [ˈtəʃtu] ~ [ˈtəʲʃtu] 'text'
loja [ˈlɔʒə] ~ [ˈlɔʲʒə] 'boutique'
longe [ˈlõʒə] ~ [ˈlõʲʒə] 'far'[8]

1.3 Vowel diphthongization

In the examples of prevocalization cited so far, prevocalization onglides were inserted between triggering consonants and the preceding vowels, which might have created a

[7] The fact that /ʃ, ʒ/ are phonetically alveolopalatals is noted in Mateus and d'Andrade (2000:11 ft 4).

[8] Giangola (2001) notes that this type of prevocalization is alien to Brazilian Portuguese, observing that pronunciations like *cereja* [səˈrəʲʒə] 'cherry' in European Portuguese correspond to Brazilian [seˈreʒə] (7 fn. 3). Nevertheless, in his discussion of Brazilian poetry Camara (1977) notes rhymes of the type *beijo* 'kiss' / *desejo* 'desire' and *baixo* 'low' / *riacho* 'brook' (99ff) which clearly indicate that prevocalization of at least the palatal sibilants /ʃ, ʒ/ is not entirely alien to Brazilian Portuguese.

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superficial impression that the emergence of the onglides is unrelated to the consonants and may be due to vowel diphthongization instead. This impression may appear to be strengthened by the consideration that vowel-onglide sequences may be close enough to underlying diphthongs to rhyme with them; for example, in Brazilian Portuguese poetry, *luz* /luS/ 'light' may rhyme with *azuis* /azujS/ 'blue (pl.)' (Camara 1977:86). Nevertheless, a number of independent considerations suggest that this initial impression is erroneous and that, rather than being the source of the onglides, diphthongization of vowels in these conditions is an accidental spin-off of prevocalization. The necessity to distinguish prevocalization from vowel diphthongization has been pointed out by the majority of researchers who examined the former process, including Grammont (1933:221f), Ritter (1934:180f), Anglade (1961:12), Pellegrini (1961:117), Reinhardt (1970:102), Foley (1975:198-99), Marcato (1983-84), and Tuttle (1991:46f), among others. The concrete evidence for the consonantal origin of the prevocalization onglides comes from a variety of sources including, but not limited to, native speakers' intuitions about whether the glide "belongs" to the consonant or the preceding vowel,[9] the compensatory character (in the sense of Mascaró 1985) of the vowel diphthongization, as well as phonetic and phonological comparisons of vowel-onglide combinations with the corresponding phonemic diphthongs in languages that have both.

The first two arguments are made use of by Reinhardt (1970) in his analysis of /S/-prevocalization in Brazilian Portuguese. Reinhardt notes that, in the absence of any

[9] Cf. a discussion of this problem in Biedrzycki (1963:30ff).

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additional evidence, there may be two ways of formulating this process cited below as (a) and (b).

(a) /S/ → [^{j}S] / ^{l}V ___ #
(b) /^{l}V/ → [^{l}V^{j}] / ___ S#

He then argues that only the formulation in (a) provides an adequate description of /S/-prevocalization based on the fact that "native speakers often feel that *the [i] is part of the s*" (italics his), and the fact that, when the final [S] is dropped, "the [i] remains as the sole phonetic representative of the /S/". The second of Reinhardt's arguments comes close to formulating the notion of "compensatory diphthongization" in the sense of Mascaró (1985). Mascaró introduces this notion in order to account for a set of synchronic processes in Majorcan Catalan where underlying sequences /VɲC/ and /VncC/ surface as [VjnC], and underlying sequences /VʃC/ and /VtʃV/ as [VjC]. Mascaró views the insertion of the glide as a compensatory feature which "compensates for the loss of the palatal . . . by keeping stable . . . the tier that specifies properties of point of articulation", whether the original palatal is deleted, as is the case of underlying /VncC/, /VʃC/, or /VtʃV/, or shifts its primary constriction, as is the case of underlying /VɲC/ (1985:132). It might also be added that the glide simply cannot be derived from vowels preceding the consonants in question: if one were to interpret the processes of Brazilian Portuguese and Majorcan Catalan as vowel diphthongizations, it would be

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difficult to derive the palatal glide from nonfront vowels; in addition, depalatalization of the palatal nasal in Catalan would remain unexplained.[10]

Phonetic differences between vowel-onglide sequences resulting from prevocalization and underlying diphthongs include lesser duration of the prevocalization onglides as compared to the offglides of diphthongs, as noted for Ring Irish by Breatnach (1947:59f) and for the Romanian of Moldova by Turculeț (1975). As a natural consequence of this fact, prevocalization onglides, but not diphthongal offglides, are easily dropped in either extremely fast or extremely careful speech, as noted for Brazilian Portuguese by Albano (2001:90). The duration of vowel-onglide combinations is also said to be closer to that of the monophthong vowels than to that of the diphthongs (cf. Mascaró 1985:133 for Majorcan Catalan). Despite these helpful observations, however, the majority of studies that report durational differences between vowel-onglide sequences and diphthongs fail to provide measurable data; a notable exception to this general state of affairs is Lehiste's (1965) comparative study of palatalization onglides and diphthongal offglides in Estonian.

Standard Estonian possesses palatalized coronals /t^{j}, s^{j}, l^{j}, n^{j}/ which contrast with their nonpalatalized counterparts /t, s, l, n/. Numerous studies report that Estonian palatalization is signaled primarily by a prominent [i]-like onglide to the palatalized

[10] It should perhaps be noted here that this compensatory characteristic of prevocalization was clear to earlier researchers as well. Meyer-Lübke (1890), for example, supplies the following commentary to prepalatalization of /ɲ/ in French forms like *-ains*, from Latin *-aneus*: "La consonne palatale, devant une dentale suivante, devient elle-même une simple dentale, puis cette partie de l'articulation palatale est compensée d'une certaine manière en ce que la palatalization affecte la voyelle: au lieu d'une voyelle pure, on obtient une voyelle palatalisée" (219).

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consonant; this onglide may be perceived as segmental by speakers of languages in which this type of palatalization is unknown (Lehiste 1965:136). Furthermore, while the initial part of the consonant itself is palatalized, its final part is depalatalized and subject to coarticulatory influence of the following vowel (cf. Eek 1973:31).[11] When the palatalization onglide combines with preceding vowels, there result surface diphthongs. Lehiste (1965) reports that the overall duration of the vowel-onglide sequences is considerably lower than the duration of the (near-)homophonous diphthongs. The durations quoted in her study for the nucleus /a/ before /s^{j}, l^{j}/ and those for the diphthong /ai/ before /s, l/, averaged across the two speakers examined, contrast as 245/405 ms (cf. Table 2 in Lehiste 1965). The duration of the palatalization onglide (averaging 77 ms across the two speakers) is half-way between vowel transitions towards nonpalatalized consonants (averaging 47 ms) and the second element of the (near-)homophonous diphthongs (averaging 145 ms) (Lehiste 1965:140, 151).[12] These data indicate that vowel-onglide combinations differ both from the monophthongs and from the diphthongs, and that prevocalization onglides are shorter than the offglides of underlying diphthongs.

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[11] Palatalization in Estonian thus differs from palatalization in Russian, where the palatalization gesture is applied to the final part of the consonant and the transition between the consonant and the following vowel. (Derkach et al. 1970, for example, report that when the [i]-like transition between the palatalized fricatives and the following vowels is removed, the fricatives are perceived by native listeners as nonpalatalized in 80% of cases.)

[12] A related point of contrast between vowel-onglide sequences and diphthongs is differences in the quality of the preceding vowel allophone: for example, /a/ has a higher second formant before a palatalized consonant than in the diphthong /ai/. Related to this is the fact that the diphthong /ai/ involves two steady states corresponding to each of its elements, while the sequence [a^{j}] is continuous (Lehiste 1965).

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Differences in the phonological behavior of vowel-onglide sequences and the corresponding diphthongs may be demonstrated from Huaraz Quechua and Majorcan Catalan. In Huaraz Quechua, the underlying diphthong /aj/ has surface realization [e:] whereas a combination of /a/ and a prepalatalization onglide is realized on the surface as [aj] (Swisshelm 1971:5-6), cf.:

Underlying nucleus	Surface form	Gloss
/aj/, as in /ˈpajʎa/	[e:], as in [ˈpe:ʎa]	'only he'
/a/, as in /ˈpaʎa/	[aj], as in [ˈpajʎa]	'thanks'

In Majorcan Catalan, /ɲ/ is prevocalized in preconsonantal coda while simultaneously shifting its point of articulation (Mascaró 1985), cf. /aɲs/ [aʲns], the plural of /aɲ/ [aɲ] 'year'. The phonetic diphthongs that result from prepalatalization differ in their behavior from the underlying diphthongs in two respects. First, while underlying word-final sequences of diphthong plus sonorant develop an epenthetic schwa, the corresponding phonetic diphthongs deriving from prevocalization fail to do so, cf.:

Underlying nucleus	Surface form	Gloss
/aw/, as in /fawns/	[ˈfawnəs]	'fauns'
/a/, as in /aɲs/	[aʲns]	'years'

Second, whereas underlying sequences of the type /VjCs/ are simplified on the surface to [Vjs], similar phonetic sequences arising as a result of prevocalization are not affected (Mascaró 1985:140f), cf.:

Underlying sequence	Surface form	Gloss
/VjCs/, as in /ˈvujtˈsens/	[Vjs], as in [ˈvujˈsens]	'eight hundred'
/VCs/, as in /aɲs/	[VjCs], as in [ajns]	'years'

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An additional piece of evidence on the phonology of prevocalization is supplied by Djabugay (Australian), where the palatal nasal is prevocalized when underlying but fails to do so when derived from the alveolar nasal through assimilation. This fact serves to distinguish words that otherwise would be (nearly) homophonous (Patz 1991:253f), cf.:

/baɲɟi/	[baʲɲɟi]	'wife's brother'
/baʲnɟil/	[baɲɟil]	'woman'

In this respect Djabugay differs from Majorcan Catalan, which makes no distinction between the underlying and derived palatal nasals for the purposes of prevocalization.

The combined evidence of the phonetic and phonological differences between vowel-onglide sequences and underlying diphthongs indicates that the source of the prevocalization onglides is to be sought in the consonant rather than the preceding vowel. At the same time, it shows that vowel-onglide sequences are synchronically ambiguous: although phonetically diphthongal, they function phonologically as monophthong allophones of the corresponding simple vowels. This inherent ambiguity determines the possibility of their development into both monophthongs and phonemic diphthongs, the particular path of development being language-specific and often dependent on prosodic factors. For example, the historical phonologies of French and Greek include instances of the creation of falling diphthongs due to onglides deriving from prevocalization. By contrast, the results of umlaut and breaking in various Germanic languages (provided the former can ever be shown to have involved a prevocalization stage) tend to be monophthongal.

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1.4 Simple and complex consonants

A cross-language investigation shows that both plain and secondarily modified consonants may be prevocalized. Prevocalization of simple consonants typically acts on segments in weak syllable positions and is best seen as a type of consonant lenition, as its major surface result is partial reduction of the consonant's primary stricture. Prevocalization of secondarily modified consonants acts as a synchronic enhancement of the secondary articulations, while in diachrony it is associated with their loss. Additionally, prevocalization of complex consonants may result in weakening of their primary stricture.

Stricture reduction accompanying prevocalization of plain consonants may be effected in a variety of ways. The simplest outcome is diminished duration of the primary stricture, which appears as simple surface prevocalization. This is the case of /S/-prevocalization in Brazilian Portuguese: Albano (1999) shows that under prevocalization the fricative portion of the /S/ is shortened while the vowel transition towards the fricative is simultaneously lengthened. Shortening of the consonantal stricture also accompanies prevocalization of geminates and is interpreted as degemination. In cases of a more extreme lenition, the primary stricture of the prevocalized consonants may undergo further modifications in the direction of greater weakening. Thus, obstruents may undergo fricativization or gliding; oral stops may become unreleased; sonorants may become glided or devoiced. Especially interesting is prevocalization of palatal stops or the lateral as it is often accompanied by their alveolarization. One instance of this kind cited above is the shift from *jn* to *jn* Majorcan

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Catalan; alveolarization of the palatal nasal is best interpreted as lenition as it involves both reduction in the surface area of primary contact between the tongue and the palate and a reduction in duration: palatals, articulated with the tongue blade, are inherently longer than the corresponding alveolars articulated with the more mobile tongue tip.[13] Alternatively, palatal stops may lose occlusion and weaken to fricatives or palatal glide (cf. dialectal Polish /koɲ/ [ko$\overset{\cdot}{\text{j}}$] 'horse'). The consonant may also lose its oral component altogether, which results in a combination of prevocalization and debuccalization (cf. *ç* > *jh* in many Mon-Khmer languages); the equivalent change in nasals involves velarization, cf. *ɲ* > *jŋ* in some varieties of Romance (Recasens et al. 1995:270).[14] Finally, the consonantal component may be deleted, at which point prevocalization becomes indistinguishable from straight vocalization, especially if records of earlier stages are lacking. Table 1 below gives an overview of the degrees of primary stricture lenition that may accompany prevocalization.

[13] Alveolarization of a palatal is also necessarily accompanied by the lowering of its F2 value, which likewise may be interpreted as consonant lenition, due to the weakening (in this case, centralization) of its secondary vocalic component. In the lenition of velarized consonants, this is paralleled by a shift from a secondary articulation of high back quality to that of schwa-like resonance; for such an interpretation of sonorant lenition in Irish and Scottish Gaelic cf. Ó Dochartaigh (1979:475ff, 482, 491f). Of related importance are the facts of Estonian where the degree of palatalization (measured as the width of lateral contact) depends on the degree of consonant quantity: higher degree of consonant quantity correlates with stronger palatality of the consonant (Eek 1973:25).

[14] The notion of nasal velarization being equivalent to debuccalization in oral consonants is discussed at length in Trigo (1988).

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Table 1. Degrees of primary stricture lenition in prevocalized consonants

Degrees of primary stricture lenition	Example of a consonant before prevocalization	Surface result after prevocalization
reduced duration	s	js
degemination	lʲː	jlʲ
lack of release (oral stops)	t	e̯tˀ
voicing (obstruents)	p	e̯b
devoicing (sonorants)	n	e̯n̥
reduced duration and area of contact	ʎ	jl
fricativization	tʃ	jʃ
gliding	ɲ	jj̃
debuccalization	ç	jh
deletion	tʃ, s, ʎ, ɲ, ç	j

Since prevocalization of plain consonants is motivated by their decomposition in weak syllable positions, it is relatively independent from the quality of the preceding vowels. Prevocalization of complex consonants, on the other hand, incorporates the effects of acoustic and articulatory assimilation of the preceding vowels towards the secondary articulation of the consonant. Prevocalization of secondarily modified consonants seems to be motivated, on the one hand, by the tendency of the secondary (= vocalic) articulation to move closer to the nucleus, and on the other, by the tendency of the preceding vowel to assimilate to the consonant's vocalic coarticulation. Due to the importance of vocalic transitions in prevocalization of this type, an important role is assumed by the degree of articulatory and acoustic distance between the consonant's secondary articulation and the preceding vowel. Prevocalization of secondarily modified consonants often functions in synchrony as an auditory enhancement of the corresponding secondary articulation (e.g., in Estonian), but it may also be an indicator of the synchronic tendency towards its loss (e.g., in Polish). Phonemicization (Hock's

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1991 "segmentalization") of the prevocalization onglides may lead to a complete loss of the secondary articulation, a type of development amply documented in the histories of many languages, including French and Greek.

The existing formal analyses of prevocalization reflect the division of this process in the two broad types just outlined. Andersen (1972), Wetzels (1993), and Wetzels and Sluyters (1995), who all address prevocalization of plain consonants, interpret it as partial vocalization of the primary stricture, formally represented as the creation of a segment contour with opposite values of the feature [vocalic]. Gussenhoven and Weijer (1990), who concentrate primarily on complex consonants in the history of English, analyze prevocalization as diphthongization of preceding vowels. Both types of analyses offer important insights into the workings of prevocalization; however, their inherent shortcomings make them unsuitable as a general explanation of this process. The relevant details of these approaches and the difficulties encountered by each are briefly summarized in Section 1.5.

1.5 Theoretical approaches to prevocalization

Previous literature offers three theoretical approaches to prevocalization, two of which are summarized here. The first analyzes prevocalization as partial vocalization of the consonant's primary stricture, and the second views it as diphthongization due to vowel assimilating to the following consonant. The partial vocalization account is applied to plain consonants in Maxakalí, and the diphthongization account to (mostly) complex consonants in the historical phonology of English, which may be partly responsible for

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their divergence. The third approach, briefly outlined in Albano (1999), utilizes the gestural framework of Browman and Goldstein and will be addressed in the next chapter.[15]

1.5.1 Prevocalization as partial vocalization of primary stricture

1.5.1.1 Andersen (1972)

This overview begins with Andersen's (1972) well-known analysis of diphthongization not for its value for the study of prevocalization as such, but rather for its importance as a precursor of later analyses of prevocalization in terms of contour creation. The particular phonological process examined by Andersen is the treatment of /ɲ/ in dialectal Polish: when in coda, it may be realized as [n̯ɲ] or [j̯j̯], with the possibility of further simplification of [n̯ɲ] to [ɲ] and of [j̯j̯] to [j], cf.:

(a) *słońce*	/swoɲtse/	[swoj̯ɲtse]	>	[swoɲtse]	'sun'
(b) *koń*	/koɲ/	[koj̯j̯]	>	[koj]	'horse'

In accordance with his general view of diphthongization "as a linear arrangement within a single segment of different distinctive-feature values" (1972:42), Andersen analyzes the process in (a) as a sequence of a primary diphthongization with respect to the feature [compact] whereby the final portion of the segment becomes [-compact] (which

[15] Optimality theory (= OT, cf. Prince and Smolensky 2004), the dominant current model of phonological theory, has so far not included prevocalization within its sphere of discussion. The theory operates with a set of hierarchically ordered constraints whose ranking differs from language to language, accounting, among other things, for context-dependent differences in a word's form. The closest thing in OT-related literature to the present concerns is Gafos' (2002) analysis of aspects of phonology of Colloquial Moroccan Arabic, couched in terms of constraints on the temporal coordination of articulatory gestures. Gafos' emphasis on the temporal dimension of speech processes makes it possible for that work to serve as a model for future OT-type analyses of prevocalization (cf. also Goldstein et al. 2006:237).

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presupposes an intermediate stage /ɲ/ → [ɲn]), and a secondary diphthongization with respect to the features [vocalic] and [consonantal] whereby the initial portion of the segment becomes [+vocalic, -consonantal] (which entails the change [ɲn] → [i̯n]). While this analysis fits the surface facts, it makes no reference to lenition entailed by the change from a strong palatal nasal to a "diphthong" both elements of which are weaker and/or much more vocalic than the original segment. Also, the actual intermediate stages of /ɲ/-prevocalization attested in the dialects indicate that changes involved in the lenition of the nasal are best analyzed in the order opposite to that proposed by Andersen, namely, as a sequence of prevocalization (/ɲ/ > [i̯ɲ]) and either vocalization or alveolarization of the consonantal component of /ɲ/ ([i̯ɲ] > [i̯i̯] or [i̯n]). While the first of these steps reduces the duration of the primary stricture, the second step further weakens the palatal stricture through gliding or alveolarization.

Subsequent to Andersen (1972) several important analyses of plain consonant prevocalization were inspired by an unusual type of synchronic allophony in Maxakalí, the only surviving member of the Maxakalí subfamily of Macro-Jê spoken in Minas Gerais, Brazil (Rodrigues 1999). Maxakalí is unusual in that under the appropriate conditions, all of its consonants with oral occlusion are subject to partial or complete vocalization. This results in a rare synchronic allophony between fully articulated, prevocalized, and fully vocalized realizations of consonants. The data of Maxakalí are of central importance to the study of prevocalization in general, and before proceeding

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to an account of the formal analyses it is first necessary to provide a brief overview of the relevant details of Maxakalí phonology.

1.5.1.2 Maxakalí

The segmental phonology of Maxakalí is mostly known through a descriptive study by Gudschinsky et al. (1970) that formed the point of reference for all subsequent analyses of Maxakalí phonology. The phonemic system of Maxakalí includes the following segments:

Consonants

p	t	c	k	ʔ h
m	n	ɲ	ŋ	

Vowels

i, ĩ			ɯ, ɯ̃
	e, ẽ	o, õ	
		a, ã	

All Maxakalí consonants, with the exception of laryngeals, can be realized as a series of allophones ranging from strong closure without prevocalization, to various degrees of prevocalization combined with various degrees of weakened closure, to syllabic vocoids without closure. Each allophone is conditioned by the position of the consonant in the syllable and the surrounding segments, and is most prominent in stressed or lengthened

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syllables (Gudschinsky et al. 1970:82). The basic allophones of oral stops occur in any syllable onset, and those of nasal stops in utterance-initial syllable onsets before nasal vowels (see Table 2). Before a nonhomorganic consonant and in absolute final position, each stop[16] is represented by an allophone that consists of a nonsyllabic prevowel[17] and the weakened stop closure.[18] Before a homorganic consonant, the stops completely lose their occlusion, and the nonsyllabic prevowel develops into a syllabic vowel of the same quality.[19]

The above picture is somewhat complicated by the development of the transition glides [j, j̃, ɣ, ŋ, w, w̃] between the nuclear vowels /i, ĩ, a, ɯ, ɯ̃, o, õ/, respectively, and the respective syllabic realizations [ə, ə̃, i, ĩ] of the alveolar stops /t, n, c, ɲ/. The development of these transitions creates an additional environment for the syllabic allophones of the coronal stops: after transition glides, they become fully syllabic not only before homorganic consonants, but in other positions as well. The full range of allophones of oral stops in Maxakalí is summarized in Table 2.

---

[16] With the possible exception of /ŋ/, which is said to prevocalize but rarely.

[17] The exact phonetic nature of the prevowels is unclear from the description in Gudschinsky et al. (1970) who limit themselves to stating that they are nonsyllabic (78 fn. 3). Clements (1991:8), using Gudschinsky et al. (1970) as his source, interprets prevowels as "extra-short unstressed" vowels. Building on this, Wetzels and Sluyters (1995:124f) hypothesize that prevocalized allophones of Maxakalí stops have the same overall duration as the basic stop allophones, suggesting that the duration of the prevowel eats into that of the occluded portion of the stop.

[18] Stops in utterance-final codas are described as having various degrees of closure weakening manifested as implosion, frication, voicing, and/or reduction of length. Specifically, /-t/ may be realized as [tˀ ~ θ ~ ɐ̯], /-p/ as [pˀ ~ ɸ ~ b], /-k/ as [kˀ ~ x ~ ɣ ~ ɯ̯], /-n/ as [n ~ n̥ ~ ø], /-ŋ/ as [ŋ ~ ŋ̊], and /-c/ as [ʃ ~ ç ~ j].

[19] Gudschinsky et al. (1970:79) observe that fully vocalized consonants differ from underlying vowels in that they cannot be stressed, and in that, impressionistically, they are "lighter" than the least stressed underlying vowels.

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Table 2. Allophones of oral and nasal stops in Maxakalí

	Basic allophones	Prevocalized allophones	Vocalic allophones	
		Before a nonhomorganic consonant or utterance-finally	After a transition glide	Before a homorganic consonant
/p/	[p]	[ɤ̯p]	—	[ɤ]
/m/	[m]	[ɤ̯̃m]	—	[ɤ̃]
/t/	[t]	[ə̯t]	[ə]	[ə]
/n/	[n]	[ə̯̃n]	[ə̃]	[ə̃]
/c/	[(t)ʃ]	[jʃ ~ jç ~ jtʲ]	[i]	[i]
/ɲ/	[ɲ ~j̃]	[j̃]	[ĩ]	[ĩ]
/k/	[k]	[ɯ̯k]	—	[ɯ]
/ŋ/	[(ŋ)g]	[ɯ̯̃ŋk]	—	[ɯ̃]

Examples of prevocalized and vocalic realizations of stops in each relevant environment are given below (Gudschinsky et al. 1970; Wetzels and Sluyters 1995).

(1) Prevocalized allophones before nonhomorganic consonants or utterance-finally:

/p/	/nɯʔcip/	[nɯʔʃiɤ̯p]	'full of'
/m/	/mĩhĩm/	[mĩhĩɤ̯̃m]	'wood'
/t/	/tapet/	[tapɛə̯t]	'paper'
/n/	/tõmãn/	[tõmãə̯̃n]	'tomato'
/c/	/-cecka/	[ʃɛjʃka]	'big'
/ɲ/	/mãʔãɲ/	[mãʔãj̃]	'alligator'
/k/	/tihik/	[tihiɯ̯k]	'man'
/ŋ/	/ɲĩmãŋ toc/	[ɲĩmãtɯ̯̃ŋktojc]	'long handle'

(2) Vocalic allophones after transition glides:

/t/	/mit/	[^{m}bijə]	'sound of jaguar's steps'

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/n/	/ɯhɯn/	[ʔɯhɯŋə̃]	'woman'
/c/	/pohoc/	[pohowi]	'arrow'
/ɲ/	/tec ŋõɲ/	[tejgõw̃i]	'storm cloud'

(3) Vocalic allophones before homorganic consonants:

/p/	/-keppa/	[kæʏpa]	'before'
/m/	/mĩm pe/	[mĩʏ̃pe]	'bed'
/t/	/mattɯk/	[mbaɔtɯx]	'toad'
/n/	/kõmãn nõʔõm/	[kõmãə̯n nõʔõm]	'another co-godmother'
/c/	/coc cecka/	[ʃoiʃæjʃkaʔ]	'big tooth'
/ɲ/	/mãʔãɲ cecka/	[mãʔãĩʃɛjka]	'large alligator'
/k/	/kɯcakkɯk/	[kɯʃaɯkɯx]	'capybara'
/ŋ/	/pɯtɯcnãŋ kɯtɯt/	[pɯtɯʏiçnãtũ̃ k-]	'old bird'

1.5.1.3 Reighard (1972) and Clements (1991)

One of the most intriguing questions raised by Maxakalí prevocalization is correspondences between the stops and their prevowels and vocalic allophones. As can be seen below, while the (pre)vocalic allophones of palatal and velar stops correspond to their places of articulation, those of labial and alveolar stops fail to do so:

[j ~ i] /c/	[ɯ̯ ~ ɯ] /k/
[j̃ ~ ĩ] /ɲ/	[ɯ̯̃ ~ ɯ̃] /ŋ/
[ə̯ ~ ə] /t/	[ʏ̯ ~ ʏ] /p/
[ə̯̃ ~ ə̃] /n/	[ʏ̯̃ ~ ʏ̃] /m/

This poses a major problem for those analyses of Maxakalí (pre)vocalization which interpret it as (partial) vocalization of the consonantal stricture. This line of analysis was initiated by Reighard (1972), the first study of Maxakalí prevocalization to appear after

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the publication of Gudschinsky et al. (1970). Working within the *SPE* framework, Reighard is able to relate palatals and velars to their prevowels and vocalic allophones on the basis of the shared values of the features [high] and [back]. However, alveolars which are (redundantly) [-high, -low, -back] should generate a [e] instead of the attested [ə]; Reighard gets around this problem by assuming that [ə] is an underlying [e] (although the reasoning behind this assumption is never made explicit, and nothing in the presentation of Maxakalí phonology in Gudschinsky et al. 1970 seems to warrant such an analysis). The remaining obstacle is to derive the vocalic allophones of labials, which are [+back, -round], while the featural composition of labials requires their prevowel and vocalic allophones to be [-back, +round], corresponding to a phonetic [ø]. In order to derive the correct (pre)vowel, Reighard introduces two redundancy rules, the first of which requires round vowels in Maxakalí to be universally [+back]:

$$\begin{matrix} V \\ [+labial] \end{matrix} \rightarrow [+back]$$

The second of Reighard's rules makes unstressed preconsonantal vowels [-labial] (and hence [-round]):

$$\begin{matrix} V \\ [-stress] \end{matrix} \rightarrow [-labial] / ___ C$$

However, as pointed out by Wetzels and Sluyters (1995:123), the second rule above is inadequate in that it fails to properly delimit the class of vocoids derived through (pre)vocalization. In the form in which the rule is stated, it requires all unstressed preconsonantal [o]s to be unrounded, a generalization that contradicts the data.

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The same objection applies to an identical rule assumed by Clements (1991:117). In addition, Clements (1991) assumes [+high] specification for labials, which complicates his analysis by an additional rule making rounded vowels redundantly nonhigh (the motivation for his decision to consider labials [+high] is left unexplained). [ə] generated by the alveolars is treated by Clements as the default vowel carrying no information as to the feature analysis of the alveolars.

Although the *SPE*-style analyses by Reighard and Clements may be observationally adequate, they are unnecessarily complicated when it comes to the treatment of labials and alveolars. Neither analysis is able to suggest a reason for the deviant behavior of these classes of segments as opposed to the palatals and velars, which also behave as a group. In addition, each analysis includes unsubstantiated assumptions about the feature composition of certain segments, and both crucially rely on a rule which does not properly separate (pre)vowels from the corresponding underlying vowels. The value of these analyses is therefore not so much in what they have accomplished, but in the problems that they have helped bring to light.

1.5.1.4 Wetzels and Sluyters (1995)

Wetzels and Sluyters (1995), the most extensive work on Maxakalí (pre)vocalization, likewise build their analysis around (partial) vocalization of consonantal stricture. The main advance of their analysis as compared to those of Reighard (1972) and Clements (1991) is their view of prevocalization as contour creation, or formation of a vocalic subsegment linked to the timing slot of the consonant. Just as the previous two analyses,

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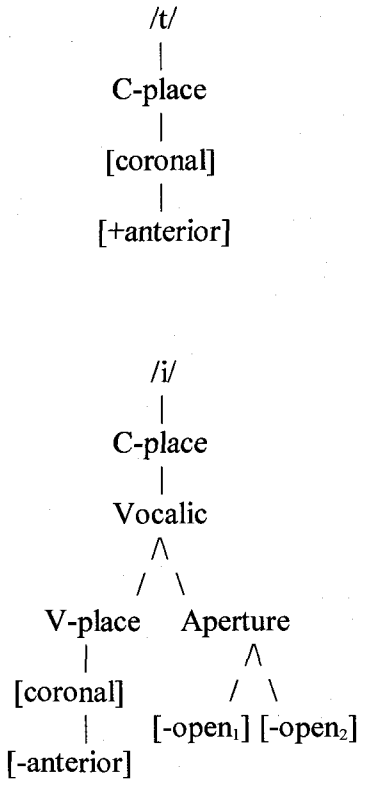
however, their solution runs into the problem inherent in this approach, namely, in how to derive the place specifications of the prevowels and vocalic allophones of the labial and alveolar stops.

Wetzels and Sluyters' analysis is carried out within the version of autosegmental phonology originally proposed in Clements (1985) and Sagey (1986) and further developed in subsequent work by Clements (e.g., 1991a and 1991b), and Clements and Hume (1995). Briefly, this model, known as feature geometry, assumes hierarchical organization of phonological primes beginning with the units of timing (CV tier), proceeding through segmental units (Root tier) and intermediate structural nodes that organize related features (Oral Cavity, C-place, Vocalic, V-place), and ending with individual feature values. Identical units are assumed to be located on their respective tiers; thus, the CV tier corresponds to subsyllabic units of timing and/or functional positions in the syllable. Clements and Hume (1995) assume a unified approach to place of articulation in consonants and vowels, which means that, depending on whether it is placed under the C- or V-place, [+labial] may define labial consonants or round vowels, [+coronal] comprises coronal consonants and front vowels, and [+dorsal] may refer to velar consonants or back vowels. The degree of openness in vowels is taken care of by means of an aperture node located underneath V-place; the number of the degrees of openness dominated by the aperture node depends on the number of height distinctions in the language. Another feature of the Clements-Hume model relevant to the present discussion is their conception of secondary vocalic features, which are formally represented as a Vocalic node under the C-place node of a consonant. This model is

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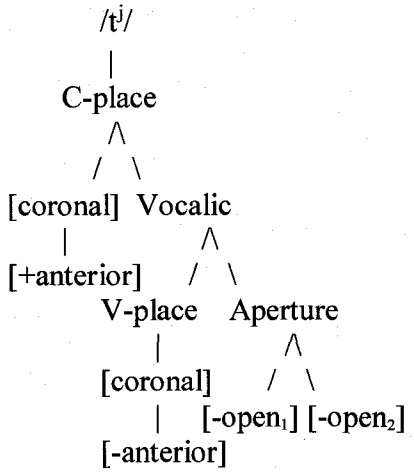
illustrated below through partial representation of the segments /t/, /i/, and /t^{j}/ (information and nodes not relevant to the discussion that follows have been omitted).[20]

Figure 1. Partial feature-geometric representations of /t/, /i/, and /t^{j}/



[20] These representations assume three degrees of height, hence the specification [-open$_1$, -open$_2$] for /i/ and the vocalic component of /t^{j}/.

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Wetzels and Sluyters (1995) adopt the same unified approach to place of articulation in consonants and vowels. Following Clements' (1991) analysis of Maxakalí, they also assume that the (pre)vowel [ə] is underlyingly unspecified for place of articulation. With these details in mind, the feature specifications for the oral segments of Maxakalí assumed by Wetzels and Sluyters are as presented in Table 3.

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Table 3. Feature specifications of Maxakalí segments

Consonants

	p	t	c	k
C-place	+labial	+coronal +anterior	+coronal -anterior	+dorsal

(Pre)vowels

	ɤ	ə	i	ɯ
V-place	+dorsal		+coronal	+dorsal
Aperture	$-open_1$ $+open_2$	$-open_1$ $+open_2$	$-open_1$ $-open_2$	$-open_1$ $-open_2$

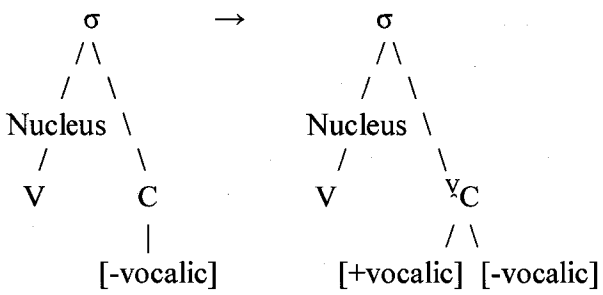
Underlying vowels

	o	e	i	ɯ	a
V-place	+labial +dorsal	+coronal	+coronal	+dorsal	+dorsal
Aperture	$-open_1$ $+open_2$	$-open_1$ $+open_2$	$-open_1$ $-open_2$	$-open_1$ $-open_2$	$+open_1$ $+open_1$

Based on the consideration that the duration of the prevocalized allophones of stops is the same as that of fully articulated stops, Wetzels and Sluyters formally represent prevocalization as the creation of a contour with opposite values of the feature [vocalic] (1995:125ff; cf. Figure 2).

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Figure 2. Formation of the vocalic root in Maxakalí stops in coda

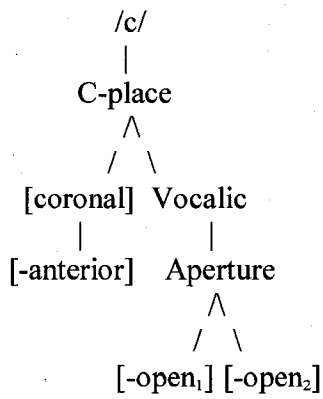


The vocalic root created as a result of the process in Figure 2 is expected to share all of its features with those of the consonant.[21] Wetzels and Sluyters derive the height of the prevowels from the (redundant) height specifications of the consonants ([+high] for palatals and velars, [-high, -low] for alveolars and labials). As shown in Figure 3, they do so by explicitly specifying the consonants in coda for height. Specification of consonants for the height features entails the presence of the Vocalic node; this constitutes a departure from the Clements and Hume (1995) model which assumes that the Vocalic node is underlyingly present only in secondarily modified consonants, as shown in Figure 2.

[21] With the exception of the features [+continuant] and [+voiced] which are universally predictable for vowels (cf. Wetzels and Sluyters 1995:125).

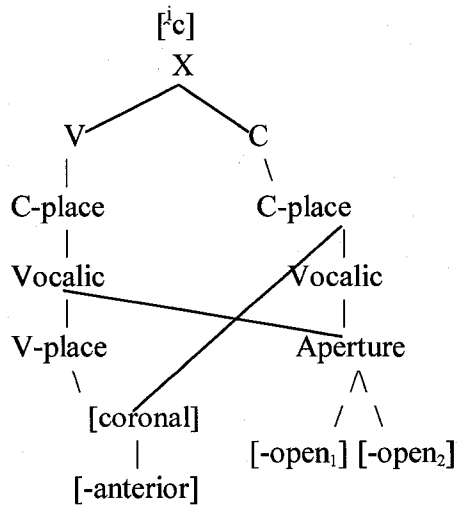
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Figure 3. Maxakalí consonants in coda



As shown in Figure 4, the height of the (pre)vowel is then straightforwardly derived from the height features of the consonant. Place of articulation of (pre)vowels corresponding to palatals and velars (i.e., [i̯] and [ɯ̯]) is also straightforwardly derived from those of the consonants.

Figure 4. Derivation of the prevowel of /c/



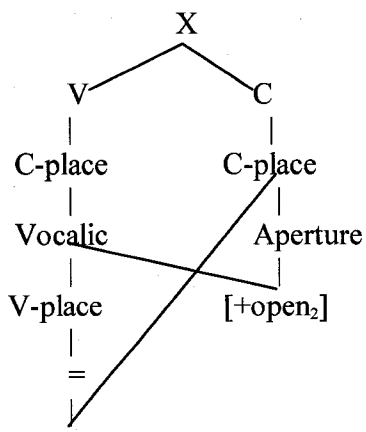
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However, if the vocalic subsegment were to inherit the place features of labials and alveolars, this would generate a [o] and [i], respectively, instead of the attested [ʏ] and [ə]. This problem is dealt with by Wetzels and Sluyters in two steps. First, they introduce a redundancy rule, equivalent to the corresponding rules of Reighard (1972) and Clements (1991) above, which requires labial vowels to be simultaneously [+dorsal]:

labial → dorsal / [+vocalic]

Second, they suggest that place features shared by the two roots of a contour segment are delinked from the vocalic root in case the segment as a whole is characterized by the feature [+open$_2$] (this is shown in Figure 5). This way the authors are able to derive correctly a mid back (pre)vowel from the labial stops and a placeless schwa from the alveolars (1995:127f).

Figure 5. Derivation of prevowels of labials and alveolars



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Wetzels and Sluyters' derivation of the (pre)vowels [ɤ] and [ə] (and also [ɤ̃] and [ə̃]), even though producing the correct results in terms of feature specifications of the (pre)vowels, is problematic. As the authors themselves note, there is no obvious reason for the existence of a rule connecting loss of place features to the height specification of a (sub)segment. This consideration forces them to conclude that this rule "captures a linguistic fact specific to Maxakalí" (Wetzels and Sluyters 1995:128). As such, this solution cannot be used as the basis for a general analysis of prevocalization, whose aim is to account for the facts in a way that would be applicable to other languages.

It will be shown in subsequent sections that the difficulty of deriving the place features of the (pre)vowels [ɤ] and [ə] is due to the inadequacy of the approach rather than to an exceptional behavior of labials and alveolars. It will be shown that, far from being an isolated phonological fact confined to a single language, failure of the prevowels to correspond to the primary place of articulation of triggering consonants not produced with the tongue body is a general characteristic of prevocalization. It will also be shown that, far from being an obstacle to a formal analysis, the "exceptional" behavior of the labials and alveolars of Maxakalí provides a path to the correct solution once the assumption that prevocalization is partial vocalization of the primary stricture is abandoned.[22]

[22] It should be noted that in his earlier analysis of Maxakalí, Wetzels rejects (pre)vocalization of primary constriction as an explanation of the Maxakalí facts (1993:55). Wetzels' earlier solution is examined in more detail in 2.3.2.2.

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Before proceeding with the analysis, it is necessary to consider briefly the vowel diphthongization solution to prevocalizations in the history of English proposed in Gussenhoven and Weijer (1990). As was explained at length in 1.3, vowel diphthongization is best seen as an accidental result rather than the source of prevocalization onglides. In the next section we will consider in more detail why the facts of English addressed in Gussenhoven and Weijer (1990) are better explained if prevocalization, rather than vowel diphthongization, is assumed to have been the underlying process.

1.5.2 Prevocalization as vowel diphthongization

Gussenhoven and Weijer (1990) analyze three processes that took place at different periods in the history of English: Old English breaking, prelabialization of [x^w] in Middle English, and pre-/r/ breaking in Early Modern English.

"Old English breaking" is the label applied to a prehistoric process dated around the 3rd-4th century A.D., which consisted in the development of a back glide between the front vowels /i, i:, e, e:, æ, æ:/ and tautosyllabic /l, r, x, w/.[23] It is often assumed that the conditioning environment of breaking involved the vocalic feature [+back] (e.g., Lass and Anderson 1975:83; Lass 1983), the elements responsible for this environment being the velar constriction of /x/, the velar component of /w/, and velarization (or pharyngealization) of /l/ and /r/ in preconsonantal allophones. The immediate result of

[23] See Luick (1921:138ff); Moore and Knott (1942:127f); Campbell (1959:54ff). Not all authorities include the vocalic developments before /w/ under the label of breaking (cf. discussion in Luick 1921:138-39 n. 3).

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breaking was the insertion of a (labio)velar glide after the vowel whose length was not affected. In Early Old English, the glide is spelled <u>, which suggests initial diphthongal shapes [i(:)u], [e(:)u] and [æ(:)u] in place of the former monophthongs. Subsequently, diphthongs resulting from breaking, in common with diphthongs from other sources, were subject to height harmony that produced [e(:)o], [e(:)o] and [æ(:)a], respectively, in West Saxon (Campbell 1959:117ff; Jones 1989:34ff), cf.:

healf	'half'
bearn	'child'
seah	'knife'
hweowol	'wheel'

Prelabialization of [x^w] took place in Middle English and consisted in the insertion of a (labio)velar glide before a [x^w] after short and long stressed back nonhigh vowels (Luick 1921:424; Mossé 1952:28; Nieuwint 1981), cf.:

Old English		Middle English
[ax]	>	[aux]
[a:x]	>	[oux]
[o(:)x]	>	[oux]

The vowel length distinction was neutralized throughout; the neutralization was in favor of a short vowel when Middle English [x^w] corresponds to Modern English [f], and in favor of a long vowel when the fricative was deleted. Long vowel outcomes are shown below for the purposes of illustration.

Old English		Middle English	
tahte	>	*taughte*	's/he taught'
da:h	>	*dough*	'dough'
dohtor	>	*douhter*	'daughter'
bo:hte	>	*bouhte*	's/he bought'

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Finally, pre-/r/ breaking took place in Early Modern English and consisted in the insertion of a schwa glide between the long nonlow vowels /i:, e:, o:, u:/ or the diphthongs /aɪ, aʊ/ and the following prepausal, preconsonantal, or prevocalic /r/. It is commonly assumed that the feature responsible for pre-/r/ breaking was the secondary pharyngeal constriction of /r/. Schwa insertion was accompanied by laxing of the preceding vowels and deletion of the rhotic when in coda; /ɔə̯/ (and in part /ʊə/) were later monophthongized to [ɔ:] (Wells 1982:213ff), cf. Table 4.

Table 4. Pre-/r/ breaking and /r/-deletion

beer	*chair*	*more*	*sure*	*fire*	*tower*
bi:r	tʃe:r	mo:r	ʃu:r	faɪr	taʊr
bi:ə̯r	tʃe:ə̯r	mo:ə̯r	ʃu:ə̯r	faɪər	taʊər
bɪə̯	tʃɛə̯	mɔə̯	ʃʊə̯	faɪə	taʊə
–	–	mɔ:	ʃɔ:	–	–

Gussenhoven and Weijer (1990) analyze all three of the above processes as assimilations of vowels to the following consonants in terms of backness, height,[24] or roundness; the assimilation is formalized as spreading of the relevant feature from the consonant to the vowel. Their analysis is couched in feature-geometric terms and pursues the narrow goal of testing the accuracy of the feature geometry model with respect to its prediction that it is possible to distinguish between two kinds of spreading processes, spreading of a single feature (shown in (a) in Figure 6 below), and spreading of the V-place node dominating the relevant feature (shown in (b)). In either case, the

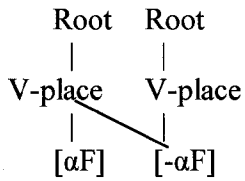
[24] The authors assume spreading of [+low] in the case of pre-/r/ breaking.

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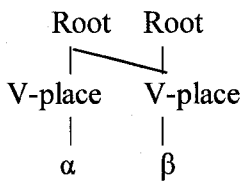
consonant-vowel interaction involves only the V-place nodes of the segments involved, which, in consonants, codify secondary articulations (velarization in the case of Old English breaking, pharyngealization in the case of pre-/r/ breaking, and rounding in the case of [x^w] prelabialization).[25]

Figure 6. Feature Spreading versus V-Place Spreading in Feature Geometry

(a) Feature Spreading



(b) V-Place Spreading



The authors argue that Old English breaking is an example of the process in (a), which creates a contour of opposite feature values on the vowel, while pre-/r/ breaking and prelabialization of [x^w] are instances of the process in (b), termed by them pre-articulation. In either case, their solution essentially involves diphthongization of the vowel preceding the consonant from which the feature is spread, whether achieved through the spreading of a single feature or that of the V-place node. Gussenhoven and

[25] In order to avoid spreading from the C-place of the consonant to the V-place of the vowel in the case of consonants devoid of secondary articulations (e.g., Old English /x/), the authors assume "feature projection", or copying of a feature from C-place to V-place within the segment (316).

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Weijer's analysis thus crucially depends on an assumed interaction between the vocalic features of consonants and preceding vowels, a theoretical position which continues the line of inquiry that originated from the earliest philological descriptions of these phenomena. Nevertheless, the view that the nature of these processes as assimilatory is unable to account for all of the relevant details as, for example, it makes no reference to the fact that all three involve consonants in weak syllable positions, fail to include stronger consonants (e.g., stops), and have as one of their major outcomes decomposition of the triggering consonants. These considerations make it more likely that rather than being motivated by vowel-consonant assimilation all three processes are driven by consonant lenition, as suggested in much recent literature including Jones (1989) and Broadbent (1999).

Another consideration suggesting that these processes are not analyzable as vowel assimilations is the fact that (nearly) identical processes are repeated in the history of English in contexts in which assimilation is ruled out by the nature of the vowel-consonant pairs involved. For example, prelabialization of /x/ [x^w] after stressed back vowels in Middle English is matched by prepalatalization of /x/ ([x^j] or [ç]) after stressed [e(:)] during the same period (Luick 1921:424; Mossé 1952:28), cf.:

Old English		Middle English	
[eç]	>	[eiç]	
seh	>	*seigh*	's/he saw'
[e:ç]	>	[eiç] > [īç]	
ne:h	>	*neigh*	'near'

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Here, a mid front vowel acquires a palatal offglide in the context of a following [-back] segment, which excludes assimilation as a possible cause. Similarly, in late Old English there is a breaking-type diphthongization before the voiced palatal fricative [j] (orthographically <g>), which affects front vowels and for which assimilation would not be the most natural explanation, either (Jones 1989:55), cf.:

West Saxon Old English	10th-century Northumbrian Old English
<frægn> 's/he asked'	<fraign> [fræijn]
<hæg> 'meadow'	<heig> [heij]

Instances of breaking-type diphthongizations abound in historical English phonology, and their combined evidence suggests that they require a combination of a stressed vowel and a following (typically) tautosyllabic fricative, liquid, nasal or glide in order to take place. As shown by Jones (1989), consonants triggering breaking-type diphthongizations are all relatively weak: liquids, glides, nasals and fricatives are all located near the vowel end of the sonority scale and are relatively vowel-like in their acoustic and articulatory makeup. An analysis that simply spreads [+dorsal] from a consonant to preceding [-dorsal] vowels (which Gussenhoven and Weijer assume in the case of Old English breaking) makes no reference to this fact as it is unable to account for the lack of such spreading when the consonant is a velar stop [k] or [g]. What the processes addressed in Gussenhoven and Weijer (1990) have in common is their double outcome which consists in strengthening of a stressed, hence already strong, vowel through diphthongization, and concomitant weakening of an already weak consonant through decomposition and possibly deletion. Therefore, the moving force behind Old English breaking and other breaking-type processes is best understood as a combination

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of these two factors: strengthening of the vowel in a strong position and weakening of the consonant in a weak position. Of course, strengthening of a stressed vowel can be achieved by means other than diphthongization, for example, through vowel lengthening or vowel epenthesis. As a result of one of these processes (i.e., prevocalization, vowel lengthening, or vowel epenthesis) the vowel space preceding the triggering consonant is strengthened by assimilating a new vocalic element. In prevocalization, the vocalic element added to the stressed nucleus takes the form of a prevowel with the properties of the vocalic component of the consonant, in vowel lengthening it is identical to the vowel, and in vowel epenthesis it may surface as a copy of an adjacent vowel or a schwa. In many instances prevocalization, vowel lengthening and/or vowel epenthesis exist in complementary distribution; this complementarity is amply attested, e.g., is the history of English,[26] and is addressed elsewhere in this work (see 5.4).

1.6 Summary and conclusion

This chapter has introduced the phonological phenomenon of prevocalization, and provided phonetic and phonological evidence for it being distinct from vowel diphthongization and a descriptive account of the most essential synchronic and diachronic aspects of this process. It also showed that prevocalization may affect both simple and secondarily modified consonants, and outlined the major features of

[26] Not surprisingly, the phonological framework where the complementary relationship between these processes has received most attention is dependency phonology, which was developed primarily with English historical phonology in view (Anderson and Jones 1977; Anderson and Ewen 1987; Jones 1989).

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prevocalization of each of the two consonant classes. A close examination of previous theoretical work has shown that the approach to prevocalization that views it as partial vocalization of primary stricture is unable to correctly generate prevowels triggered by consonants not produced with the tongue body. The approach that treats prevocalization as vowel-consonant assimilation, on the other hand, is unable to explain why this process applies to weak consonants, preferentially in coda, while failing to target stronger consonants and syllable positions. In the next chapter it will be shown that a gestural account of prevocalization is better able to cope with the areas which are problematic for the approaches reviewed here.

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2. Prevocalization in articulatory phonology

2.1 Introduction

It was shown in the preceding chapter that the approaches to prevocalization treating it as partial vocalization of the consonants' primary stricture encounter difficulties when it comes to prevowels that differ in place of articulation from the triggering consonants. In addition, the generative treatment cannot handle the subsegmental nature of the prevowels and is forced to represent prevocalization as the insertion of phoneme-sized units. The feature-geometric approach of Wetzels and Sluyters (1995), where the prevocalized allophones of consonants are represented as internally complex, or contour, segments, can handle the subsegmental character of the prevowels; however, derivation of place of articulation of prevowels generated by consonants not produced with the tongue body is problematic.

The main claim of this chapter is that prevocalization involves surfacing of the secondary (or vocalic) component of consonants rather than simple vocalization of their primary (or consonantal) stricture, as assumed by most previous analyses.[27] To express this claim in precise terms, I use the gestural model of Browman and Goldstein (1986 et seq.). The main motivation for adopting this model is the fact that it is in general well suited to account for low-level phonological processes, and the fact that the analysis of prevocalization proposed here relies primarily on articulatory considerations. The gestural model crucially differs from most other models of phonology in that it incorporates the temporal dimension in its analytical apparatus; by assuming that the

[27] With the exception of Wetzels (1993), for which see section 2.3.2.3.

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fundamental units of phonological organization are the dynamic gestures rather than the static segments or features, it is not only able to provide a unified view of prevocalization across all classes of consonants, but also to account for the subsegmental character and place of articulation of the prevowels, and to offer a principled explanation for the surface order of the subelements of a prevocalized consonant.

The discussion is organized as follows. After a brief summary of the key assumptions and representational conventions of articulatory phonology, section 2.2 argues that the key evidence for the internal mechanism of prevocalization is provided by the behavior of secondarily modified consonants. Section 2.3 constitutes the main body of the chapter. It defines the notion of vocalic and consonantal gestures (V- and C-gestures) discussed in earlier literature in the context of liquids, and argues that prevocalization of complex liquids is the surface result of a prosodically driven retiming of their V- and C-gestures which may be accompanied by simultaneous shortening of the C-gestures. This analysis is then extended to prevocalization of secondarily modified consonants in general, as well as that of plain consonants. A separate discussion establishes the existence of a vocalic component in plain consonants (their V-gesture), defined as the position of the tongue body (and, where applicable, lips) accompanying the production of the consonant. Specifically, it is argued that while for tongue body consonants the position of the body of the tongue is determined by the consonant's primary constriction, for (at least some) plain labial and tongue tip consonants the position of the tongue body has a definite target that needs to be

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specified on a language-specific basis. Section 2.4 provides a brief summary of the main points.

2.2 Gestures in articulatory phonology

The research program known as articulatory phonology was put forth for the first time in Browman and Goldstein (1986) and gradually refined in later publications by the same researchers (1988, 1989, 1990, 1992, 1995, 2000); selected aspects of the model have also been discussed and developed elsewhere (e.g., Byrd 1996; Gafos 1999, 2002; Albano 2001; Goldstein et al. 2006). The fundamental assumption of this approach is the view of linguistic structures in terms of coordinated articulatory movements, called gestures, which refer to the formation of constrictions somewhere in the vocal tract. Each constriction is specified as to its location and degree by a set of parameters called vocal tract variables. Oral gestures consist of paired tract variables referring to the location of the constriction in the vocal tract, and the degree of the constriction. For example, a gesture articulated by the tip of the tongue is described by the values of the constriction location (CL) and the constriction degree (CD) of the articulator (in this case, TT = tongue tip). CL specifies the place of articulation and may take on the values [labial], [dental], [alveolar], [postalveolar], [palatal], [velar], [uvular],[28] and [pharyngeal]. The CD variable makes the categorical distinctions [closed] (assigned to stops), [critical] (assigned to fricatives), and [open] (assigned to vowels and approximants); in order to encode height distinctions in vowels, [open] may be further

[28] But cf. Goldstein's (1994:238) observation that uvular fricatives may require both tongue body and tongue root constrictions.

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subdivided into [mid], [narrow], and [wide] (cf. Browman and Goldstein 1986 et seq.; Gafos 1999:12f; Albano 2001:54f). The glottal (GLO) and velic (VEL) gestures are specified by the constriction degree alone, given that the notion of constriction location does not apply to these articulators (see Table 5).

Table 5. Vocal tract variables

Articulators	Tract variables
Lips, jaw	LP (lip protrusion), LA (lip aperture)
Tongue tip, tongue body, jaw	TTCL (tongue tip constriction location), TTCD (tongue tip constriction degree)
Tongue body, jaw	TBCL (tongue body constriction location), TBCD (tongue body constriction degree)
Tongue root, epiglottis[29]	TRCL (tongue root constriction location), TRCD (tongue root constriction degree)
Velum	VEL (velic aperture)
Glottis	GLO (glottal aperture)

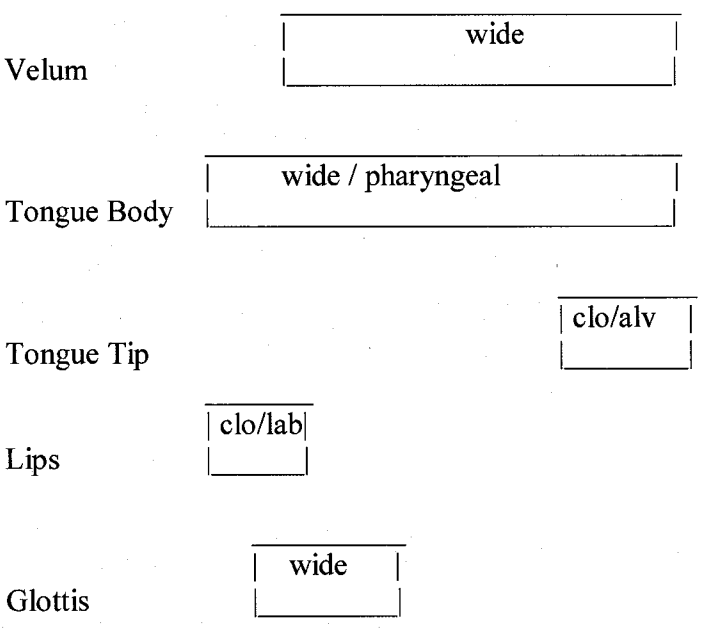
In order to represent the temporal extent of gestures and their coordination with one another, Browman and Goldstein have introduced two-dimensional diagrams, so-called gestural scores, in which gestures are shown as boxes located on separate tiers corresponding to the different articulators. In these diagrams, the time dimension, or periods of activation of the individual articulators, is shown by the length of the boxes. A schematic gestural score in Figure 7 (based on Browman and Goldstein 1992:158), which represents the sequence [p^hæ̃n] 'pan', exemplifies this notation. In this gestural score, the vowel gesture executed by the tongue body begins simultaneously with the lip closure gesture of the labial stop. The velum opening gesture associated with the nasal

[29] This articulator set is posited, e.g., in Browman and Goldstein (1989) and Goldstein (1994).

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precedes its closure gesture, and overlaps considerably with the preceding vowel gesture bringing about allophonic nasalization of the vowel. The opening of the glottis is coordinated with the lip closure gesture in such a way as to account both for the lack of voicing during the labial closure and for the aspiration at its release.

Figure 7. Gestural score for the sequence [p^hæ̃n] 'pan'

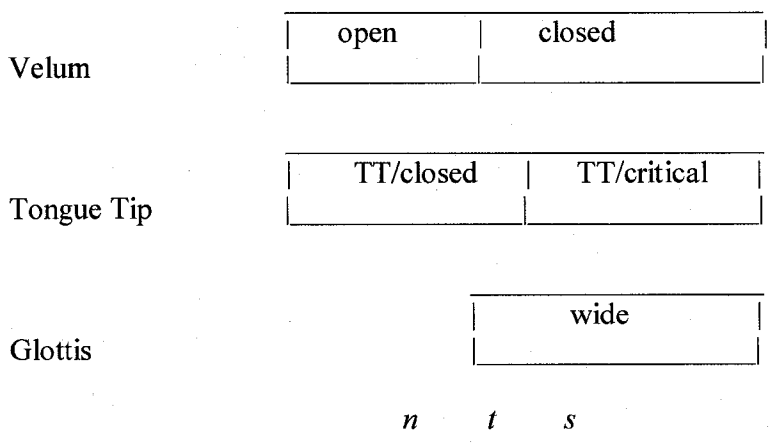


An important area of application of articulatory phonology has been those processes in which the dynamic character of gestures provides a clear descriptive and analytical advantage over approaches operating with static features and/or segments. Among such processes are allophonic variation, consonant-vowel coarticulation, and various fast-speech phenomena like consonant deletion, insertion, and assimilation. In articulatory phonology all such phenomena are analyzed by means of two processes, reduction in

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the magnitude of the gestures (resulting in smaller or shorter gestures), and temporal overlap between gestures (cf. Browman and Goldstein 1990:369; Byrd 1996:141f). The model does not allow deletion, modification, or insertion of gestures that are not already present in the gestural score (Browman and Goldstein 1992:173). Browman and Goldstein's approach is especially well suited to account for low-level phonological processes and may be exemplified by means of Figure 8 below representing a possible analysis of the "insertion" of intrusive stops between nasals and fricatives in English (cf. Albano 2001:58). As shown in Figure 8, the percept of an intrusive voiceless alveolar stop between the nasal and the fricative in *prince* [prĩnts] results directly from temporal overlap between the tongue tip closure gesture of [n] and two gestures associated with the following fricative, the closure gesture of the velum and the opening gesture of the glottis.

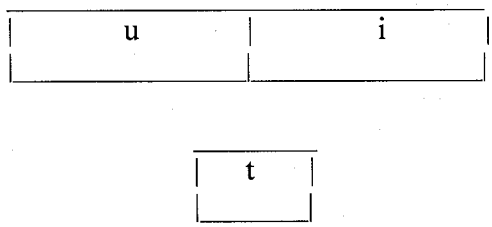
Figure 8. Partial gestural score for the sequence [nts] of [prĩnts] 'prince'



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Another element of the articulatory phonology model which is especially relevant in the context of prevocalization is the separation of articulations into consonantal and vocalic, with a distinct functional tier corresponding to each articulation type (cf. Browman and Goldstein 1990:351ff). Beginning with the now classic study by Öhman (1966), vowel articulations have been conceptualized as articulatorily contiguous and serving as a background to consonantal articulations, which are seen as superimposed on the continuous vowel articulations (cf. references in Coleman 1992:34). Browman and Goldstein (1990) capture this notion in their model by separating articulations into C- and V-tiers; the difference between theirs and other frameworks which make a similar distinction[30] is that the units of representation are gestures. Tier separation and articulatory contiguity of vowel articulations may be schematically captured as in Figure 7, which contains a gestural score for a hypothetical VCV utterance [uti] (cf. Gafos 1999).

Figure 9. Schematic representation of the sequence [uti]



[30] E.g., the CV phonology of Clements and Keyser (1983).

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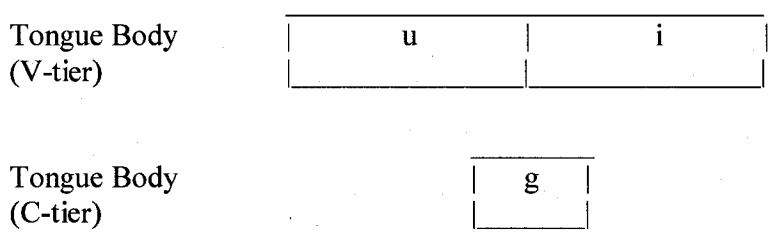
Separation of articulatory gestures into functional tiers is directly relevant to an analysis of prevocalization since the V-tier may be used to represent not only discrete vocalic articulations, as in the sequence [uti] above, but also the vocalic articulations associated with the secondary features of consonants. Such a representation is consistent with the view of secondary articulations as vowel-like articulations superimposed on the consonants' primary constriction (e.g., Ladefoged 1993:230); one famous antecedent for a unified treatment of the two articulation types in phonology is Chomsky and Halle's (1968) description of secondary articulations with a set of articulatory parameters traditionally reserved for vowels. Representing secondary articulations on the V-tier recognizes their contiguity with segmental vowels, which is consistent not only with their phonetic implementation but also with their phonological behavior. For example, in languages like English or Swedish tongue body consonants do not interfere with vowel-to-vowel coarticulation in VCV sequences; but in Russian, consonants which include contrastive tongue body coarticulations (palatalization or velarization) block it (cf. Öhman 1966). Keating (1985) models the coarticulatory resistance of Russian consonants in VCV sequences by placing their tongue body coarticulations on the vowel tier. Using the representational apparatus of articulatory phonology, this difference in the behavior of tongue body consonants in English and Swedish, on the one hand, and secondarily modified consonants in Russian, on the other, may be schematically captured as in Figure 10. As shown in (a), the vowel articulations are contiguous and free to interact given that the tongue body articulation of the consonant is located on a separate tier. In (b), the contiguity, and hence the possibility of

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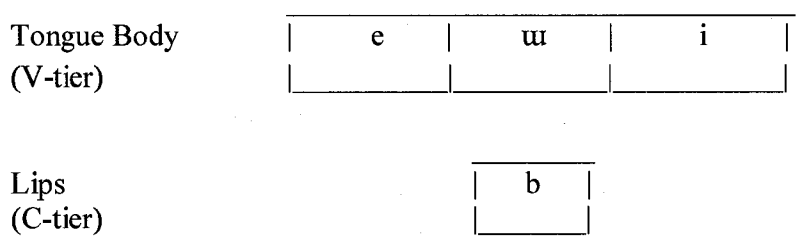
interaction, of the segmental vowel articulations is interrupted by the velarization gesture of a plain (hence, phonetically velarized) /b/ located on the same tier.[31]

Figure 10. V-to-V coarticulation in English/Swedish and Russian

(a) V-to-V coarticulation is present: sequence /ugi/ [ug^{j}i] in English or Swedish



(b) V-to-V coarticulation is absent: sequence /ebi/ [ɛb^{ɯ}i] in Russian



The possibility of representations similar to the ones in (b) above is important for an analysis of prevocalization in a number of respects. First, as will be established below, prevocalization crucially involves the vocalic rather than the consonantal component of consonants, and the availability of a representational device which allows separating the

[31] Although the segment has no independent status within the model of articulatory phonology, gestures may co-occur in segment-sized slots (cf. McMahon and Foulkes 1994-95).

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vocalic features from the consonantal within the traditional consonant-level units is essential for a proper analysis of this process. Second, separation of articulations into the vocalic and consonantal tiers allows two tongue body (and two labial) tiers in the representation, one corresponding to vowels and secondary articulations (V-tier), and another to tongue body and labial consonants (C-tier). This separation into the V- and C-tiers affords an easy way of representing both distinctive and nondistinctive consonantal coarticulations, and to distinguish, e.g., between plain labials and velars, on the one hand, and labialized labials and velarized velars, on the other. The possibility of making such distinctions is essential as it may help to account for potential differences in the prevocalization patterns of these segment groups. These features of the representation also allow for a uniform account of prevocalization across all consonant types, including tongue body consonants, plain consonants not produced with the tongue body, and secondarily modified consonants. Finally, the representation in (b) also directly accounts for interaction between the vocalic components of consonants and preceding vowels, implemented on the surface as vowel diphthongization or lengthening and representing the single most obvious surface manifestation of prevocalization.

Before proceeding with the gestural analysis of prevocalization, it is first necessary to substantiate the claim that this process involves the vocalic rather than the consonantal component of consonants. The clearest evidence for the correctness of this assumption emerges from prevocalization of secondarily modified consonants, in which the consonantal and the vocalic components as a rule can be easily teased apart. In

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languages discussed up to this point, prevocalization has targeted plain consonants; nevertheless, it is with the secondarily modified consonants that this process is reported most frequently.[32] As an example of a language with productive prevocalization of this type we may cite Modern Irish, which is especially rich in consonantal on- and offglides. This feature is so characteristic of the contemporary language Irish that most dialect grammars include a chapter on "transitional glides" or otherwise incorporate a description of consonantal on- and offgliding in their phonology sections. The transitional glides in Modern Irish accompany the phonemic opposition between plain and palatalized consonants. Leaving aside the precise surface realization of the palatalization contrast, palatalized (slender) consonants in dialect after dialect are said to trigger the appearance of front on- and offglides, and plain (broad) consonants trigger the appearance of central to back on- and offglides. The most relevant characteristic of the transitional glides in the present context is the fact that their frontness or backness is independent of the primary stricture of the consonant and is determined exclusively by the quality of the secondary articulation. Thus, Erris Irish attests [ə̯] and [j] onglides from vowels to consonants: [ə̯] occurs between /i:, e:/ and following velar or velarized consonants and between back vowels or the diphthong /əu/ and a following velarized /r/,

[32] The reasons cited for the prevocalization of secondarily modified consonants differ from language to language. In some cases, they lie in language-specific details of the realization of secondary articulations: for example, the phonemic palatalization in Estonian and other closely related Finno-Ugric languages is realized primarily as an onglide to the palatalized consonant. Another reason may be the tendency towards the loss of the secondary articulations: for instance, massive depalatalization during the Gallo-Roman period of French was accompanied by a productive forward segmentalization of the onglide to the palatalized consonants. In languages without secondary articulation contrasts, prevocalization of consonants with secondary articulations, such as liquids, may signal consonant weakening.

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and [j] occurs between back vowels or the diphthong /əu/ and following palatalized consonants, cf. [dri:ˠxdə] *draoidheacht*, [bʲi:ˠf] *bíodh*, [ku:ˠr] *cubhar*, [su:ʲlʲ] *súil*, [mʲəuʲrʲ] *meabhair* (Mhac An Fhailigh 1980:45ff). In West Muskerry Irish, the onglide [ə̯] emerges after long front vowels and diphthongs ending in [i] and velarized labials, velars and dentals, cf. [bʲi:ˠn] *bíonn*, [ki:ˠrə] *caora*, [ne:ˠfə] *naomhtha*, [fəiˠb] *fadhb*, [be:ˠxəs] *buidheachas*, [ʃi:ˠl] *síol*; [j] in the same dialect emerges after back vowels and diphthongs ending in [u] and following palatalized consonants, cf. [u:ʲlʲ] *ubhaill*, [ʃauʲkʲ] *seabhaic*, [ko:ʲrʲ] *cóir* (Ó Cuív 1975:52f).

The behavior of secondarily modified consonants in Irish and other languages with contrastive coarticulations sheds important light on the mechanism of prevocalization since here the relationship between the prevocalization onglides and the consonants' vocalic component is at its clearest: prevocalization onglides generated by secondarily modified consonants always have the properties of the secondary articulation, and never those of the primary articulation, of the consonant. To illustrate this with a schematic example, while one may find instances of a [pʲ] developing a prevocalization onglide of the [j] quality (i.e., $p^j \rightarrow jp^j$), there are no instances of it developing a prevocalization onglide of the [w] quality (i.e., $p^j \rightarrow wp^j$). If prevocalization could indeed be analyzed as partial vocalization of the consonant's primary stricture (as assumed in some earlier analyses discussed in 1.5), the latter, but not the former, would be the expected outcome. The absence of such outcomes in the actual data is a clear indication that in

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analyzing prevocalization the focus needs to be shifted from the consonants' primary stricture to their secondary, or vocalic, component.

If the behavior of secondarily modified consonants may be taken as indicative of the mechanism of prevocalization as a whole, it appears that this process is best analyzed as the articulatory anticipation of the vocalic or secondary articulation gesture of the consonant with respect to its consonantal or primary articulation gesture. Before considering the possibility of such an approach to prevocalization in general, however, it is first necessary to examine more closely the notion of the vocalic and consonantal gestures, which up to this point have been used in a more or less impressionistic manner. It is also necessary to investigate the motivation behind the changes in the mutual timing of the component gestures of secondarily modified consonants that result in surface prevocalization. There are at present few studies which address these questions in detail; nevertheless, a fair amount of data relevant to both questions may be gleaned from experimental investigations of the coordination of intrasegmental gestures in liquids. Most studies addressing intrasegmental gesture coordination have been carried out using the liquids of American English, but in some cases additional languages have been investigated as well. The results of these studies that are relevant to our analysis of prevocalization are discussed in the following section.

2.3 V- and C-gestures

This section begins by examining the notion of the vocalic and consonantal gestures (V- and C-gestures) of consonants. It is then argued that, regardless of their place of

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articulation or status as a plain or secondarily modified consonant, all consonant-sized units[33] include both types of gestures. In secondarily modified consonants, the V-gesture uncontroversially corresponds to the consonant's secondary articulation, while in plain consonants the V-gesture coincides with the primary constriction in tongue body consonants and takes on a range of values in labial and tongue tip consonants. It is further argued that prevocalization is the surface result of a prosodically driven retiming of the constituent gestures of consonants whereby the V-gesture tends to be executed closer to the preceding nucleus while the C-gesture stays in the syllable margin and may be subject to shortening. The original impetus for this approach to prevocalization has been provided by the behavior of syllable-final liquids, in American English and other languages, which is reviewed in detail immediately below.

2.3.1 Complex consonants

The liquids of American English are complex segments in that they include a secondary tongue dorsum articulation supplementing their primary tongue tip constriction. Long before the existence of a tongue dorsum constriction in liquids received experimental confirmation, the connection between the vocalic coarticulations of liquids and back vowels, in English and other languages, has been known from the facts of synchronic and historical phonology. For example, a connection between the vocalic coarticulation of English /r/ and a schwa emerges from the schwa realization of /r/ in nonrhotic dialects, while that of dark /l/ and a back vowel is clear from the results of /l/-

[33] With the exception of laryngeals (see 4.3).

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vocalization, realized as a back glide somewhere in the [o̯] ~ [ɣ̯] ~ [ɔ̯] ~ [u̯] ~ [ɯ̯] range. Until the experimental study by Gick et al. (2002), however, it was unclear whether the tongue dorsum gestures of liquids in American English were to be equated with the vocalic gestures of any specific vowels. Gick et al. (2002) have shown that, at least for some dialects of American English, the tongue dorsum gestures of /l/ and /r/ may be definitely identified with those of the vowels /ə/ and /ɔ/, respectively.

In an earlier study, Sproat and Fujimura (1993) investigated the mutual timing of the component gestures of /l/ in different syllable positions. In the context of their study, they suggested an important distinction between the component gestures of /l/ by categorizing its tongue tip gesture as consonantal and its tongue dorsum gesture as vocalic. With reference to /l/, the terms "vocalic" and "consonantal" referred to the degree and manner of constriction of each gesture. More generally, however, consonantal (or C-gestures, using a broader and more handy term introduced in Gick 2003) were defined by Sproat and Fujimura as "those that produce an extreme obstruction in the mid-sagittal plane", and vocalic gestures (or V-gestures, cf. Gick 2003) as "those gestures that do not produce an extreme obstruction; furthermore vocalic gestures may actually involve the opening of a channel as in the case with velum lowering" (1993:304). In this way, the original definition of the V-gestures comprised both vocalic articulations, such as the tongue dorsum gesture of /l/, and nonvocalic articulations, such as lowering of the velum.

The reason for adopting such a broad definition of the V-gestures was to bring out the parallelism in the timing relationship between the component gestures of /l/ and that

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obtaining between the labial closure and the velum lowering gestures of /m/ discovered earlier by Krakow (1989). Sproat and Fujimura's experimental investigation showed that pre- and postvocalic allophones of American English /l/ systematically differ in the timing of their component gestures. Specifically, in prevocalic allophones the tongue dorsum gesture is simultaneous with or slightly follows the tongue tip gesture, and in postvocalic allophones it significantly precedes it (cf. X-ray data in Browman and Goldstein 1995:23).[34] Sproat and Fujimura's (1993) results on the intragestural timing of American English /l/ were replicated by Browman and Goldstein (1995) and Gick (2003); the latter study also includes an investigation of /l/ in intervocalic contexts. American English /l/ is not unique in showing these intragestural timing effects: for example, Gick et al. (2001) show similar effects for /l/ in Mandarin, Korean, and Quebec French; observations in Bondarko (1967) seem to point to a similar effect in Russian, while those in Lowman (1932) clearly indicate the same effect in Albanian. Gick et al. (2006:63) show that Beijing Mandarin /r/, which is similar to American English /r/ in being composed of a consonantal tongue anterior raising gesture and a vocalic tongue root backing gesture, likewise shows a positive lag between its component gestures in postvocalic allophones.

[34] The articulatory difference between pre- and postvocalic /l/ is accompanied by acoustic differences. While the distinction is only allophonic in RP and other varieties of English, diachronic evidence from other language areas indicates that it may become phonemicized. This kind of development took place, e.g., in different Romance languages; according to an elegant summary by Nörrenberg (1932:298 fn. 30): "Doch wird *l* in vielen Sprachen vor Vokal heller gesprochen als nach Vokal; vgl. die verschiedene Aussprache in engl. *let* u. *tell* und die Entwicklung von ital. *piano – altro*, franz. *plain – autre*, span. *llano – otro* < lat. *planum – alterum*."

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The above findings indicate that, at least in some languages,[35] the constituent gestures of liquids in weak syllable positions tend to be executed sequentially rather than simultaneously. The linear order of the vocalic and consonantal gestures is dictated by the syllable position of the liquid: when in a (typically, post-stressed) coda, the wider of the two gestures (= V-gesture) tends to move closer to the (strong) nucleus, while the narrower of the two gestures (= C-gesture) stays in the syllable margin. Another fundamental fact that needs to be mentioned in the present context is that in weak syllable positions generally associated with prevocalization consonants independently attest weakening of their closure, or in other words, their C-gesture. This is indicated, e.g., by the data in Browman and Goldstein (1995) who report that the closure gesture in the oral consonants of English is subject to word-final weakening; while the weakening is most perceptible in coronals (the specific cases reported in that study are /l/, /n/, and /t/), it is also attested in labials and velars.[36] The weakening of the C-gestures may be effected as reduction of their magnitude (i.e., tightness of stricture), or their duration. The latter fact is especially relevant here because various sources indicate that prevocalizing consonants frequently shorten their C-gesture. Shortening of C-gestures in the context of prevocalization is indicated, e.g., by prevocalization of geminates, that become degeminated as a result, and by that of palatal stops, which reduce their duration through alveolarization of the primary stricture. Temporal

[35] Experimental evidence suggests that such gesture coordination patterns are language-specific: for example, the data from French and languages of Australia indicates that in those languages the velum lowering and oral closure gestures of nasals in coda are apparently coordinated synchronously (Goldstein et al. 2006:236).

[36] For additional data on syllable-final weakening see Krakow (1999).

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reduction of consonantal stricture has also been discussed in the context of prevocalization of Maxakalí stops by Wetzels and Sluyters (1995:125), while the spectrogram of a prevocalized /S/ in Brazilian Portuguese reproduced in Albano (1999) clearly shows a significant temporal reduction of the closure phase of the sibilant.[37] Combining these two independently established facts, it is easy to see that if the observed retiming of the constituent gestures of a liquid is coupled with simultaneous shortening of the C-gesture, the lag between the gestures may become sufficiently long to result in the percept of an onglide to the liquid.[38]

From the above discussion surface prevocalization of liquids seems to emerge as the result of two prosodically motivated processes, retiming of the constituent gestures of liquids, and simultaneous shortening of the C-gesture. The V-gesture of the consonant shifts closer to the (typically, stressed) nucleus and strengthens it by increasing its overall duration, while the C-gesture stays in the syllable margin and may be subject to reduction in magnitude or duration. At this point in time, there comes into being an allophone of the liquid which may be described as prevocalized. From a synchronic point of view, prevocalized allophones always seem to alternate with fully articulated and/or fully vocalized allophones, which contributes to their existence being frequently unperceived. In addition, they are synchronically unstable and tend to be replaced by less ambiguous articulations, such as fully vocalized allophones that come into being

[37] It is conceivable that prevocalization-related shortening of the consonant's C-gesture may be motivated by additional factors, such as the tendency to keep constant the duration of the rime. Thus, Gick and Wilson (2001, 2006) report that the duration of rimes of the [Vjl] type in English remains constant despite the "insertion" of an "excrescent schwa".

[38] Ohala (1992) cites 60 milliseconds or more as the length of such onglides to laterals (333).

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when the weakened C-gesture is completely erased, or at least reinterpreted as such by the listeners. Frequently the two-step process that has led up to the resulting situation, namely, prevocalization and deletion of the liquid's C-gesture, remains unrecorded and the surface result is analyzed as straight "vocalization". Despite the general unavailability of historical records indicative of prevocalization, however, in some cases we are fortunate enough to possess early materials in which the prevocalized allophones are recorded in writing and/or remarked upon by the contemporary grammarians. As an example of diachronic evidence of this kind may be cited such forms as *faulse* 'false', *waulken* 'walk' and *taulken* 'talk', widely attested in fifteenth- and sixteenth-century English and generally assumed to reflect a type of pronunciation current at the time of writing (cf. Jespersen 1922:289ff; Ritt 1997:535). This interpretation is based to a large extent on the testimony of contemporary grammarians: Salesbury (1567), for example, provides the following commentary: "For in these wordes and such other in Englyshe, domage, heritage, language, ashe, lashe, watch, calme, call, *a* is thought to decline toward the sound of these diphthonges *ai*, *au*, and the wordes to be read in thys wyse, domaige, heritaige, languaige, aishe, waitche, caul, caulme" (quoted in Parry-Williams 1923:210).

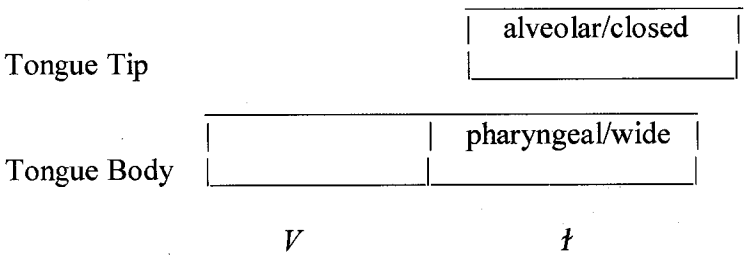
The above account of liquid prevocalization may be schematically captured by means of the gestural scores in Figure 11 which show the articulatory events leading to the percept of an onglide to, and vocalization of, a dark [ɫ]. The phonetic quality of the onglide, shown here as [ə̯], is language- or dialect-specific and determined by the phonetic quality of the coarticulation. The figure in (a) shows mutual sliding of the

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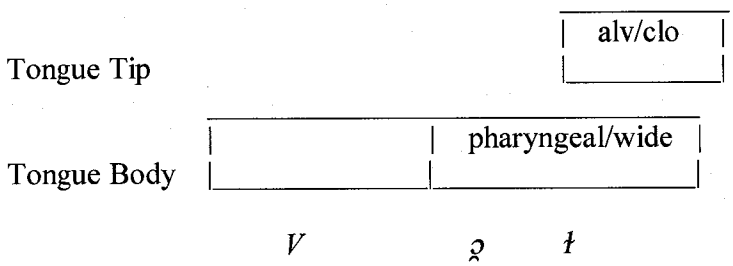
constituent gestures of [ɬ], but as yet no reduction of the liquid's C-gesture. In (b), mutual realignment of the gestures of [ɬ] is accompanied by shortening of the C-gesture; this stage corresponds to the existence of a prevocalized allophone. In (c), the C-gesture of [ɬ] is no longer recoverable synchronically, which corresponds to full vocalization of the liquid.

Figure 11. Dark [ɬ] (pre)vocalization

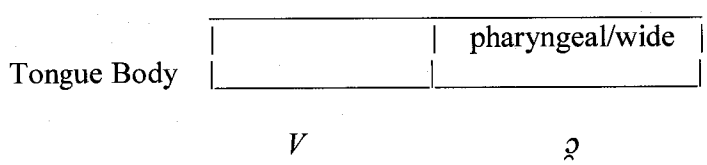
(a) Fully articulated [ɬ]



(b) Prevocalized [ɬ]



(c) Vocalized [ɬ]



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The location and degree of constriction of the vowel preceding the liquid in Figure 11 have been left blank on purpose. If the vowel is articulatorily removed from the V-gesture of [ɬ], the latter will be perceived as a diphthongal offglide, and if the two are articulatorily close, the onglide will be interpreted as increased duration of the nucleus. In the case of extreme articulatory incompatibility between the vowel and the V-gesture of [ɬ], the onglide may surface as a compromise, as an articulatory overlap between the two gestures. This seems to be the case of the so-called excrescent schwa, or a schwa-quality onglide to [ɬ] that surfaces in rimes of the shape [iɬ] or [Vjɬ] in American (and other varieties of) English (cf. Gick and Wilson 2006). There, the overlap between the narrow palatal constriction of the [i] or [j] and the wide pharyngeal constriction of the V-gesture of [ɬ] seems to result in an articulatory compromise between the constriction locations and degrees of the conflicting positions of the tongue body.

The choice of liquids to introduce the above analysis of prevocalization has been dictated by the availability of recent work on the syllable-based effects on their production. Nevertheless, there is much evidence to indicate that the above scenario represents the general mechanism of prevocalization and, as such, is applicable to all classes of consonants. The evidence includes the fact that, in common with liquids, other consonants also tend to prevocalize in weak syllable positions and typically after a stressed nucleus; the fact that their prevocalization likewise may be shown to result from retiming of segment-internal V- and C-gestures; and the fact that, in addition to liquids, consonants most frequently affected by prevocalization are fricatives, palatals,

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and secondarily modified consonants, in other words, consonants whose vocalic component is especially prominent.

It is natural to consider secondarily modified consonants first, as in their case the concept of V-gestures is especially transparent. More precisely, velarization is usually associated with a [ɯ]-like, palatalization with a [i]-like, pharyngealization with a [ɑ]-like, and labialization with a [u]-like tongue body position, even though it is well known that the implementation of secondary articulation gestures admits of a great deal of individual and dialectal variation, e.g., due to prosodic and/or segmental context (cf. Kelly and Local 1989; Bauer 2001). The analysis of prevocalization as forward sliding of the consonant's V-gesture predicts that velarized consonants will produce [ɯ]-like, palatalized consonants [i]-like, pharyngealized consonants [ɑ]-like, and labialized consonants [u]-like vocoids. The actual data, examined in detail in the next two chapters, bears out these predictions, while also exhibiting language- and context-specific variation in the degree of height, frontness or retraction, and tenseness of the prevocalization onglides. Just as in liquids, prevocalization of secondarily modified consonants typically takes place in weak syllable positions and often only after a stressed nucleus, and involves temporal sliding of the secondary articulation gesture of the consonant with respect to its primary constriction gesture. The temporal realignment of consonant-internal gestures is captured, among other considerations, by the various labels applied to this process in the specialist literature: for example, prepalatalization of palatalized consonants in dialectal Polish goes by the name "asynchronous

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palatalization", with the term "asynchronous" referring to the lack of synchronization between the palatalization gesture and the primary articulation gesture of the consonant (Koneczna 1965:148ff). Prevocalization of secondarily modified consonants sometimes results in weakening of their C-gesture and strengthening of the preceding vowel space through lengthening or diphthongization. In addition, prevocalization of this type frequently entails loss of the secondary articulation, at least from a diachronic perspective, which may be construed as a type of consonant lenition as well.

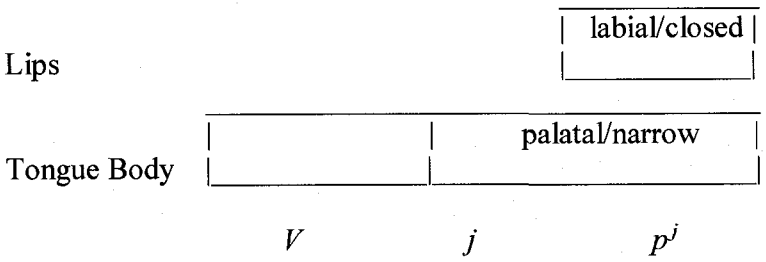
From the formal point of view, prevocalization of secondarily modified consonants is identical to that of complex liquids presented in Figure 11, except that the surface results attest a considerable degree of variation. Figure 12 below presents schematically three attested scenarios of prevocalization of secondarily modified consonants: prevocalization without loss of the secondary articulation on the consonant, prevocalization with partial loss of the secondary articulation, and prevocalization with complete loss of the secondary articulation. /p^{j}/ has been used as convenient shorthand for a consonant at any place of articulation with any type of secondary articulation. The figure in (a) shows the type of prevocalization in which the consonant's secondary articulation is preserved. A concrete example of this prevocalization type is attested in the Romanian of Moldova, where prevocalizing consonants preserve the secondary palatalization. The scheme in (b) shows the prevocalized allophone of a consonant with partial loss of the secondary articulation. A concrete example of this type is the realization of phonemic palatalization in Estonian, where palatalized consonants begin with a subsegmental onglide, are followed by the palatalized initial portion of the

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consonant, and conclude with its depalatalized final portion, subject to coarticulation from the following vowel. In gestural terms, the coarticulation ability of the final portion of the consonant directly follows from the failure of the palatalization gesture of the tongue body to extend to the end of the closure phase of the C-gesture. The figure in (c) shows the prevocalized allophone of a consonant in which the secondary articulation gesture has been lost. A concrete example of this type is the realization of palatalized labials in Irish and Scottish Gaelic, in which the distinctive palatalization is signaled by the palatal onglides from preceding vowels and offglides to following vowels rather than by a palatalization gesture during the labial closure.

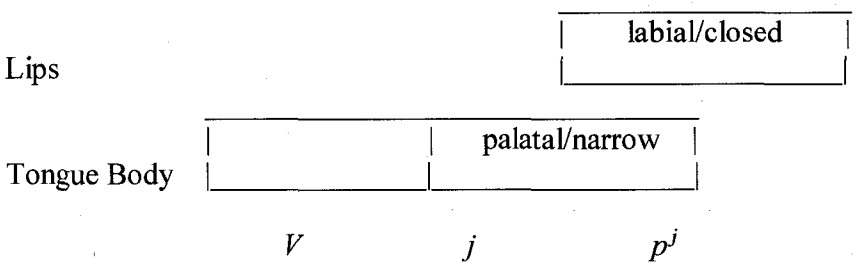
Figure 12. Prevocalization of secondarily modified consonants

(a) Prevocalization without loss of secondary articulation

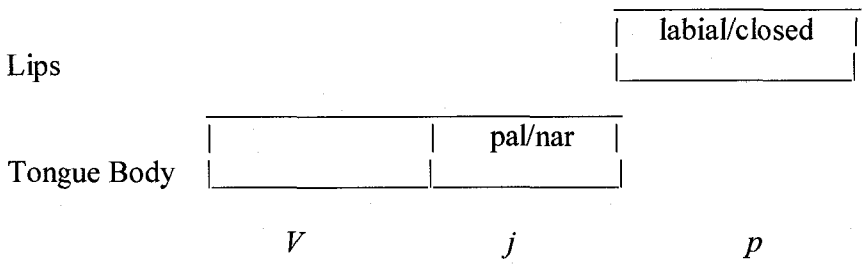


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(b) Prevocalization with partial loss of secondary articulation



(c) Prevocalization with complete loss of secondary articulation



The preceding discussion has suggested that prevocalization of consonants with secondary vocalic articulations may be viewed as the result of two related processes, temporal sliding of the consonant's V-gesture in the direction of the preceding nucleus, and, possibly, simultaneous shortening of its C-gesture. The next subsection investigates the possibility of extending this scenario to prevocalization of plain consonants.

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2.3.2 Plain consonants

The analysis of prevocalization presented above crucially relies on temporal realignment of the V- and C-gestures of prevocalizing consonants. The existence of V-gestures in the consonants considered so far – complex liquids and secondarily modified consonants in general – did not require demonstration; this, however, is not the case with plain consonants, which are generally regarded as devoid of a vocalic component. In order to extend the above account of prevocalization to plain consonants, it is therefore first necessary to demonstrate that the production of plain consonants involves a vocalic component. The discussion below demonstrates this by considering first the production of consonants whose primary constriction leaves the body of the tongue (relatively) free for coarticulation (i.e., labials and nonpalatal coronals), and then the production of consonants whose constriction involves the tongue body as the primary articulator (i.e., palatals, velars, uvulars, and pharyngeals).

2.3.2.1 Labials and nonpalatal coronals

As already indicated, Sproat and Fujimura (1993) and later researchers included lowering of the velum in their definition of vocalic, or V-gestures. For the purposes of the present analysis, however, it will be more convenient to restrict the notion of a V-gesture to those component gestures of consonants that may be interpreted as vocoids when the accompanying C-gesture is removed (e.g., when a consonant is vocalized). As no vocoids are produced exclusively by means of the velum lowering gesture, it would not be included among the V-gestures. The tongue dorsum gesture of dark [ɬ], on the

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other hand, which produces a back vocoid when the tongue tip constriction is removed, would count as a V-gesture. Likewise, the raising and fronting gesture of the tongue body accompanying the production of a [pʲ] would count as a V-gesture: if the lip closure is removed, this gesture will be interpreted as a high front vocoid. Thus, for the purposes of the following discussion, the V-gesture may be defined as position of the tongue body (and, in relevant cases, also lips) accompanying the C-gesture of the traditional consonant-level units, which may be interpreted as a vocoid when the corresponding C-gesture is partially or completely removed. When defined in this way, V-gestures may be argued to accompany the production not only of complex consonants, in which they are (mostly) articulatorily distinct from the C-gestures, but also that of plain consonants.

From the viewpoint of the mobility of the tongue body during their production, plain consonants may be divided into two broad classes: consonants whose constriction is produced with the tongue body (including tongue dorsum and root), and consonants produced with the tongue tip or lips. The first class includes palatals (i.e., palatoalveolars, alveolopalatals, and palatals proper), velars, uvulars, and pharyngeals; and the second class includes nonpalatal coronals (dentals, alveolars, and retroflexes), and labials. In the production of tongue body consonants, there is no movement of the body of the tongue separate from that which produces the main constriction. In the production of labials and tongue-tip coronals, on the other hand, the whole or a considerable part of the tongue body is free to assume a variety of coarticulatory shapes. From this point of view, the resonance quality of this second class of consonants always

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depends in part on the coarticulatory shape of the tongue body, defined above as the consonant's V-gesture.

As a concrete example of tongue-body coarticulation in plain coronals we may consider the allophones of the lateral in RP. RP has two principal allophones of the lateral known as clear /l/, used before vowels and /j/, and dark /l/, used finally and preconsonantally.[39] Clear /l/ involves "a raising of the front of the tongue in the direction of the hard palate (in addition to the tongue-tip articulation)" (Jones 1964:176). The tongue-body position of clear /l/ can also become more central: "when the clear **l** is followed by **ɑ** or **ɔ**, it tends towards a 'neutral' **l** with the resonance of **ə**" (Jones 1964:176 ft 8). On the diagrams illustrating Jones' description, it may be clearly seen that the tongue body position of the "neutral" [l^{ə}] is intermediate between the fronted tongue body position of [l] and the retracted one for [ɬ]. The description in Jones (1964) suggests that, from an articulatory point of view, there is no difference between the V-gestures corresponding to phonemic coarticulations and the V-gestures accompanying the production of plain coronals (and labials): in both cases, the position of the tongue body plays a role in the production of the consonant, and may sometimes play a role in the phonology as well.

A concrete instance of the phonological relevance of tongue body position in plain coronals comes from an investigation of the relative resistance of German and Catalan /l/s to vocalic coarticulation (Recasens et al. 1995). German /l/, which is of the clear

[39] For an alternative statement of the distribution cf. Rubach (1996).

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type, is seen in standard terms as produced with a single apicoalveolar gesture and lacking active tongue dorsum control; phonologically this is interpreted as an absence of secondary articulation. Catalan /l/ is of the dark type, which is usually interpreted, both phonetically and phonologically, as a combination of two lingual gestures, a primary apicoalveolar gesture and a secondary dorsovelar or dorsopharyngeal gesture. At the outset of their study, Recasens et al. (1995) hypothesized that the existence of a defined tongue dorsum gesture in Catalan /l/ would make the lateral in that language more resistant to coarticulation from adjacent vowels than German /l/ which, lacking a tongue dorsum target, would be more liable to coarticulation. However, the results of their study have shown that the /l/ in German is just as resistant to coarticulation as the /l/ in Catalan. In a later study, Recasens et al. (2005) established that the clear /l/ of Valencia Catalan has more coarticulatory sensibility than the dark /l/ of Majorcan Catalan. The difference between the coarticulatory behavior of clear /l/s in German and Valencia Catalan seems to indicate that only the former lateral has a tongue dorsum target which is underlyingly specified. This difference in the coarticulatory resistance of German and Valencia Catalan /l/s, both described as "clear", may be taken as evidence that the internal composition of plain consonants has to be determined on a case-by-case basis, and only after their phonological behavior has been taken into account.

A number of additional considerations indicate that it is necessary to specify the tongue body position of (at least some) plain consonants. The idea that the phonological effects of noncontrastive tongue body coarticulations are far more widespread than generally recognized is not new (cf. Carter 2003:238), but few studies attempt to

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correlate the phonological behavior of plain consonants with such noncontrastive coarticulations. One exception is Ní Chiosáin and Padgett's (1993) proposal that consonant-vowel interactions involve only the vocalic features of consonants, or their "inherent V-place". More relevant to the present analysis are the conclusions reached by Flemming (2003) who investigates the patterns of fronting and retraction of vowels in the context of adjacent coronal consonants. Specifically, Flemming questions the claim, taken for granted in much of the feature-geometric literature, that the fronting of vowels next to coronal consonants provides evidence for spreading of the feature [coronal] from the consonants to the vowels (cf. Clements 1991; Hume 1994; Clements and Hume 1995, among others). Flemming provides cross-linguistic evidence that only coronals with a fronted tongue body may condition vowel fronting, while those with a retracted tongue body condition vowel retraction. His evidence comes from the retraction of vowels in the context of pharyngealized coronals in Arabic, velarized coronals in Russian, and retroflexed coronals in many languages of Australia. Modification of vowels under the influence of adjacent coronals is interpreted by him as the result of assimilation in tongue body position, which he correlates directly with the nature of the coronal constriction by claiming that dentals, alveolars and palatoalveolars are preferentially produced with a fronted, and retroflexes with a retracted tongue body (2003:336f).

The fronted tongue body position assumed by Flemming may not be justified in the case of dentals, as various researchers have noted a persistent connection between dental articulation and tongue body retraction. Recasens and Espinosa (2005), for

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example, report a persistent connection between dentality and darkness, on the one hand, and alveolarity and clarity, on the other. In Erris Irish, and in Irish English, dentals are phonetically dark; in conservative Irish English, alveolars are said to be dentalized before dark /r/, which is interpreted as assimilation in tongue body position (Harris 2006). However, the major contribution of Flemming's (2003) study is not in the exhaustiveness of detail on tongue body position in coronals, but rather in assembling a rich body of evidence for "relatively detailed, redundant phonological representations", and a proposal to represent tongue body position in coronals "even where it is highly redundant" (337). The tongue body positions distinguished by him are [front], [central], and [back], with the feature [front] assigned to plain coronals with a fronted tongue body and palatalized consonants, and the feature [back] assigned to retroflexes and velarized consonants. Plain fronted coronals are distinguished from palatalized consonants at the level of phonetic detail, with the coarticulation of the former identified by Flemming with a [ɪ] and that of the latter with a [i] so that plain fronted coronals receive the representation [front, -tense] and palatalized consonants [front, +tense].

The feature [central] is left by Flemming unexplored. The phonological relevance of a centralized tongue body coarticulation, however, is warranted by a number of phonological facts, which include (pre)vocalization of alveolars in Maxakalí, realized as [ə, ə̃]. The (pre)vowels generated by the Maxakalí alveolars contrast in their tongue body features with (pre)vowels generated by all other consonants; in this way, the

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phonological contrast between the palatal and nonpalatal coronals, for example, is maintained even after their full vocalization.[40] There are other languages in which a centralized tongue body setting may be exploited in phonologically relevant ways, e.g., in coronals and in labials. Moreover, there is evidence that a centralized tongue body setting can contrast both with fronted and with retracted tongue body positions. For example, Scottish Gaelic dialects typically oppose plain, velarized and palatalized sonorants (Ternes 1973:25ff). An instrumental investigation of this three-way opposition, focused on the laterals of the Bernera dialect of Scottish Gaelic, has resulted in the lowest second formant values for /lˠ/, the highest for /lʲ/, and intermediate for the plain /l/, which is consistent with the high back, raised front, and central positions, respectively, of the tongue body (Ladefoged 1998 et al.:15ff). A similar three-way opposition seems to have obtained in Old Irish which is said to have distinguished three separate consonant "qualities", palatal, velar, and neutral (Thurneysen 1946:55).

More often, the centralized tongue body position is found within the allophonic range of one of the more extreme tongue body positions. As already mentioned, RP has two principal allophones of the lateral liquid, clear [l] and dark [ɫ]; when followed by the low back vowels /ɑ, ɔ/, the tongue-body position of the clear allophone becomes

[40] Independently from their behavior under prevocalization, the retracted tongue body specification of Maxakalí alveolars is also suggested by their backing influence on the adjacent /a/. The retracted tongue body position of the anterior coronals suggests that they may be dental rather than alveolar, at least in some of their realizations. A connection between a centralized tongue body position and dentality is also implied by the data from certain Kurdish dialects of Iraq. MacKenzie (1961) reports that, e.g., in the Suleimani dialect dental stops /t, d/ are realized in some positions as a (semi)vocalic schwa described as a "half-close, central continuant, accompanied by a raising of the tip of the tongue towards the position of realization of a dental stop" (8).

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centralized, producing a schwa-like resonance (Jones 1964:176 ft 8).[41] In Albanian, clear /m, n/ are said to contrast with dark /m^{ɯ}, n^{ɯ}/ (Lowman 1932). While the dark /m^{ɯ}/ is described as being produced with the back of the tongue raised toward the soft palate, the clear /m/ is described by Lowman as nonpalatalized and requiring "that the front of the tongue be held not so low nor in so relaxed a manner as is customary in making the English [m], or the ordinary [m] used in most languages" (1932:277). In the same language, dark /n^{ɯ}/ is phonetically dental, whereas clear /n/ is alveolar and "only neutrally clear" (277); elsewhere in his article Lowman notes that "the vocalic resonance of [n] is not especially clear" (288). In Cois Fhairrge Irish, on the other hand, the schwa-resonance is within the allophonic range of the velarized tongue body position. In that variety, dental stops [t̪, d̪] are said to vary in their degree of darkness from a central [ə]-resonance word-finally after front vowels to a back [u]-resonance word-finally after /u(:)/; in other positions they have the resonance quality of a half-open retracted [ɘ] (De Bhaldraithe 1945). Here, the quality of the coarticulation gesture of the dentals is clearly derived from partial overlap with the vocalic gesture of the preceding vowel.

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[41] So-called 'European' /l/ also shows variation between the fronted and centralized tongue body positions, even though the more fronted specification appears to be more widespread. This is indicated by numerous diachronic developments and synchronic alternations; for instance, (pre)vocalization of [l] in many European languages often results in a palatal glide, cf. Old Italian spellings such as *ailtri* 'altri', the regular change *pl, bl, fl* > *pj, bj, fj* in Italo-Romance and *ult* > *ujt* in Spanish, and realization of /l/ in coda as /j/ in dialectal Italian (cf. *aitə* 'alto') and Spanish (*aigo* 'algo'), and Viennese German ([pujt] *Pult*). In Romance languages, /l/ could also cause palatalization of preceding velars, cf. Latin *oculu* > **okʲlu* > *oʎo* (Portuguese), *oʒo* (Old Spanish). In contemporary Slavic languages with phonemic palatalization, the nonpalatalized /l/ may be realized as a range of allophones between a [l^{ɯ}] and a [l^{ɘ}] (cf. Dalewska-Greń 1997:79ff).

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The preceding discussion makes it clear that, quite independently from the recognized contrastive secondary articulations such as palatalization and velarization, tongue tip coronals are always coarticulated in the sense of being produced with a specific position of the body of the tongue whose phonological relevance may not be immediately apparent.[42] The ability of tongue tip coronals to take on a variety of distinct colorings has to do with the specifics of their articulation which leaves a considerable part of the tongue free to assume a variety of coarticulatory shapes. There is a considerable body of evidence suggesting that the difference between coarticulations associated with the plain tongue tip consonants and the "canonical" secondary articulations is gradient and noncategorical. This fact is emphasized, e.g., by Kelly and Local (1989) in their discussion of nondistinctive coarticulations broadly defined as "those features of consonants which are the auditory concomitants of configurations of the oral cavity and/or movements of the tongue other than those which relate to the principal articulation" (72-73). The coarticulation categories discussed by Kelly and Local include palatalized, clear, half-clear, central, half-dark, dark, and velarized. While the first and the last of these correspond to the notions of palatalization and velarization as codified in the IPA, "clear" seems to refer to a front vowel resonance such as the one normally associated with clear /l/ (perhaps identifiable with

[42] In addition to consonant-vowel interactions, noncontrastive coarticulations may become phonologically relevant in the context of consonant-consonant interactions as well. For example, alveolars are known for their frequent palatalizing influence on adjacent consonants. Thus, in the history of various Romance languages a variety of word-medial consonants got palatalized before the following /t, d, s, n, l, r/. While the palatalization of coronals in this position (cf. Occitan *pèiro* 'stone' < Latin *petram*) may be open to alternative interpretations, that of velars (*-kt-* > *jt* as in Portuguese *seis* 'six' < *sex*) and labials (*-ps-* > *-js-* as in Old Provençal *caissa* 'box' < *capsam*) seems to warrant an interpretation in terms of consonant palatalization due to assimilation to the fronted tongue body position of the following alveolar.

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Flemming's [ɪ]), while "half-clear", "central" and "half-dark" seem to map out the central range, as suggested by the authors' use of a schwa in their notation of these coarticulations.

The coarticulatory capacity of the tongue body is even more pronounced in labials, in whose case the tongue is not involved in primary articulation. Secondary articulation contrasts on labials may include palatalization versus velarization (as in Russian /p^j, b^j, m^j, f^j, v^j/ versus /p, b, m, f, v/ [$p^ɯ$, $b^ɯ$, $m^ɯ$, $f^ɯ$, $v^ɯ$]). In Irish and Scottish Gaelic, palatalization contrast in labial consonants seems to be instantiated as one between the centralized and retracted tongue body settings: as in Russian, palatalization in this language area has brought about the velarization of nonpalatalized consonants; unlike in Russian, however, this also triggered depalatalization of the palatalized labials so that the phonemic contrast /p^j/ : /p/ has become to be realized phonetically as [p] : [$p^ɯ$]. During the pronunciation of the slender [p] in dialects that have depalatalized the /p^j/ the tongue is said to be in a "neutral" position (cf. Oftedal 1963:71ff), perhaps to be interpreted as centralized.[43]

Although in the preceding discussion the coarticulatory tongue shapes accompanying the production of coronals and labials have for convenience been referred to as [i]-, [ə]- or [ɯ]-like, it is, in fact, unclear whether they can be matched with any of the specific vowels in the languages concerned, or with any identifiable segmental vocalic gestures. In fact, one major source of evidence for the identity of the

[43] Cross-language studies show that a centralized tongue-body setting is the default option with labial articulations (more on this in Chapter 6).

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coarticulatory tongue shapes are the outcomes of prevocalization which indicate, for example, that the tongue body setting accompanying the production of labials in Maxakalí (roughly) corresponds to a [ɤ]-quality vocoid. It is interesting to note that the importance of the coarticulatory tongue shapes in plain consonants was clear to phoneticians as early as the end of the nineteenth century (cf. Rosapelly 1898a, 1898b). Rosapelly's study has more than an antiquarian interest as it provides an interesting support for the importance of tongue body position in plain labials, and its main points are briefly summarized in the Appendix.

This subsection has reviewed evidence for the phonological role of noncontrastive coarticulations in plain coronal and labial consonants. It has been shown that plain alveolars are typically associated with a fronted to central tongue body position, and dentals with a central to back tongue body position, and that the labials described in the literature as "plain" may be associated with a central or back tongue body position. If the analysis of prevocalization developed in the context of complex consonants is applied to plain coronals and labials, it would predict that the alveolars would produce prevocalization onglides of the front to central quality, while the dentals and labials would produce prevocalization onglides of the central to back quality. As will be shown in detail in the next two chapters, these predictions are borne out by the data, subject to surface variation due to language- and context-specific conditions.

Since plain labials and coronals are gesturally specified as to their tongue body position, the analysis of prevocalization in their case will not formally differ from the analyses of secondarily modified consonants in Figures 11 and 12. A concrete example

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of a gestural analysis of prevocalization of this type of consonants will be presented below in connection with the proposed account of the facts of Maxakalí.

2.3.2.2 Tongue body consonants

This section looks at the vocalic component of tongue body consonants, and proposes a gestural account of prevocalization of both subclasses of plain consonants. Consonants produced with the tongue body differ from tongue-tip coronals and labials in that they lack a tongue body movement separate from that which produces the consonant's primary constriction. For this reason, the location of the V-gesture in these consonants coincides, in terms of gestures (or features in featural accounts), with the primary constriction of the consonant. In articulatory phonology this may be modeled as simultaneous presence of two tongue body tiers in the representation, C- and V-tiers.

The necessity to recognize the existence of a vocalic component in tongue-body consonants emerges from a number of phonological facts unrelated to prevocalization; in addition, and at a purely conceptual level, the gestural structure of tongue body consonants just outlined has a number of antecedents. One obvious antecedent is the decision of Chomsky and Halle (1968) to apply the vocalic features [high], [low], and [back] to the phonological description of tongue body consonants (305f). It will be recalled that the only other way in which these features are used in the consonantal context in that work is with reference to secondary articulations. Another antecedent for the present treatment of tongue body consonants is the "inherent V-place theory" of Ní

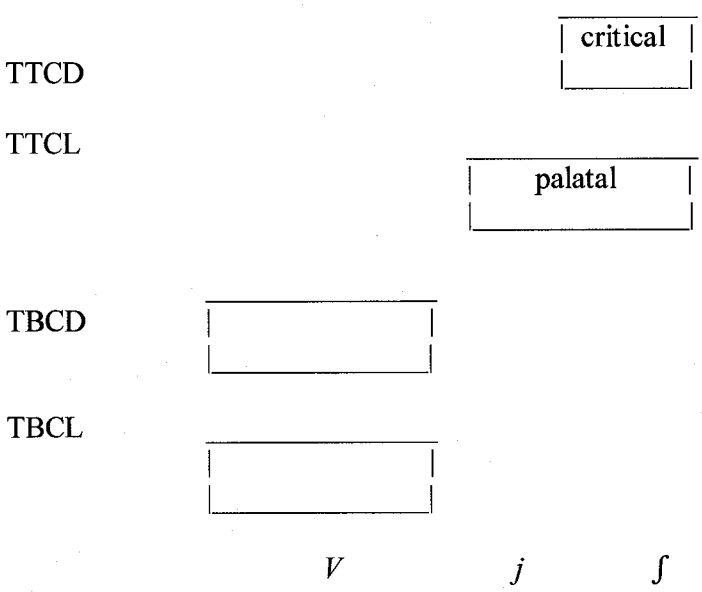
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Chiosáin and Padgett (1993) who analyze consonant-vowel interactions as involving exclusively the vocalic features of both types of segments. This claim is formally expressed by the authors as the existence in plain consonants of an "inherent V-place" corresponding to their primary stricture. The notion of the inherent V-place is rationalized by them as the presence of vowel-like onsets and offsets that necessarily accompany consonantal constriction (13f). The third important antecedent is provided by early work in feature geometry that makes use of the device of "feature projection", defined as copying of the C-place articulator under the consonant's V-place node. The sole purpose of this device is to make it possible for plain consonants to interact with adjacent vowels. Feature projection is made use of, e.g., in Gussenhoven and Weijer's (1990) analysis of Old English breaking: while the liquids and /w/ have a secondary dorsal node by virtue of being underlyingly complex, the velar fricative /x/ acquires such a node through feature projection. Finally, it may also be mentioned that palatals and uvulars are often treated as segments with secondary articulations: even though each of these consonant types is usually seen as produced with one active articulator at a single place of articulation (cf. Recasens et al. 1995), they are often treated as internally complex on phonological grounds. For example, palatal consonants are viewed as "inherently palatalized" by Hume (1990) and as simultaneously coronal and dorsal by Keating (1988) and Lahiri and Evers (1991); while uvulars are analyzed as simultaneously dorsal and pharyngeal by Elorrieta (1991) and McCarthy (1994), among many others.

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Prevocalization of consonants produced with the tongue body appears at first blush to allow two interpretations. One of these does not make use of these consonants' vocalic component and is briefly discussed by Albano (1999) in connection with /S/-prevocalization in Brazilian Portuguese. Albano's solution involves treating the CL and CD defining the articulation of the sibilant as differing in duration. The essence of her solution consists in temporally offsetting the (critical) constriction of the sibilant whereby the prevocalization onglide emerges as partial vocalization of its stricture, as schematically shown in Figure 13.

Figure 13. Prevocalization of Brazilian Portuguese /S/[44] (Albano 1999)

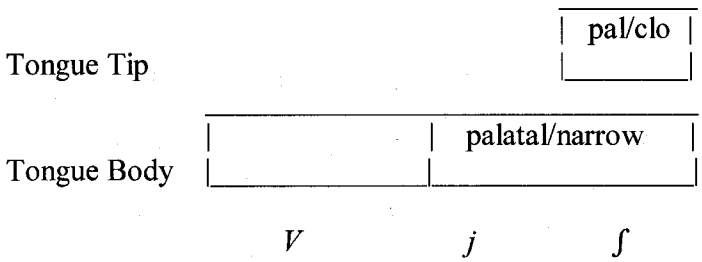


[44] Albano assumes the palatoalveolar realization of /S/; the articulatory information for the preceding nucleus has been left blank.

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In Albano's solution, the constriction degree and constriction location of the sibilant are treated as mutually independent variables. This represents an innovation in one technical aspect of the current model of articulatory phonology, namely, its treatment of the two variables as interdependent and sharing the same duration. While similar extensions to the model have been suggested elsewhere (cf. Gafos 1999:45f), it is also possible to avoid treating the two variables as independent by assuming that prevocalizing tongue body consonants possess a vocalic tongue body tier, in the manner explained previously. This way, the temporal reduction will affect the paired CD and CL variables specifying the articulation on the C-tier simultaneously. Introducing the V-tier in the representation of tongue body consonants also makes their prevocalization formally identical to that of all other consonant types, cf. Figure 14.

Figure 14. Prevocalization of Brazilian Portuguese /S/



The analysis of prevocalization of plain tongue body consonants proposed here predicts that the prevocalization onglides to these consonants would always coincide with their primary place of articulation. In other words, palatals are expected to develop prevocalization onglides of the [i] quality, velars are expected to develop

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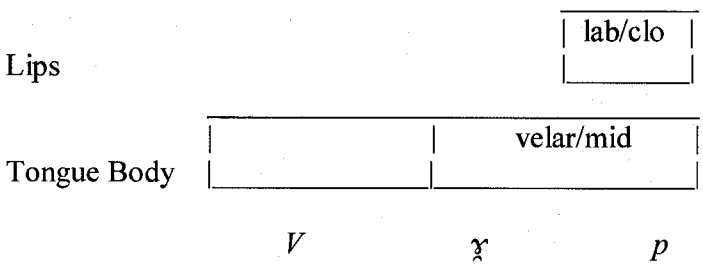
prevocalization onglides of the [ɯ] quality, and uvulars and pharyngeals are expected to develop prevocalization onglides of the [ɔ]/[ɑ] quality. It will be shown in Chapter 4 that this prediction is borne out by the data, with surface modifications introduced by language- and context-specific conditions such as gesture overlap with adjacent vowels and constraints on the surface realization of the cross-linguistically rare pharyngeal glide (discussed in detail in 4.3).

The proposed gestural solution to prevocalization should also be able to account for the facts of Maxakalí, which inspired most earlier analyses of this process. It is easy to see that by assuming a V-tier in their underlying representation, analyzing prevocalization of labials and alveolars becomes straightforward as each of these consonants is now represented as articulatorily associated with a specific tongue body gesture. Based on the quality of the prevowels, the tongue body gesture of the labials may be specified as retracted and nonhigh, and that of the alveolars as nonhigh and centralized. Prevocalization of Maxakalí labials and alveolars may be schematically represented as in Figure 15.

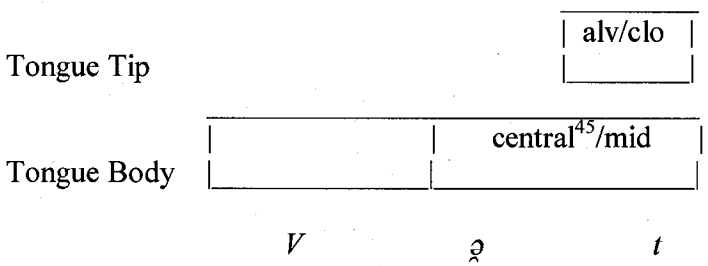
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Figure 15. Prevocalization of Maxakalí labials and alveolars

(a) Prevocalization of /p/



(b) Prevocalization of /t/



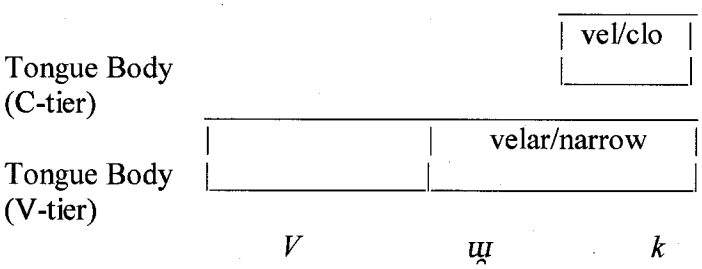
As shown in Figure 16 below, prevocalization of tongue body consonants is assumed to involve two tongue-body tiers. In strong syllable positions, the V-gesture of the tongue body consonants is unperceived as it is completely "covered" by the C-gesture; however, in environments that favor temporal reduction of the C-gesture, the V-gesture becomes partially uncovered and may become segmental when the C-gesture is removed.

[45] The current set of CL values ([labial], [dental], [alveolar], [postalveolar], [palatal], [velar], [uvular], [pharyngeal], cf. Gafos 1999:12) does not allow a straightforward way of deriving central vocoids, hence the provisional [central] gesture has been used.

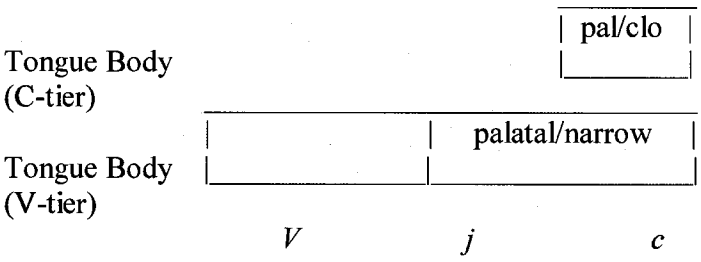
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Figure 16. Prevocalization of Maxakalí palatals and velars

(a) Prevocalization of /k/



(b) Prevocalization of /c/



The above solution successfully accounts for prevocalization of all consonants in Maxakalí because it makes no reference to the consonants' primary constriction, or C-gesture. It is interesting to note that a solution that is similar in spirit to the gestural account proposed here is discernible in the original publication on Maxakalí phonology by Gudschinsky et al. (1970). The authors' view, which appears in the closing paragraph of their paper, is worth quoting in full (1970:88):

> "By convention, consonantal articulation is described in terms of points on the rigid structures of the mouth (the lips, teeth, alveolar arch, palate, etc.), whereas vowel articulation is described in terms of the position of the flexible tongue (front, central, back). If the consonants of Maxakalí were re-arranged in terms of tongue position, the order would be palatal, alveolar, labial, velar. In the palatal consonants, the tongue is thrust forward so that the closure is made by the blade

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> of the tongue against the palate; increasing the aperture without changing horizontal tongue position produces front vocoids. In the alveolar consonants, the tongue is retracted so that the closure is made by the tip of the tongue against the alveolar ridge; increasing the aperture without changing horizontal tongue position produces central vocoids. In the labial consonants, the inactive tongue is in a relatively lax back position; increasing the aperture without changing the tongue position produces mid back vocoids. In the velars, the tongue is retracted so that closure is made by the back of the tongue against the velum; increasing the aperture without changing horizontal tongue position produces back vocoids."

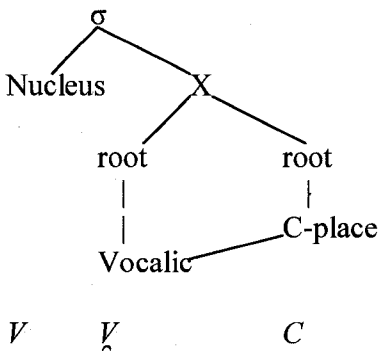
As seen in this passage, Gudschinsky et al. relate the quality of Maxakalí (pre)vowels to the position of the tongue body during the consonantal closure, while at the same time appearing to imply that the tongue body position is determined by the primary stricture of the consonants. A similar approach to Maxakalí (pre)vocalization was adopted by Wetzels (1993),[46] who analyzed Maxakalí consonants as endowed with secondary articulations, and prevocalization as contour creation in which the features of the prevowels are derived from the secondary articulations. Based on the featural composition of the prevowels, he specified the secondary articulations of labials and coronals as mid back and mid central, respectively, and derived the secondary articulations of the palatal and velar consonants from their primary articulations via feature projection;[47] the remaining portion of the analysis, presented in Figure 17 below, is identical to that of Wetzels and Sluyters (1995).

[46] This work became available to me after my analysis of prevocalization was completed.

[47] This device, extensively used both by Wetzels (1993) and Gussenhoven and Weijer (1990), is apparently introduced in an unpublished work by Clements, which is unavailable to me at the time of writing.

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Figure 17. Maxakalí prevocalization as contour creation



Wetzels' (1993) analysis is on the whole compatible with the gestural analysis proposed here as both crucially rely on the specification of the tongue body position in plain consonants. Nevertheless, an analysis in gestural terms seems preferable in that it can better account for the phonetic variation in the outcomes of prevocalization, and in that under such an analysis prevocalization naturally follows from language-specific facts of syllable organization: in this way, the primes of phonological analysis become directly linked to higher units of phonological organization, e.g., syllables. In addition, prevocalization, a dynamic and gradient process located midway between the phonetics and phonology, is better addressed by the gestural model, whose stated purpose is to bridge the traditional gap between these two approaches to the sound structure of language.

2.4 Summary and conclusion

The main proposal put forth in this chapter is that surface prevocalization is due to a prosodically motivated realignment of intrasegmental gestures of consonant-level units,

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whereby the consonant's V-gesture comes to anticipate its C-gesture. The original impetus for this analysis was provided by studies of syllable-position effects on the timing of the constituent gestures of liquids; the proposed account was then extended to other secondarily modified consonants, plain labial and tongue-tip consonants, and plain tongue-body consonants. In order to maintain a uniform account across all consonant classes, it was demonstrated that consonants in general are characterized by a vocalic component, or V-gesture. In plain labial and tongue-tip consonants, the V-gesture was shown to be identical in nature to that of the canonical secondary articulations. In the case of tongue body consonants, it was shown that the existence of a V-gesture is warranted not only by their behavior under prevocalization, but also by a number of independent phonological considerations examined by other researchers. After the existence of an internal vocalic component was demonstrated for all types of consonants, the proposed gestural account of prevocalization turned out to be uniform across all classes of consonants.

Factual evidence supporting the proposed analysis is provided in the next two chapters in the form of case studies of prevocalization in individual languages. The discussion is organized around the quality of prevowels resulting from prevocalization, with front prevowels discussed first. As already noted in the preceding discussion, the proposed analysis of prevocalization makes certain predictions as to the quality of the prevocalization onglides expected to be triggered by each consonant class; these predictions are summarized for easy reference in Table 6 below.

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Table 6. Expected quality of prevocalization onglides

Type of consonant	Expected quality of prevocalization onglide
C^{j}	i ~ e
$C^{ɯ}$	ɯ ~ ə
C^{w}	u ~ o
$C^{ɑ}$	ɑ ~ a ~ ə
alveolar	i ~ ɔ
dental	ə ~ ɯ
labial	ə ~ ɯ
palatal	i ~ e
velar	ɯ ~ ə
uvular	ɔ ~ ɑ ~ a ~ ə
pharyngeal	ɑ ~ a ~ ə

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3. Front prevowels

3.1 Introduction

This chapter and the next together provide a cross-language investigation of prevocalization patterns which supplies the primary data and factual support for the gestural analysis of this process advanced in this work. From an expository point of view, the most natural way to organize the discussion was by focusing on the quality of the onglides resulting from prevocalization. The most frequent of these is the front-quality onglide, due in part to the relative frequency of palatalization as compared to other secondary articulations. Consequently, the present chapter focuses on prevocalization onglides of a front quality, and the discussion of onglides of other types is deferred to Chapter 4.

Apart from palatalized consonants, front onglides may be triggered by two other segment types, palatals and alveolars. The relevant feature that the three consonant types (palatalized, palatal, alveolar) have in common is fronted position of the tongue body responsible for the fronted quality of the consonants' V-gesture. In environments that trigger retiming of the consonants' constituent gestures and weakening of their C-gesture, the advanced setting of the tongue results in the percept of prevowels of a fronted quality. In the available sources such prevowels are most frequently notated as [i] or [j], but synchronic descriptions point to a great deal of variation as to their actual phonetic implementation, which includes varying tenseness, duration, height, and the degree of frontness.

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In some of the languages and dialects about to be discussed prepalatalization is attested as a living process, while in others its existence can only be inferred from orthographic evidence, subsequent developments of the affected segments, typological considerations, and/or the evidence of neighboring dialects or closely related languages. Nevertheless, the synchronic and diachronic data are presented here side by side as prevocalization is assumed to involve the same underlying mechanism regardless of the type of available evidence. The two types of data are thus essentially complementary and their simultaneous presentation has the advantage of allowing a more complete understanding of prevocalization by highlighting its different stages. Thus, while descriptive treatments show prevocalization in its synchronic functioning, diachronic treatments throw light on the subsequent development of the triggering consonants and, where applicable, of the adjacent nuclei.

Presentation of the data is organized as follows. After a brief discussion of the areal distribution of contrastive secondary palatalization, Section 3.2 investigates prepalatalization of contrastively and contextually palatalized consonants in a representative number of languages and language families. It also discusses the possibility of a prepalatalization stage in umlaut, especially the Germanic *i*-umlaut. Section 3.3 focuses on prepalatalization of palatal consonants and draws a categorial distinction between their prevocalization language-internally and their decomposition in language contact situations. Section 3.4 concentrates on prevocalization of the alveolar consonants noted in the literature most frequently: /l/, /r/, /s/, and /n/. I show that some of the diachronic changes involving these consonants traditionally viewed as straight

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vocalization represent instead a sequence of two processes, prevocalization and deletion of the alveolars' C-gesture. I also argue that language-internal vocalic prothesis in initial *sC*-clusters has all the earmarks of prevocalization, in the sense adopted in this work. In the case of the alveolar nasal and nasal consonants in general, a distinction is drawn between prevocalization and pre-nasal diphthongization of vowels. Section 3.5 briefly summarizes the main points.

3.2 Palatalized consonants

3.2.1 Preliminary considerations

3.2.1.1 Prepalatalization in synchrony and diachrony

As noted in 1.4, from an articulatory point of view prevocalization of secondarily modified consonants represents the sum effect of vowel assimilation to the secondary articulation of the following consonant and anticipation of the secondary articulation gesture with respect to that of the consonant's primary constriction. Synchronic prevocalization of coarticulated consonants is generally subphonemic: prevocalization onglides do not surface as separate phonemes,[48] and the secondary articulation of the consonant is generally preserved. The phonological status of the secondary articulation does not affect prevocalization: allophonically modified consonants are as likely to prevocalize as those in which the secondary articulation is phonemic. One previously reported synchronic function of prevocalization is as a perceptual enhancement of the corresponding (phonemic) secondary articulation; in some cases it functions as its

[48] Nevertheless, with some speakers prevocalization may become lexicalized in certain lexical items (cf. Turculeț 1975).

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principal cue. This is true, for example, of prepalatalization in Estonian and prelabialization in Seri, both of which are signaled primarily by onglides to the consonants.

In diachrony, prevocalization of secondarily modified consonants most often results in phonemicization of the onglides, especially as offglides of diphthongs, while the triggering consonants generally lose the corresponding secondary articulation. Case studies assembled in this section indicate that, from a diachronic perspective, loss of the secondary articulation may be the main motivation behind prevocalization of secondarily modified consonants. This becomes especially clear when examining languages whose phonological tendencies during the prevocalizing period are known reasonably well: in such languages, the tendency towards elimination of the corresponding secondary articulation can usually be established on independent grounds. Thus, in Proto-Greek, Gallo-Roman and the majority of West and South Slavic languages, the recorded periods of prepalatalization coincide with the historical stages at which these languages undergo depalatalization. To give one concrete example, Proto-Greek is commonly reconstructed with a series of palatalized consonants that originated mostly from combinations with the following palatal glide. Palatalized consonants in Proto-Greek are assumed to have been systemically unstable and tending towards depalatalization (cf. Sheets 1976:119). Their depalatalization took several distinct forms that included coronalization (**tj*, **t^{h}j*, **kj*, **k^{w}j*, **k^{h}j*, **k^h^{w}j* > *s(s)* ~ *t(t)*; *dj*, *gj*, *g^{w}j* > *zd*), consonification of the palatal offglide (**pj*, **p^{h}j*, **bj* > *pt*), and loss of secondary palatalization accompanied either by compensatory lengthening (**lj*, **rj*, **nj*

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> *ll*, *r:/r̄r*, *n:/n̄n*) or prepalatalization (**rj*, **nj* > *jr*, *jn* / *a*, *o*__). When examined against this background, Proto-Greek prepalatalization is best viewed as one of the strategies that allowed the language to eliminate a secondary articulation which, for reasons yet to be understood, was not tolerated by its phonemic system (cf. Stang 1957; Diver 1958; Sheets 1976).[49]

3.2.1.2 Contrastive and contextual palatalization

Palatalization as a secondary articulation is usually defined as a supplementary tongue gesture superimposed upon a labial, dental, alveolar, or postalveolar consonantal constriction. Such a definition is traditionally used to distinguish secondary from primary palatalization, which refers to processes whereby noncoronal consonants becomes coronal[50] and nonpalatal coronals becomes palatal. The above definition of palatalization is also commonly used to distinguish secondarily palatalized consonants from palatals, for which the hard palate serves as the primary place of articulation. However, this definition is not entirely adequate in that it fails to include those cases of secondary palatalization that involve a change in the location of the consonant's primary constriction l. Among these are the palatalized counterparts to the alveolar affricates and fricatives in Polish and Bulgarian, or the palatalized version of the Slavic retroflex fricative /ʂ/, all of which are articulatorily alveolopalatal (cf. Hamann

[49] One of the reasons cited for the loss of phonemic palatalization in Proto-Greek is the loss of the palatal glide phoneme (cf. Sheets 1976:119 fn. 2).

[50] In the most recent phonological literature, however, there is a tendency to refer to such processes as coronalization.

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2004:64). In such cases, the analysis of palatalization as a primary or a secondary phonological feature is based on its functioning in the phonemic system of a language rather than its articulatory implementation. If the palatals as a group are opposed to consonants at other places of articulation, they are analyzed as palatal segments: this is the case, for example, of those West and South Slavic languages in which palatals are systemically opposed to the sets of labial, dental, and velar consonants. If, on the other hand, the palatals are synchronically derivable, e.g., from palatalized coronals or velars and continue to function in the phonology as their palatalized counterparts, they are interpreted as secondarily palatalized segments. This is the case, e.g., of Polish alveolopalatals [ɕ, ʑ, tɕ, dʑ, ɲ] which function phonologically as the palatalized counterparts of /s, z, ts, dz, n/, respectively.

An important type of languages attesting synchronic prepalatalization is languages with contrastive palatalization. Jakobson (1930) and, more recently, Stadnik (2002) showed that contrastive palatalization is an areal feature of northern Eurasia that occurs in a significant number of Turkic, Uralic, Slavic, Baltic and Celtic language families, Romani, Romanian, and Japanese. Not surprisingly, many case studies of prepalatalization assembled in this section are supplied by representatives of these (groups of) languages. It is interesting to note, though, that neither Jakobson nor Stadnik discuss reconstructed systems like Proto-Greek or Gallo-Roman, for which specialists posit a full series of palatalized articulations. It is possible that their non-inclusion of these languages has to do with an implicit assumption that there is a qualitative difference between quintessential palatalizing systems like those of the modern East

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Slavic languages and reconstructed palatalizing stages of modern essentially nonpalatalizing languages. Such, for instance, is the position of Malmberg who believes that secondary palatalization reconstructable for Gallo-Roman was essentially noncontrastive (1962:91 fn. 14):

> "Pour une raison dont il reste encore à rendre compte, le système n'a pas pu accepter des consonnes palatales du type [r^{j}], [d^{j}z^{j}], [s^{j}t^{j}], et la palatalité nécessaire pour maintenir l'equilibre du mot a dû se réaliser autrement (a "sauté par dessus" la consonne). Ceci ne veut pas dire que les consonnes n'aient pas été palatalisées. Au contraire. Il serait impossible qu'elles ne l'aient pas été. Seulement ça n'a pas été une palatalité distinctive, à laquelle le système, pour une raison ou une autre, a répugné, mais une palatalité combinatoire ("allophonique")".

A similar position is adopted by Nandriş (1952) who, moreover, divides linguistic systems into three types differing in their tolerance towards palatalization as a secondary feature of consonants. According to Nandriş, languages of the palatalizing type include palatalization as a systematic structural trait, and languages of the nonpalatalizing type lack such a distinction altogether. In addition, there is a third type of languages which, although essentially nonpalatalizing, may develop secondary palatalization as a transitory historical stage (143). It is in this last group that he places earlier stages of the Romance languages; following his logic, Proto-Greek would belong here as well. It is unclear to what extent Malmberg's and Nandriş' views can be substantiated, and for the purposes of the present study no distinction needs to be made between synchronically attested systems with secondary palatalization like those of the

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East Slavic languages, and reconstructed systems with secondary palatalization like those of Gallo-Roman or Proto-Greek.[51]

Languages without phonemic palatalization may attest allophonic palatalization which, unlike its phonemic counterpart, is not subject to any genetic or areal limitations. Languages with allophonic palatalization frequently attest prepalatalization, which closely follows in its degree of manifestation the relative strength of the palatalizing environment. Prepalatalization is reported most frequently with consonants followed by a palatal glide, which is the strongest palatalizing environment cross-linguistically (attributed by Straka 1965:131 to its articulatory tenseness). The next prepalatalization environment with respect to its frequency is consonants adjacent to front vowels, with frequency of occurrence roughly following the order *i* > *e/ɛ* > *æ* and the direction of consonant-vowel assimilation following the general direction of the assimilatory processes in a language. Adjacent front rounded vowels and the glide, as well as following alveolar consonants represent the least frequent palatalizing environment

[51] The historical phonologies of Greek and the Romance languages are instructive in the way they show the cyclic character of the palatalization/depalatalization processes. Thus, while Proto-Greek is reconstructed with a full set of palatalized consonants, Ancient Greek is a nonpalatalizing language that preserves many traces of earlier palatalization, including prevocalization-induced diphthongs. Modern Greek, in marked contrast to the immediately preceding stage and in parallel to Proto-Greek, shows a distinct tendency towards palatalization. And, interestingly, both Proto- and Modern Greek display prepalatalization as an accompanying process. The Romance area attests the cyclicity of the palatalization/depalatalization processes equally well. While Classical Latin is clearly nonpalatalizing, dialectal and Vulgar Latin attest the beginnings of palatalization, which culminated in the development of a series of palatal or palatalized articulations in Gallo-Roman. Gallo-Roman palatalization left many traces, among them prevocalization-induced diphthongs, in the phonology of Old French. Modern French preserves only a few of the palatal(ized) phonemes of Gallo-Roman; however, the contemporary language, in parallel to Gallo-Roman, has a strong tendency towards contextual palatalization which, just as in Gallo-Roman, is of the assibilating kind.

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(although this statement perhaps need to be revised when the consonant preceding alveolars is a /n/).

The most frequent environment for contextual palatalization in the above list is consonants followed by a palatal glide. In diachrony, prepalatalization of such consonants is often accompanied by deletion of the palatal glide and, if the consonant suffers no additional changes, there results a surface transposition of the elements of the original sequence (i.e., Cj > jC). In diachronic treatments, especially when intermediate stages are undocumented, such processes usually give rise to two mutually exclusive interpretations: an interpretation by means of simple metathesis, and an interpretation by means of prevocalization (which goes by various names such as "epenthesis", "attraction", and "anticipation"). In order to arrive at a correct interpretation of the facts in such cases it is first necessary to consider the place of the process in the overall phonological tendencies of the language at the time of its operation; the purpose here is to see whether this process is isolated at a given historical period, or whether it forms part of some more general tendency. Based on the evidence assembled here, prevocalization of secondarily modified consonants is usually not an isolated phenomenon: it tends to affect all or most of the relevant consonants, and it also tends to cooccur with either umlaut or depalatalization. Metathesis, on the other hand, is mostly a sporadic process and does not necessarily correlate with other phonological processes in a language. These facts may be used as a criterion for distinguishing between metathesis and prepalatalization in unclear cases. If, for example, it is known that during the period at which the transposition in Cj sequences was taking place the

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language was in the process of losing secondary palatalization, the transposition is more likely to represent prevocalization than metathesis. If, on the other hand, the language shows no evidence for depalatalization or umlaut, or if the transposition in Cj sequences affects only one or a small number of consonants, the process in question is more likely to be metathesis. This approach to resolving the controversy may be illustrated with the following example involving Semitic.

In Mandaic Aramaic, there is a process whereby earlier sequences of nasal plus palatal glide end up in a reversed order after a /a/, cf. /sanja:/, /samja:/ > /sajna:/ 'ugly', /sajma:/ 'blind'. Brockelmann (1908), whence this data is quoted, also notes parallel developments in Arabic and Ethiopian. In Arabic, this concerns original sequences of labials, sibilants and liquids followed by *w*, which end up transposed, cf. *ʔašwāb* > * *ʔawšāb* > *ʔawbāš* 'mob'. In Ethiopian, *r* followed by *w* also ends up visibly metathesized, cf. *ʔarwē* > *ʔawrē* 'wild beast' (Brockelmann 1908:279f). The intermediate stages between the input forms of consonant plus glide and the output forms of glide plus consonant in Mandaic are not available for inspection, and both prepalatalization (Brockelmann 1908:279; Nöldeke 1964:24; Macuch 1965:122f; Malone 1971:412 fn. 67) and metathesis (Semiloff-Zelasko 1973) have been proposed as the mechanism. Now, Mandaic has no independent evidence for consonant palatalization and, on the contrary, has evidence for other productive processes of metathesis. Based on these considerations, it is preferable to analyze the change *nj/mj* > *jn/jm* as metathesis. The change *rw* > *wr* in Ethiopian, on the other hand, calls for a

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different analysis since that language has independent evidence for prelabialization of labialized velars and uvulars, cf. *ḥaq*^{w}*ē* >* *ḥawq*^{w}*ē* > *ḥawqē* 'loin' (Brockelmann 1908:278f). For this reason, the change *rw* > *wr* in Ethiopian is perhaps best interpreted as prevocalization of *r*, which is allophonically rounded by the labial glide.

The sketches that comprise this subsection include only those instances of prepalatalization in Cj sequences which conform to the above criterion. In what follows, prepalatalization of phonemically palatalized consonants is considered first, and that of allophonically palatalized consonants last; the breadth and depth of coverage in each individual case is determined mostly by the availability of sources.

3.2.2 Contrastively palatalized consonants

3.2.2.1 Finno-Ugric

The Finno-Ugric subbranch of Uralic includes several languages with phonemic palatalization. In some of these, palatalization is concentrated in the initial portion of the consonant and the [i]-like transition preceding it; the rest of the consonant is nonpalatalized and subject to coarticulation from a following vowel (Eek 1973:31). The languages and dialects in which a palatal onglide to the consonant serves as an important acoustic cue for the identification of secondary palatalization include Estonian, East Finnish, Karelian, Livonian, and Saami.

Standard Estonian has palatalized coronals /t^{j}, s^{j}, l^{j}, n^{j}/ derived from the positional allophones of /t, s, l, n/ before a following [i] or [j], synchronic or reconstructed. A [i]-like onglide has long been known to be an important perceptual correlate of

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palatalization in that language (cf. Harms 1962; Lehiste 1965:136; Hint and Paunonen 1984:283). Forms cited in Harms (1962:18, 28ff) seem to imply that the onglide is especially prominent after the back vowels /u, o, o̰, a/; it is reported to be so salient that nonspeakers interpret it as segmental. Lehiste (1965) showed in an experimental study that combinations of vowels with prepalatalization onglides differ from the corresponding phonemic diphthongs in terms of length: the length of underlying diphthongs is considerably greater than that of the glided allophones of monophthongs, with the offglides of diphthongs approximately twice as long as prepalatalization onglides. Lehiste also showed that Estonian diphthongs involve two steady states with a separate target for the offglide, while glided monophthongs before palatalized consonants are continuous.[52] Lehiste's study is especially significant in providing phonetic support for the widely noted phonological differences between the glide-like allophones of vowels before palatalized consonants and the (nearly) homophonous phonemic diphthongs.

Prepalatalization is also cited as a historical and/or synchronic process in Karelian, Eastern Finnish, Livonian, and Saami (cf. Itkonen 1958:xxxvi; Sammallahti 1984; Leskinen 1984:252-53; Korhonen 1984:314). In some eastern dialects of Saami, vowels of the initial syllable except *i* are fronted before consonants[53] which are palatalized by a front vowel in the following syllable; vowel fronting may be accompanied by prevocalization, cf. Wagner's (1964) notations *ā$_i$kke*, *vā$_i$lte-* (238). In Livonian, a

[52] Of related importance is Piir's (1985) conclusion that Estonian diphthongs are biphonemic.

[53] Excluding labials and the palatal glide.

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language closely related to Finnish, prepalatalization is occasionally found in complementary distribution with umlaut.[54] Historically, prepalatalization in Livonian has resulted from palatalization of postvocalic consonants by a following *j, *ĭ, or *i (cf. details in Pajupuu and Viitso 1986). Palatalized velars and labials were later depalatalized, developing a prepalatalization onglide: cf. *laigā* 'broad', *tūoimki* 'bird-cherry' < Proto-Finnic **lakja*, **tōmi-*. According to the authors, alveodental consonants did not prepalatalize, while prepalatalization of *ʒʲ(:)* in words with the stød has not yet extended to all speakers; among the alternative contemporary realizations of the affected items are [su'ʒʲ] ~ [su'iʒ] 'wolf' and [si'ʒʲ] ~ [si:'ʒ] 'coal' (Pajupuu and Viitso 1986:101).[55] Late borrowings from Latvian containing the palatal stop /c/, which took place when prepalatalization already ceased to be operative, are nevertheless adapted as [ik], cf. Latvian *brāķis* > Livonian *brōik* 'scrap'. The authors attribute this rendering pattern to active bilingualism, that caused the adaptation of recent loanwords to be based on an analogy with older borrowings.

3.2.2.2 Slavic

Slavic is divided into Eastern (Russian, Ukrainian, Byelorussian), Western (Czech, Polish, Slovak, Lower and Upper Sorbian), and Southern (Serbo-Croatian, Slovenian,

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[54] Krygier (1997), for example, notes that "in Livonian *i* or *j* in the second syllable always caused palatalisation of the preceding consonant, and in the cases where this palatalisation later disappeared, umlaut is found (or sometimes an epenthetic *i* in its stead)" (35 fn. 79; cited after Posti 1946-48:47).

[55] Note the lengthening of a preceding /i/ as opposed to diphthongization of a preceding /u/. *ʒ(:)* and *ʒʲ(:)* are my interpretations of the authors' Finno-Ugric transcriptions *ž(ž)* and *ž(ž)*, respectively.

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Bulgarian, Macedonian) branches. Although contrastive palatalization was present in Common Slavic (cf. Shevelov 1965:207ff), at present this feature is characteristic only of East Slavic and the easternmost parts of West and South Slavic (cf. Stadnik 2002:81ff). The rest of the Western and Southern Slavic languages have eliminated contrastive palatalization from their phonological systems, preserving some traces of it in the form of glides deriving from prepalatalization onglides. For example, the glide in words like *kojs* 'bone' from earlier *kost^{j}* in East Slovak developed out of a prepalatalization onglide, and earlier *l^{j}*, *n^{j}* in Slovenian depalatalized, producing a prepalatalization onglide when in coda, cf. *lajnske le̋t* 'last year', *kojn* 'horse', *kojnak* 'stable' (Broch 1910:158; Koneczna 1965:152).

Productive synchronic prepalatalization exists in areas that preserved at least traces of distinctive palatalization; these include older and dialectal Polish and Czech, Ukrainian, and East Bulgarian. For example, prevocalization of palatalized sibilants is noted in writing in Greater Poland since the second half of the fifteenth century. Early spellings of the type *oyczcza* [o^{j}tɕtsa] 'father (genitive singular)' indicate pronunciations [^{j}tɕ], [^{j}tɕts], and [^{j}ɕtɕ] of medial *-ć-* [tɕ], *-ćc-* [tɕts] and *-ść-* [ɕtɕ], respectively. The forms *ojca* [ojtsa] 'father (genitive singular)', *ojcu* [ojtsu] 'father (dative singular)', etc. in the modern standard are of dialectal origin and show deletion of the affricate: *-^{j}ćc-* [^{j}tɕts] > *-^{j}c-* [^{j}ts]. Prepalatalization of *ś*, *ź* ([^{j}ɕ], [^{j}ʑ]), whether alone or in clusters, is also notated in these early materials (Koneczna 1965:149f). In the modern literary language, prepalatalization which arose in the older medial groups *-źrz-* [ʑʐ] and *-śćs-* [ɕtɕs] has

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been preserved in *we[jz]eć* (*spo[jz]eć*, *do[jz]eć*, *prze[jz]eć*), and *še[js]et*, respectively. In present-day dialectal Polish prevocalization is noted by Broch (1910), who observes that the onglide is most salient after back nonhigh vowels /a, o/ and less so after /u/ (218). In the southern part of Greater Poland, prepalatalization affects [ɕ], [ʑ] in medial groups *-śř-* [ɕʐ], *-źř-* [ʑʐ], *-śtrz-* [ɕtʂ], *-ździ-/-źdź-* [ʑdʐ], cf. *we^{j}ździć*, *po^{j}śčotku* ~ *po^{j}sčotku*, *za^{j}źžoł*. Prepalatalization is also reported word-initially, e.g., in *^{j}(s)šoda*, *^{j}(s)šybło*; unfortunately, the source (Koneczna 1965) does not make it clear whether word-initial prepalatalization requires the presence of a preceding vowel across the word boundary. Prepalatalization at word boundaries is also attested, and accompanied by deletion of the triggering consonant, cf. *be^{j} źime* (*be^{j}* < *beź*) 'without winter', *ve^{j}že* (*ve^{j}* < *veź*) 'take (and a particle)' (Koneczna 1965:150ff). In western dialects of Polish, where the tendency towards prepalatalization is more pronounced, it may involve single consonants in addition to consonant groups, cf. *za^{j}ś*, *ta^{j}k^{j}e*, *sto^{j}g^{j}i*. In Krajna, the palatalized consonant triggers simultaneous on- and offglides, cf. *z výe[^{j}lk^{j}]am trudem*, *po[^{j}v^{j}]am*. In addition, many varieties of Polish regardless of their dialectal base (pre)palatalize /ɲ/ in coda. The C-gesture of the nasal is deleted word-finally and before fricatives but is assimilated to the following affricate or stop (Biedrzycki 1963; Koneczna 1965:151f), cf.:

hańba	/xaɲba/	[xãj̃mba]	'shame'
bańka	/baɲka/	[bãj̃ŋka]	'jar'
słońce	/swoɲtse/	[swõj̃ntse]	'sun'

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In Upper and Lower Sorbian, prepalatalization is attested before palatalized and palatal consonants, cf. *wʲečor* 'evening', *sʲedźi* 's/he sits', *zʲemja* 'earth' (Stieber 1956:17; Koneczna 1965:151). In Czech, prepalatalization is said to be especially pronounced in the north-western belt of the Las dialects, cf. *dʲevʲet* 'nine', *sʲedźeł* 'he sat' (Lehr-Spławiński and Stieber 1957:116, 119, 140); in other Czech dialects prepalatalization is noted before *š* and *ž*. Old Czech spellings indicate that prior to their simplification, the relevant medial consonant clusters were prepalatalized (*naykrayššieho* for *naykrašširho*, *zaizzen* for *zažžen*) (Koneczna 1965:151). Broch (1899) notes prepalatalization in a rural Ukrainian dialect from Hungary (26). He observes that the prepalatalization onglide in that dialect differs in quality depending on the location of stress and the degree of aperture in the preceding vowel. The onglide is said to be the strongest after open stressed vowels, especially /a/, in a position before voiceless dental stops. In unstressed syllables, the onglide is either weak or absent. Broch also comments on the unstable nature of prepalatalization and the difficulty for determining precise rules for its appearance (1899:36ff). Finally, Mladenov (1929:136, 154) cites forms like *koʲn* ~ *koʲń* 'horse', *nóʲkʲ* 'night', *imáʲne* 'name', *kaméʲne* 'stone' for *koń*, *nókʲ*, *imanje*, *kamenje* in the Bulgarian dialects of Macedonia. He also quotes Middle Bulgarian *tajta* 'thief (accusative singular)' corresponding to Old Church Slavic *tatĭ* (nominative). Forms with prepalatalization after /u/, such as *druʲkʲ*, *kùʲkʲa* and *sindùʲkʲ* are cited in Mirchevŭ (1901).[56]

[56] The Bulgarian dialects of Macedonia are adjacent to Romanian, which attests prepalatalization in

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3.2.2.3 Greek

In Greek, prepalatalization forms part of the complexes of phenomena known as the first and second compensatory lengthenings. The latter process will be discussed in 3.4.3; the discussion here is limited to prepalatalization that forms part of the first compensatory lengthening.

The first compensatory lengthening resulted from the treatment of intervocalic clusters of sonorant plus yod. One type of change, shared by all dialects, looks superficially like metathesis and has been viewed as either prevocalization or metathesis by different scholars. The change in question may be schematized as *Sj* > *jS* / *{a, o}* __, where *S* = /r, n, w(?)/, cf.:

**daw-jo:* > **daj-wo: (?)* > *dajjo:* 'kindle'[57]
**phan-jo:* > *phajno:* 'show'
**g^{w}ṃ-jo:* > **g^{w}an-jo:* > *bajno:* 'go'
**mor-ja* > *mojra* 'lot'

If the sonorant preceding the palatal glide is a /l/, transpositions are attested only in Cypriote and Elean, while in the rest of the dialects the lateral is geminated, cf. Cypriote *a-i-lo-ne* /ajlo:n/ 'other (genitive plural)' and *a-pe-i-lo-ni* /apejlo:ni/ 'Apollo (dative)' with Attic-Ionic ἄλλων, Απόλλωνι (Lejeune 1972:156; Rix 1976:61; Panagl 1981:329).[58] In the case of vowels other than /a, o/, the result is a compensatory

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similar conditions. Close geographical proximity and similarity of prepalatalization in the two languages have led Ivănescu (2000) to advance an areal explanation for the phenomenon (407, 417).

[57] The details of the development of the *wj sequence are unclear, and in what follows the discussion is limited to /n, r, l/.

[58] Unless ιλ and λλ are simply different ways of spelling palatalization, which would imply that depalatalization of the lateral did not involve prevocalization (cf. Lejeune 1972:156 n. 1).

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lengthening of the preceding consonant (in Lesbian and Thessalian) or vowel (everywhere else), cf.:

**klin-jo:* > *klin:o:* (Lesbian/Thessalian), *kli:no:* (elsewhere) 'tend'
**awer-jo:* > *aer:o:* (Lesbian/Thessalian), *ae:ro*: (elsewhere) 'lift'

Both metathesis and prevocalization have been suggested in order to account for the first set of the above changes. Metathesis is posited, among others, by Lejeune (1972), Kiparsky (1967), and Steriade (1982), and prevocalization through an intermediate palatalized stage of the sonorants by Danielsson (1903), Wetzels (1985), and Hock (2003a). Kiparsky (1967), for example, has the palatal glide and *h (< *s) behave as a natural class metathesizing across the class of sonorants /l, r, m, n, w/. Building on Kiparsky's analysis, Semiloff-Zelasko (1973) suggests that the glide metathesis is driven by the attraction of the glide toward the vowel nucleus. In doing so, she exploits the concept of sonority hierarchy suggesting that "the preferred syllable position for glides as a class is immediately following the vowel" (1973:71f). However, in order for this explanation to work it has to be shown that /h/ is more sonorous than each of the sonorants /l, r, m, n, w/ that it is said to displace from their position next to the vowel (evidence for this is not provided in the paper). In addition, this analysis takes it for granted that the glides in Greek are higher on the sonority scale than both liquids and nasals; however, Woodard (1994) shows, on the basis of a study of Greek historical phonology and the spelling practices of the syllabic scripts, that liquids in Greek are more sonorous than glides, which makes the sonority-based explanation of the metathesis doubtful.

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A prevocalization analysis is to be preferred over the metathesis account for at least two reasons. Firstly, Proto-Greek consonants underwent an across-the-board palatalization when followed by yod. At the earliest stages accessible to observation Greek is no longer a palatalizing language, but it preserves traces of palatalization in the form of coronal articulations in the case of velar and coronal obstruents, and hardening of the palatalization offglide in the case of $*p^j$. In the case of /r, n/, on the other hand, what is available for observation is an apparent transposition of segments after /a, o/ and compensatory lengthening after other vowels. The most economical explanation of these facts is that the sonorants were likewise palatalized, that and the sequences /jr, jn/ as well as the compensatory lengthenings represent the outcome of their depalatalization. When viewed against the background of other simultaneous depalatalization processes of Proto-Greek, prepalatalization simply becomes a specific strategy of palatalization loss restricted to a subset of sonorants. Secondly, a prepalatalization account is better able to cope with the fact that the glides are not attested after vowels other than /a, o/. The greater articulatory distance between these nonfront vowels and the palatal onglide lead to a greater perceptual saliency of the onglide, which ultimately results in its phonologization (cf. Wetzels 1985; Hock 2003a). The failure of the transpositions *rj > jr and *nj > jn to occur in the case of other preceding vowels receives no satisfactory explanation if one assumes metathesis instead.

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3.2.2.4 Romance languages

3.2.2.4.1 Preliminaries

The history of the Romance languages is shaped to a significant extent by the palatalizing tendencies inherited from Vulgar Latin. Inscriptional evidence indicates that at the beginning of the Christian era Latin had the beginnings of palatalization, especially in velars before front vowels and in clusters *lj*, *nj*, *-cl-*, *-gl-*, *-ct-*, *-cs-* (Nandriş 1952; Straka 1965:118). Later on, palatal glides that resulted from desyllabification of *e*, *i* in hiatus as well as from the groups *dj* and *gj* palatalized preceding *k*, *l*, *n*, *t*, *s*, *r*. Towards the period of the individual Romance languages, these developments created a whole series of palatalized or palatal[59] articulations. For instance, the palatal(ized) inventory of Gallo-Roman given by Pope (1934) includes k^j, g^j, t^j, d^j, s^j, l^j, n^j, r^j (see 3.2.2.4.3).[60] Some researchers believe that the Proto-Romance consonant system was based on an opposition of phonemic palatalization (Burger 1955; Petrovici 1956) and have compared the rise of palatalization in Romance with the parallel development in Slavic (e.g., Burger 1935:136; 1955:26; Straka 1965:118ff). Although the existence of a systematic opposition of phonemic palatalization in Proto-Romance is not accepted by all experts, the existence of a palatalizing stage in the histories of the individual Romance languages is indicated by numerous traces in the form of palatal articulations,

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[59] The surface phonetics of these reconstructed segments is a matter of conjecture; some of the segments notated here with a secondary palatalization may have been palatal.

[60] According to Straka (1965), the order of palatalization in Gallo-Roman was as follows: (1) velars [k, g / __ j > k, g / __ e, i > k, g / __ a], (2) alveolars [n, l / __ j > t / __ j > s / __ j > r / __ j], (3) labials / __ j (132).

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hardening of the palatalization offglide, and/or diphthongs that arose out of vowel combinations with prepalatalization onglides.

Prepalatalization in the Romance area is frequently attested in clusters *-rj-*, *-s(s)j-*, labial plus yod (Lausberg 1965:386ff), and in clusters of stops followed by coronal consonants. In some clusters of the latter type the initial consonant did not undergo fronting, which resulted in the development of a labiovelar glide. For example, much of South Italy attests the development *ŋn* > *wn* (Keller 1943), Portuguese has *kt* > *wt*, and Occitan *pt*, *ps* > *wt*, *ws*. In other instances, the initial member of such clusters was fronted prior to its (pre)vocalization, cf.:

pt > *jt*	*captivu* > Old Provençal *caitiu*, Old French *chaitif* 'captive'
ps > *js*	*capsa* > Old Provençal *caissa* (Straka 1949)[61]
kt > *jt*	*lacte* > French *lait*, Portuguese *leite* 'milk'
gd > *jd*	*frig(i)du* > Ligurian *freidu* 'cold'
gr > *jr*	*flagrare* > Portuguese *cheirar* 'smell'
ŋn > *jn*	*regnum* > Portuguese *reino* 'kingdom'
tr > *jr*	*patre* > Ligurian *paire* 'father'
dr > *jr*	*quadru* > Ligurian *kairu* 'square-shaped rock'

A frequent case in the above category is (pre)palatalization of velars, which is also widely attested outside of Romance (e.g., in Italic, Germanic, Celtic, and Indo-Iranian). *kt* > *jt* and similar developments involving velars are widely regarded as resulting from a sequence of fricativization, fronting/palatalization, and gliding of the velar, i.e., *kt* > *xt*

[61] There exist various explanations for the palatal glide outcome of the labial in *pt* and *ps* (cf. Straka 1949). It is possible that, prior to its vocalization, the labial stop was palatalized by the following sibilant (the palatalizing effect of /s/ on adjacent consonants is considered in detail below). Cf. also [pejsi] for *Pepsi* in Mexican Spanish.

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> *çt* > *jt*.[62] The gliding stage is directly attested in various languages, cf. popular and dialectal Spanish *aición* (*acción*) 'action', *doitor* (*doctor*) 'doctor' (Cuervo 1898:302ff; Delattre 1966:189f). The fronted or palatalized stage of the fricativized velar is also confirmed by spelling evidence, cf. use of <h> for /(h)i/ in Old Persian (Kent 1950:14f) or <h> for /i/ in Siculian (Schmoll 1958:26ff) and Old English (Campbell 1959:21, 181). Consequently, the literature on the development of *-kt-* and similar clusters rarely addresses the alternative possibility of a development through prevocalization and deletion of the velar, i.e. *kt* > *çt* > *ʲçt* > *jt*. Nevertheless, such a development is not only possible but also plausible in view of the synchronically attested Tsakonian forms like oʲχτó [oʲçto], as well as prepalatalization and subsequent deletion of /x/ after stressed front vowels in Middle English, cf. Old English *feohtan* > Middle English *feighte(n)* 'to fight' (more on this development in 3.2.3.5).[63]

3.2.2.4.2 Balkan Romance

Palatalization, whether phonemic or allophonic, is frequently accompanied by prevocalization in the Balkan Romance languages. The phonological system of standard

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[62] Planta (1892:356ff); Meiser (1986); Meyer-Lübke (1925-26:644f); Pope (1934); Azaretti (1977:92).

[63] A group of existing theories do suggest prepalatalization as the source of the glide; all of them presuppose that such clusters first developed an intermediate palatalized or palatal stage. Thus, Thomsen (1875) suggests an initial development *kt* > **tʲtʲ*; Meyer-Lübke (1925-26) makes *kt* first develop to **tʲ*, from which he derives the actual Romance forms through depalatalization (**tʲ* > *t*) or coronalization (*tʲ* > *tʃ*); Lausberg (1965) reconstructs a very similar intermediate form to account for Spanish *leche* 'milk': *lacte* > [*lajtʃe] > *leche* (381 fn. 71); and Tillander (1963) proposes an intermediate palatal stage in the development of *-ks-/-sk-* in the Iberian languages, basing his theory in part on the evidence of early spellings like *possedeixes* that seem to indicate the palatalization onglide, and *possedeisse* that apparently indicate depalatalization of an earlier palatal (79).

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contemporary Romanian, for instance, shows the effects of two chronologically distinct periods at which prepalatalization was operative. The first of these is Proto-Romanian, in which most consonants of Balkan Latin underwent palatalization before front vowels or a palatal glide (*k*, *t* > *ts* ~ *tʃ*; *d* > *dz* ~ *dʒ*;[64] *g* > *dʒ*; *s* > *ʃ*; *n* > n^j; *l* > l^j). In the case of labials *p*, *b*, *f*, *v*, *m*, the palatal glide remained after preceding unstressed vowels but is reflected as a diphthongal offglide after stressed vowels (Coteanu 1969:200; Nandris 1963:222; Ivănescu 2000:208), cf.:

*ˈscupio > scuip
ˈhabea(t) > *abja > aibă
diˈffamia > defaimă
ˈcoefea > coif
but: appropiˈare > apropia

A number of researchers explain the development of the palatal glide before labials in Romanian and/or parallel developments in other languages by suggesting that labials are more resistant to palatalization than lingual consonants (cf. Hock 1991, 2003b); and similar claims have been made with respect to /r/ (Hall 2000). The arguments advanced in support of this view include relatively low frequency of palatalized labials and rhotics as compared with that of palatalized nonlabials and nonrhotics, and frequent failure of both labials and rhotics to undergo synchronic palatalization. If palatalized, both classes of segments are quickly depalatalized, e.g., by losing the secondary palatalization, spirantization (cf. /r^j/ > [ʒ] in Polish) or coronalization (cf. *p* > *ʃ* in French *sache* 'that I know', from Latin *sapiam*), as well as pre- and postpalatalization

[64] The type of the affricate is determined by the preceding vowel (Coteanu 1969:203).

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(cf. /p^{j}/ > [pj] ~ [jp] in Modern Irish and Scottish Gaelic. Under this view, palatalization in Cj clusters in Gallo-Roman French, for example, is believed to have affected the lingual consonants to the exclusion of labials (Pope 1934:129). By the same token, the above outcomes in Romanian are viewed as simple metatheses.

However, it is much more economical to assume that in languages with an across-the-board palatalization both labials and rhotics are palatalized with the rest of the consonants and then undergo a fairly rapid depalatalization. This assumption is suggested by cross-linguistic evidence on the dynamics of the palatalization/depalatalization processes. For example, in Tocharian A, present-day Polish, and Irish and Scottish Gaelic, all of which possess contrastive palatalization, labials do not have a simultaneous secondary palatalization gesture. Nevertheless, it is clear that in each of these languages palatalized labials were present in the system at an earlier stage: traces of labial palatalization are preserved in the form of synchronic alternations, influence on adjacent segments, and/or pre- and postpalatalization (cf. 3.2.3.3).

An additional reason to think that the above developments in (Daco-)Romanian represent depalatalization of labials rather than simple metathesis is the fact that after the split of Proto-Romanian, labials were palatalized before a [i] or [j] in the individual Balkan Romance languages, and the fact that the newly arisen palatalized labials were rather unstable. In three of the languages, the palatal offglide of the labials is hardened, or consonified, which sometimes leads to elimination of the labial gesture (Macrea 1936-38 et seq.; Vasiliu 1968:119f; Caragiu 1975:162ff), cf.:

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$p > pk^j \sim pt^j \sim pt\int \sim k^j \sim t\int$
$b > bg^j \sim bd^j \sim bd\mathfrak{z} \sim g^j \sim d\mathfrak{z}$
$m > mn^j \sim n^j$
$f > x^j \sim \int \sim s$
$v > \dot{z} \sim \mathfrak{z} \sim z$

This process, traditionally known as palatalization of labials (*palatalizarea labialelor*), apparently started out as fricativization of the labial offglide; written evidence indicates that it began around the sixteenth century and first applied to [f^j] (cf. Rosetti 1926:90f; Vasiliu 1968:121).[65] The process is completed (i.e., presents no intermediate stages) in Aromanian, but it is still ongoing in Megleno- and Daco-Romanian.[66] Forms like /kopil/ [kopk^{j}il] 'child' and /bine/ [bg^{j}ini] 'well' (Caragîu 1975:162), where palatalization of the labial is induced by a front vowel, clearly show that the object of consonification here is a palatalization offglide rather than a segmental palatal glide. Incidentally, parallelism with the development of labial plus yod clusters in Gallo-Roman (see next section) makes it likely that developments of the type *rubeu(m)* > **rubju* > **robdʒe* > *rouge* 'red' in that language likewise involved consonification of palatalization offglides rather than of segmental palatal glides, as traditionally assumed (cf. related discussion in Hock 2003b). More relevant to the issue at hand is the fact that consonification of the offglides proves the unstable nature of palatalized labials in

[65] Earlier dates for the palatalization of labials have also been suggested (cf. Ivănescu 2000:338). As similar phenomena are found in southern Italian, Rhaeto-Romance, Provençal, French, Neo-Greek of Chios, and sporadically also in Albanian and Dalmatian (Sala 1970:29; Lazarou 1986:167), there have been attempts to explain palatalization of labials in Balkan Romance as a result of substrate influence (see references in Sala 1970:28 fn. 27).

[66] Cf. Rosetti (1926:111ff), Macrea (1936-38), Vasiliu (1968:120f), Sala (1970:29), Caragiu (1975:231, 272f), Lazarou (1986:167). Istro-Romanian does not attest palatalization of labials.

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Balkan Romance. Given the cyclic nature of the palatalization/depalatalization processes and the periodicity of the phonological tendencies in a language, it is possible to conjecture not only that Proto-Romanian had palatalized labials, but also that they were unstable and depalatalized relatively quickly through prepalatalization.

The second wave of prevocalization is the source of the falling diphthongs *aj*, *ej*, *ɨj*, *oj*, *uj* in standard and dialectal Romanian, and Aromanian (Rosetti 1978:404f). Judging by the forms most frequently cited in the literature, the affected consonants include /k, n, r/ (in Aromanian, also /l/) followed by /e, i, j/ and preceded by a stressed vowel. Apart from the forms with the diphthong [ɨj] (orthographically *âi)* that form part of the standard language, most of the examples cited in the literature are from dialectal Daco-Romanian (Caragiu 1975:172; Rosetti 1978:404ff; Ivănescu 2000:406ff), cf.:

ɨj	*câine* 'dog', *pâine* 'bread', *mâini* 'hands'
aj	*căraire* 'road', *cântaire* 'singing, song', *grainiță* 'border'
ej	*ureik^{j}e* (= *ureche*) 'ear'
oj	*oik^{j}* (= *ochi*) 'eyes'
uj	*minuine* 'marvel', *păduire* 'woods', *genuiche* 'knee'

This prevocalization type is attested in Daco-Romanian since the beginning of the sixteenth century, while in other Balkan Romance languages it appears to be more recent (for early documentation and discussion of the chronology cf. Rosetti 1926:84f, 143f and 1947:558ff). Ivănescu (2000:407, 412) connects this wave of prevocalization in Romanian with the parallel development in the Bulgarian dialects of Macedonia.

In present-day Daco-Romanian, prepalatalization is generally considered characteristic of Muntenia, Oltenia, and southern Transylvania (cf. Nandris 1963:90).

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Turculeț (1975) documents prepalatalization in the north-west of Moldova, where it seems to have arisen independently. According Turculeț (1975), prepalatalization was registered in thirteen out of the forty-one localities that were interviewed. In all of them, prepalatalized pronunciation is not generalized in the speech of the community or even the individual speakers; one and the same word can be produced by the same speaker with or without prepalatalization, the choice being apparently unpredictable. Among the factors that may influence the choice of a particular variant Turculeț cites phrasal stress, intonation, and tempo of speech. Individual speakers may generalize the prepalatalized pronunciation of certain words which would then constitute the "standard" pronunciation of those words in their idiolects. In over half of the examples, the prepalatalization onglide was registered as weak or hardly perceptible. In careful speech, for instance when the speaker concentrated on the answer, the onglide did not surface at all. The speakers were not conscious of prepalatalization in their own speech, nor noticed it in the speech of others. Among the prepalatalizing consonants and consonant groups Turculeț notes k^j, g^j, x^j, b^j, f^j, $ʃ$, $ʒ$, $tʃ$, t^j, d^j, n^j, l^j, r^j, *-tr-*, *-str-*, and *-nk*.[67] While the majority of the consonants in this list are either palatal or phonemically palatalized, some are contextually modified by the following *e*, *i*, *j*. Consonant groups *-tr-*, *-str-* are palatalized before the final *i*, and the nasal of the word-final group *-nk* is deleted after prevocalization. Prevocalization is attested both word-medially and across word boundaries, cf. *mă duc eu* /mə duk jew/ [mə du^{j}k^j jew] 'I am going'. In contrast to the dialects of Oltenia, for which the sources cite *a*, *e*, *o*, *u* (and only rarely *i*) as the

[67] b^j and f^j also appear in one example each.

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vowels that may precede prepalatalizing consonants, in north-western Moldova any vowel may be affected, in the order of frequency *u*, *o*, *a*, *ă* [ə], *e*, *i*, *â* [ɨ]. While phrasal stress is cited as a factor that may condition or disfavor prepalatalization, lexical stress is not cited as a conditioning factor; in this, Moldova Romanian is in sharp contrast to the other dialect areas for which lexical stress on the preceding vowel is cited as an important conditioning factor. Among other factors conducive to prepalatalization in north-western Moldova, Turculeț (1978) notes a strongly fronted articulation of k^j and g^j characteristic of the dialect, and a generally relaxed pronunciation of vowels together with their lengthening under phrasal stress.

3.2.2.4.3 Gallo-Romance

Consonant palatalization before the following palatal glide and, to a lesser extent, front vowels is characteristic of at least some varieties of present-day French; historically, consonants were likewise palatalized in these environments as well as before coronal consonants. In Gallo-Roman (5th-9th centuries A.D.), this tendency resulted in the creation of a series of palatal and palatalized articulations, of which only /ʒ, ʃ, ɲ/ and, regionally, /ʎ/, persisted into Modern French. In contrast to languages like Estonian or Russian, in which the main acoustic correlate of palatalization is a [i]-like on- or offglide, respectively, consonant palatalization[68] in present-day varieties of French is signaled primarily by assibilation, or affricated burst (Gendron 1966:120ff;

[68] Specifically, of the voiceless stops /p, t, k/.

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Belakovskaja 1973). Gallo-Roman palatalization included among its results assibilated realizations of earlier groups [kj, gj, tj, dj] and thus also was of the assibilating kind.[69]

According to Pope (1934), Gallo-Roman coronals were palatalized by the following palatal glide, and velars by the following palatal glide, front vowels, or coronal consonants (121ff). Fouché (1961) assumes that palatalization of velars resulted in palatal articulations, and that of coronals in palatalized ones (908f). Palatalization did not last long and the majority of the newly formed palatal and palatalized segments were depalatalized already by the end of the Gallo-Roman stage. Palatal stops were first affricated and then fricativized, while palatalized coronals were prepalatalized. Subsequently, the palatalized coronals were depalatalized, and the palatal onglide merged with the preceding /i/ and formed a diphthong with other vowels. Prepalatalization processes in Gallo-Roman may be summarized as r^j, s^j, z^j, t^j, d^j, n^j > *jr*, *js*, *jz*, *jt*, *jd*, *jn* (Pope 1934:92). Prepalatalization-related developments of the Gallo-Roman and some later periods of French are detailed below in (a) through (d).

(a) Velars and consonant groups ending in velars before front vowels:

k	> *jz* / __ {*i, e*}	*vicinu* > *voisin* 'neighbor'
	> *js* / __ {*i, e*}*t*	*jacet* > **dʒia*^{j}*st* > Old French *gist* 'lies'
	> *jts* / __ {*i, e*}#	*nuce* > Old French *noiz* 'nut'
sk	> *js* / __ {*i, e*}	*nascere* > Old French *naistre* 'be born'
ng	> *jn* / __ {*i, e*}	*plangere* > *plaindre* 'complain', *longe* > *loin* 'far'

[69] This is yet another example of a cyclic application of a phonological tendency inherent in a linguistic system (cf. Fouché 1961:908ff).

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(b) Velars and consonant groups ending in velars before coronals:

k	> *j* / __ *t*	*vectura* > *voiture* 'car'
	> *j* / __ *s*	*coxa* > *cuisse* 'thigh'
g	> *j* / __ *d*	*frigidu* > *froid* 'cold'
k, g	> *j* / __ *r*	*lacrima* > *lairme* 'tear'
k, g	> *j* / __ *l*	*oculu* > **ojlo* > Old French *ueil* 'eye'
nk	> *j* / __ *t*	*sanctu* > *saint* 'holy'
ŋ	> *j* / __ *n*	*pugnu* > *poing* 'fist'

Changes in (b) above are frequently interpreted as direct vocalization of the velars, which are assumed to have been fronted before coronals and then weakened (cf. Fouché 1961:714, 816 and the discussion in 3.2.2.4.1).[70] If this were an isolated development in French, in the absence of information on the intermediate stages it would have been impossible to decide whether it was due to prevocalization and subsequent deletion of the consonantal gesture of the velars, or their direct vocalization (De Chene and Anderson 1979, for example, discuss the two possibilities as equally plausible: 510 fn. 5). However, a comparison with the processes in (a) suggests that a prevocalization account is at least marginally more attractive than an account in terms of a direct vocalization. Both processes began with fronting of velars and had as their outcome diphthongization of preceding vowels. In (a), the modified velar persisted when in onset but was deleted when in coda (cf. Old French *gist* 'lies', *naistre* 'be born' > Modern French *gît*, *naître* with *voisin* 'neighbor'). If the process in (b) is interpreted along the same lines, deletion of the velar can be readily explained by its location in a coda. The two processes are thus parallel and it would be more economical to assume that the

[70] Velar palatalization and gliding before the following coronals also took place in the Brittonic languages (cf. Jackson 1953:404ff), and the corresponding development in French is often seen as the result of Celtic substrate influence (cf. Tagliavini 1972).

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same mechanism was at work in both. Incidentally, fronting of velars before coronals, together with their weakening and prevocalization, are not cross-linguistically uncommon; one concrete example of this is the form oʲχτό (phonetically [oʲxʲto] or [oʲçto]) 'eight' cited by Pernot (1894:86).

(c) Coronals and consonant groups ending in coronals *t, d, s, r, (n, l)* before a palatal glide:

t	> *jz / V __ jV*	*potione* > *poison* 'poison'
	> *js / V __ #*	*palatiu* > *palais* 'palace'
st, sk	> *js / __ j*	**postius* > *puis* 'then', **piscione* > *poisson* 'fish'
(n)s	> *jz / __ j*	*mansione* > *maison* 'house'
ss	> *js / __ j*	**grassia* > *graisse* 'fat'
d	> *jt / __ j*	*medietate* > *moitié* 'half'
r	> *jr / __ j*	*coriu* > *cuir* 'leather'
pr, br	> *jvr / __ j*	**copriu* > *cuivre* 'copper', *ebriu* > **ieivre* > *ivre* 'drunk'
(s)tr	> *j(st)r / __ j*	*ostrea* > Old French *uistre* 'oyster'

Intervocalic /ɲ, ʎ/ are believed to have prevocalized in Gallo-Roman, but the prevocalization onglide merged with the palatal rather than forming a diphthong with the preceding vowel (Fouché 1961:257, 917), cf. *palea* > **paʲʎa* > *paille* 'straw', *montanea* > **mongaʲɲa* > *montagne* 'mountain'.[71] Prepalatalization of intervocalic /ʎ, ɲ/ is inferred from the failure of the preceding stressed /a, e, o/ to diphthongize (Zinc 1986:100) and from the existence of prevocalization onglides in dialectal and Early Old French. Some of the prevocalized forms have become part of Standard French; among these are *cuillère* 'spoon' (< *cochlearium*) and the prevocalized variant of the suffix

[71] But cf. Straka (1965:145) who assumes no prepalatalization in this case.

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–agne that appears in *chataîgne* 'chestnut' (< *castanea*), *musaraigne* 'shrew' and *araignée* 'spider' (the last two are from *araigne* 'spider' < *aranea*, cf. Pope 1932:161f; Fouché 1961:387, 919). Palatal /ɲ, ʎ/ are generally prevocalized when word-final or preconsonantal (Fouché 1961:919, 924), cf.:

consiliu > *conseil* 'advice' (regionally [-ejl])[72]
balneu > **ban^{j}* > *bain* 'bath'
nuntiu > Old French *noinz* 'messenger'

(d) Traditional accounts of Gallo-Roman palatalization assume that the labials failed to be palatalized by the following palatal glide (Pope 1934:129). This view derives primarily from the development of internal clusters of labials plus yod, which appears on the surface as consonification of the yod with deletion of the labial, cf.:

pj > * *ptʃ* > *ʃ*[73]	*appropiare* > *approcher* 'approach'
bj > **bdʒ* > *ʒ*	*rubeum* > *rouge* 'red'
vj > **vdʒ* > *ʒ*	*cavea* > *cage* 'cage'
mj > **ndʒ* > *ʒ*	*simium* > *singe* 'monkey'

Despite the traditional view, and as has been argued above for a similar situation in Balkan Romance (3.2.2.4.2), it is more economical to assume that labials were palatalized with the rest of the consonants at the time of the across-the-board consonant palatalization in Gallo-Roman.[74] As argued by Hock (2003), palatalized labials may be

[72] The phonetic, rather than orthographic, nature of the diphthong spelled <ei> is shown by its subsequent development to /oi/ (Rheinfelder 1968:116).

[73] Intermediate stages are warranted by the evidence of closely related Occitan and Romanian, as well as cross-linguistic evidence.

[74] This is posited, e.g., by Fouché (1961:925).

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more marked than palatalized coronal or dorsal consonants, which may be the reason for their speedy elimination from the phonemic system of Gallo-Roman. The marked status of palatalized labials may be connected with the fact that the palatalization gesture is executed separately from the primary constriction gesture. This opens the possibility of asynchronous realizations of the two gestures, which may eventually result in pre- or postpalatalization, with the possibility of offglide consonification.[75] Palatalization of labial consonants and their subsequent removal from the language's phonology are securely attested in Tocharian A, Polish, and Irish and Scottish Gaelic, providing important typological support for the assumption that Gallo-Roman labials passed through a palatalized stage as well. In addition, Hock (2003) argues that consonification of the palatal glide in historical labial-yod sequences is restricted to palatalization offglides and does not apply to a segmental yod. If cross-linguistically valid, this piece of evidence provides a welcome support for the existence of palatalized labials in Gallo-Roman.

Clusters of consonant plus yod in learned and semi-learned vocabulary that entered Old French after the palatalizing period also show interchange of the segments, variously analyzed as metathesis (Pope 1934:230, Fouché 1961:942) or prevocalization (Straka 1965:145 ft 55). Examples include the following:

lj > *jl*	*apostolje* > *apostoile* 'apostle', *oleum* > *huile* 'oil'
nj > *jn*	*Antoniu* > *Antoine*
rj > *jr*	*memorje* > *memoire* 'memory'
ƒj > *jf*	*graphiu* > Old French *graife* 'graft'

[75] Such a view is held, e.g., by most Slavists with respect to the "epenthetic lʲ" (cf. Stieber 1979:75).

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Since these words entered the language after the palatalizing period was over, a metathesis analysis of the adaptation pattern is more attractive than an analysis in terms of prevocalization. The reason for the metatheses is probably to be sought in the conditions in which the borrowing of such words took place. Known as learned or semi-learned borrowings, these and similar items were introduced into the language through the medium of writing by individuals who may be defined as bilingual in Romance and Latin, and who apparently based their adaptations of such borrowings on analogy with words like *cuire* 'leather' which, although of popular origin, preserved a transparent connection with their Latin sources.[76]

The developments in Occitan are in general similar to the ones outlined above for French. In Old Provençal, Latin *k, g, t, d, l, n* were coronalized or palatalized by the following primary or secondary yod, and labial plus yod combinations resulted in a consonified palatal glide. *Rj* and *s(s)j* combinations show prevocalization (*ss* may be original or the result of mergers in clusters *(n)ks, ps, sk, st, kt*) (Appel 1918: 86ff; Ronjat 1932), cf.:

bāsiāre	>	*baisa* 'kiss'
**ecclesia*	>	*glèiso* 'church'
**bassiare*	>	*baissar* 'lower'
**angustia*	>	*angoissa*, etc[77]

Latin *tj* and [k] before front vowels also occasionally develop into *jz* (medially) and *js* (*jtz*) (finally), cf. *raizo(n), pais* < *pace* 'peace', *ratione* 'reason' (Ronjat 1932:88f).

[76] It will be recalled that a similar situation obtains in Livonian with respect to recent borrowings from Latvian (cf. 3.2.2.1).

[77] Gascon has similar treatment of consonant plus yod clusters (Rohlfs 1970:159ff).

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Latin [kt], [ks] and Germanic *ht* gave *jt* ~ *jch* ~ *ch* according to the dialect, cf. *factu* 'deed' > *fait, facho* (Ronjat 1932 2:171, 184).

Prevocalization is also noted as a living process in present-day Provençal dialects. For example, Entraunes (Alpine) Provençal has a series of palatalized consonants [t^j, d^j, s^j, l^j, r^j, n^j, k^j, g^j] that result from assimilatory palatalization in contact with front vowels and palatal or palatalized consonants. This dialect is said to attest prepalatalization "à l'état de variation vivante" (Blinkenberg 1939). Blinkenberg describes prepalatalization onglides as being perceptually mid-way between a [i] and a [j], and prepalatalized pronunciation as the norm in some lexical items and optional in others. Blinkenberg's examples include [adret^j ~ adre^{j}t^j], [estryt^j ~ estry^{j}t^j], and [trut^{j}ɔ ~ tru^{j}t^{j}ɔ]. In the Eastern Provençal dialect of the village of Beuil palatalized consonants [t^j, d^j, s^j, l^j, n^j] are optinally prepalatalized "à la tension ou à la détente", cf. [pun^j ~ pu^{j}n^j] 'fist', [seri^{j}ed^{j}ɔ ~ seri^{j}e^{j}d^{j}ɔ] 'cherry' (Blinkenberg 1948). In one word, /rul^j/, the prepalatalization onglide has become lexicalized. Word-medially, no prepalatalization is attested with /n^j/, cf. [pun^{j}es] 'fists'.

In archaic Ligurian dialects, stressed vowels (except [i]) receive a palatal offglide when followed by the Latin clusters *rj*, *lj*, *sj*, and *tj* (> *sj*) (Azaretti 1972). The diphthongs are preserved in the dialect of Ventimiglia, cf.:

> *riparia* > *rivaira* 'torrent'
> *basiu* > *baiʒu* 'kiss', *ceresea* > *çereiʒa* 'cherry', *Ambrosiu* > *Ambröiʒu*
> *palatiu* > *paraiʒu* 'palace', *pretiu* > *preiʒu* 'price'

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These developments are to be analyzed as prevocalization rather than metathesis given that posttonic [i] has a similar effect on these consonants, and posttonic [i] or [e] on the original [k], cf.:

ad quasi > *ascaiʒi* 'almost', *basilikón* > *baiʒaricò* 'church'
heri > **eiri* > *iéiri* (initial *i-* through the influence of Italian *ieri*)
pace > *paiʒe* 'peace', *dece* > *deiʒe* 'ten'
voce > *vuʒe* ~ *vuiʒe* 'voice', *cruce* > *cruʒe* ~ *cruiʒe* 'cross'[78]

Archaic and Ventimiglia Ligurian also have diphthongs in which the palatal offglide comes from the initial consonant of Latin clusters *kt*, *gd*, *tr*, and *dr*, cf. *lacte* > *laite* 'milk', *pectu* > *péitu* 'chest', *quadru* > *kairu* 'square-shaped rock', *frigidu* > *freidu* 'cold'. As in the case of similar development in French (see 3.2.2.4.3), it is possible that the development of the glide is the result of prevocalization.

A couple of additional processes in Ligurian are of interest. In Old Genoese, a palatal glide is epenthesized before plurals ending in *–ni*, and possibly also those ending in *–li* and *–ri*. Diphthongs of this origin are only partially preserved in Modern Genoese, but better so in Ventimiglia Ligurian (Azaretti 1977:21, 33, 39, 54, 99):

christiani > Old Genoese *chrestiàin* > Modern Genoese *crestién* 'christians'
(cf. Ventimiglia Ligurian *crestiài*)

patroni > Old Genoese *patrùin* > Modern Genoese *patruĩn* 'masters'
(cf. Ventimiglia Ligurian *patrùi*)

Finally, as Italian and Romanian, Ligurian attests gliding of word-final [s], cf. *stas* > *(ti) stai* 'you (sg.) are', *nos* > *nui* 'we', *sex* > *sei* 'six' (more on this in 3.4.2).

[78] Ventimiglia Ligurian eliminates diphthongs with the closed vowels, but they are preserved in other dialects.

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3.2.2.4.4 Ibero-Romance

In Portuguese, prevocalization arose mainly in clusters of consonants followed by a Vulgar Latin yod. Clusters involving velars and most coronals produced coronal articulations, as follows: kj > [s], gj > [ʒ], tj > [s] ~ [z], dj > [s] ~ [ʒ], l > [ʎ], n > [ɲ]. Similar clusters of labials, /r/ and /s(s)/ resulted in a surface transposition of segments accompanied by palatalization in the case of /s(s)/ (Huber 1933:61, 103, 115; Williams 1962), cf.:

*sj > jʒ / V__V[79]	*basiu* > *beijo* 'kiss', *phasiolu* > *feijoo* 'bean'
*ssj, *psj, *sc > jʃ / V__V[80]	*russeu* > *roixo*, **capsiu* > *queixo* 'jaw-bone',
*rj > jr / V__V[81]	*primariu* > *primeiro* 'first'
*pj > jp / V__V[82]	*sapia(m)* > *saiba* 'that I know'[83]
*bj > jb / V__V[84]	*rubeu* > *ruivo* 'red-haired'
*mj > jm / V__V[85]	*calumnia* > *coima* 'fine'
*vj > jv / V__V[86]	*pluvia* > *chuiva* 'rain'
*fj > jf / V__V[87]	*cuphia* > *coifa* 'cap'

[79] The preceding vowel is a stressed or pretonic Vulgar Latin *a*.

[80] The preceding vowel may be a stressed Vulgar Latin *a*, *ō* or a pretonic *a*.

[81] The preceding vowel may be a stressed Vulgar Latin *a*, *ē*, *e*, *ō*, *o* or a pretonic *a*, *ē*, *ō*.

[82] The preceding vowel may be a stressed or pretonic Vulgar Latin *a*.

[83] The resulting diphthong *aj* rather than the more usual *ej* has been variously explained. One of the suggested explanations assigns these forms to a late working of this rule, another to the southern origin of *aj* as opposed to the northern *ej* (see summary in Williams 1962:29).

[84] The preceding vowel may be a stressed Vulgar Latin *a*, *ō*.

[85] The preceding vowel may be a stressed Vulgar Latin *a*, *ō*.

[86] The preceding vowel may be a stressed Vulgar Latin *a*, *ō* or an unstressed *a*.

[87] The preceding vowel may be a stressed Vulgar Latin *ō*.

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If the consonant in question was the final member of a cluster, the yod was deleted, cf. *atriu* > *adro* 'forecourt of a church' (Huber 1933:116).

In addition to transpositions involving the palatal glide of Vulgar Latin origin, Old Portuguese attests similar developments involving a secondary palatal glide arisen through the fall of intervocalic *d, g, l, n*, cf. *comedo* > **comio* > *coimo* 'I eat'. In later borrowings from Latin, some transpositions are attested in popular and dialectal Portuguese (Williams 1962: 34, 36, 82f), cf.:

daemoniu > *demónio* > *demoino* 'devil' (popular)
memoria > *memória* > *memoira* 'memory' (popular)
historia > *história* > *histoira* 'history' (popular)
Tīmotheu > *Timóteo* > *Temóito* (dialectal/Algarvio)

To summarize: transposition of consonant-yod sequences in Portuguese depends both on the nature of the consonant and on that of the preceding vowel. In the case of a preceding *e, ē* it is attested only with the *rj* sequence, while in the case of other consonants it is attested only after a stressed or pretonic *a* or a stressed *o*, as follows:

rj > jr / á, a, ẽ, é, ē, õ, ó, ō __
ssj, vj > jʃ, jv / á, õ, a __
sj, pj > jʃ, jp / á, a __
bj, mj > jb, jb / á, õ __
fj > jf / õ __

Similar transpositions are attested in the historical development of Galician (García de Diego n.d.:26, 56), cf.: *sapiam* > *saiba* 'that I know', *pluvia* > *choiva* 'rain', *calumnia* > *calomia* > *coima* 'fine', etc.

The development of internal consonant plus yod clusters in Spanish is similar to that found in Portuguese and Galician: *kj, tj* assibilate (*puteu, ērīciu* > *pozo* 'well', *erizo*

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'hedgehog'); *dj*, *gj*, *nj*, *lj*, *ssj* produce palatal articulations (*radiāre*, *fāgea*, *vinea*, *aliu*, *russeu* > *rayar* 'scratch', *haya* 'beech', *viña* 'vineyard', *ajo* 'garlic', *rojo* 'red'); *pj*, *bj*, *mj* mostly stay unchanged (cf. *apiu* > *apio* 'celery') but show surface transposition (with subsequent monophthongization of the root vowel) in a handful of verb forms including *sepa* 'that I know' < *sapia(m)*. *-sj-* and *-rj-* show surface transposition of the consonant and the glide, cf. *beso* 'kiss' < *basiu* and *cuero* 'leather' < **coiro* < *coriu* (Wireback 1997:33f; Penny 2002:53).

As in the case of other Romance languages with similar developments, transposition of segments in sequences of labials, /r/, /s/ and the palatal glide in Spanish and Portuguese received two alternative interpretations. One interpretation looks upon them as metatheses (e.g., Malkiel 1969; Penny 2002); the explanation invoked to justify this solution usually focuses on the alleged failure of these segments to palatalize. This explanation, though it might legitimately be argued for in the case of labials (cf. Wireback 1997) and /r/ (cf. discussion in 3.2.2.4 and Wanner 1989), does not hold water in the case of /s(s)/.[88] It also does not address the fact that these transpositions are sensitive to the nature of the preceding vowels, as noted above for Portuguese. The role played by the preceding vowels becomes clearer if one adopts the alternative explanation, namely, that such transpositions really result from prepalatalization of palatalized consonants. The fact that the preceding vowels are nonhigh and, except in the case of /rj/, also nonfront, is easily explained by their articulatory distance from the

[88] The metathesis analysis is rejected by Wanner (1989) who, nevertheless, also assumes that the labials and *r* are not affected by palatality (41ff).

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palatal onglide. The prepalatalization analysis is straightforward in the case of /s(s)j/ in Portuguese and Galician, where the outcomes remain palatalized suggesting a developmental sequence like *bāsiu* 'kiss' > *basʲo > *baʃo > *baiʃo > *beijo* (cf. similar interpretations in Thomsen 1875:113 fn. 2 and Tillander 1963:80). In Spanish, depalatalization in this and similar items may have occurred at the stage *basʲo, producing *bajso > *beso*; the form *beiso* is attested in western Leonese (Tillander 1948-49:91).[89] In the case of *rj*, the existence of a palatalized [rʲ] is indirectly confirmed by its development in Italian where the result is sometimes a [j], cf. Italian *aia* < *area* 'area' with Portuguese *eira* (and French *aire*) and Italian *cuoio* < *coriu* 'leather' with Portuguese *coiro* (and French *cuir*). Although there is no comparable external evidence for the prior existence of palatalized labials, their palatalization is suggested by systemic considerations, as outlined in 3.2.2.4.2.[90]

3.2.2.5 Insular Celtic

The Celtic family of languages is represented by the now extinct Continental Celtic (Lepontic, Gaulish, Celtiberian), and the Insular Celtic languages, the majority of which

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[89] In order to accommodate the palatal outcome of [sj] in Portuguese, Tillander suggests depalatalization of an earlier [ʃʲ] in Spanish and Western Leonese, i.e., *basiu* > *baʃo* > *bajʃo* > *bejʒo* (Portuguese) > *bejso* (Leonese) > *beso* (Spanish) (1948-49:91). However, an alternative interpretation in the form of quick depalatalization of [sʲ] in Spanish and Leonese (i.e., *sj* > *sʲ* > *js*) as opposed to its further palatalization to [ʃʲ] in Portuguese (i.e., *sj* > *sʲ* > *jʃ*) is at least equally plausible.

[90] Transpositions of consonant plus yod of the type *aigro* for *agrio* and *naide* for *nadie* in popular and dialectal Spanish are best viewed as metatheses (cf. discussion in 3.2.2.4.3 and Delattre 1966:204).

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survive. The latter branch consists of Goidelic (Irish and Scottish Gaelic, and the extinct Manx) and Brittonic (Welsh, Breton, and the extinct Cornish) (Macaulay 1996).

Continental Celtic is known from sparse epigraphic evidence, place names, and borrowings into better known languages. One suspected instance of prevocalization in the whole of the Continental Celtic corpus has been cited for Hispano-Celtic, where the orthographic <i> in the dative-locative singular *ToCoiTei* (Botorrita I A4, 10) and the genitive singular *ToCoiTos* (Botorrita I A1, 10) were analyzed by Schmidt (1986) as "epenthetic".[91]

Unlike Continental Celtic, most Insular Celtic languages securely attest prevocalization as a diachronic process, a synchronic phenomenon, or both. For example, many dialects of present-day Irish have distinctive palatalization, which is accompanied by the secondary features of lip-spreading as well as consonantal on- and offglides, which are said to be especially prominent with palatalized labials and after long back vowels. On- and offgliding of secondariliy modified consonants is such a salient phonological feature of contemporary Irish that most dialect grammars include a section on "transitional glides", or otherwise incorporate a description of such glides in their phonology chapters. In some dialects, the palatalization contrast in labials is expressed differently than in the lingual consonants in that, in palatalized labials, the front of the tongue is not raised towards the hard palate but stays in a neutral position. In this case, the accompanying transitional off- and onglides become the only mark of their palatalization (Oftedal 1963:71ff).

[91] Eska (1996) contends that the orthographic <i> is susceptible of alternative explanations.

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Examples of on- and offglides may be cited from Erris Irish, which has palatal onglides of varying phonetic quality between back vowels or the diphthong *əu* and palatalized consonants, cf. [su:^{j}l^{j}] *súil*, [m^{j}əu^{j}r^{j}] *meabhair*. Onglides to final consonants closing a stressed syllable are reported to be more prominent than onglides to the same consonants in a nonfinal position (Mhac An Fhailigh 1980:45ff). In West Muskerry Irish, there are palatal offglides between palatalized consonants and back vowels, and palatal onglides between back vowels and diphthongs ending in *u*, and palatalized consonants (Ó Cuív 1975:52-53), cf. [u^{j}l^{j}] *ubhaill*, [ʃau^{j}k^{j}] *seabhaic*, [ko:^{j}r^{j}] *cóir*. In Cois Fhairrge Irish, onglides to final consonants are more prominent than onglides to intervocalic consonants. Interestingly, the description notes not only on- and offglides between palatalized consonants and back vowels, but also onglides between long stressed front vowels and diphthongs ending in a front vowel, and final palatalized consonants, cf. [m^{j}i:^{j}n^{j}] *mín*, [m^{j}ai^{j}r^{j}] *meadhair* (De Bhaldraithe 1945:43-45). The differing phonetic quality of the palatal on- and offglides and its dependence on the adjacent nuclei can be appreciated from a description of the transitional glides in Tourmakeady Irish. In that dialect, the onglide to palatalized consonants after /u(:), o(:), a(:), uə/, and the offglide between a palatalized consonant and the following u(:) have a [i]-quality, cf. [s^{ij}u:^{j}l^{j}] *siobhail*, [kruə^{j}x^{j}ə] *cruaiche*, [ma^{j}x^{j}] *maith*, [do:^{j}b] *dóib*. The offglide from a palatalized consonant to the following /o(:), a(:)/ is, however, of the [e]-quality, cf. [b$^{j\varepsilon}$o:] *béo*.

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The palatalization contrast of contemporary Irish has its roots in the three consonant "qualities", or secondary articulations, of Old Irish: the [i]-quality or palatalization, the [a]-quality or velarization, and the [u]-quality or labialization (Thurneysen 1946:54ff). The existence of the different consonant qualities is inferred primarily from their effect on the adjacent vowels, which often takes the form of prevocalization onglides spelled <i> and <u> (in traditional terminology, i- and u-epentheses). The phonetic nature of the vowels <i> and <u> is sometimes questioned and they are interpreted as a mere graphic device to indicate secondary modification of the consonants; for example, this is the interpretation that these vowels receive in Pedersen (1909) and Lewis and Pedersen (1937). However, even a cursory comparison of the spelling conventions associated with the epenthetic vowels in Old Irish and the synchronic behavior of the "transitional glides" in the modern dialects suggests that the two entities should be recognized as part of the same phenomenon. In Old Irish, a palatalized consonant or group of consonants conditions the insertion of a graphic <i> after vowels or diphthongs except /i, i:, oi, oe, ai, ae/. Thurneysen interprets this vowel as a glide, and it fits the description of a prepalatalization onglide on several counts: (i) in that its combinations with preceding vowels are never confused with underlying diphthongs so that the syllable remains light, (ii) in that its presence is tied to lexical stress (it is consistently marked after stressed vowels and considerably less so after the unstressed ones), (iii) in that it occurs only before consonants in coda, and (iv) in that it is not notated after high front vowels and diphthongs ending in front offglides, which can be easily explained if the epenthetic vowel is phonetically real. In addition, Vendryes (1906-08) notes that *a*

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changes to *o* before an epenthetic <i>, just as it does in the underlying homophonous diphthong (395). Examples of the <i> onglide in Old Irish include *clainde* [klan^{j}d^{j}e] 'children (genitive)' and *doirsib* [dor^{j}s^{j}iβ] 'doors (dative)'. Labialized consonants condition the insertion of a graphic <u> after /a, e, i/, cf. *daum* [da^{w}m^{w}] 'ox (dative)'. This vowel also fits the description of a prevocalization onglide in that it is notated after stressed vowels in coda, and in that it is found only after nonlabial vowels -- in other words, in an environment where a rounded glide would be most salient. When the labialized consonant is found in an intervocalic position, <u> is mostly omitted (Thurneysen 1946:57). In addition to onglides to palatalized and labialized consonants, Old Irish also notates offglides from palatalized consonants to the following word-final /a, o, u/, which in consequence are written <ea, eo, iu>, cf. *toimseo* [tom^{j}s^{jj}o] 'measure (genitive)'. These offglides are just as sensitive to lexical stress as the onglides: when the final syllable is separated from the stress by at least one other syllable, they are omitted from writing (Thurneysen 1946:61f).

Among the Brittonic languages Welsh has the most evidence for prepalatalization, described in the traditional grammars under the heading of "*i*-affection". Jackson (1953) distinguishes between "final affection, which is caused by *ī* or *i̯* in a late British final syllable, and internal affection, which is caused by Pr[imitive] W[elsh-]C[ornish-] B[reton] *i*, *ī* or *i̯* . . . in what had been a British internal syllable" (579). While in some cases the "affection" is realized as fronting of the stressed vowel, in others it surfaces as prepalatalization, with the original vowel invariably realized as *e*. Word-finally,

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prepalatalization takes place in the case of nonhigh back vowels /a, o/ when the trigger of consonant palatalization is a palatal glide, and in the case of /a/ when the palatalization trigger is a *ī*; word-medially, it takes place in the case of /e, a, o/ when the trigger of palatalization is a palatal glide, as follows:

Final affection realized as prepalatalization

-aCī(-) > -eiC (or yC)	*brani > brein
-aC̭i- > -eiC	*aḽos > eil
-oC̭i- > -eiC (or yC)	Ambrosius > Emreis ~ Emrys

Internal affection realized as prepalatalization

-aC̭i- > -eiC-	*mab̭on > meibion
-oC̭i- > -eiC-	*oḓon > eidion
-eC̭i > -eiC-	*pencerḓeid > penceirdieid

Relatively few instances of prepalatalization are noted in Cornish and Breton, which has led Jackson (1953, 1967) to conclude that the degree of palatalization in these languages is weak by comparison with Welsh. Old Cornish has a few instances where the result of *i*-affection is spelled with a palatal offglide; this infrequent spelling is always attested before dentals, primarily dental continuants (Jackson 1953:592f). In Breton, the result of *i*-affection is spelled *ei* before Middle Breton *-z* (< Old Breton *-d̄*, *-th*, *-s*). Jackson (1967) lists only three securely attested instances of prepalatalization in the latter language: *aḽos > *eil* 'other', *taḽio- > *teil* 'dung', and *sas̭io- > *heiz* 'barley' (158).

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3.2.3 Contextually palatalized consonants

Allophonic palatalization may be triggered by a variety of adjacent segments, although the most efficient palatalizing environment seems to be when a single consonant is followed by a palatal glide. Contextually palatalized consonants are just as frequently subject to prepalatalization as those in which palatalization is phonemic. Languages discussed in this section include those in which palatalization is conditioned by the following palatal glide or front vowel (this includes a number of older and present-day Indo-European languages, Zapotec, Hausa, Sorowahá, Yucuna, and Uradhi), those in which palatalization is conditioned by preceding front vowels (including South American English, Irish English, and English-based creoles), and those in which it is conditioned by the following coronal consonants (including Tocharian A and Tsakonian Greek).

3.2.3.1 Iranian

Graphic prevocalization is one of the most characteristic features of Avestan orthography. The two kinds of prevocalization in Avestan are referred to in the traditional literature as *i*- and *u*-epentheses; the former describes the appearance of an orthographic *i* in front of certain consonants followed by *i*, *ī*, *e*, *ē*, or *ii* [j], and the latter refers to an orthographic *u* in front of a *r* followed by *u*, *ū*, or *w*.[92]

[92] *U*-epenthesis is taken up again in Chapter 4.

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I-epenthesis is most frequent word-medially; word-initially, it is restricted to the historical *θi̯ and *ri̯, cf. *iθiiejah-* 'solitude', *irista-* 'damaged' (cognate with Vedic *tyájas*, *riṣṭá-*) (initial **re-*, **θe-* are not attested in Avestan). Word-medially, not all consonants are equally capable of triggering *i*-epenthesis, and not all of the possible vocalic triggers can create an equally powerful conditioning environment. Specifically, mid front vowels are weaker triggers of *i*-epenthesis than high front vowels and the palatal glide. Thus, while *i*-epenthesis across dentals is regular before *i*, *ī*, *ii*, it is sporadic before *e*, *ē*; final *-e* does not induce *i*-epenthesis when the nucleus preceding the consonant is the diphthong *aē*; in the ending *-ne* *i*-epenthesis is notated only after *a*; and in certain sequences, for example *-ēte* and *-ēθe*, *i*-epenthesis is unattested.

Medial *i*-epenthesis is attested with dentals, labials *p*, *b*, *β*, as well as the clusters *n̥t*,[93] *rm*,[94] and Old Avestan *db*; examples include Old Avestan *bara*^{i}*tī* 's/he bears', *a*^{i}*bī* 'to, towards', and Young Avestan *ka*^{i}*ne* 'maiden' (cf. Vedic *kanyā*), *kiriie*^{i}*ti* 'it is being done'. *I*-epenthesis in the sequences *ti*, *tī*, *tii*, *θi*, *θī*, *θii* and *ði*, *ðī*, *ðii* is notated regardless of the quality of the preceding vowel; this includes anaptyctic *ə* (the full set of possible preceding vowels is thus *a*, *ā*, *e*, *ē*, *o*, *u*, *ū*, *ə*, *ə̄*). When the preceding vowel is a *i*, it develops into a schwa under the influence of the epenthetic *i*, and merges with

[93] Cf. present participles with the suffix *-n̥t-* cited in Swennen (1995). The symbol *n̥* is used for the nasals **n*, **m* word-medially preceding a stop, and sometimes word-finally before an enclitic (Testen 1997:586).

[94] Two forms are cited in Narten (1967-68):247.

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the homophonous sequence arising in the same environment from an anaptyctic schwa (Hoffman and Forssman 1996:53), cf. the following Old Avestan forms:

ānəi̯ti- 'immobility (?)' < *ən-i̯ti- < *an-iti-
āskəi̯ti- 'escort, retinue' < *āsk^{əj}ti- < *ā-skti-

di, dī, dii always yield *i*-epenthesis except in cases of a clear morpheme boundaty, as in Late Avestan *ā-diδaiia.*

Clusters capable of triggering *i*-epenthesis include *ṇt, rm,* and Old Avestan *db.* The vowels *a, e* regularly admit *i*-epenthesis in front of the ending *-ṇtī*; two words show *i*-epenthesis in front of the sequence *rm.* Old Avestan word-initial cluster *db-* is represented graphically as <*daib*>, which is assumed to be the result of the development **dbi* > **dəbi* / **dabi* > *daib.* Word-medially, this sequence is written without the *i*-epenthesis.

In front of *p, i*-epenthesis is attested only in the preposition *aipī* 'over, across, during; after' and in compounds which include *aipī* or *an-aipī* 'NEG-*aipī*' as the first member. *β* regularly triggers *i*-epenthesis when it precedes a *i* or a *ii.* In front of *b, i*-epenthesis is attested regularly in Old Avestan and in Old Avestan borrowings in Young Avestan. It is also regularly attested in the *b*-cases of *a*-stems, both in Old and in Young Avestan, cf. *-aēibiia, -aēibiiasca, -aēibiiō, -aēibiš* < Indo-Iranian **-aibjā,*-aibjas, *-aibiš.* Other stems ending in a vowel do not attest *i*-epenthesis in these case-forms.[95]

[95] It is interesting to note that all the cases of *i*-epenthesis across *b* cited in De Vaan (2003) as Old Avestan borrowings in Young Avestan also take place after a preceding *a* (552).

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There are no clear indications on the influence of stress on the Avestan epentheses. De Vaan (2003) believes that the absence of epenthesis in the vocative singular of *hāuuane / hāuuani-* 'deity of haoma-preparation' may be due to the initial stress of the vocative (550). The initial development of Old Avestan *db-* may also be due to the presence of a secondary stress in that position. The addition of the clitic *-ca* 'and' to words ending in *e*, *i* may block *i*-epenthesis in the case of a preceding *t*, *ṇt*, *d*, *p*; the palatalizing effect of *r*, however, is not affected. It is possible, however, that the stress-pulling properties of *-ca* may have nothing to do with the obstruction of *i*-epenthesis since it is not impeded by the addition of *-ĭmcă*, *-ĭščă*, *-iiaca*, and *-iiâsca*. De Vaan (2003) accounts for the absence of *i*-epenthesis before *-ca* by suggesting that consonants became depalatalized before a following palatal.

Avestan *i-* and *u*-epentheses are sometimes interpreted as a graphic device indicating allophonic palatalization or labialization of the consonant by the following vocoids; this view is expressed, e.g., by Morgenstierne (1973:57). One of the reasons for such an interpretation is the fact that the epenthetic vowels are disregarded in metrical structure. However, as already discussed elsewere in this work, this fact alone does not provide evidence for or against the surface reality of the prevowels since their failure to occupy a metrical position is consistent with their synchronically subsegmental status. Furthermore, under the assumption that the epenthetic vowels serve as mere indicators of secondary modifications of consonants, it is hard to explain

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why word-medial palatalization affects most of the labials and dentals while leaving out the velars. Also, if *i*- and *u*-epentheses are, indeed, a simple notational device to indicate secondary modifications of consonants, it is unclear why the notation of secondary articulations should be influenced by the nuclei that precede the affected consonants. That this is true of *i*-epenthesis has already been noted above; word-medial *u*-epenthesis is attested only when the preceding vowel is a *a*, *ā*, or *ə*. And there is another feature of *i*-epenthesis that makes it look like an instance of prevocalization, namely, the fact that its application is lexically selective. Thus, *i*-epenthesis across *p* is attested only in the preposition *aipĭ* 'over, across, during; after' and in compounds with *aipĭ* or *an-aipĭ* as the first member. Similarly, *i*-epenthesis across *b* in Young Avestan is attested only in the *b*-cases of *a*-stems. Additional characteristics of the Avestan epentheses that make them look like prevocalization rather than simple consonant modification are their sensitivity to morpheme boundaries and/or the location of stress. Such sensitivity can be easily situated within the context of prevocalization but is harder to explain if the epentheses are taken as consonant modifications. Finally, when the vowel preceding an epenthetic <i> is a *i*, it develops into a schwa, merging with the homophonous sequence arising in the same environment from an anaptyctic schwa. Although this may well be a case of simple dissimilation between a segmental [i] and the secondary articulation of a consonant, parallelism with the development of [ɪ] in English loanwords in Welsh discussed in 3.4.2.6 below suggests that this development is best taken at its face value, i.e., [iʲ] > [əʲ].

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Prevocalization is attested in other Iranian languages as well. Among West Iranian languages that attest this process Eilers (1974) includes Middle and New Persian, Kurdish and Baluchi; he also believes that *i*-epenthesis accounts for the <ai> diphthongs in the Old Persian adjective *yāumainiš* 'powerful (?)' and preposition *paišiyā* 'before, in front of' (< **patyā*).[96] An East Iranian language with well-documented prevocalization is Ossetic. Based on the treatment of Proto-Iranian *a before prevocalizing consonants, Cheung (2002) posits two prepalatalization periods in the history of Ossetic: in the earlier period, the vowel-onglide sequence *a^{j} merges with Proto-Iranian *ai and is reflected in modern Ossetic as *i* or *e*, and in the more recent period, the same sequence produces modern Ossetic *æj* (117). According to Cheung, early prepalatalization applied to the sequences *V(C)Ci, *V(C)C̯i, and *V(C)CiC, and the later prepalatalization occurred in the clusters *C̯i that arose from earlier *CV̯i. Early prepalatalization affected *ð, ʁ, v, x (from Proto-Iranian *d, *g, *b, *x/k, respectively), n, s, w. Prepalatalization is not attested in consonant clusters containing *š or l, or in clusters of more than two consonants. The coronal consonants *t^{j}, *θ^{j} and *r^{j} were palatalized to *ts^{j}, ts and l, respectively, without triggering prepalatalization. The later of the two prepalatalizations applied to r, l, n, f, and w (Cheung 2002:117).

[96] Cf. also Kent (1950:45). Eilers' interpretation is not accepted by all researchers; for an alternative explanation of the Old Persian forms cf., e.g., a review of Eilers' article by Rüdiger Schmitt in *Wiener Zeitschrift für die Kunde des Morgenlandes* 68 (1976:254-56).

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Turning back to Avestan, it would appear that the above facts all point to the surface reality of the preconsonantal onglides. The onglides must have been salient enough to warrant a notation in the orthography; not being segmental, however, they did not contribute to the metrical structure. The relatively short duration of such onglides has already been mentioned: thus, the palatal onglide in Majorcan Catalan is described by Mascaró (1985) as very short and not affecting the length of the preceding nucleus (134, 145 fn. 12), and pre-/s/ onglide in Welsh does not count as a syllable in the earliest recorded poetry (3.4.2.6). Lastly, the interpretation of Avestan *i*- and *u*-epentheses as prevocalization is also urged by the presence of prevocalization in other Iranian languages, which indicates that this is a common tendency of secondarily modified consonants in that language family. Even though this last piece of evidence is circumstantial at best, it does make the prevocalization account of the Avestan vowel epentheses preferable in that it agrees with the well-known fact that closely related languages tend to undergo similar structural developments. Thus, even though the existence of prevocalization in other Iranian languages does not prove that this feature was present in Avestan, it nevertheless makes the presence of this feature in Avestan more likely on typological grounds.

3.2.3.2 Early Latin and the Romance varieties of North Italy

Archaic Latin inscriptions attest several cases of prepalatalization. Of the various datable instances cited by Wachter (1987), the author himself considers certain only early Praenestine *painiscos* 'little Pan' and *poimilionom* 'dwarf (genitive plural)'

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(Classical Latin *pūmiliō*, from Greek Πυγμαλίων) (143, 164ff, 265). Other attested forms include *creisita* and *leigibus* from Praeneste and *iuilio* from Rome (160, 258, 348).[97] Wachter cites a parallel case of prepalatalization in Oscan, namely, β[ρα]ιτηις for a *i*-stem *brāt-* (272). Vine (1993) uses the evidence of these and similar inscriptions in his reexamination of the historical change *ē* > *ī* in noninitial syllables before *-i-* in the next syllable, proposed for a handful of forms in earlier treatments. After eliminating several of the forms, he explains the remaining items, in which the development in question is of the shape *-īli-* < *-ēli-*, through prevocalization, with an intermediate stage **-ē^{j}li-* (Vine 1993:101f).[98]

A somewhat similar development is noted in some of the Romance varieties of North Italy where the masculine plural marker *-i* induces prepalatalization in stems ending in a nasal or a nasal followed by another consonant; the consonant intervening between the nasal and the /i/ may be depalatalized after palatalization has been transferred to the nasal. This prepalatalization pattern has left traces in contemporary Ligurian, cf. *kai* 'dogs', *kristiai* 'christians' with Standard Italian *cani*, *christiani* (Devoto et al. 1974:24). Other northern forms cited in the literature include *töjnt* 'many', *köjmp* 'fields', *pójnt* 'bridges', *fójnč* 'mushrooms', Istrian *barcoin* 'balconies' (Standard Italian *tanti*, *campi*, *ponti*, *fungi*, and *balconi*, respectively) (Tuttle 1991:47

[97] Less certain is *caepa/cēpa* 'onion' adduced as an example of prevocalization in Latin by Eilers (1974:290) who derives it from Hesychius' κάπια 'garlic'.

[98] Meiser's (1998:69) suggestion that the change in question may be connected with the location of stress is not mutually exclusive with the prevocalization solution. I would like to thank Brent Vine for bringing this alternative account to my attention.

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fn. 61). Tuttle (1992:130 fn. 22) suggests the following phases of decomposition of a word like **bottoni* 'buttons': *botoɲ*, *botojn*, *botoj̃*, *botoj* (cf. also Rohlfs 1966:15). Forms attested in older sources, such as Old Venetian *fainti* 'babies' (Standard Italian *infanti*); Old Piedmontese *boyn* 'good (masculine plural)' and *larroyn* 'thieves' (Standard Italian *boni*, *ladroni*); Old Ligurian *quainti* 'how many (masculine plural)', *graindi* 'large (masculine plural)' (Standard Italian *quanti*, *grandi*); and Old Genoese *graindi*, *fainti*, *cristiain*, indicate that this prepalatalization pattern is at least several hundred years old (Rohlfs 1966:15, 43f, 418f). In Old Piedmontese, the addition of the masculine plural ending also causes prevocalization of /l/, cf. *tajl* 'such (masculine plural)', *quajl* 'which (masculine plural)' (Standard Italian *tali*, *quali*). In the old dialect of Asti, final *-i* causes prepalatalization of /p/, cf. *drayp* (Standard Italian *drappi*). Some of the modern dialects have likewise extended this pattern to a wider range of consonants, cf. Valsesia Piedmontese *divöjt* 'devout (masculine plural)', *gait* 'cats', and *raim* 'branches', (Standard Italian *devoti*, *gatti*, and *rami*, respectively).

3.2.3.3 Tocharian

Proto-Tocharian consonants were palatalized before front vowels and *j* (Ringe 1996:101-17). Palatalized labials and /r/ seem to have been depalatalized fairly rapidly, but the existence of the palatalized stage of, e.g., labials is assured by their effect on the adjacent consonants and vowels (Pinault 1989:II.2.2.1), cf.:

A *s̲älyp*, B *s̲alype* 'grease, unguent' < Proto-Tocharian *s^{j}äl^{j}p^{j}ă < *selpes-os (genitive)

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B *mit* 'honey' < Proto-Tocharian *m^{j}æt(u) < *medhu

Tocharian A is less palatalizing than Tocharian B; for instance, it depalatalizes Proto-Tocharian *p^{j}, *m^{j}, *w^{j}, while Tocharian B palatalizes its *w^{j} to *j* and secondarily develops *p^{j}, m^{j}, k^{j}*, and *ts^{j}*. Palatalization contrasts in the two languages are given below for comparison (Pinault 1989:II.2.1):

Tocharian A

p	t	ts	k	s	m	n	r	l	w
-	c	ś	ś	ṣ	-	n^{j}	-	l^{j}	j

Tocharian B

p	t	ts	k	s	m	n	r	l	w
p^{j}	c	ś	ś	ṣ	m^{j}	n^{j}	-	l^{j}	j
		ts^{j}	k^{j}						

Given the existence of a phonemic palatalization in the Tocharian languages, it is interesting to note that the only reported case of prepalatalization there involves a phonemically nonpalatalized segment, namely, the nasal in the noninitial sequence **ans* in Tocharian A, the less palatalizing of the two languages. For example, the genitive of ancient thematic nouns in Tocharian A is *-es* (cf. *yukes*, nominative *yuk* 'horse') which, according to Pinault (1989), presupposes the following developmental sequence: Proto-Tocharian **-a-nsă* > **-a^{j}nsă* > **-ajns(a)* > *-es* (III.1.3.2). The sequence *-ns- that arose language-internally was treated the same way, cf. third plural present endings *-nd >

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*-ænd^{z} > *-ænz > *-æ^{j}nz > *-æ^{j}z > *-i* (athematic) and *-o-nd > -œnd^{z} > *-œnz > -œ^{j}nz > *-œ^{j}z > *-e* (thematic) (Pinault 2004).

3.2.3.4 Tsakonian Greek

Much of the descriptive literature on Modern Greek includes observations on the existence of prepalatalization in contextually palatalized consonants. The majority of accessible sources, however, provide little more than a cursory overview of the phenomenon. For example, Thumb (1910) cites a number of the relevant forms from southern Maina; the forms include βάιζει for βάζει 'he puts', μωιρή for μωρή (an interjection), μάιτα for μάτια 'eyes', ποιδά for ποδιά 'apron', κουλλούϊρα for κουλλούρια 'pretzels', ειδάητσε for εδιάβηκε 'he went', παιδά [pejðá] for παιδιά 'children', ειννιά [ejnʲá] for εννιά 'nine', φίδα for φίδια 'snakes', and ναι διορδώσου for να διορθώσω 'that I improve' (8). From these examples it may be inferred that prepalatalization in Maina affects the coronals *t*, *d*, *dz*, *r*, *n* word-internally and in tightly bound syntactic groups, and does not depend on the location of stress or the quality of the preceding vowel. Another cursory study is Lockwood's (1952-53), where prevocalization is mentioned as being typical of the Athens-Piraeus district (97f). In forms cited by Lockwood prepalatalization takes place between open vowels and the following palatal stops, in native words as well as borrowings, word-internally and across word boundaries. It is said to be independent of stress, though the onglide is

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more prominent when the preceding vowel is stressed. Lockwood's examples include the following forms:

κουταλάκι	[kutaˈlaʲci]	'tea-spoon'
Τάκης	[ˈtaʲcis]	'Takes' (proper name)
θέλω να κοιμηθω	[ˈθelo naʲcimiˈθo]	'I want to go to bed'
σοκάκι	[soˈkaʲci]	'alley' (< Turkish *sokak* 'street')
φαλάγγι	[faˈlaʲɟi]	'tarantula'

Against this general background Pernot's relatively comprehensive sketch of prevocalization in the Tsakonian dialect is especially welcome (1984, 1907). According to Pernot, prepalatalization in that dialect is triggered by consonants palatalized by the following [i] or [j]. After triggering prepalatalization, the palatal glide is optionally deleted, cf. ξόι̯δα (< ξόδι̯α), βαι̯θού (< βαθι̯ού) (Pernot 1894:85). Stress or quality of the preceding vowel do not affect prevocalization (82). The consonants most likely to trigger prepalatalization are θ and δ, which are said to prepalatalize even in the speech of those informants who generally avoid prepalatalized pronunciations. In the speech of the oldest generation of speakers available to Pernot's observation, prepalatalization of χ, σ, τ, and the clusters ρδ and ρθ is also attested, albeit more sporadically (82f). The following examples are cited as characteristic of the speech of Lenidi (except for the palatal onglide itself, which is indicated by a raised ʲ, Pernot's transcription is used throughout):

θ / __ [i], [j]	αʲθί, αʲθι̯α., κοʲθίνι (for κοφίνι)
δ / __ [i], [j]	παπαʲδία, πεʲδι̯ου (for παιδιοῦ), πόʲδι, κουʲδί (for κλουβί)
χ / __ [i]	αʲχινέο (εχῖνος)
σ / __ [i]	δροʲσίσις (for δροσίσης)
τ / __ [i]	χαʲτίρι

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ρδ / __ [j]	κα^{j}ρδ̯ιά
ρθ / __ [i]	κα^{j}ρθί (for καρφί)

In the speech of Tyro, prepalatalization affects a greater range of consonants than in that of Lenidi. As in Lenidi, however, the affected consonants are mostly coronal.[99] Especially interesting are prepalatalizations of consonant clusters and at word boundaries in tightly bound syntactic groups. The following examples illustrate consonants and consonant groups that may be prepalatalized in Tyro but not in Lenidi:

ρ / __ [i]	φανά^{j}ρι, Μα^{j}ρία, ξιφτέ^{j}ρι
φ / __ [i]	τα^{j}φιλακί (τη φυλακή)
στ / __ [i]	α^{j}στία (αστεῖα)
(σ)τρ / __ [i]	[tsena^{j}trí̯a] καὶ ηένα τρία, [pa^{j}striká] παστρικά
γδ / __ [i]	[para^{j}g^{j}d̯imata] παραδίγματα

In addition to prepalatalization triggered by [i] and [j], Pernot cites forms like o^{j}χτό [o^{j}çto] and μα^{j}ντανέ [ma^{j}nta^{l}ne], where it is triggered by coronal consonants (1894:86). The special interest of these forms lies in their providing a concrete synchronic evidence for prevocalization as a possible stage preceding complete vocalization of preconsonantal nasals and velars (cf. 3.2.2.4.3).

Pernot observes that the heaviest users of prepalatalized forms are the oldest women available to his observation; this allows him to correlate the peak of popularity of this pronunciation type with the generation born around 1833. In spite of this, the overall popularity of prepalatalized pronunciations at the time of his fieldwork was such that Pernot predicted that prepalatalization would eventually become automatic in this

[99] The form τα^{j}φιλακί looks like an exception; however, it is possible that φ in this word stands for θ, cf. κα^{j}ρθί for καρφί cited earlier.

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dialect. Or, in his own words: "Dans un bref délai s'établira la règle phonétique suivante: *tout ɪ ou i̯, précédé d'une consonne prépalatale ou d'un groupe de consonnes prépalatales, se répercute sous forme d'ɪ dans la syllabe précédente*" (1894:83f).

3.2.3.5 English

In English, prepalatalization is often triggered by velars palatalized by adjacent, more often preceding, vowels. Prepalatalization of velars fronted by preceding front vowels is also widely noted for Germanic languages in general. Thus, Old Norwegian texts show epenthetic glides after *e* before *ng*, *nk*, cf. *dreingr* 'boy' (< *drengr*); a similar development is attested in Faroese (Lockwood 1952-53:97). In English, the development of a glide between stressed vowels and the following velar fricative has already been noted. The frontness of the fricative and that of the onglide is determined by that of the preceding vowel: when the vowel is back, the onglide is of a velar quality, and when the vowel is front the onglide is palatal, cf. Middle English *feighte(n)* 'to fight' from Old English *feohtan* (Luick 1921:424; Mossé 1952:28; Lass and Anderson 1975:195ff).

Another instance of the same tendency in English is due to palatalization of /k, g, ŋ/ next to /a/ in Middle / Early Modern English. Depending on the syllable position of the velar, its palatalization is reflected as a palatal on- or offglide. Long abandoned in metropolitan English, this pronunciation has survived in the peripheral varieties like

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Irish English, Southern American English,[100] and English-based creoles of the Atlantic, all of which resulted from British colonial expansion during the seventeenth and subsequent centuries. The exact phonetic nature of Middle / Early Modern English /a/ and the chronology of its fronting, which caused palatalization of adjacent velars, are not entirely clear (Wyld 1936:196ff). Subsequent to its fronting, the vowel was retracted leaving behind a residue in the form of palatal on- or offglides (Harris 1987:275; Holm 1993:342f and 1988:133f). Palatalization of velars and the presence of the glide are noted by seventeenth-century orthoepists, with the earliest of such records, that of [gj] in *guarded*, going back go 1617. Glided pronunciation of the palatalized velars was general in the eighteenth century: around 1800 postvelar glides were said to set off "the polite conversation of London from that of every other part of the island" (Wyld 1936:310). Post- and prevelar glides went out of use in mainstream English by the end of the nineteenth century (Dobson 1968:952; Patrick 1999:89), but were preserved in the Midlands dialects of British English where the process is believed to have had its origin (Orton and Dieth 1962-71:124, 143). It is thought that it was from there that this pronunciation spread to Ireland, south Atlantic coastal states of the United States, and the English Creoles of the Atlantic (Kurath and McDavid 1961:175; Holm 1988:134; Patrick 1999:89-91).[101]

[100] Postvelar glides are reported to be especially characteristic of eastern and central Virginia and of the Low Country of South Carolina.

[101] According to Harris (1987), the palatalizing tendencies of Irish and West African substrata may have contributed to the preservation of this trait in these contact varieties (275).

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In Irish English, a strong palatal onglide between /a/ and word- or syllable-final velars is reported by Milroy (1981) in *back*, *track*, and *thanks*.[102] The onglide is said to be so prominent that the /a/, phonetically [a^{I}] or [æ^{I}], tends to be interpreted as diphthongal by speakers of the varieties of English where this pronunciation type is unknown (Ó Baoill 1997:78). Postpalatalization of initial velars is noted in many dialects of Ulster in words such as *car*, *cabbage*, *calf*, and *garden*. Centralized or retracted realizations of /a/ make the adjacent on- and offglides more prominent, cf. more conservative pronunciations [bɛg] 'bag', [bæ:ŋ] 'bang' with the more innovating [bä:^{j}g], [bä:^{j}ŋ] noted by Harris (1985:52). In the American South, /k, g, ŋ/ after any front vowel may condition the insertion of a palatal glide (Sledd 1966:32f). Examples, cited from various locations in Texas, North Carolina, Kentucky, Alabama, and Georgia, include *thing*, *bag*, *rang* [æ^{j}]; *lack* [a^{j}]; *egg*, *keg*, *neck* [ɛ^{j}]; and *big*, *blink*, *cling* [i] (Wilson 1934:211; Stanley 1936:17; Cearley 1974:36).

English-based creoles for which the existence of palatalized velars has been mentioned in the literature include the following Atlantic varieties: Jamaican, Guyanese, Belizean, Bay Islands English, Barbadian, Gullah, Sranan, Saramaccan, and Ndjuka.[103] While all of these varieties are described as having postvelar glides, the existence of prevelar glides has been noted only for Jamaican, Barbadian, and Bay Islands English. In all these, prevelar glides are lexically selective. In Jamaican, their

[102] Before /g, ŋ/ the pronunciations may vary from [ɛ:g/ŋ] to [ɛ:ɪg/ŋ] to [a:ɪg/ŋ] (Milroy 1981:75).

[103] Turner 1949:24; Cassidy and Le Page 1967:lviii; Alleyne 1980:59; Warantz 1983; Holm 1993:324; Greene 1999:28f; Patrick 1999.

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occurrence is etymological in *bag* [bajg], *flag* [flajg], *hag* [hajg], *ragging* [rajgin] and *lags* [lajgz], and analogical in *prog* [prajg ~ prag] and *dog* [dajg ~ dag] (Cassidy and Le Page 1980:lii). Cassidy and Le Page (1980) note that the glided pronunciation of these lexical items is considered archaic even in the broad dialect, and is getting replaced by the more usual /a/, except in formulaic expressions like *old hag* [wol hajg], a frequent character in folk tales (Le Page 1957:383 fn. 1).[104] For the Bay Islands English of Honduras, Ryan (n.d.:130) cites only eight words with the prevelar glide; these include [bajk] *back*, [blajk] *black*, [ajks] *ask* and [bajg] *bag*; the last item has also been mentioned for Barbadian (Ryan n.d.:130; Warantz 1983:84 n. 5; Holm 1988:134 and 1989:481).

3.2.3.6 Other languages

Van Dantzig (1929) notes that the alveolar consonants in Modern Dutch are palatalized under the influence of a heterosyllabic palatal glide citing words like *mandje*, *hoentje*, *hondje*, *huisje* as examples. Palatalization is manifested in part as prepalatalization; this results in diphthongized pronunciations of preceding vowels which are said to be acoustically and articulatorily halfway between a monophthong and a diphthong. The onglide is reported to be very slight or lacking after *e*, *ie*, *i*, *ee*, *aa*, *ɔ*, *oo*, *u*, and significant after the back nonhigh vowels *a*, *ŏ*, *oe*. It is also reported to be more salient in the case of an intervening *n* than in the case of a *t*.

[104] Wells (1973) does not mention prevelar glides in the Jamaican speech of London.

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In Hausa, a Chadic language of Nigeria, velars are palatalized before front vowels /i/ and /e/. Kraft and Kraft (1973) note that after /a:/ palatalization of velar stops may be realized as an onglide to the stop, cf. /za:kì:/ [za:ʲkì:] 'lion' (184).

In Guelavía Zapotec, an Otomanguean language of Mexico, word-final glides /j/ and /w/ cause prepalatalization and prelabialization, respectively, of the preceding consonant. Significantly, /i/ and /u/ do not have this effect on the preceding consonant (Jones and Knudson 1977:170), cf.:

/dzɨnj/	[dzɨ:inj]	'honey'
/bekw/	[bɛˀɛukw̥]	'dog'

In Quiaviní Zapotec, there is prepalatalization before final /Cj/ clusters. Prepalatalization does not occur when the cluster moves to the onset position, e.g., due to the addition of vowel-initial clitics, cf.:

telebisyoony	[-ooʲnj]	'TV-set'
x-telebisyoony-a	[-oo-nja]	'my TV-set'

Pam Munro (p. c.) observes that native speakers tend to notate the palatal glide in words like *Jwaa(y)ny* 'Juan', which might indicate the beginnings of phonologization of the prepalatalized pronunciation.

Sorowahá, an Arawá language from Amazonas, Brazil (Dixon 1999), has prepalatalization of consonants before rising diphthongs of the shape /jV/. This is said to be the only synchronic source of the phonetic falling diphthongs in Sorowahá (Suzuki 1995:342), cf.:

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/adiei/	[áaʲdʲéʲ]	'a woman's name'
/ania/	[áaʲnʲá]	'a man's name'
/uniaua/	[úuʲnʲawá]	'a woman's name'

Yucuna, an Arawak language from the state of Amazonas, Colombia, is reported to have free variation between /s/ and /h/ in approximately 90% of the words where one of these phonemes occurs. When there is a /e/ after the [s] realization of one of these fricatives, there is a palatal onglide after a preceding /a/, cf. /wahé/ [waʲsé] 'new' (Schauer and Schauer 1967:71).

In Djapu (Australia), velars /k, ŋ/ are prepalatalized when followed by /j/ (Morphy 1983:19). In Uradhi, /a/ is fronted and often receives a palatal offglide when followed by a laminopalatal or the cluster /kj/. When followed by the cluster /kw/, it may receive a labial onglide (Crowley 1983:318), cf.:

/akjun/	[æʲkjun]	'camp'
/akwaṉumu/	[aʷkwaṉumuŋ/	'dingo'

3.2.4 Prepalatalization and palatal umlaut

Umlaut refers to an assimilatory modification of a (typically stressed) vowel in the direction of a following segment. The type of umlaut of immediate interest to the present study concerns assimilation in place of articulation, for instance, fronting or front-raising of vowels under the influence of a following coronal or palatalized consonant, or that of a consonant or group of consonants followed by a front vowel or palatal glide. The set of conditioning environments that produce palatal umlaut is thus almost identical to the set of conditioning environments the produce prepalatalization,

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but it is important to stress that the two processes differ as to their mechanism and immediate results. As discussed in 1.4, the primary defining element of prepalatalization is gestural decomposition of a weakly positioned consonant in the case of plain consonants, and synchronic enhancement of secondary articulations in the case of complex consonants. Although the portion of the vowel to the immediate left of the consonant may be assimilatorily modified in the direction of the consonant's V-gesture, this assimilation is epiphenomenal to prevocalization and its effect is completely incorporated in the prevowel. This constitutes the principal difference between prevocalization and umlaut, since the principal element of the latter process is the assimilatory movement of the vowel, which starts at its beginning rather than its end portion. As a consequence of the underlying differences in the mechanism, the immediate result of prevocalization is a surface diphthong, while that of umlaut is a modified monophthong.

Prevocalization and umlaut frequently coexist in the same language or in a closely related group of languages. For example, Celtic, Germanic, and Iranian languages all attest both umlaut and prevocalization. Sometimes the two processes exist in complementary distribution, cf. Krygier's (1997) observation that "in Livonian *i* or *j* in the second syllable always caused palatalisation of the preceding consonant, and in the cases where this palatalisation later disappeared, umlaut is found (or sometimes an epenthetic *i* in its stead)" (35 fn. 79; cited after Posti 1946-48:47). Part of the reason for this co-occurrence is similarity of the conditioning environment. Another possible reason may lie in the general direction of the language's assimilatory processes: for

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instance, Pernot (1894) correlates prepalatalization in Tsakonian Greek with the general tendency of that dialect toward anticipatory assimilation. Finally, it the diphthongal products of prevocalization become monophthongized, which is the case in many documented cases of prevocalization, it becomes difficult to distinguish them from direct vowel modifications due to umlaut. In such cases, distinguishing between the two processes is possible only when the intermediate diphthongal stages are attested. Such seems to be the case, e.g., of the three Nakh languages (Chechen, Ingush, Bats) in which a *a(:)* was modified by a following *w* or *i*. This process has led to diphthongization of /a(:)/ in Bats and its front-raising in Chechen and Ingush (Desheriev 1979:181), cf.:

a(:)	>	Bats *aj*
	>	Chechen *ɛ*, *e(:)*
	>	Ingush *ɛ(:)*, *e*

Desheriev (1979, 1964) interprets the diphthongal outcome in Bats as an archaism, and hence as evidence for prevocalization as the mechanism responsible for the surface umlaut in all three languages.

An interpretation of surface umlaut along the same lines suggests itself even more forcibly in the case of the Mixean languages of Mexico. There, umlaut is caused by palatalized consonants which arose from clusters with the following palatal glide (Wichmann 1995:167). Diphthongal and fully umlauted stages of the affected vowels may coexist in the same language or closely related languages. For example, [ui] ~ [i] and [oe] ~ [e] are the results of the umlauting /u/ and /o/, respectively, in the Tepuxtepec dialect (Wichmann 1995:22, 142). In the Atitlán dialect of Midland Mixe,

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the results of umlauting /o/ and /u/ are [oö] and [uü], respectively, as against /e/ and /i/ in most other dialects (Wichmann 1995:36). In Tutla, umlaut under the influence of the following palatalized noncoronals is realized as vowel diphthongization, cf. *cuiik*^{j} 'orange', *c-*^{j}*üiik*^{j} 'his orange' (Bickford 1984:27). When this evidence is combined with the fact that palatalized consonants in Mixean tend to be signaled by palatal onglides,[105] it becomes more than plausible that the attested vowel modifications are due to prepalatalization followed by optional monophthongization of the results, rather than to umlaut in the strict sense.

The language area in which the connection between prepalatalization and umlaut has been discussed most extensively is Germanic. In the Germanic languages, with the sole exception of Gothic, stressed vowels of the root were front-raised under the influence of a high front vowel or palatal glide in the suffix, a process known as *i*-mutation or *i*-umlaut. For example, in Old English all vowels except *ǣ*, *ē*, *i*, *ī* were front-raised when *i*, *ī* or *j* occurred in the following unstressed syllable. As a result of this change, /æ, a, o/ > /e/, /u, ū/ > /y, ȳ/, /ā/ > /ǣ/, /ō/ > /ē/, /io/ > /ie/, /īo/ > /īe/, and /ēa/ > /īe/; examples include **sættjan* > *settan* 'set', **fulljan* > *fyllan* 'fill', and **dohtri* > **dæhtri* > *dehter* 'daughter' (Moore and Knoot 1942:129; Campbell 1959:71ff). The view of *i*-umlaut as a direct front-raising of the umlauting vowel is complicated by the fact that manuscripts in nearly all early Germanic languages contain numerous instances

[105] Hoogshagen (1984:4) and Hoogshagen and Halloran (1993:343), e.g., mention this as characteristic of Coatlán Mixe, and Haitsma and Haitsma (1976:10) note that in the Mixe dialect of San José El Paraíso palatalized consonants trigger both on- and offglides "with a noticeable 'i' quality".

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of diphthongal spellings of the umlauted vowels. Thus, the digraphs *ei*, *oi*, *ui* are frequently used to spell "normal" Old English *e*, *oe*, *y*, and the more archaic of the Old High German documents sometimes indicate the *i*-umlaut of *a* by means of digraphs *ai*, *ae*, *ei*, and that of *u* as *iu*, *ui*. The existence of the digraph spellings provided an impetus for the interpretation of the early stages of *i*-umlaut as the epenthesis of a palatal glide into the umlauting syllable, and to comparisons with Indo-European languages in which the evidence for this type of development is more explicit. Antonsen (1961) traces the beginnings of this interpretation to a suggestion made by Karl W. Heyse (1838), who saw the initial products of *i*-umlaut as diphthongal and compared Germanic forms with French developments of the type Latin *gloria(m), rationem* > *gloire* 'glory', *raison* 'reason'. Jacob Grimm, discussing umlaut in the third edition of his *Deutsche Grammatik* (1840), drew attention to the difference between Sanskrit *patis*, *taruna* and Avestan *paitis* 'lord', *tauruna* 'youth', on the one hand, and the parallel differences between Gothic and Old High German forms, on the other. A comparison between the diphthongal spellings in Old High German and the parallel orthographic phenomena in Avestan and Old Irish was also made by Friedrich Wahlenberg, who proposed, e.g., that an umlauted short [a] in Germanic first developed into a diphthongal short [ai], which was later modified to [ei] or [e] (1855:19, 23).

The epenthesis theory of *i*-umlaut, as it came to be known, incorporates as its component the suggestion that umlaut proceeded via the consonants separating the root vowel from the vowel of the ending or the derivational affix. The first proponent of this (= *mouillierung* or *palatal*) theory was Wilhelm Scherer, who sought evidence for it in

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Old High German spellings of the type *airin*, *muillen*, *scoina* (Antonsen 1961:149ff). Scherer's theory was based on a comparison with the front-raising influence of palatalized consonants on preceding vowels in Lithuanian and Slavic. He suggested that the onglides combined with preceding vowels to produce diphthongs, while the triggering consonant lost its palatalization. To illustrate the loss of palatalization, Scherer drew upon Greek *kteînō* and *phtheírō*, from **kténjō* and **phthérjō*, respectively.

The palatal theory of umlaut is in direct competition with the anticipation theory, or the view of umlaut as a distant vowel-to-vowel assimilation (cf. Sommerfelt 1934). Proponents of the latter theory based their objections to the palatal theory in part on the alleged absence of consonant palatalization in Germanic. This charge led to a series of experimental works investigating allophonic palatalization in Germanic, dialectal studies presenting evidence for intermediate stages of umlaut in Germanic, and studies demonstrating earlier presence of palatalized consonants in Germanic languages based on internal evidence and loanwords from and into Slavic and Romance (Antonsen 1961:380ff). Independently from their contribution to the issues at stake, the first two types of studies are valuable in providing additional data on Germanic prevocalizations. Thus, Van Dantzig's (1929) study on prepalatalization in Modern Dutch diminutives, briefly outlined in 3.2.3.5, owes its existence to this debate, and so does Götlind's (1927) description of an ongoing change in some Swedish dialects of southern Västergötland, where a *a* tends to become fronted before the following *gg*. In parts of the area, the outcome is *ä*, cf. *vägga*, *bägge* for *vagga*, *bagge*, whereas in other localities it is a^j or $ä^j$, cf. *va^{j}gga*, *bä^{j}gge*. Götlind interprets the diphthongal

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pronunciations a^j, $ä^j$ as intermediate stages in the development from a monophthongal *a* to monophthongal *ä*. He also applies his theory to the runic forms *sairawidaR* and *eirilaR* and to Old High German *aigi* 'discipline' and *airin* 'floor', concluding that these forms likewise contain intermediate stages of *i*-umlaut (cf. Antonsen 1961:339f).

A new element was introduced in the discussion of *i*-umlaut by Eduard Sievers (1924, 1925, 1929). Briefly, in analyzing Old Norse poetry, Sievers noticed that his pronunciation of *æ*'s and *œ*'s depended on the phrasal stress, and concluded that these phones were diphthongal under a *Steigton* and monophthongal under a *Fallton*, suggesting that the products of *i*-umlaut could be mono- or diphthongal depending on the prosodic conditions. In his own words (Sievers 1929:16):

> "... wenn ich in den betreffenden Versen (annähernd) diphthongisches *ai*, *oi* spreche, so komme ich zu dem Schluss, dass in der Sprache Bragis der *i*- (bzw. *R*-) umlaut von *ā*, *ō* bei Falltonigkeit zu der Stufe *æ*, *œ* bei Steigtonigkeit aber zu einer Stufe von ungefähr **ai*, **oi* geführt habe. Historisch und phonetisch aber lässt sich ein solches Verhältnis, wie mir scheint, nun so verstehen, das der Umlaut in diesen Fällen überhaupt nicht direkt von *ā*, *ō* zu *æ*, *œ*, sondern zunächst zu einer (epenthetischen) Zwischenform **āi*, **ōi* geführt habe, die später einmal durch innere Assimilation monophthongiert wurde. (...) Ist alles das richtig, so bekommen wir hier auf altnordischem Gebiet eine volle Parallele zu den entsprechenden Erscheinungen im Altenglischen ..."

Sievers' theory was accepted by Max Förster, who suggested that during the transitional period from the non-umlauted to fully umlauted forms there was a fluctuation between the original unaffected forms, diphthongized forms, and umlauted variants, the choice of the particular variant being determined by sentence stress ("Melodieführung im Satzzusamenhange"). According to him, during this transitional period the name of the

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river Thames, for instance, would have been rendered *Taimis or *Tæmis under a *Steigton*, and *Tamis under a *Fallton* (Förster 1921 et seq.).[106] [107]

Although the epenthesis and palatal theories are cited in some standard textbooks (cf. Brunner 1942:68f; Campbell 1959:17f), many arguments were advanced against them in the course of time. Without necessarily suggesting that *i*-umlaut involved prepalatalization, it is nevertheless worth noting that the two main objections to these theories cited in Krygier (1997) are easily reconcilable with the prepalatalization theory. In Krygier's own words, these are "(i) why is there almost no trace of the postulated palatalised consonants (especially non-velars) both in old texts and in modern dialects?, and (ii) why is the behaviour of the root vowel and epenthetic glide clusters different from that of regular diphthongs?" (1997:36). One possible answer to the first objection is that palatalization, being an allophonic feature, would not necessarily be recorded in writing. Moreover, prepalatalization that produced modern umlauted vowels may have been accompanied by depalatalization of the triggering consonants, which would account for the lack of palatalized reflexes in the modern languages. The second objection ceases to be an issue when it is recalled that gliding monophthongs produced

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[106] Förster seeks further evidence for the diphthongal stage of *i*-umlaut in the form of early spellings, placenames, and borrowings from and into Old English. Thus, he equates the place name *Loidis*, recorded by Bede, with Modern English *Leeds*, and posits the development from Old Welsh **Lodissā* > **Lodis* > Old English *Loidis* > **Lædes* > * *Ledes* > Modern English *Leeds* (1922) (see also Antonsen 1961:332f). From among the other Early Old English forms recorded in Bede with digraph spellings Förster cites *Oiddi* for later *Eddi*, *Cōifi* for *Cēfi*, and *Bōisil* for *Bǣsil*, *Bēsel*. Additional instances of prevocalization cited in this works are the geographic names *Campaina* and *Puille* < Latin *Campania* and *Apulia*.

[107] Despite Förster's support, Sievers' theory of a prosody-driven distribution of the products of *i*-umlaut failed to gain general acceptance; nevertheless, the fact that prevocalization frequently depends on sentence-level prosody, noted by various researchers, lends this theory a fair amount of plausibility.

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by prevocalization do, in fact, differ from underlying diphthongs both phonetically and phonologically (see 1.3). In sum: while the above data and discussion do not necessarily prove that prevocalization played a role in Germanic *i*-umlaut, they do indicate that such an interpretation is at least plausible from a general typological standpoint.

3.3 Palatals

3.3.1 Preliminary considerations

Palatals of all manners of articulation are among the most frequently prepalatalizing consonants. The term "palatal", as used here, comprises three types of segments: palatals proper ([ç, ʝ]), palatoalveolars [tʃ, dʒ, ʃ, ʒ], and alveolopalatals [c, ɟ, ɲ, ʎ, tɕ, dʑ, ɕ, ʑ].[108] Although palatals are produced by a single articulator at a single place of articulation (cf. Recasens et al. 1993), some previous researchers have treated them as "inherently" palatalized based on their phonological behavior.[109] Among the arguments adduced in favor of this interpretation is the fact that palatals often serve as synchronic and/or diachronic results of palatalization of nonpalatals: this is true, e.g., of Polish, where the palatalized counterparts of /ts, dz, s, z, n/ are /tɕ, dʑ, ɕ, ʑ, ɲ/, respectively. Another argument is that palatalized alveolopalatals and palatalized palatals (i.e., palatals proper) are disallowed. Furthermore, no language is reported to contrast palatals (in the narrow sense) with palatalized velars, or alveolopalatals with palatalized

[108] Cf. Hall (1997:50) on phonological distinctions between these categories, and Ladefoged and Wu (1984:267) on the difficulty of drawing precise boundaries between adjacent phonetic categories, including palatoalveolars and palatals.

[109] But see Keating and Lahiri (1993) for a similar conclusion based on a phonetic study.

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palatoalveolars; in fact, alveolopalatals [tɕ, dʑ, ɕ, ʑ] and palatalized palatoalveolars [tʃʲ, dʒʲ, ʃʲ, ʒʲ] are reported to be articulatorily indistinguishable. These and related facts have led, e.g., Hall (1997) to interpret alveolopalatals as palatalized (palato)alveolars, and palatals as palatalized velars (50ff).

In a gestural approach to intrasegmental composition of consonants, the notion of "inherent palatalization" of palatals may be straightforwardly modeled as palatal specification of the consonants' V-tier (it will be recalled that in tongue body consonants, the place specification of the V-tier is identical to that of the C-tier). In strong prosodic positions, the V-gesture of the palatal will tend to be synchronized with or lag behind the C-gesture, while in weak positions associated with prevocalization, the C-gesture will tend to get shortened, partially uncovering the V-gesture which will surface as a subsegmental palatal vocoid.

The tendency of palatal consonants to prevocalize in suitable environments is especially pronounced in stops, which are described as articulatorily tense and long (Straka 1949:20 fn. 1; 1965:121ff), but at the same time represent the least stable manner of articulation in the palatal area. Recasens (1996:4) notes that the lack of stability in palatal stops is due to the difficulty inherent in forming a complete closure at the hard palate; as a result, the dorsopalatal contact of palatals is often reduced in positions favoring articulatory reduction. One frequent strategy for reducing palatal contact is spirantization of palatal stops; in cases of interest here, such reduction is achieved through a combined action of prevocalization and alveolarization, gliding, or

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debuccalization of the palatal's C-gesture.[110] In cases of a more extreme lenition, the C-gesture of the palatal is deleted altogether.

Data bearing on the prevocalization of palatals comes from a variety of sources including historical records that reflect pronunciations current at the time of writing, observations of early grammarians and orthoepists, contemporary dialect studies and descriptive grammars, and specialized treatments of loanword phonology. The combined evidence of these sources indicates that, synchronically, the C-gestures of palatals tend to remain palatal under language-internal prevocalization, while the prepalatalization onglide remains subphonemic. In diachrony, the onglide becomes phonemicized while the C-gesture of the original palatal may be alveolarized, debuccalized, or deleted. When words containing palatals are borrowed into languages that lack the corresponding articulations, the palatals are often unpacked into sequences of a palatal glide and the corresponding alveolar.

3.3.2 Catalan, French, Portuguese

In Majorcan Catalan, [ɲ, ʃ, tʃ] prevocalize preconsonantally; no difference is made between the underlying palatal nasal and that derived by assimilation to the following /c/.[111] Prevocalization is accompanied by alveolarization of the nasal and deletion of the

[110] It is possible that alveolarization of the consonantal component of palatals may be an automatic result of reduction in their duration (cf. data on the relative length of the alveolar and palatal nasals cited in Tuttle 1991:57ff).

[111] It will be recalled that prepalatalization of [ɲ] in Majorcan Catalan differs in this respect from the equivalent process in Djabugay, where only the underlying palatal nasal is subject to prevocalization (see 1.3).

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other palatals; although the prevocalization stage is not directly attested for the sibilants, it is nevertheless assumed by Mascaró (1985), who provides the following data:

/tronc/ [ˈtroŋc] 'log'	[ˈtroʲns] 'logs'[112]
/aɲ/ [ˈaɲ] 'year'	[ˈaʲns] 'years'
[əˈceʃ] 'this, that'	[əˈceʲ(s)s] 'these, those'
[dəˈzitʃ] 'wish'	[dəˈziʲ(s)s] 'wishes'

As discussed in 1.2, surface diphthongs that result from the above prevocalization behave differently from the corresponding underlying diphthongs. For instance, whereas an underlying *VjCs* sequence is simplified to *Vjs* in the case of underlying diphthongs, it is preserved in the derived diphthongs, cf. *vújséns*, not **vújtséns* 'eight hundred' (from *vújt* 'eight' plus *séns* 'hundreds') versus *ájns* 'years' (Mascaró 1985:140f).

In European Portuguese, the palatals /ʎ, ɲ, ʃ, ʒ/ [ʎ, ɲ, ɕ, ʑ][113] are subject to synchronic prepalatalization; the conditioning environment is described as the location after stressed nonfront vowels. This gives rise to alternative pronunciations of the type *malha* [ˈmaʎə ~ ˈmaʲʎə] 'mesh', *caixa* [ˈkaɕə ~ ˈkaʲɕə] 'box', *loja* [ˈlɔʑə ~ ˈlɔʲʑə] 'boutique', *unha* [ˈuɲə ~ ˈuʲɲə] '(finger)nail' (Barbosa 1965:64). The diphthongized vowels behave as allophones of the corresponding monophthongs and fail to contrast with them in ways typical of underlying diphthongs. Prepalatalization is not affected by vowel nasalization, cf. *longe* [ˈlõʒə ~ ˈlõʲʒə] 'far' (Barbosa 1965:101). Prepalatalization

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[112] The deletion of [c] in a three-consonant cluster /ncs/ is predictable, cf. *vént* 'wind', *sólc* 'furrow' with *véns* 'winds', *sóls* 'furrows' (Mascaró 1985:135).

[113] According Mateus and d'Andrade (2000), /ʃ, ʒ/ are phonetically alveolopalatals (11 fn. 4).

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of palatals is described by Giangola (2001) as alien to Brazilian Portuguese (7 fn. 3); nevertheless, rhymes of the type *beijo* /ˈbɛjʒɔ/ 'kiss' - *desejo* /dɛˈsɛʒɔ/ 'desire' and *baixo* /ˈbajʃɔ/ 'low' - *riacho* /riˈaʃɔ/ 'brook', mentioned in Camara (1977:99ff), clearly indicate that at least the palatal sibilants /ʃ, ʒ/ may prepalatalize in that variety as well.

Palsgrave (1530) notes in his discussion of the pronunciation of French vowels: "Also all wordes in the frenche tong which in writtyng ende in *age* shall in redyng and spekyng sounde an *i* between the *a* and *g*, as though that *a* were this diphthong *ai*: as for *langáge*, *heretáge*, *ságe* . . . they sounde *langwaige*, *heritaige*, *waige* . . . and so of all suche lyke excepte *rage*" (quoted in Parry-Williams 1923:210). The existence of the above prepalatalization of [ʒ] is confirmed by French borrowings in Breton, in which the onglide is notated in writing, and by rhymes of the type *sage / ferai ge* and *Marie ay ge / marriage* in literature of the period (cf. Meyer-Lübke 1913:91). During the fourtheenth and fifteenth centuries, this phenomenon was noted in Parisian French (Meyer-Lübke 1890:218). A parallel tendency was noted in contemporary English, as discussed immediately below.

3.3.3 English

Prevocalization of palatals, especially of the voiceless palatal fricative /ʃ/, runs in periodic cycles through a great part of the known history of English. For example, in the course of the thirteenth century prevocalization of /ʃ/ after /a, e/ was reflected in the diphthongized spellings like *aische* 'ashes', *aischen* 'to ask', *fleisch* 'flesh', and *freissch*

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'fresh'; such spellings disappear in the course of the next two centuries (Luick 1921:422f). Luick (1921) also notes prepalatalization of /dʒ/ after /a/ in Early Middle English as reflected in *aige* 'age' and *wayge* 'wager', among others (487f). Salesbury (1567) describes prepalatalization onglides to /tʃ, dʒ, ʃ/ as follows: "For in these wordes and such other in Englyshe, domage, heritage, language, ashe, lashe, watch, calme, call, *a* is thought to decline toward the sound of these diphthonges *ai*, *au*, and the wordes to be read in thys wyse, domaige, heritaige, languaige, aishe, waitche, caul, caulme" (quoted in Parry-Williams 1923:210). Prepalatalization onglides are reflected in the English words borrowed during that period into Welsh and Breton; the prevocalizing sibilants in them could be preceded by other consonants, primarily /n/ and /r/. In both Welsh and Breton, the vocalic nuclei preceding the sibilants are notated as full diphthongs, which indicates that they were salient enough to be perceived as segmental. The following examples of the relevant English borrowings in Welsh illustrate the onglides with different sibilants and distinct preceding vowels (cited after Parry-Williams 1923:210ff and 1914:65ff):

	Welsh		English
[dʒ]	*oraens, orains, oreins, orates, oraits*	<	*orange*
	cabaitsch	<	*cabbage*
	caets	<	*cage*
[tʃ]	*braens*	<	*branch*
[ʃ]	*brwiss*	<	*brush*
	bernais, fernais[114]	<	*varnish*

[114] Note that when [ɪ] is combined with the onglide to the following sibilant, there results a diphthong spelled <ai> (cf. *malais* < *malice* and *bernais/fernais* < *varnish*). This development is interesting in view of an apparently identical development in Avestan, whereby a high front vowel is notated as a schwa before an "epenthetic" <i>, cf. *ānəiti-* 'immobility (?)' < **ən-iiti-* < **an-iti-*. An alternative response is found in the American Midlands, where the affected vowel is tensed instead, cf. *fish* [fiʃ] (Hock 2003).

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[s]	*kwyset*	<	*gusset*
	malais	<	*malice*

Prepalatalization of palatals after /a/ and /ɛ/ also occurs in present-day Irish English, cf. *smash* [sma^{j}ʃ]. Harris (1987) notes that the onglide is more prominent after the more retracted and lowered realizations of the /a/ and /ɛ/, respectively (273). Prepalatalization of /ʃ/ is also well documented in the American Midlands and the South (cf. Lass 1985:184; Hock 2003a). Prevocalization phenomena of Southern American English were introduced into the range of problems discussed here by Sledd (1966), who, as far as can be ascertained, was the first to explicitly compare them to the breaking and umlaut phenomena of the earlier historical stages of English. Sledd's analysis and the rich dialectal data provided in his study are worth a detailed mention.

Sledd distinguishes the following two prevocalization environments (1966:36ff):

(1) Between any vowel or diphthong and word-final or preconsonantal liquid there develops a glide which is palatal before clear /r/ and /l/ and velar otherwise. /r/, normally dark, becomes clear between a stressed /ə/ and an immediately following consonant (/ə/ here being the result of a merger of Middle English /ɪ/, /ɛ/, /ʊ/ before final and preconsonantal /r/, cf. Wells 1982:199ff), and between /o/, /ɔ:/, /ɑ:/ and a following palatal consonant. /l/, normally dark, becomes clear when adjacent to a palatal glide. In certain positions, prevocalization of liquids is followed by their deletion: this is the explanation provided by Sledd for the southern shape of words like *bird* [bəjd], in which preconsonantal /r/ was palatalized after a stressed /ə/,

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prevocalized, and subsequently deleted; and *porch* [pojtʃ], where /r/ was palatalized before the following palatal, prevocalized, and deleted.[115]

(2) Between any stressed lax vowel and any prevocalic liquid or any consonant except dental or alveolar there develops a glide which matches the articulatory properties of the consonant. Palatal and palatalized consonants[116] condition the appearance of a palatal onglide, which combines with preceding vowels to produce gliding monophthongs. Examples of palatal prepalatalization, culled from Sledd (1966) and other dialect descriptions, include the sampling below.

[ɑʲ]/[ɔʲ] / __ ʃ, tʃ, dʒ	*wash, Washington* (Ocracoke, North Carolina; Howren 1962: 166); *wash, bosh, watch, dodge* (Atlanta, Georgia; Sledd 1966: 32);
[ʌʲ] / __ ʃ, (n)tʃ, dʒ	*Russian, hush* (Ocracoke, North Carolina; Howren 1962:167); *gush, hush, much, such* (Atlanta, Georgia; Sledd 1966:32); *judge, brush, mush, mushrooms, lunch, not much* (Louisville, Kentucky; Howren 1958:80-81); *brush, flush, much, touch* (Wilson 1934:212); *hush* (East Central Alabama; McMillan 1946:41);
[ʊʲ] / __ ʃ	*bush, push* (Ocracoke, North Carolina; Howren 1962:167); *push, I'm bushed* (Louisville, Kentucky; Howren 1958:86); *bushel, cushion, wish* [ʊʲ] (Wilson 1934:212);
[ɛ] > [ɛʲ] /__ dʒ	*hedge* (Louisville, Kentucky; Howren 1958:71);
[ɛ] > [eʲ]/__ ntʃ	*bench* [bẽintʃ] (East Texas; Stanley 1936:15-16);
[æʲ] / __ ʃ, (n)tʃ	*batch* (West Texas; Cearley 1974:36); *smashed, ashes* (Ocracoke, North Carolina; Howren 1962:166); *blanch* (Wilson 1934:212); *ashes, mash* (Louisville, Kentucky; Howren 1958:77); *cash, branch* (East Central Alabama; McMillan 1946:37f, Wells 1982 3:532);

[115] Words with such a (pre)vocalized /r/ current in other parts of the American South include *burn, burst, murmur, dirt, twirl, earth, heard, jerk, work, world, Brandenburg,* and *disturb* (quoted from Wilson 1934:212; McMillan 1946:55f, 57, 62; Howren 1958:64; and Wells 1982 3:548).

[116] The palatalized consonants include /k, g, ŋ/ after front vowels (discussed above).

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[ɪ] > [i] / __ (n)tʃ *itch, inch*
[i] > [i̝] / __ tʃ *beach, speech* (Wilson 1934:211)

Researchers dealing with the emergence of a palatal glide before palatal consonants in the Southern United States hesitate between assigning the glide to the consonant or the preceding vowel; when the latter alternative is considered, the emergence of the glide is ascribed to vowel tensing. The tensing hypothesis is strongest in the case of /æ/, which may be glided not only before palatals, but in other environments as well, subject to regional and social variation:[117] thus, in addition to forms like *cash, batch* and *branch* cited above, the glided realization of /æ/ is noted in words like *path, grass, half, halve, ask, last, pan, aunt, panther, pantry, land, ram, bat, rat, bad,* and *cap,* in which the glide seems to be due to vowel diphthongization.[118] The tensing analysis of [æ̝] is even more plausible in light of the fact that in the speech of East Central Alabama described by McMillan (1946:37ff, 51f), for example, /æ/ is not diphthongized before the palatal consonants /tʃ, dʒ, ʒ/, which would be expected if the appearance of the palatal glide were due to prevocalization, while diphthongizing before the nonpalatals /f, v, ŋ, g, n, θ,

[117] The gliding of /æ/ seems to have had a long history in American speech, as shown by the following half-literate spellings from seventeenth-century town records from eastern Massachusetts (Orbeck 1927:22f): *paith, raythe*r, *(schoole)maister, grain* 'grand', *haynge* ~ *hainge, saick* 'sack', and *substainshall.*

[118] Wilson (1934:212); Stanley (1936:18f); McMillan (1946:37f); Trager and Smith (1957:25); Howren (1958:100); Wells (1982:532).

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s, ʃ/.[119] However, even if vowel tensing could be considered the source of the glide in the case of /æ/ (or even all lax front vowels),[120] it would be difficult to invoke the same explanation in the case of nonfront vowels as this will leave the palatal quality of the glide unexplained. In the latter case, prevocalization of the palatals is more likely as an explanation for the presence of the glide.

It is interesting to note that even in Southern American English, with its fairly extensive prevocalization, the most frequent trigger of prevocalization seems to be the palatoalveolar sibilant /ʃ/. Howren (1958), for example, reports that in Louisville, Kentucky, the diphthongal allophone of /ʌ/ regularly occurs before /ʃ/ regardless of the pitch pattern, while before /tʃ/ it occurs only when there is an accompanying change of pitch on the vowel, citing as examples *lunch* (with a [ʌʲ]) and *lunchbox* (with a [ʌ]) (81). The only examples of prepalatalization after [ʊ] that appear in the above sources also invariably take place before /ʃ/ (e.g., Howren 1958:86).

3.3.4 Other languages

Many synchronic language descriptions offer additional evidence for the prevocalized realization of palatals. Thus, Jackson (1960-61) notes that word-final /ɲ/ in the

[119] Wells (1982) reports that the [æʲ] realization of /æ/ is becoming contrastive in the speech of some speakers, distinguishing such minimal pairs as *can* [kæʲn] 'container' and *can* [kæn] 'be able', and *halve* [hæʲv] and *have* [hæv] (536).

[120] And, possibly, also of the [u] realization of /ʊ/ in the Midlands forms of the type *push* [puʃ] cited by Hock (2003a).

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Plougrescant dialect of Breton may be realized with a palatal onglide (the full set of palatals in that dialect consists of /ɲ, ʎ, ʃ, ʒ/) (379). In Afrikaans, prepalatalization has been noted in the diminutive suffixes *-(n)tje* [(ɲ)c] and *-(n)dje* [(ɲ)ɟ]. Although it is said to take place only after back vowels, in the examples cited in Hooper (1937) and Lockwood (1952-53) it is also noted after the front vowels *œ* and *e*, cf.:

groot [xro:t]	'large'	*grootje* [xro:ʲci]
voet [fut]	'foot'	*voetje* [fuʲci]
munt [mœnt]	'coin'	*muntje* [mœʲnci]

Westermann and Ward (1933:56) note that in Nuer, a Nilotic language of southern Sudan and Ethiopia, palatal stops /c, ɟ/ are prevocalized after open vowels. Many examples of prevocalization of syllable- and word-final palatals are offered by Mon-Khmer languages. For instance, Smalley (1954:220) reports that, in Srê, palatals /c, ɟ, ɲ/ color following and especially preceding vowels, except high front vowels, with a palatal offglide "varying in prominence with the distance of the vowel from a palatal position"; cf. /pwac/ [pwaʲc] 'meat', /ntruɲ/ [ntruʲɲ] 'jump down'. In Ngeq, word- and syllable-final palatals are preceded by a prepalatalization onglide, which in addition tends to raise the glided high vowels /ua/, /ia/[121] (Smith 1973:79), cf.:

/luas/ 'to release'	[lueʲɕ]
/buaɲ/ 'to heap up'	[bueʲɲ]
/biac/ 'to be poor'	[bieʲc]

[121] /ua/, /ia/ are described in the source as high vowels followed by a low central offglide.

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In Katu, all palatal consonants are said to have a [i]-like onglide word-finally (Wallace 1969). In Stieng, palatals are preceded by an [i]-like onglide syllable-finally (Haupers 1969:134). Comparable statements are also found in the descriptive studies of Rəlɔm, Pacŏh and Chrau (Blood 1976:7f; Watson 1964:142; Thomas 1971:35).

Txukuhamẽi, spoken in the Brazilian states of Para and Mato Grosso, has four palatals /tʃ, ʒ, ɲ, j/ two of which, /tʃ/ and /ɲ/, are prepalatalized syllable-finally (Stout and Thomson 1974:163), cf. [ɲo^{jd}ɲ][122] 'urubu', [bo^{j}tʃ] 'arrive'. In Huaraz Quechua, /ʎ/ may be variably realized as [ʎ] or [^{j}ʎ] in noninitial allophones, cf. /aʎpa/ [ˈa^{j}ʎpa] 'ground', /paʎa/ [ˈpaʎa ~ ˈpa^{j}ʎa] 'thanks' (word-final /ʎ/ is not attested). /a/ followed by a prevocalization offglide surfaces as [a^{j}] while an underlying diphthong [aj] is realized as [e:], cf. /pajʎa/ [ˈpe:ʎa] 'only he' vs. /paʎa/ [ˈpa^{j}ʎa] 'thanks' (Swisshelm 1971:5f, 17). Texmelucan Zapotec prevocalizes palatoalveolars /c, ʒ, ʎ, ɲ/ (Speck 1978:19). Morpheme boundaries, as well as the height, rounding, length, laryngealization, nasalization or tone of the preceding vowel do not affect prepalatalization (Charles Speck, p. c.), cf.:

/lac/	[la^{j}c^{h}]	'flat'
/laʒ/	[la^{j}ˈc]	'liver'
/naɲ/	[na^{j}ˈɲ]	'inside'
/mbaʎã/	[mba^{j}ˈʎã]	'my compadre'

[122] The pre-stopped allophone of the nasal is conditioned by the preceding oral vowel.

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In Ayoquesco Zapotec, the etymological sequence -**ni* has produced [-jɲ], cf. **na+ pani* 'alive' > *nàbayn*, **tani* 'hill' > *dàyn* (MacLaury 1970:100, 141ff; reconstructions cited after Kaufman 1994). The existence of an intermediate palatal stage [ɲ] representing an earlier **ni* is suggested by the parallel treatment of /ɲ/ in Spanish loanwords, cf. *escaño* 'bench with a back' > Ayoquesco *škǎyn*.

Many examples of palatal prevocalization come from languages of Australia. The majority of Australian languages have a rich system of coronals which includes two sets of apicals (apicoalveolars and retroflexes) and two sets of laminals (laminodentals and laminopalatals). Most cases of prepalatalization reported from languages of Australia involve laminopalatals. For example, long vowels /i:, a:, u:/ in Kayardild are said to acquire a palatal offglide before laminopalatals; Evans (1995a:59f) reports that the tendency is the most marked after /a:/, but is also possible after /u:/ or /i:/ before /c/, cf.:

/ba:c/	[ba:ʲɟ]	'bite'
/du:cind/	[ˈdo:ɟɪnt ~ ˈdo:ʲɟɪnt]	'younger brother'
/di:c/	[ˈde:ɟ ~ ˈde:ʲɟ]	'sit'

In Margany and Gunya, laminopalatal consonants frequently have an onglide after stressed nonfront vowels. Alternatively, or simultaneously, the vowel may be fronted and raised (Breen 1981:288), cf.:

/maji/	[mǽʲjɪ]	'bread'
/baɲɟaɽa/	[bɛ́ʲɲɟɐɽɐ]	'pine'
/gabuɲ/	[gɐ́buʲɲ]	'egg'

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In Djapu, both long and short vowels are front-raised next to /j/ and the laminopalatals /c, ɲ/, and back-lowered next to /w/. Of the three vowels, /a/ shows the widest range of allophonic variation. The following realizations are especially interesting (Morphy 1983:19):

/a/	>	[aʲ]	/ __ c, ɲ
/i/ [ɪ]	>	[i]	/ __ c, ɲ, j
/u/	>	[ʊʲ]	/ __ c, ɲ, j

In Mantjiltjara, the palatal glide is prevocalized after /a/; the laminoalveolar stop /c/ is also sometimes prepalatalized in this environment (Marsh 1969:139), cf.:

/maji/	[mɑji ~ mɑʲji]	'food'
/pacala/	[pɑɟɑlɑ ~ pɑʲɟɑlɑ]	'bite it!'

In Pintupi, /j/ is prepalatalized after /a/ (Hansen and Hansen 1969:157), cf.:

/wajunpuwa/	[wɑʲjunpʊwʌ]	'pare (it)'

In Yaraldi, laminals /c, ɲ, ʎ/ have a distinct palatal onglide after /a/; the onglide is often notated as segmental in early records on the language, cf. <kainyani> and <kainyi> for /kaɲuni/ 'youth' and /kaɲi/ 'urine' (McDonald 2002:96, 104). In Ngalkbun, all vowels except /i/ have glided allophones when followed by the laminopalatals /c, ɲ/ (Sandefur and Jentian 1977:70), cf.:

/e/	/bojeɲ/	[bɔjɛʲɲ/	'big'
/a/	/macun/	[mɑʲcun]	'carpet snake'
/o/	/kocno/	[kɔʲcno]	'his head'
/u/	/nakomtuc/	[nʌkɔmtuʲc]	'little boy'

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Finally, Djabugay prepalatalizes its underlying palatal nasal. This distinguishes it from the apical nasal /n/, which is realized as laminopalatal before a laminal stop. This helps distinguish words that otherwise would be (nearly) homophonous (Patz 1991:253f), cf.:

banydji	[baʲɲd̪ʲi]	'wife's brother'
bandjil	[baɲd̪ʲil]	'woman'

The data from the languages of Australia is especially interesting in showing prepalatalization of the palatal glide. Elbert (1974) notes that in Puluwat, a Trukic language of Micronesia, an initial /j/ also develops a [e]-like onglide, cf. [ᵉjóó] 'yes'. The ability of the palatal glide to prevocalize seems to indicate that, in common with other tongue body consonants, its underlying representation includes both C- and V-gestures, the latter of which surfaces in language-specific environments that favor prevocalization. This contradicts the position of Gick (2003), who finds no evidence for the bigestural composition of /j/ in American English. While it is possible that the underlying gestural composition of the palatal glide may turn out to be language-specific, it is also clear that more experimental and phonological research is needed to uncover its internal structure.

3.3.5 Unpacking of /ɲ/, /ʎ/ in loanwords

Unpacking of palatal articulations in loanwords into sequences of a palatal glide and an alveolar coronal warrants a separate mention as, although its outcome may parallel that of language-internal prevocalization, the two processes differ with respect to their motivation. As outlined in 1.4, prevocalization of palatals is motivated by consonant

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lenition in weak prosodic positions. Unpacking of palatals in loanwords, on the other hand, is motivated by the comparative rarity of some types of palatal articulations (for example, stops) in the world's languages: rather than being an instance of consonant lenition, it represents one possible strategy of loanword adaptation. Languages that do not have a comparable articulation type may deal with palatals in the borrowed vocabulary in a variety of ways: in some instances, the palatal component is eliminated altogether, and in others, it is preserved in the form of a segmental glide. The position of the glide with respect to the original consonant is usually determined by the syllable position of the latter, with the result Cj frequently obtaining in onsets and jC in codas.[123]

Some of the most frequently cited cases of palatal unpacking in loanwords have to do with the palatal nasal and lateral. Thus, these were unpacked into [jn] and [jl] in French loanwords in Middle English, which has resulted in falling diphthongs in the recipient language, cf. *Spaine* 'Spain', *joinen* 'to join, bind', *(a)vailen* 'to make useful' (Luick 1921:449). /ɲ/ is also unpacked in the sixteenth-century French loanwords in Breton, cf. *cigoing* < *cigogne* 'stork' and *compaignun* < *compagnon* 'companion' (Parry-Williams 1914:69). In the central dialects of Basque, earlier /ɲ/ and /ʎ/ of native and foreign origin underwent depalatalization resulting in [jn] and [jl], cf. such native words as *leinu* ~ *leñu* 'lineage, tribe' and *zeinha* ~ *zeña* 'cross oneself', and such Spanish borrowings as *oilo* 'hen' (< *pollo*), *boteila* ~ *botoila* 'bottle' (< *botella*), *bainu*

[123] Some languages seem to attest only one of these shapes (cf. Andersen 1972).

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~ *mainu* 'bath' (< *baño*) and, more recently, *painuelo* 'handkerchief' (< *pañuelo*) and *koinak* 'cognac' (< *coñac*).[124]

3.4 Alveolars

Prevowels resulting from prevocalization of alveolars take the surface form of palatal onglides of varying height, frontness, tenseness, and duration. The discussion below focuses on clear liquids, /s/, and /n/, which emerge from the literature as the most frequent triggers of prevocalization.

3.4.1 Liquids

3.4.1.1 Clear /l/

According to Delattre (1965), the tongue body position of clear /l/s "is in the region of front vowels, perhaps nearest an [ɛ]" (102); nevertheless, prevowels triggered by clear /l/s surface as a high front [i] or [j]. Thus, syllable-final /l/ may be prevocalized or fully vocalized in Sardinian; the process is at the most advanced stage preceding labials, less so before the following velars and /tʃ/ (Contini 1987:370ff). Full vocalization of /l/ may induce gemination in the following obstruent, which in turn may cause deletion of the prevocalization onglide. Various stages and outcomes of this process are presented in Table 7, which shows the varying dialectal shapes of /ˈkulpa/ 'fault', /ˈalva/ 'beard', and /ˈmaltʃu/ 'male'.

[124] Michelena (1976:195f), Echaide (1976:164), Trask (1997:146ff), Hualde (1991:119), Hualde and Ortiz (2003:28).

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Table 7. /l/-prevocalization in Sardinian

Underlying form	Prevocalization	Liquid deletion (± obstruent gemination)	Onglide deletion
/ˈkulpa/	[ˈkuʲlpa]	[ˈkujppa]	[ˈkuppa]
/ˈalva/	[ˈaʲlva]	[ˈaj(v)va]	[ˈavva]
/ˈmaltʃu/	[ˈmaʲltʃu]	[ˈmajʃu]	[ˈmaʃu]

Prevocalization of clear /l/ is also attested across word boundaries: thus, Contini (1986) cites /l/-(pre)vocalization as one of the most conspicuous external sandhi of contemporary Sardinian, cf. [sɔj ˈbɔɛzɛ] 'the cows', where [sɔj] is the realization of the plural definite article /sɔl/ before words beginning with consonants, especially labials. While in some localities the vocalization of the lateral is complete, others attest the prevocalized stage [ʲl], as in [sɔʲl ˈbɔɛzɛ] (Contini 1986:548 fn. 56).

The Sardinian data above shows that it is perhaps unnecessary to make a categorical distinction between prevocalization of liquids and those cases in which the liquids are vocalized without a documented prevocalization stage. Examples of the latter process include realizations of words like *cuerpo* 'body' and *algo* 'something' as [kwejpo] and [ajgo], respectively, in many colloquial varieties of Spanish; of words like *Pult* as [pujt] in Viennese German; and possibly such diachronic outcomes as Latin *vulture(m)*, *multu(m)* > Spanish *buitre* 'vulture', Portuguese *muito* 'much'. In the last case, and in diachrony in general, the existence of a prevocalization stage can only occasionally be inferred from older spellings, concurrent developments, or on the basis of a comparison with similar facts of genetically related languages with similar phonological tendencies. One specific historical change where the existence of a prevocalization stage may be inferred in such a roundabout way concerns the treatment of preconsonantal /l/ in some

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Romance varieties of Italy noted in Rohlfs (1966). While in some areas the liquid seems to have directly vocalized producing diphthongs with preceding vowels (compare *vuipe* 'fox', *aiburo* 'tree' with Standard Italian *volpe*, *albero*), in others this development resulted in gemination (and occasional voicing) of the consonant following the lateral: compare *una oitta* ~ *vòidda*, *aittro* with Standard Italian *una volta* 'once', *altro* 'other' (Rohlfs 1966:344ff). Without the evidence of Sardinian, the development in these varieties could have been analyzed as direct vocalization. However, comparison with the Sardinian data makes the prevocalization analysis more attractive than an analysis based on simple vocalization, not only because of the clear parallelism between the two cases, but also because such an account is able to provide a straightforward explanation both for the diphthongal offglide and for the gemination of the following obstruent. Under the prevocalization account, this development may be accounted for by a combination of leftward sliding of the lateral's V-gesture, responsible for the front onglide (VlC > V^{j}lC), and overlapping of its C-gesture with the gestures appropriate to the following consonant, responsible for their increased duration (and, where applicable, voicing). The prevocalization analysis is also supported by seventeenth-century spellings which include some of the forms that have been used here for the purposes of exemplification, e.g., *ascoiltare* 'hear', *ailtro* 'other', and *voilta* 'time' (Kolovrat 1923:253).

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3.4.1.2 Clear /r/

Clear /r/ frequently prepalatalizes in Germanic languages, including English. Thus, in some areas of the American South, underlying /ər/ is realized preconsonantally as [əj], cf. *bird* [bəjd]. The source of the onglide here is clear /r/ characteristic of this and some other English dialect areas (Harris 1994:259, 2006).[125] The glided allophone of the rhotic alternates with vowel length (Wells 1982:545). Prevocalization, rather than direct vocalization of the rhotic is assumed here by Sledd (1966). Southern American English is also known for pronunciations like [pojtʃ] 'porch'; Sledd (1966) explains them as resulting from palatalization of the rhotic before palatals, followed by prevocalization and deletion.

Prevocalization of clear /r/ has also been described for some dialects of Swiss German. Thus, Vetsch (1910:87) mentions prepalatalization of /r/ after a preceding /e/; among his examples are [vɛʲrə] 'forbid' and [ʃpɛʲrə] 'obstruct' (cf. German *wehren, sperren*).[126] Following prepalatalization, the rhotic may be deleted, as, for example, in the family name *Herschi*, realized as [hɛjʃɪ]. If prevocalized forms were not documented in closely related dialects, the realization of /r/ in *Herschi* could have been interpreted

[125] I.e., the pronunciation of /r/ in the Ottawa Valley, heavily influenced by Irish, is described as follows: "[the /r/] has a quite different tongue-position from the typical Canadian pronunciation: a high front position which gives an [i]-like quality to the consonant" (Pringle and Padolsky 1981:345).

[126] Dialects which prepalatalize /r/ may prepalatalize other alveolars as well, including /n/, as in [tsɛʲj] 'teeth' and /d/, as in [grɛʲdə] 'gerade machen' (German *Zähne, gerade*). The same dialect also prepalatalizes /ʃ/, cf. [ɛʲʃ] 'ash-tree' (German *Esche*) (Vetsch 1910).

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as straight vocalization; however, presence of such forms in the neighboring dialects suggests that this form went through a prevocalization stage as well.

The situation in Swiss German provides evidence, albeit indirectly, that the development of the [j] realization of clear /r/ in Southern American English also involved a prevocalization stage. Although the existence of this stage is deduced by Sledd (1966), it is not directly attested, and the evidence from a closely related language goes a long way toward filling this gap. This observation is based on the well-known tendency of closely related languages to undergo similar developments at various structural levels; among frequently cited examples of this kind specifically involving Germanic is umlaut, which developed in the individual languages after their separation (cf. Zhirmunski 1960:310). Prepalatalization of liquids, in this case of clear /r/, appears to be yet another tendency that the histories and synchronic phonologies of Germanic languages display as a group (see also Howell 1991).

3.4.2 /s/

/s/ is one of the most frequent triggers of prepalatalization. For example, the /s/ phoneme of Ngeq (Mon-Khmer), described as a dental sibilant, patterns together with palatals with respect to prepalatalization, cf. /luas/ [lue^{j}h] 'to release' (Smith 1973:19). Straka (1949) notes that in an especially relaxed pronunciation of some Czechs there is an audible transitional glide between /s/ and a preceding /a/ (38 fn. 3). In various dialects of Ancient Greek, /s/ developed a sporadic prepalatalization onglide in preconsonantal coda; the development is said to have been especially frequent in

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Aeolic, but it is also attested elsewhere. Thus, Schwyzer (1953) cites Lesbian παλαίσταν (cf. Attic παλαστή) and Boeotian αιστεα (cf. Homeric άστεα), among other forms (276). In Brazilian Portuguese, the archiphoneme /S/ is prepalatalized in coda both in dialects in which it is realized as a palatoalveolar sibilant, and in dialects in which it is realized as a dental sibilant (cf. 1.2). At present, prevocalized allophones of /S/ freely alternate with fully vocalized allophones, and unless the tendency for full vocalization is checked, this may eventually result in the replacement of the sibilant with the palatal glide.

The situation in Brazilian Portuguese is privileged in that prevocalization of /S/ is relatively well documented. If this had been an instance of a diachronic change, with no documented intermediate stages between the original /S/ and /j/, it may have been interpreted as straight vocalization. This consideration makes it likely that the vocalizations of this sibilant in the historical development of other dialects of Portuguese and other Romance languages also may have involved prevocalization, rather than straight vocalization of the sibilant, as is usually assumed. Examples of such developments include, e.g., *uj dias* (< *os dias*) 'the days' in Madeira Portuguese, *laj bèloj bakós* (< *las bèlos bakós*) 'the beautiful cows' in Occitan, *noi* (< *nos*) 'we' in Italian and Romanian, and [ʃoj] (< French *chose* [ʃoz]) 'thing' in Haitian (Michel 1948-50; Guiter 1948-50; Straka 1964; Tinelli 1981; Stein 1984). In some cases, prevocalized forms are, in fact, attested through older spellings and/or dialectal evidence. For example, final *-s* in Occitan may have surface realizations [s], [z], [j], or [ø] depending

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on the dialect,[127] but the prevocalization stage is attested both through older spellings, such as *leis* for *las*, and dialectal pronunciations, such as [laʲz] for *les*, noted by Ronjat (1932:277ff).

The above analysis of /s/-vocalization in the (Western) Romance languages is warranted not only by comparison with the genetically and typologically related Brazilian Portuguese, but also on the basis of a related change in the languages themselves whereby /s/ developed a vocalic onglide word-initially before /p, t, k/. This phonological tendency, common to all Western Romance languages, had its roots in Vulgar Latin, where the onglide was spelled with a [i] or [e]; examples in the modern languages include Spanish [espexo], Brazilian Portuguese [(i)speʎu], Catalan [espiʎ] 'mirror', French [ekɔl] 'school' and [eskalje] 'stairs' (Freitas and Rodrigues 2003). The development of a vocalic onglide by word-initial preconsonantal /s/ was related to the prevocalization of word- and syllable-final /s/ by Millardet, who viewed the two phenomena as different manifestations of the same underlying process (1910:143). Millardet's basic insight may be substantiated by viewing both processes as progressive weakening of /s/ in a syllabic coda. A coda analysis of /s/ in the initial *sC*-clusters is widely adopted, for instance, for European Portuguese, where the sibilant is assumed to occupy the coda position of an initial syllable with a left-adjacent nucleus (cf. Kaye 1991). The nucleus may be empty (e.g., in the speech of Lisbon, where *escola* 'school' is phonetically [ʃkɔlɐ]), or it may be realized as a surface vowel: thus, in Braga it

[127] The precise details of this gradual weakening vary from one dialect to the next, but, as a rule, /s/ is said to weaken everywhere except before voiceless stops (cf. Ronjat 1932:275ff; Guiter 1948-50; Michel 1948-50).

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surfaces as [i], [ɨ], or [e] (Mateus and Andrade 2000; Freitas and Rodrigues 2003). In gestural terms, the pre-/s/ onglide may be interpreted as the V-gesture of /s/: such an analysis would account for the diversity of its surface manifestations in terms of their height and the degree of frontness, which can be related to the language-specific realization of the orthographic <s>. Such an analysis predicts, among other things, that the pre-/s/ onglide will be initially subsegmental; this is borne out by the available data. For example, the pre-/s/ onglide in Welsh fails to count as a syllable in the oldest verse (Jones 1931:26; Jackson 1953:528), and speakers of Modern Western Armenian fail to perceive a similar onglide in their language as a segmental vowel (Bardakjian and Thomson 1977). Subsequently, the vocalic onglide may develop into a full vowel phoneme in the same way as postvocalic prevowels may develop into diphthongal offglides. The C-gesture of the sibilant, staying in a coda, may be subject to further weakening, which may take a variety of forms including gliding, debuccalization, or deletion. For example, forms like *eitable* 'stable' < *stabulum* in Occitan suggest a sequence of prevocalization and gliding, i.e., /s/ → [^{e}s] → [es] → /ej/ (Allières 1964:75). In some varieties of Spanish, the C-gesture of /s/ is debuccalized, suggesting a sequence of prevocalization and debuccalization, i.e., /s/ → [^{e}s] → [es] → [eh]. In French, the C-gesture of /s/ is deleted altogether: /s/ → [^{e}s] → [es] → /e/. Obvious parallelism between the behavior of /s/ in the initial *sC*-clusters and its behavior in other syllable-final positions suggests that both may be accounted for by means of the same

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underlying process, namely, prosodically driven retiming of the sibilant's component gestures.[128]

Another issue that may be briefly addressed here is the reason behind the frequent prepalatalization of /s/. Various researchers have attempted to explain the tendency of /s/ to (pre)vocalize with a vocoid of a palatal quality by assuming an intermediate stage [ç]; thus, the existence of a palatalized stage is assumed by Rousselot (1891) and Allières (1954). This approach may be justified by the frequently attested tendency of /s/ towards "spontaneous" palatalization in coda (cf. Cooper and Cooper 1966; Smith 1973 for such a development in Mon-Khmer). However, there also exists a strand of research where the passage from /s/ to a palatal glide is explained by reference to the articulatory and acoustic properties of the sibilant without recourse to intermediate shifts. For example, Millardet (1910) sees the vocalization of /s/ as relaxation of its constriction ("une sorte de relâchment de l'[s] que la langue a articulée incomplètement", 142), and essentially the same explanation is offered by Straka (1949:38 fn. 3).

Among the phonetic and phonological characteristics of /s/ that may help explain its tendency towards (pre)vocalization are its palatalizing properties, its frequent connection with palatals in consonant systems, and its connection with the subtype of palatalization known as assibilation. Thus, Bhat (1978) notes that sibilants and

[128] Prothetic vowels are also attested in the initial *sC*-sequences in loanwords into languages in which such sequences are disallowed. It is unclear to what extent this development is comparable to the language-internal developments treated here; for the present, this issue will have to remain an open question. (Cf. 3.3.5 for a comparable problem involving the unpacking of palatals in loanwords versus their prevocalization language-internally).

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affricates with sibilant release are among the apical consonants which are most susceptible to palatalization: in addition to palatalizing spontaneously, they also palatalize in a greater number of environments than other apical consonants (71). Sibilants may also pattern in the phonology together with palatals, function as allophones of palatals, and be replaced by palatals in language contact situations. Thus, the dental sibilant /s/ patterns with the palatals in Ngeq (Mon-Khmer); in various languages of South America, [s] and/or [z] function as allophones of /j/ (Jensen 1999:135); and in Fitzroy Crossing Children's Pidgin, English /s/ and /z/ are rendered by palatal consonants, cf. /can/ < *sun*, /cuwim/ < *swim*, /tracit/ < *trousers* (Fraser 1977:175, 186). A subtype of palatalization known as assibilation consists in the [s]- or [z]-like realization of the fricative noise of the palatalized consonant. Assibilation is one of the distinguishing characteristics of synchronic palatalization, e.g., in Quebec French (Belakovskaja 1973:36). In some Bantu languages, palatalization of labial fricatives has led to the development of doubly articulated labio-alveolar fricatives, cf. *f, β* > $\widehat{fs}$, $\widehat{\beta z}$ (Ponelis 1974:48). Carried to its logical end, this subtype of palatalization results in sibilant realization of the affected consonants, cf. French *ciel* [s-] and Spanish *cielo* [s-] 'sky' < Latin *caelum* [k-] (Bhat 1978:56ff). Assibilation is especially interesting in the present context in view of the recent claim that, at least in the case of coronal stops, it results from reinterpretation of the brief period of turbulence taking place at the release of a stop into the following (front) high vocoid (cf. Kim 2001; Hall et al. 2004). The relevance of this claim to prevocalization lies in its establishing a concrete acoustic and articulatory connection between [s], [z] and the front high vocoids. One way this

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connection surfaces in the phonology is in the ability of /s/ to induce palatalization or fronting of adjacent segments: thus, the palatalizing effect of /s/ on adjacent vowels is documented by Michel (1948-50), and its effect on adjacent consonants includes palatalization of a preceding /n/ discussed immediately below.

To summarize: the ability of /s/ to prevocalize with a front onglide seems to go hand in hand with those of its acoustic and articulatory properties which determine its propensity towards spontaneous palatalization, and fronting or palatalization of adjacent segments. These properties of /s/ may be due in part to the fronted tongue body position of the sibilant, and in part to its acoustic properties. The Romance data reviewed in this section show that prevocalization of coda /s/ may take place not only word-medially and finally, but also word-initially: the similarity of the conditioning environment and results of prothesis in the initial *sC*-clusters and prevocalization of /s/ in other positions allows both phenomena to be explained by means of the retiming of the sibilant's constituent gestures.

3.4.3 /n/

Among the consonants frequently palatalized by a following /s/ is the alveolar nasal; as a result of such palatalization, the nasal often develops a prepalatalization onglide. If, in addition, the nasal loses its oral occlusion, this development is often interpreted as surface vocalization. For example, /s/ had a palatalizing effect on the preceding nasal in Lesbian Greek (cf. Grammont 1948; Pellegrini 1961; Foley 1975; Wetzels 1985). This development affected the primary word-final /ns/, and /ns/ of secondary origin where

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the sibilant developed out of earlier *-*ts*, *-*tj*-, and **t* before *i*, cf. primary active third plural endings *-*a-nti*, *-*e-nti*, *-*o-nti* > Lesbian -*ajsi*, -*ejsi*, -*ojsi* (Lejeune 1972:129ff). Other dialects attest compensatory lengthening in the same environment, cf. Lesbian παῖσα versus Doric and Ionic πᾶσα, all from earlier *παντϳα. Grammont (1948) attributes the palatalizing effect of /s/ on the preceding nasal to its articulation: "En lesbien la nasale perd sa nasalité devant -*s*, et ses vibrations, palatalisées par l'-*s*, qui était vraisemblablement articulé avec la pointe de la langue levée vers la partie antérieure du palais, subsistent après la voyelle sous forme d'*i* deuxième élemént du diphtongue" (150).

Schwyzer (1953:287) and Pellegrini (1961) both draw a parallel between the above development in Lesbian Greek and that in the Visperterminen dialect of Swiss German. In that dialect, /n/ was deleted before fricatives while nasalizing the preceding vowel, which was in addition either lengthened or diphthongized (Wipf 1910:44ff; Anderson 1981:516f). Judging by the data presented in Wipf (1910), lengthening affects the high vowels and, in one word, also /a/, cf.:

un > *ũ:* / __ *ʃ, f, s*	*xũ:ʃt* 'art', *fũ:f* 'five', *rũ:s* 'ditch'
in > *ĩ:* / __ *ʃ, ç*	*fi:ʃter* 'dark, dark-colored', *trĩ:çu* 'to drink'
ün > *ĩ:* / __ *ʃ,f*	*ĩ:ʃ* 'us', *frnĩ:ftig* 'sensible, prudent'
an > *ã:* / __ *f*	*sã:ft* 'easily'

When /n/ occurred between /e/ or /a/ and coronal fricatives, the preceding vowel was diphthongized, cf.:

an > *ãj* / __ *s, ʃ, ç*	*Hãisi* < *Hansi* (*Hans*, diminutive), *xãjʃt* 'you (sg.) can', *ãjço 'butter'*, *bãjç* 'bank'

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en > *ẽj* / __ *ʃ, ç* *dẽjçu* 'to thank', *pfẽjʃter* 'window'

The Visperterminen German data are interesting in several respects. First, the diphthongization of /a/ and /e/ shows that the nasal was fronted or palatalized before the following coronal fricatives. The diphthongal offglide in this case may represent what was, originally, a prevocalization onglide. After these nonhigh vowels, which are articulatorily removed from it, the palatal onglide was reinterpreted as a diphthongal offglide. Second, this set of data shows complementary distribution of vowel diphthongization and compensatory lengthening. While the former is confined to nonhigh vowels, the latter takes place in two types of cases: in one word, it affects /a/ before /f/, and elsewhere it affects high vowels. It appears that the lengthening, rather than diphthongization, of /a/ before /nf/ indicates the failure of the nasal to be palatalized in that position: prevocalization of a nonpalatalized nasal would have produced an onglide of a central or back quality, which would have been reinterpreted as an added vowel length. In the case of high vowels, vowel lengthening is probably due to vowel merger with the palatal onglide. The secondary status of vowel lengthening is clear in the case of /u/, since the diphthongal stage is preserved in *xujʃt* 'you (sg.) come' (Wipf 1910:46). A typological parallel from Midlands American English (Hock 2003a) provides additional support for the correctness of this analaysis: while after the nonhigh vowels prepalatalization of /ʃ/ is realized as an onglide to the sibilant (cf. *mash* [mæʲʃ], *gosh* [gɔʲʃ], *measure* [mɛʲʒər]), it may surface as vowel tensing after the high vowels (cf. *fish* [fiʃ], *push* [puʃ]).

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In many varieties of Romance, there is a tendency for a (nasalized) glide to develop between a stressed /a/ and a following nasal, whether syllable-final or intervocalic. When the tongue body position of the nasal is fronted, the glide is of a front quality, and when it is retracted, the glide is retracted as well. For example, the areas of Rhaeto-Romance that favor palatalization of final velars attest forms like *sa^{j}ntç* 'blood', whereas for nonpalatalizing areas Gartner cites forms like *ma^{w}n* 'hand', *sa^{w}n ~ sa^{w}ŋk* 'blood', and *ufá^{w}nts* 'children' (1910:133f). Ritter (1981) notes similar forms in Samnaun Romansh, e.g., *maun* 'hand', *ufaunt* 'child', *launa* 'wool', and *fəntáuna* 'spring' < *manu(m)*, *infante(m)*, *lana(m)*, and *fontana(m)*. Tuttle (1991) cites an interesting morphophonemic alternation in a variety of Italo-Romance in which the singular forms ending in /n/ have a phonetically weakened form of the nasal, possibly a [ŋ], whereas in the plural forms the influence of the plural morpheme /s/ has helped preserve the alveolar articulation of the nasal.[129] Both allophones of the nasal were subject to prevocalization, producing onglides that correspond to the tongue body position of each allophone, cf. singular *maun* 'hand' versus plural *mains* 'hands' (Tuttle 1991: 47 fn. 62). Other instances of nasal prevocalization include Old French forms like *faim* 'hunger', *pain* 'bread' < *fame*, *pane* (Meyer-Lübke 1890:227ff; Marcato 1983-84:211; Tuttle 1991:47).[130] In early thirteenth century Anglo-Norman, the spelling *an*

[129] It is also possible that, rather than preserving the earlier phonetic value of the nasal, the sibilant may have had a fronting influence on it.

[130] Interestingly, only originally intervocalic nasals prevocalize in French, while those in coda fail to do so (cf. Meyer- Lübke 1890:227).

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began to be replaced by *aun* which, at least originally, represented a diphthongal pronunciation: the diphthongal spelling is preserved in such borrowings, done during the Middle English period, as *aunt* (Old French *ante* < *amita*) and *launch* (Anglo-Norman *lanche* ~ French *lance*), but no longer in *example*, *ancestor*, or *plant* (for which *exaumple*, *auncester*, and *plaunt* were attested earlier) (Wyld 1927:132; Pope 1932; Jespersen 1922:108ff; Fouché 1961:374). It is sometimes held that the Anglo-Norman spelling <au> reflects nasalization of the vowel rather than a diphthong. However, the phonetic reality of the diphthong is borne out by the shape of the English borrowings in Welsh, where it is rendered as such, cf. *dance, example* > *dawns*, *exawmpyl* (Parry-Williams 1923:199ff). Diphthongal pronunciation of a prenasal [a] in sixteenth-century continental French is also noted, e.g., in place-names like *Normaund* and *Aungers* (Fouché 1961:374). Onglides to intervocalic onsets are attested in many Romance varieties of Italy, cf. *lajna* ~ *laŋna* 'wool' in Cairo Montenotte Ligurian (Parry 1984),[131] *kambàjna* 'bell' < Latin *campana* in Abruzzese (Rohlfs 1966:47; Marcato 1983-84:210); and *lo^{w}na* < *luna* 'moon', *ve^{j}n* > *vinu* 'wine' in Bolognese (Hajek 1991).

The development of diphthongs as a result of nasal prevocalization is usually distinguished from simple diphthongization of nasalized vowels. A sharp line between the two is drawn by Grammont (1933:221f), Richter (1934:180f), Pellegrini (1961), Anglade (1961:12), Marcato (1983-84), and Tuttle (1991:46f), among others; Meyer-Lübke (1890) also remarks on "le son furtif palatal qui se développe facilement entre

[131] Both glides were subsequently "hardened" to produce a [ŋ] (cf. Hajek 1991).

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une voyelle nasale et une *n* dentale" (228). Some of the typical cases of nasalized vowel diphthongization are the realization of nasalized vowels in Portuguese, cf. *bem* [bẽj] 'well', *bom* [bõw̃] 'good', *um* [ũw̃] 'one', [pãw̃] 'bread'; in Canadian French, cf. *différence* /diferãs/ [dᶻɪferaə̃s], *panse* /pãs/ [põws]; or in Charleroi Walloon, cf. *Jean* [ʒãj̃] (Gendron 1966:99; de Reuse 1985; Hajek 1991:273; Giangola 2001:100ff). The researchers who make a distinction between prevocalization and diphthongization of nasalized vowels, however, do not offer an objective criterion that would enable an automatic separation between the two. One criterion that may be suggested here is the relative chronology of glide insertion and loss of the nasal: if the glide emerges before the loss of the nasal, it is possible for it to be the result of prevocalization, while if the appearance of the glide postdates the loss of the nasal, the process in question is to be analyzed as vowel diphthongization. Another possible criterion is the quality of the inserted glide, which will reflect the tongue body position of the nasal in the case of prevocalization, and the place of articulation of the vowel in the case of diphthongization. In practice, however, the distinction is not so easily drawn, in part because of the conservative nature of writing systems, which may preserve the nasal long after it has been lost in the pronunciation, and in part because of the possibility of vowels influencing the place of articulation of a following nasal. In such cases, vowel diphthongization and prevocalization may become indistinguishable, and further research is needed in order to find a reliable way to distinguish between the two.

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3.4.4 Malay and Tibeto-Burman

The preceding sections have focused on instances of prevocalization affecting a single alveolar consonant in a language. In the examples assembled in this section, prevocalization serves as a systematic response to coda weakening. One instance of such a development is provided by the northern dialects of Malay, in which word-final consonants were subject to an across-the-board weakening. The results of this process are dialect-specific, and those of interest here concern final /s/ and /l/ in the Kedah dialect, which prevocalizes both alveolars. After their prevocalization, /s/ is debuccalized and /l/ is deleted. A comparison of the Kedah forms with those of Standard Malay, where the original situation is preserved, illustrates this development (Teoh 1988:214ff; Trigo 1991:124ff):

	Standard Malay	Kedah Malay	
-s > -^{j}s > jh	*balas*	*balajh*	'finish'
	bagos	*bagojh*	'good'
	habes	*habeh* (*eh* < *ejh*)	'to finish'
-l > -^{j}l > j	*batal*	*betaj*	'cancel'
	pikol	*pikoj*	'carry'
	sambel	*sambe:* (*e:* < *ej*)	'while'

A somewhat similar series of changes is noted by Michailovsky (1975) in Tibetan and some Tibeto-Burman languages of Nepal, where word-final consonants seem to have been first prevocalized and then deleted. In Sunwar and the Lahul dialect of Tibetan, prevocalization onglides became phonemicized as diphthongal offglides, resulting in such correspondences as *as*, *os*, *us* : *ai*, *oi*, *ui* between Written and Lahul Tibetan, cf.:

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Written Tibetan	Lahul	
ras	*rai ~ rē*	'cloth'
spos	*poi*	'incense'
lus	*lui*	'body'

In the Khawa dialect of Bantawa, word-final /n/ and /t/ [t$^{?}$] were prevocalized and then deleted, leaving behind nasalization and glottalization, respectively. The glide, which became phonemic, is attested after all vowels except /i/. The Khawa forms below are contrasted with those of the more conservative Dilpa dialect:

Dilpa	Khawa	
set	*sei?*	'he kills'
sat	*sai?*	'he spins thread'
dot	*doi?*	'he begs'
tshut	*tshui?*	'he sends'
puɯt	*puɯi?*	'he bathes'
phen	*pheĩ*	'he undoes'
nan	*naĩ*	'he rests a load'
ton	*toĩ*	'he pushes'
ruɯn	*ruɯ̃*	'he scours'

In other dialects, for example Lhasa Tibetan, the correspondences with Written Tibetan are of the kind *as*, *os*, *us* : *ɛ*, *ɔ*, *y*, which may be interpreted as monophthongization of earlier diphthongs. Although this is not explicitly argued for by Michailovsky, prevocalization rather than straight vocalization would be a more attractive analysis of the above changes since the prevocalization stage is attested in some related languages, cf. the following forms from Taungyo (Burmish) (Javkin 1979:51):

**nat*	>	*nait*	'spirit'
**-wat*	>	*-wait*	'clothes'

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3.5 Summary and conclusion

This chapter has provided a representative selection of case studies of prevocalization in which the onglides are of a front quality. It was shown that, in accordance with predictions made in 2.4, front onglides are triggered by the three consonant types characterized by fronted position of the tongue body: palatalized consonants, palatals, and alveolars.

This chapter has also discussed some synchronic and diachronic aspects of prevocalization in connection with several related or competing processes. Specifically, it has been argued that many diachronic developments traditionally analyzed as straight vocalization may be reanalyzed as a sequence of two processes, prevocalization and deletion of the consonants' C-gesture. It has also been shown that language-internal vowel protheses in initial *sC*-clusters are analyzable as prevocalization. Principled ways of distinguishing between prevocalization of nasal consonants and nasal vowel diphthongization, and between prevocalization and metathesis in cases of apparent Cj > jC transpositions, have been suggested. Finally, a separate discussion briefly examined the relationship between prevocalization and umlaut, and the possibility of positing a prevocalization stage in Germanic *i*-umlaut.

The following chapter completes the present survey by focusing on cases prevocalization in which the onglides have the velar, labial, or pharyngeal quality.

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4. Nonfront prevowels

4.1 Introduction

In the preceding chapter it was shown that prevowels with the properties of front vocoids are triggered by consonants with a fronted position of the body of the tongue. It was also shown that, from the viewpoint of prevocalization, it is irrelevant whether the tongue body position is associated with the consonant's secondary articulation (as in palatalized consonants), or with its primary stricture (as in palatals and alveolars). The present chapter will endeavor to demonstrate the same for prevowels of other types by exploring the properties of consonants by which they are triggered.

The chapter is organized as follows. Section 4.2 examines prevocalization patterns of velars, velarized consonants, labials, and labialized consonants. Case studies assembled here indicate that prevowels triggered by velars, velarized consonants and plain labials are of a central to back quality, with consonant-specific variations in the degree of retraction, and prevowels triggered by labialized consonants are retracted and rounded. Section 4.3 examines prevocalization of uvulars, pharyngeals, and pharyngealized consonants. Evidence assembled in this section supports the view of Pulleyblank (1980 et seq.) that there is a need to recognize the existence of a pharyngeal glide as an independent entity, and suggests that, e.g., so-called "excrescent schwa" in present-day American English and elsewhere is an instance of this glide. Section 4.4 documents two patterns in the prevocalization of retroflexed consonants: prevocalization with a retroflex onglide, and prevocalization with a pharyngeal onglide. While the latter pattern concerns pharyngealized retroflexes and is entirely predictable,

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the former leads to a suggestion that retroflexion may in some cases be viewed as a secondary modification rather than as a place of articulation, as currently assumed by the IPA. Such an analysis of retroflexion is also supported by its prosodic properties, well known from Sanskrit and various languages of Australia. Section 4.5 briefly summarizes the main results.

4.2 Labial(ized) and velar(ized) consonants

4.2.1 Dentals

In his study of interaction between vowels and coronal consonants, Flemming (2003) considers all anterior coronals to have a fronted tongue body. However, he demonstrates this only for the alveolars: in fact, it is well known from experimental studies and the phonological behavior of anterior coronals that the acoustic and articulatory properties of alveolars, on the one hand, and those of dentals and interdentals, on the other, are quite distinct. The difference is determined only in part by differences in their consonantal constriction, and a significant portion of their distinct properties is contributed by the tongue body position specific to their production. For example, in Albanian, dental /t, d/ and interdental /θ, ð/ are reported to be distinctly dark. Clear /n, l/ in the same language are alveolar, while dark /n^{w}, l^{w}/ are dental (Lowman 1932). In Malayalam, which contrasts dental /t̪/ and alveolar /t/, the second formant is considerably higher for the alveolar than for the dental stop; Stevens et al. (1986) correlate this with a more fronted tongue body position of the alveolar. They argue that

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the production of dental and interdental consonants is facilitated by a more backed tongue body position, and that of apicoalveolars by a more fronted tongue body position; the mechanical basis of this connection is also urged by Harris (2006). A persistent connection between dentality and darkness, on the one hand, and alveolarity and clarity, on the other, is also reported in a more general way by Recasens and Espinosa (2005).

The connection between the position of the tongue body and the type of coronal constriction is a two-way relationship: just as the type of coronal constriction may determine the position of the tongue, so the position of the tongue may determine, or bring about a change in, the type of coronal constriction. For example, dark /l/ is reported to be preferentially produced with a dental or dentialveolar constriction, and clear /l/ with an apicoalveolar one (Recasens and Espinosa 2005:4). In some Irish dialects, the palatalization contrast in coronals is expressed as a combination of tongue body position and the type of coronal constriction, with plain coronals having a dental and their palatalized counterparts an alveolar realization (cf. Breatnach 1947; Búrca 1958); moreover, retraction of the tongue body may result in dentalization of the alveolars. Thus, Harris (1985) reports that in conservative Irish English the alveolar coronals /t, d, n, l/ are productively dentalized before the following dark /r/ (211ff), interpreting the dental realization of the alveolars as their assimilation to the tongue body position of the dark rhotic (cf. also Harris 2006).

Given the differences in the preferential tongue body position of the two classes of the anterior coronals, one would expect dentals and interdentals to generate prevowels

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of a central to back quality, depending on the degree of retraction of the tongue, and alveolars with prevowels of a front quality. This expectation is borne out by the data (and has been demonstrated for the alveolars in 3.4). Thus, Lowman (1932) reports that final /ð/ in Albanian may (pre)vocalize with a "dark glide which gives a distinct impression of [ɫ]" (287). In West Muskerry Irish, dentals have a [ə]-like onglide after long front vowels and diphthongs ending in a palatal offglide, cf. *slaodán* [sl̪əi̯ˠˈdɑn] (Ó Cuív 1975). In Ring Irish, dentals, in common with other plain consonants, prevocalize after front vowels /i(:), e(:)/ with a central to back onglide described by Breatnach (1947) as [ə] or [ɯ].[132] A schwa-like onglide developed by dentals is also noted for Erris Irish (Mhac An Fhailigh 1980) and Cois Fhairrge Irish (De Bhaldraithe 1945). The latter work also contains valuable observations on the realization of the dental stops [t̪, d̪], which correlates with variations in the quality of the prevowels. The degree of darkness of the dentals is said to vary from a central [ə]-like resonance word-finally after front vowels to a back [u]-like resonance word-finally after /u(:)/; in other positions the dentals have the resonance quality of a half-open and retracted [ə].

With the above data in mind, it is interesting to return briefly to prevocalization of nonpalatal coronals in Maxakalí. As will be recalled (1.5.1.2), Maxakalí /t, n/ (pre)vocalize with (pre)vowels of a schwa-quality, i.e., /t/ > [ə̯t] ~ [ə] and /n/ > [ə̯n] ~ [ə̃]. In addition, Gudschinsky et al. (1970) note that anterior coronals condition the most

[132] [ɯ] is the quality of the onglide when there is a front vowel following the triggering dental or velar.

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retracted allophone of /a/. The centralized quality of their (pre)vowels and their backing effect on /a/ both point to a relatively back tongue body position in these coronals, which in turn argues for their dental or perhaps dentialveolar realization, even though their articulation is described in Gudschinsky et al. (1970) as alveolar. However, before more first-hand data on Maxakalí phonology is available, this suggestion must remain tentative.

4.2.1.1 [ɬ]

The propensity of dark or velarized [ɬ] towards vocalization hardly requires exemplification (cf. Kolovrat 1923; Straka 1940). In many attested cases, the vocalization of this liquid is described as direct, i.e., involving no prevocalization stage. Prevocalization of [ɬ], however, is also attested, both synchronically and in diachrony, and it may be surmised that at least some historical processes traditionally seen as vocalization may have involved prevocalization as well.

An important generalization behind the reasons for the synchronic prevocalization of [ɬ] is contained in Sproat and Fujimura (1993) and Browman and Goldstein (1995), who report on differences between the pronunciation of American English /l/ in syllable-initial and syllable-final positions. As discussed at length in 2.3, the production of American English /l/ involves two gestures, one involving the tongue tip and providing a greater constriction (C-gesture), and the other involving the tongue dorsum and providing a wider constriction (V-gesture). The generalization reported in the above works consists in the fact that the component gestures of /l/ are inherently asynchronous

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in that the V-gesture tends to occur closer to the nucleus. In syllable-final positions, this results in its preceding the C-gesture, and if the lag between the two gestures is above the threshold of auditory perceptibility, the initial, uncovered, portion of the V-gesture will be perceived as an onglide to the liquid (cf. 2.3.1). The C-gesture of the liquid is simultaneously weakened and may become inaudible (Browman and Goldstein 1995). When the weakened C-gesture of the liquid is interpreted by the listeners as nonexistent, the /l/ may be considered "vocalized".

The above account of /l/-vocalization explains earlier spellings, which may reflect the successive stages of this process. Thus, in Medieval French, Latin preconsonantal [ɬ] was prevocalized to [wɬ] after [a, o, ɔ, e, ɛ] (Bourciez 1967:187). While some specialists view this development as straight vocalization, numerous spellings of the type *chevaulx* 'horses' and *royaulme* 'kingdom', etc., have led some Romance scholars to propose prevocalization as an intermediate stage (cf. Kolovrat 1923:89ff). The "prevocalization first" standpoint is evident, e.g., in Fouché's (1961) two-stage account of this process which includes the stages of "vocalisation partielle" and deletion of the lateral (corresponding to prevocalization and deletion of the lateral's C-gesture, respectively) (856ff).[133] By the same token, Fouché notes that the very concept of "vocalization" loses its explanatory power (1961: 858):

> "Il est d'ailleurs probable qu'il ne s'agit pas ici, à proprement parler, de vocalisation. Pris entre l'*u̯*, résultant de la vocalisation partielle et la consonne

[133] Fouché (1961) believes that the lateral loss first affected words in which it preceded a prepausal consonant (e.g., **ault* 'high (masculine singular)' < *altu*) and only later those in which it preceded syllable onsets (e.g., **aulte* 'high (feminine singular)' < *alta*) (858).

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implosive finale, *ł* a dû disparaître purement et simplement."

A prevocalization analysis of the above development in French is also urged by the evidence of a comparable development in the Rhaeto-Romance area; among the relevant Rhaeto-Romance forms may be cited *kault* 'hot', *awłt* 'high', and *fawłs* 'false', from *calidu(m)*, *altu(m)*, and *falsu(m)*, respectively (Fouché 1961:856).

Another /l/-prevocalization that is relatively well documented took place in fifteenth-century English: the affected liquid was in final or preconsonantal coda after a /a/ or /o/ (Jespersen 1922:289ff). The sequence of changes /al/ > [awl] > /ɔ(l)/ accounts, e.g., for the present-day pronunciation of words like *chalk*. A "lingering pronunciation of /l/" was still attested at the beginning of the nineteenth century, as witnessed by the following quotation: "I never hear him *tawlk* without feeling ready to worship him" (Carlyle with reference to Coleridge; cited in Jespersen 1922:292).

4.2.2 [ŋ]

The velar nasal [ŋ] is often prevocalized with a velar or labiovelar onglide. (Pre)vocalization of this segment in Maxakalí, with the result /ŋ/ > [ṵ̃(ŋ)], has already been mentioned (1.5.1.2); in that language, it runs parallel to (pre)vocalization of /k/, producing [ṵ(k)]. The rounded onglide [w̃] is found more frequently, possibly because of its being the less marked articulation, as shown by the relative frequency of the two

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glides in the world's languages.[134] Both results, however, are entirely expected as both glides are known to productively alternate with [ŋ] in many languages. For example, in Toura (Ivory Coast) [ũ] may be realized as a syllabic velar nasal [ŋ̍] (Greenlee and Ohala 1980:288), and the syllabic nasal in Japanese may be realized as [ũ̃] (Yoshida 2003).

Many examples of [ŋ]-prevocalization are supplied by the Rhaeto-Romance area, where *ø* > ^{w} / *a* __ *n*, *ng*, *nk*, *nt*, as in *ma^{w}n* 'hand', *sa^{w}ŋ(k)* 'blood', or *ufá^{w}nts* 'children' (Gartner 1910:133f). Similar forms are noted for Samnaun Romansh by Ritter (1981), cf. *maun* 'hand', *ufaunt* 'child' < Latin *manu*, *infante*.[135] In early thirteenth century Anglo-Norman, the spelling *an* began to be replaced by *aun* which represented, at least originally, a diphthongal pronunciation; the diphthong is noted in Middle English borrowings including *aunt* (Old French *ante* < Latin *amita*) and *launch* (Anglo-Norman French *lanche*, French *lance*) (Pope 1932; Wyld 1927:132f; Fouché 1961:374). As discussed in 3.4.3, these developments are to be distinguished from diphthongization of nasalized vowels.

An interesting case of nasal prevocalization is the surface realization of nasalized vowels in Polish, notated in the traditional orthography as *ą*, *ę*. The nasal component of

[134] According to De Jong and Gyasi (2000), the phonemic inventories in Maddieson (1984) include 238 languages with a labiovelar glide and only 5 with a velar glide (683).

[135] Although in some of the above examples [ŋ] may be explained as the result of assimilation to the following velar, it is also possible to connect it with the coda weakening of nasals, cf. Caribbean Spanish [moŋtoŋ] for *montón* 'bunch' (Guitart 1982:135). Trigo (1988) argues that velarization of nasals is essentially a type of weakening and corresponds to debuccalization in oral consonants.

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these vowels is traditionally described as a [w̃] before a fricative or a pause, as a nasal stop coinciding in place of articulation with a following stop or affricate, and as vowel nasalization elsewhere. Biedrzycki (1963) argues for an analysis of nasalized vowels as underlying sequences /oŋ, eŋ/, based in part on parallelism with the behavior of coda /ɲ/, cf.:

	Before a stop or an affricate	Before a fricative or a pause
/ɲ/	/swoɲtse/ [swõ̃ɲtse] 'sun' /xaɲba/ [xã̃ɲba] 'shame' /baɲka/ [bã̃ɲka] 'jar'	/koɲski/ [kõ̃ɲski] 'horse's' /koɲ/ [kõ̃ɲ] 'horse'
/oŋ/	*kąt* /koŋt/ [kõnt] 'corner' *ząb* /zoŋb/ [zõmp] 'tooth' *bądź* /boŋdʑ/ [bõɲtɕ] 'be (2s)!'	*kąski* /koŋski/ [kõw̃sci] 'pieces' *są* /soŋ/ [sõw̃] 'they are'
/eŋ/	*pęd* /peŋd/ [pẽnt] 'flight' *zęby* /zeŋbɪ/ [zẽmbɪ] 'teeth' *będzie* /beŋdʑe/ [bẽɲdʑe] 'will be (3s)'	*rzęsa* /zẽŋsa/ [zẽw̃sa] 'eyelash' *idę* /ideŋ/ [idẽw̃] 'I go'

Biedrzycki's analysis is corroborated by the results of an experimental study reported in Dukiewicz (1995) indicating that the realization of *ą*, *ę* before a pause, and more rarely before a fricative, may consist of three rather than two elements. In the three-member realization, the initial portion consists of the vowel [ɛ̃] or [ɔ̃], the middle portion of a nasalized back vocoid, and the final portion is a nasal consonant (notated by Dukiewicz as [ŋ]) or a back nasalized glide (notated as [ũ]) (1995:33ff).[136] If *ą*, *ę* are, indeed, underlyingly /oŋ, eŋ/, the last two elements of their three-member phonetic realization

[136] Dukiewicz notes that the asynchronous realization of *ą*, *ę* is not consciously noted by Polish speakers.

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can be interpreted as prevocalization of the underlying nasal, where the nasalized back vocoid is the prevocalization onglide and the final, more consonantal element as the C-gesture of the nasal.

The realization of non-syllable-initial nasals in Japanese is similar in its basic outline to the realization of the underlying Polish /ŋ/ outlined above (cf. Yoshida 2003). Any non-syllable-initial nasal has three basic types of realizations. Before a pause, it is realized as a uvular or velar stop in the more formal, and as a nasalized version of the preceding vowel in more casual speech styles, cf.:

/kin/	[kiɴ ~ kiŋ ~ kĩi]	'gold'
/sen/	[seɴ ~ seŋ ~ seẽ]	'line'
/bun/	[bɯɴ ~ bɯŋ ~ bɯɯ̃]	'sentence'

Before affricates and stops, it is realized as a nasal stop agreeing in place of articulation with that of the following consonant, cf.:

/hon-ka/	[hoŋka]	'book-(interrogative)'
/hon-ga/	[hoŋga]	'book-(nominative)'
/hon-mo/	[hommo]	'book-also'
/hon-de/	[honde]	'book-by'

Before continuants (vowels, glides, and fricatives) it is realized as [ɯ̃] or, in more casual styles, as a nasalized version of the preceding vowel. Before a palatal glide, it is realized as [i] in the more casual speech, cf.:

/kensa/	[keɯ̃sa ~ keẽsa]	'inspection'
/tan'i/	[taɯ̃i ~ taãi]	'credit'
/sin'wa/	[ʃiɯ̃wa]	'myth'
/kon'ja/	[koɯ̃ja ~ koĩja]	'tonight'

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The point of interest here is Yoshida's (2003) analysis of the prepausal allophones [N] ~ [ŋ] as alternative realizations of the sequence [ũĩʔ]; the basis for this analysis is the fact that prepausal words in Japanese often end with an unreleased glottal stop. Yoshida views the glottal stop as an automatic consequence of the word-final position of the nasal, but we may at least consider the possibility that it may represent the outcome of debuccalization of the C-gesture of the underlying nasal.

4.2.3 Irish and Scottish Gaelic

In many dialects of contemporary Irish and Scottish Gaelic, the existing palatalization contrast has brought about velarization of plain consonants. Such velarization becomes phonemically important when it comes to contrasting plain labials with their palatalized counterparts: as the articulation of palatalized labials includes no palatal gesture of the tongue, the palatalization contrast is realized as that between plain and velarized labials (cf. Sommerfelt's 1952 analysis of Torr Gaelic). Plain velars pattern together with the other plain (i.e., phonetically velarized) consonants in conditioning transitional back glides, preferentially in contact with front vowels. Irish Gaelic data is interesting in providing details on the phonetic variation of transitional glides as a function of the syllable position of triggering consonants and/or the participating consonant-vowel combinations. For example, consonantal offglides in Erris Irish show more phonetic variation than the onglides. Velarized labials *p, b, f, m* trigger offglides of the [w] quality before a *i(:), e(:)*; velars *k, g, x, ɣ, ŋ* trigger a [o]-quality offglide before a *i(:)*;

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and dentals trigger an offglide of the [ə] quality before front vowels. Onglides, on the other hand, are invariably of the schwa quality; they are noted between *i:*, *e:* and the following velar or velarized consonants, and between back vowels or the diphthong *əu* and the following velarized *r*, cf. [dri:ᵊxdə] *draoidheacht*, [bʲi:ᵊf] *bíodh*, [ku:ᵊr] *cubhar*. The onglide to a final consonant closing a stressed syllable is more clearly perceptible than the onglide to the same consonant in a nonfinal position (Mhac An Fhailigh 1980:45ff). In West Muskerry Irish, the transitional glides between velarized consonants and front vowels are said to be more perceptible in case of the more closed vowels. The back glides are of the [w] and [ə] quality, the former occurring between velarized labials and front vowels and diphthongs beginning with such vowels (cf. [fʷe:r] *faobhar*), and the latter between long front vowels and diphthongs ending in *i* and velarized labials, velars and dentals (cf. [bʲi:ᵊn] *bíonn*, [ki:ᵊrə] *caora*, [ne:ᵊfə] *naomhtha*, [fəiᵊb] *fadhb*, [be:ᵊxəs] *buidheachas*, [ʃi:ᵊl] *síol*), and between velarized dentals and velars and front vowels, cf.[dᵊi:] *duibhe*, [kᵊe:] *caoi* (Ó Cuív 1975:52f). In Cois Fhairrge Irish, the onglides to stressed vowels are said to be more prominent than those to unstressed vowels, and the onglides to final consonants are more prominent than those to intervocalic consonants. Examples quoted by De Bhaldraithe include [ku:ʷll] *cúl* and [bauʷr] *bodhar* (1945:43ff). In Tourmakeady Irish, there are [w]-quality transitional glides between velarized consonants and a flanking *i(:)*, cf. [bʷi:ʷl]

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baoghal. The offglide between a velarized consonant and the following *e(:)*, on the other hand, is of a [o] quality, cf. [k^{g}el^{j}] *coill* (Búrca 1958:59f).

4.2.4 Back umlaut and breaking

Old English back umlaut (or "mutation"), estimated to have taken place around 700 A.D., is an example of prevelarization that may have been triggered by allophonic velarization of consonants. The immediately observable conditioning environment is the presence of a back vowel *a*, *o* or *u* in the syllable following the triggering consonants. Old English dialects differ in the degree to which back umlaut was carried through. Thus, in Kentish it occurs freely across all consonants, in Anglian it is blocked by intervening velars (*c*, *g*), while in West-Saxon it takes place only across liquids (*r*, *l*) and labials (*f*, *p*, *w*, *m*). Back umlaut is generally blocked by intervening geminates and consonant groups; examples include **ælu* > *ealu* 'ale', **hefon* > *heofon* 'heaven', **sifun* > *siofun* 'seven'; Latin *lactūca* > *leahtric* 'lettuce', *psalma* > *sealm* 'psalm', *arca* > *earc* 'ark' (Moore and Knott 1942:127f; Campbell 1959:54, 200). While back umlaut is generally viewed as unmediated interaction between vowels, it is also possible that the immediate source of the onglides was velarization of the intervening consonants by following back vowels. This hypothesis is strengthened by the similarity between the outcomes of back umlaut and those of breaking, or prevocalization of the Old English reflexes of Proto-Germanic *x, *r, *l, cf. **fællan* > *feallan* 'fall', **selh* > *seolh* 'seal', and ***sihith* > **siohith* > *siehth* 'sees' (and 1.5.2). Under the traditional interpretation,

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breaking is due to a primary or secondary velar constriction in the conditioning consonants (Campbell 1959:85f; Lass and Anderson 1975:102ff).

Other Germanic languages attesting back umlaut are Old Frisian and North Germanic (cf. Flom 1937; Awedyk 1975). In Old Frisian, the triggering consonants are [g] followed by [u] or [(ŋ)k, ŋg] followed by [w], when the preceding vowel continues Proto-Germanic *i or *e; in North Germanic, back umlaut is triggered by consonants followed by a [u] or [w], cf. Old Frisian *siunga ~ sionga ~ syonga* and Old Swedish *siunga* with Gothic *siggwan* 'sing'. Breaking before preconsonantal /x, l, r/ is also attested, cf. Old East Frisian *tsiurke* 'church', Old Icelandic *hiarta* 'heart', and Old Swedish *skiuld* 'guilt'. Old High German attests breaking before [x]; interestingly, Vetsch (1910) notes breaking of /i:, y:/ before final or preconsonantal [x] in Swiss German, cf. /liəxt/ 'light', /gliəx/ 'equal' with German *Leicht, gleich* (101f).

4.2.5 Labial(ized) consonants

Examples of labial prevocalization include (pre)vocalization of the oral and nasal labial stops in Maxakalí, realized as [ɤ(p)] and [ɤ̃(m)], respectively. This pattern of prevocalization is entirely expected based on the gestural composition of labials (see 2.3.2.1). The prevocalization onglide of plain labials in Irish and Scottish Gaelic, where these segments are phonetically velarized, is typically an unrounded schwa, cf. *maoidheamh* /mi:v/ [m^{w}i:^{ə}b], *fadhb* /fəib/ [fə^{ə}b] (Oftedal 1963:71ff).

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Seri, a language isolate from the state of Sonora, Mexico, has four vowels, /i/ [i], /e/ [ɛ ~ æ], /o/ [o ~ u], /a/ [ɑ], and three labialized consonants, /k^w, x^w, χ^w/. When a labialized consonant follows an unrounded vowel, it develops a labial onglide agreeing with the vowel in height (Marlett 1981:7). The onglide is described in Marlett et al. (2005) as "a very short round transitional vowel" (Marlett et al. 2005; Moser and Marlett 2005: 830f), cf.:

/iˈtak^w/	[iˈta^{ʊ}k^w]	'did s/he kill him/her/it?'
/taˈʔɛx^{w}k/	[taˈʔɛ^{ʊ}x^{w}k^w]	'Tiburon Island'
/ˈkik^w/	[ˈki^{ʉ}k^w]	'the one who killed him/her/it'

In Guelavía Zapotec (Mexico), there is prelabialization and prepalatalization before the final cluster of consonant plus glide, cf. /bekw/ [bɛ^{ʔ}ɛukw̯] 'dog'. Prevocalization is absent in case of a following /i/ or /u/ (Jones and Knudson 1977:170).

In Sesotho (Bantu), nonlabial consonants are labialized when followed by *w* so that "the w-element precedes, persists through, and follows the articulation of a labialized consonant". "Before a consonant there is a silent *w*, while after the consonant there is an audible *w*" (quoted by Roux 1981:375 after Kunene 1961:120). In languages comprising the Suto-Chuana group of Bantu, round vowels /u, o, ɔ/ induce allophonic labialization of preceding consonants, which can be anticipated as an onglide. The only example of prelabialization cited in Tucker (1929) is /ŋku e kxholo/ [ŋku e^{ʊ} kxholo] 'the sheep is big' (30).

In Uradhi (Australia), /a/ may receive a rounded offglide when followed by the cluster /kw/ (Crowley 1983:318), cf. /akwaṉumu/ [a^{w}kwaṉumuŋ/ 'dingo'.

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Avestan has the process of *u*-epenthesis, which refers to (orthographic) insertion of a *u* in front of an initial or medial *r* followed by *u*, *ū*, or *w*. The result of *u*-epenthesis of **a* is represented by the graphemes <aur>, <aour> (after a labial) and <our> (following the *u*-mutation of **a*); that of **ā* by <āur>; and that of **r̥* by <əur>. *U*-epenthesis in front of an initial *r is spelt <ur>, and the inserted orthographic <u> does not count for the purposes of meter. Examples of *u*-epenthesis include *dāuru* 'wood', *pouru* 'full', and *uruθman-* 'growth' (De Vaan 2003:561-62; Hoffmann and Forssman 1996:51ff).

As in the case of prevocalization associated with the presence of a palatal glide after the consonant (see 3.2.3), there are instances of surface transposition of labial glide and the preceding consonant that are ambiguous when it comes to interpreting their mechanism. Thus, Portuguese has developed *ô*, *ou* out of *o*, *a*, respectively, in several of its perfects in which the final stem consonant was followed by a labial glide; according to Williams (1962), this development is attested across *p*, *b*, *c*, *t*, *s*, and *x* (88). The same change is also attested in Spanish, which regularly simplified the resulting diphthongs (Cornu 1904-06:962; Huber 1933:50; Penny 2002:225), cf.:

Portuguese	Old Spanish	Gloss	Latin
ouve	*ove*	'I had'	*habui*
soube	*sope*	'I knew'	*sapui*
prougue	*plogue*	'I liked'	*placui*

Both Portuguese and Spanish also attest this development in words of dialectal origin, cf. dialectal Portuguese *aug(u)a* 'water', *iug(u)al* 'equal', *euga* 'mare' < *aqua(m)*, *aequale(m)*, *equa(m)*, and the river name *Vouga* < *Vacua* (Huber 1933:93; Williams

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1962:30). The adjoining Spanish dialects attest *a^{w}gua* 'water', *ye^{w}gua* 'mare' < *aqua*, *equa*; similar forms are also documented in Judeo-Spanish (Krüger 1914:230).

4.2.5.1 [w]

The labiovelar glide may prevocalize with a velar onglide, which may or may not be rounded. The phonological status of [w], as a diphthongal offglide or a consonant, appears to be of no importance to its role as a trigger of prevocalization, and the difference in onglide rounding probably has to do with the relative phonological status of the velar and labial constrictions of the triggering glide (cf. Gick 2003). Thus, theories of Old English breaking that include /w/ as one of the conditioners appeal to the velar constriction of this glide (cf. Gussenhoven and Weijer 1990). In the historical development of French, a diphthongal [w]-offglide that resulted from vocalization of /l/ in preconsonantal coda prevocalized with a schwa-like onglide after a [ɛ],[137] producing triphthongs spelled <eau>, phonetically something like [ɛəw]. Subsequent accent shift from [ɛ́əw] to [ɛə́w] produced different outcomes, including the standard [o] and dialectal [jaw], [jo], and [ja], cf. Vulgar Latin *aucellu(m)* > *oiseau* 'bird' and Old High German *helm* > *heaume* 'helmet'. This development in French seems to find an exact parallel in the development in the Chengzhou dialect of Mandarin, where a schwa-like transition breaks the sequence of a high front vowel [i] and a suffixal [u] (Lin 1989):

[137] The sequence [ɛw] that resulted from this development is generally analyzed as a diphthong (cf. Fouché 1961:319; Lausberg 1965:264f).

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	ua + u	>	uau	'socks'
	ia + u	>	iau	'sprout'
but:	pi + u	>	pi^{ə}u	'nose'[138]

An example of prevocalization of a consonantal *w* may likewise be drawn from historical French phonology, cf. the outcome of Latin */ak^{w}a(m)/* 'water':

*ak^{w}a(m) > *akwa >* axwa >* ajwe >* ɛwe >* ɛawe > eau*
(derivation sequence according to Pope 1934:135), or:

*ak^{w}a(m) > *awa > *aɛwa > *ɛ:wa > *ɛ:we > *eawe > eau*
(derivation sequence according to Fouché 1961:338, 644)[139]

[w]-prevocalization producing a rounded prevowel may be exemplified from Mantjiltjara (Australia), where /a/ is described as receiving a rounded offglide before /w/, cf. /kawu/ [kɑwu ~ kɑ^{w}wu] 'body' (Marsh 1969:139). The same is true of Pintupi, another Australian language, cf. /kawuŋpa/ [ka^{w}wuŋpʌ] 'cold ashes' (Hansen and Hansen 1969:157). In Puluwat, a Trukic language of Micronesia, initial /w/ before a phonetically dissimilar vowel has a slight *u*-like onglide. The onglide is not identical to an underlying /u/, although the nature of this difference is unclear from the description, cf. [^{u}wa] 'canoe' with /uwaato/ 'bring' (Elbert 1974:8). Some additional examples of

[138] It is interesting to speculate in this connection whether the spelling <aour> that reflects *u*-epenthesis of Avestan **a* after labials may represent a similar phenomenon (the reflex in all other contexts is <aur>, see De Vaan 2003:419, 561). It is just possible that the graphic <o> here may stand for a transition between the two elements of the phonetic diphthong [aw].

[139] An alternative account of the development of the sequence *ɛl* presupposes prevocalization of the *l* prior to its complete vocalization, i.e., *bɛllus > be^{a}ls > beau* (Anglade 1961:17). However, Fouché's and Pope's derivations are preferable in that they are better able to account for the parallelism between the development of words in *ɛl(l)* and the above word for 'water'. A parallel development may be cited from Gascon, where both *w* and (vocalized) *l* seem to develop a [e]-onglide after *i*, cf. *filu > hiéu* 'thread' and *rivu > arriéu* 'river bank' (Rohlfs 1970:119).

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[w]-prelabialization are cited by Millardet (1910:187f) and include Arabic *(at)tabut* [ta^{w}wuk] 'ring'.

4.3 Postvelar consonants

It has been argued by Hayward and Hayward (1989), Trigo (1991), McCarthy (1991, 1994) and others that pharyngeals, uvulars and laryngeals form a natural class, e.g., for the purposes of some phonological processes in Semitic. In other languages with rich postvelar inventories, however, laryngeals fail to pattern together with pharyngeals and uvulars: this is explicitly noted, e.g., by Bessell (1992) for Interior Salish. The data assembled here confirms the presence of a pharyngeal component in uvulars, thus allowing the inclusion of at least some uvulars[140] in the category of pharyngeal(ized) consonants, but it provides no evidence for the inclusion of laryngeals in the same class, as laryngeals do not seem to prevocalize. For example, in Maxakalí, which has the most productive synchronic prevocalization of any language examined, laryngeals /ʔ, h/ are the only consonants that fail to prevocalize. Likewise, in Nuuchahnulth (formerly Nootka), laryngeals fail to pattern with uvulars and pharyngeals for the purposes of various phonological processes including prevocalization. The failure of laryngeals to prevocalize may be understood as their lacking a vocalic component. While this result is not unexpected given their mode of production, it does offer a fresh look at the source of their uniqueness, which may be due to their being the only class of consonants

[140] The qualifier "some" is indispensable here since there are languages, e.g., some North Caucasian languages, in which plain and pharyngealized uvulars are contrastive (cf. Colarusso 1988:219ff).

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composed solely of a C-gesture. This line of inquiry, although clearly warranting further investigation, is tangential to the present purposes and will not be pursued in what follows.

From a gestural perspective, pharyngeals and pharyngealized consonants are characterized by a pharyngeal V-gesture associated with a primary or secondary articulation in the pharynx, and are expected to prevocalize with a pharyngeal, or [ɑ]-like, prevowel. In actual fact, the attested prevowels rarely have the expected [ɑ] quality, surfacing more often as a low central [a/a̰] or as a mid central [ə/ə̰]. One reason behind the rarity of a surface [ɑ̰]-onglide may be the virtual impossibility to distinguish such an entity from a voiced pharyngeal fricative, and another may be the raising and/or fronting influence from the preceding nuclei; the latter reason is also suggested by Pulleyblank (1986), cf.: "The shwa-like offglide that is the typical realization of [H] [i.e., [ɑ̰], N.O.] after other vowels can be compared to the glides in the diphthongs [aj] and [aw], which often do not reach their ideal target but are more accurately represented as [ae] and [ao]" (241). The surface realization of pharyngeal prevowels is thus in perfect agreement with cross-linguistic variability in the realization of the segmental pharyngeal glide.

Although the pharyngeal glide does not form part of the current version of the IPA, the necessity to recognize the existence of this segment as distinct from the pharyngeal fricatives /ħ, ʕ/ and parallel to the coronal /j, ɻ/ and dorsal /ɯ̟, w/ glides was urged in a number of publications by Pulleyblank (1980 et seq.). Pulleyblank regards the

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pharyngeal glide as "a pharyngeal approximant (in contrast to the pharyngeal fricatives [ħ] and [ʕ])", which represents "the nonsyllabic form of the vowel [ɑ]" (1997:195). The existence of a pharyngeal glide is suggested by two types of evidence. First, in some languages it acts as a nonsyllabic allophone of [ɑ] or as a vocalized allophone of the pharyngeal fricatives. In its function as an allophone of one of these phonemes, the pharyngeal glide may range from a tongue root [ɑ̰]-like articulation to a dorsal [ɯ̟]-like constriction (1986:235f). For example, vocalized allophones of /ʕ/ in Salish surface as [a] ~ [ɑ] and those of /ʕ'/ as [a'] ~ [ɑ'] (Bessell 1992). In Axininca Campa, /a/ surfaces as [ɯ̟] when it loses syllabicity between vowels; under similar conditions, /i/ surfaces as [j] and /o/ as [β]. Second, there are languages in which the pharyngeal glide acts as an underlying phoneme. Put differently, the phonological analyses of a great many languages are simplified when it is assumed that there is an underlying pharyngeal glide in the inventory. For example, the assumption of an underlying pharyngeal glide has allowed Bender (1968) to simplify significantly his analysis of Marshallese phonology: although uncertain as to the exact identity of the mysterious "zero consonant" in his analysis, Bender mentions that an Arabic-speaking colleague "does hear an *ayin* at the proper places" (33). Pulleyblank (2003) solves a problem in Mandarin phonology by assuming that certain types of syllables, traditionally considered open, end instead in a pharyngeal glide. The analysis of centering diphthongs and long vowels that arise after the deletion of coda /r/ in the nonrhotic dialects of English is likewise simplified if the schwa-offglide and the phoneme of length, respectively, are reanalyzed as an

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underlying pharyngeal glide; analyses along these lines are made by Bloch and Trager (1942) and Pulleyblank (1980:78 and 1986:237f), among others.

Cases of uvular and pharyngeal prevocalization assembled here provide additional evidence for the recognition of a pharyngeal glide, and for the internal structure of uvulars. One exceptionally clear instance of prevocalization resulting in a pharyngeal-quality prevowel involves surface realization of word-final /ʁ/ in German. German /ʁ/, which according to Delattre (1971, 1981) is produced midway between Arabic /ʁ/ and /ʕ/, is usually described as subject to syllable-final vocalization, with the relevant allophone variably transcribed as [ɐ], [ə], or [ʌ] (cf. Krämer 1979; Hall 1993). However, Delattre's (1971) phonetic study of the word-final allophone clearly indicates that the final rhotic is only prevocalized rather than fully vocalized. Delattre describes the production of both intervocalic and word-final allophones of /ʁ/ as involving a "circling motion" consisting of a retraction of the tongue toward mid-pharynx and its raising along the pharyngeal wall till it comes into contact with the uvula (140). The main difference between the word-final and intervocalic allophones of /ʁ/ consists in the relative duration and magnitude of the two gestures. In the word-final allophone, the retraction gesture is much longer than in the intervocalic allophone, and is both articulatorily close to, and gives a distinct auditory impression of, a [ɑ]. The uvular contact gesture of the word-final allophone is much shorter and lighter than in the intervocalic allophone: while in the latter it results in "some loud trills", in the former it produces "a very light friction sound" (Delattre 1971:140f). When the recording of

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words with final /ʁ/s was played in reverse to speakers of American English, they heard a [ɑ] followed by a light constriction ranging from uvular to velar: the words *Flur* and *wir*, for instance, were given in reverse transcription as [ʁɑul ~ xɑul] and [ʁɑiv ~ xɑiv], respectively (Delattre 1971:142).

The surface realization of /ʁ/ not only provides a rare clarity of detail regarding prevocalization of this rhotic in German: it also confirms the view that at least some uvulars may be described as dorso-radical segments (cf. Elorrieta 1991; Trigo 1996). Another rare characteristic of the prevocalization of German /ʁ/ is the quality of the prevowel, that fully reaches the pharyngeal constriction: all other pharyngeal prevowels examined herein fall articulatorily short of their target. Thus, this is the case of the prevowel triggered by word-final /ʁ/ in Kedah Malay. In that language, /ʁ/ is weakened word-finally together with the rest of the consonants; more specifically, /s, k, ʁ/ are debuccalized to /h, ʔ, ʕ/, respectively, and /l/ is deleted. Prior to their debuccalization, /s, l, ʁ/ are prevocalized. Prevocalization of /s, l/ takes place regardless of the preceding vowel and produces [jh] and [j], respectively. Prevocalization of /ʁ/ is noted only after a /i/ and results in an onglide transcribed by Teoh (1988) as [a̯] (214ff), cf.:

/malas/	[mãlajh]	'lazy'	/s/ > *[js] > [jh]
/bagus/	[bagojh]	'good'	
/lukis/	[lukeh][141]	'to draw'	
/pukul/	[pukoj]	'to hit'	/l/ > *[jl] > [j]
/pasir/	[pasia̯ʕ]	'sand'	/ʁ/ > *[a̯ʁ] > [a̯ʕ]

[141] Monophthongization and changes in vowel height are ignored here (see Teoh 1988:215 for details).

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After nonlow front vowels, or high vowels, the quality of the pharyngeal onglide is often schwa-like. For example, Allen's (1950) notes on the phonetics of Modern Eastern Armenian include the observation that there is a distinct offglide between the front vowels *e*, *i* and the following /ɬ/, realized as [ʁ] or [ɣ],[142] cf. *p^{h}iɬ* [p^{h}i^{ə}ɣ], *teɬ* [te^{ə}ʁ]. Armstrong (1964) notes the existence of "a very noticeable central glide" between /i(:)/ and the following /q, ħ/ in Somali, and Bliese (1981) comments on a schwa glide between the long high vowels /i:, u:/ and pharyngeal fricatives /ħ, ʕ/ in Afar. A nonsyllabic schwa onglide to /r/ is also noted in many dialects of Mandarin Chinese, where it develops when the diminutive suffix /r/ is added to words ending in a front vowel. Gick and Wilson (2006) describe /r/ as a pharyngealized retroflex, and Lin (1989) cites the following examples from the Beijing dialect:

	xua + *r*	>	*xuar*	'flower'
	t^{h}u + *r*	>	*t^{h}ur*	'picture'
but	*p^{h}i* + *r*	>	*p^{h}i^{ə}r*	'skin'
	y + *r*	>	*y^{ə}r*	'fish'

The schwa-quality of the pharyngeal onglide gave rise to its sometimes being accorded a special status under the label "excrescent schwa". The above examples, however, hopefully make it clear that "excrescent schwa" is simply a variety of the pharyngeal onglide that has failed to reach its target, presumably as a result of partial overlap with the gestures characteristic of preceding articulatorily distant vowels. An analysis of the "excrescent schwa" in terms of gesture overlap between the triggering consonant's V-gesture and that of the nucleus to its left derives support, e.g., from the

[142] The pronunciation of this voiced fricative is said to vary from "uvular to pre-uvular" (Allen 1950:196).

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quality of postvocalization offglides that arise between emphatic (i.e., pharyngealized or uvularized) consonants and front vowels in Arabic, where a word like /t̪iin/ 'mud' may have a centralized offglide transition to the vowel notated by Card (1983) as [i̤]. Among other languages with a productive "excrescent schwa" is Nuuchahnulth, a Wakashan language from Vancouver Island. A number of uvular and pharyngeal consonants in Nuuchahnulth develop a schwa-onglide after a preceding /i/ or /i:/, cf. /siqi:ɬ/ [sɪʔqe:ɬ] 'to cook'; prevocalization across word boundaries is also noted, cf. "Randi qitʃin" [-diʔ#qe-] (Gick and Wilson 2006; Wilson 2007; Carlson et al. 2001:277). The postvelar inventory of Nuuchahnulth includes /q, qʷ, χ, χʷ, ʕ, ħ/, and it is interesting to note that the labiovelar uvular stop which, according to recent analyses of uvulars should be analyzed as having both labial and pharyngeal coarticulations, is prevocalized in response to its pharyngeal coarticulation, cf. /t'iqʷa:sitɬ/ [t'ɪʔqʷa:sɪtɬ] 'to sit down' (Wilson 2007). It is equally notable that the offglide always occurs before consonants that involve "some sort of closure", i.e., /q/, /qʷ/, and /ʕ/, while the results before the fricative /ħ/ admit of a greater range of variation.[143]

[143] The development in the history of French detailed in 4.2.5.1 may also be seen as a case of an "excrescent schwa". As will be recalled, the development in question consists in the insertion of a centering vocoid between *ɛ* and the following preconsonantal *l*, as in Vulgar Latin *aucĕllu* > *oiseau* 'bird'. As the available historical records do not make it clear whether the schwa developed before or after the vocalization of the lateral, both derivation sequences have been cited in the literature (Anglade 1961; Fouché 1961). If the centering vocoid was inserted before the vocalization of *l*, it is possible to see it as another case of an excrescent schwa: for an exact typological parallel cf. surface forms like [tɛə̯l] 'tell' in contemporary Scots (Broadbent 1999).

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The "excrescent schwa" most frequently discussed in the literature, however, is one that develops in sequences of /i/, /j/ and the following liquid in English, cf. [fiᵊɬ] 'feel', [fajᵊɬ] 'file'.[144] Gick and Wilson (2001, 2006) show that the presence of an "excrescent schwa" in the /iɬ/ and /VjɬV/ syllable rimes does not contribute to the duration of the rime, and interpret the schwa as a physiologically based transition between the conflicting articulatory targets. However, since Gick (1999b) reports that "excrescent schwa" is temporally accommodated in part due to a reduced temporal lag between the V- and C-gestures of the liquid in coda, it becomes possible to venture the hypothesis that the "excrescent schwa" constitutes the initial portion of the liquid's V-gesture. In this case, the apparent reduction of the lag between the V- and C-gestures of the liquid receives a ready explanation as a result of a gliding, or temporally nonuniform structure of the liquid's V-gesture, whose initial portion is realized as a schwa due to partial overlap with the preceding front vowel or offglide, and whose final portion is realized as expected, i.e., as a constriction in the upper pharynx.

The problem of "excrescent schwa" is often considered together with so-called linking [r] (e.g., Halle and Idsardi 1997). Linking [r] refers to an etymological [r] in nonrhotic dialects of English which surfaces only when followed by a vowel across a morpheme or word boundary (Hay and Sudbury 2005:799), cf.:

fear [fiə]	but	*fearing* [fiəɻɪŋ]
car [kɑ:]	but	*car alarm* [kɑ:ɻəlɑ:m]

[144] Earlier accounts of "excrescent schwa" include epenthesis (McCarthy 1991) and as licensing of trimoraic syllables (Lavoie and Cohn 1999).

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The historical development of linking [r] forms part of a series of processes that took place in Early Modern English and included the apparent insertion of a schwa glide between long nonlow vowels and diphthongs /i:, e:, o:, u:, aɪ, aʊ/ and the following prepausal, preconsonantal, or prevocalic /r/ (pre-/r/ breaking, cf. 1.5.1.4). As shown in Table 8, the insertion of a schwa in nonrhotic varieties of English was accompanied by laxing of preceding vowels and deletion of the rhotic when in coda; /ɔə̯/ (and in part /ʊə/) were monophthongized in some dialects to [ɔ:][145] (Wells 1982:213ff).

Table 8. Pre-/r/ breaking and /r/-deletion

beer	*chair*	*more*	*sure*	*fire*	*tower*
bi:r	tʃe:r	mo:r	ʃu:r	faɪr	taʊr
bi:ə̯r	tʃe:ə̯r	mo:ə̯r	ʃu:ə̯r	faɪər	taʊər
bɪə̯	tʃɛə̯	mɔə̯	ʃʊə̯	faɪə	taʊə
–	–	mɔ:	ʃɔ:	–	–

At approximately the same time, the short vowels /ə, a, ɒ/ were lengthened in the same environment; the results are summarized in Table 9 (cf. Wells 1982:201ff).

Table 9. Pre-/r/ lengthening and /r/-deletion

sir	*bar*	*for*
sər	bar	fɒr
sə:r	ba:r	fɒ:r
sɜ:	bɑ:	fɔ:

[145] Giegerich (1997) reports that /ɔ:/ from historic /ɒ:r/ still contrasts with /ɔ:/ from other sources in the speech of some conservative speakers, citing [ɔə̯] in *source, sore* versus [ɔ:] in *sauce, saw* as examples. Nonmonophthongal pronunciations of the reflexes of /ɒ:r/ are also noted by Trager and Smith (1957).

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Some previous analyses have identified the above developments as part of the same process, or syllable-final retiming of the component gestures of coda /r/, together with articulatory reduction of the liquid's C-gesture (cf. McMahon et al. 1994). It has also been shown that pre-/r/ breaking is not unique to English but has parallels in Scots (Jones 1989; McMahon 1996; Broadbent 1999) and Germanic in general (Howell 1991:41ff). The basis for the distinction between the diphthongization of nonlow vowels, on the one hand, and lengthening of /ə, a, ɒ/, on the other, has been explained by differences in the articulatory distance between the affected vowels and the V-gesture of /r/, assumed to have been pharyngeal.[146] Although the surface phonetics of Early Modern English /r/ is not recoverable from direct evidence, the assumption that it included a pharyngeal component is generally accepted (cf. Pulleyblank 1986; Gussenhoven and Weijer 1990; McMahon et al. 1994), especially in view of recent discussions of a comparable constriction in American English /r/ (cf. Delattre and Freeman 1968; Catford 1983, 2001; Gick 2002).

The synchronic analysis of linking [r] has for long been a challenging problem of English phonology, and a number of solutions have been suggested, some of them couched in gestural terms. The gestural approach is taken, e.g., by McMahon et al (1994), who tentatively suggest that both the coda and the onset allophones of /r/

[146] In this connection it will be useful to quote the following testimony of Delattre concerning the fact that, in the case of German /ʁ/, the tongue does not glide through the [ɑ] position after [a] or [ə]: "After /a/ the sound or /-r/ is less audible than after /i/ or /u/", and "Except after /a/ and /ə/ the final /-r/ always glided through an obscure /a/ and ended with a very light friction sound" (1971:141). This means that after [a] and [ə] there will be a tendency, on the part of the listeners, to perceive the pharyngeal onglide to the rhotic as an added vowel length.

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deleted one of the gestures of the originally bigestural /r/. The deletion occurred in such a way that, at least in RP, the coda allophone has preserved the pharyngeal, and the onset allophone the coronal gesture or the rhotic (McMahon et al. 1994:304). Gick (1999a) offers a slightly different solution by suggesting that the coronal gesture continues to be underlyingly specified in final allophones: although obscured by coda vocalization, it surfaces when the rhotic becomes ambisyllabic before vowel-initial words or morphemes. Gick supports his analysis by the consideration that linking [r] patterns with other underlying consonants as regards the correlation of the strength of their C-gesture with their prosodic position, with the greatest magnitude in onsets, the least magnitude in codas, and intermediate strength ambisyllabically (1999a:45ff).

It is possible to reconcile the gestural analyses of linking [r] by Gick (1999a) and McMahon et al. (1994) with the analysis of the centering diphthongs and long vowels that arise after the deletion of coda /r/ as ending with a pharyngeal glide (Bloch and Trager 1942; Pulleyblank 1980:78 and 1986:237f). As will be observed, the forms in Tables 8 and 9 present a five-way surface alternation of the reflexes of an earlier /r/: (1) a consonantal [r] word-initially and intervocalically within a single morpheme, (2) a fully vocalized [ə] prepausally and preconsonantally after nonlow lax vowels and diphthongal offglides, (3) as vowel length after /ə, ɑ, ɔ/ in the same environments as the previous allophone, (4) as the sequence [ər] intervocalically within a single morpheme or when the [ə] or [:] allophones are found before a vowel at a morpheme or word

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boundary, (5) and as the sequence [:r] when /ə, ɑ, ɔ/ are found before a vowel at a morpheme or word boundary (summarized in Table 10).

Table 10. Reflexes of Early Modern English /r/

[r]	[ə]	[ər]	[:]	[:r]
rɛd, mæri	bɪəd	dʒʊəri	stɑ:t	
	fiə	fiər#ɪz		
			kɑ:	kɑ:#rɪz

If the vowel length after /ə, ɑ, ɔ/ and the schwa offglide after the nonlow vowels are analyzed as an underlying pharyngeal glide, the number of allophones will be reduced to three (Table 11).

Table 11. Reflexes of Early Modern English /r/: a revised statement

[r]	[ɑ̰]	[ɑ̰r]
rɛd, mæri	bɪɑ̰d	dʒʊɑ̰ri
	stɑɑ̰t	
	fiɑ̰	fiɑ̰r#ɪz
	kɑɑ̰	kɑɑ̰#rɪz

The basis for considering the schwa as a surface manifestation of the pharyngeal glide is consistent with the analysis of Early Modern English /r/ as containing a pharyngeal constriction, and with the presence of such a coarticulation in the schwa and the /r/ phoneme of present-day American English. It is legitimate to assume that, even though in other varieties of English, such as RP, the schwa and/or /r/ have no pharyngeal component, they would have had it at an earlier stage. An analysis of vowel length as an

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underlying pharyngeal glide, on the other hand, is warranted by its being realized as a schwa offglide dialectally and in conservative lects (Giegerich 1997).

It will be observed that the allophones of /r/ in Table 11 are similar to the consonantal allophones of Maxakalí in that the rhotic has a consonantal [r], a fully vocalized [ɑ̯], and a prevocalized [ɑ̯r] realizations. Analyzing the sequence [ɑ̯r] as a prevocalized allophone of /r/ is plausible in view of the prosodic distribution of the allophones, with the most consonant-like, hence the strongest, allophone in syllable onsets; the most vowel-like, hence the weakest, allophone in syllable codas; and the intermediate-strength, or prevocalized allophone ambisyllabically. What makes this analysis especially attractive is the fact that such a distribution closely corresponds to what is currently known about the relative strength of consonantal allophones in general, and those of linking [r] in particular: thus, findings in both McCarthy (1993) and Gick (1999a) clearly show that ambisyllabic allophones of [r] are intermediate in strength between the onset and coda allophones.

Another attractive feature of the present approach is that it is able to account for the prevocalized allophone of /r/ by appealing to the notion of optimization of the syllable affiliation of its component gestures. The present analysis implies that retiming of the constituent gestures of Early Modern English /r/ consisted in phasing its V-gesture with the preceding nucleus, and its C-gesture with the following one, thus moving the V-gesture into a coda, and the C-gesture into an onset position. This suggestion derives typological support from so-called linking [l] in Portuguese. The lateral in that language is generally described as dark syllable-finally and clear in onsets, but various experts

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note that before a vowel across a word boundary it is realized as a sequence of dark and clear allophones. Herslund (1986), e.g., describes this realization in European Portuguese as "a non-velarized syllable-initial *l* following the velarized syllable-final *l* in phrases like *sal amargo* [saɫ lɐˈmargu] 'bitter salt' . . ." (516f n. 2); and Collischonn and Costa (2003) note for Brazilian Portuguese: "Finally, we have observed in our data that sometimes vocalization may be followed by an alveolar lateral, for example *hospital[wl]existe* . . . '(the) hospital exists . . ." (35 fn. 4). The Portuguese pattern may have resulted from the affiliation of the lateral's V-gesture with the preceding nucleus and its C-gesture with the following one, the type of development suggested here for ambisyllabic /r/s of Early Modern English.

4.4 Retroflex consonants

Prevocalization of retroflexes is considered last as it is more problematic than that of the rest of the consonants. Retroflex consonants are produced by curling in the tip of the tongue while producing apical coronals. A distinction is usually drawn between sublaminal retroflexes, produced between the underside of the tongue and the postalveolar region, and apical retroflexes, produced by placing the tip of the tongue behind the alveolar ridge (cf. Hall 1997:46). Retroflex consonants cover all manners of articulation: stop, affricate, fricative, flap, nasal, and lateral. Retroflexion has a pronounced areal pattern of distribution, being common on the Indian subcontinent, in Australia, and the Pacific coast of the Americas and central Africa, and rare elsewhere.

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Acoustically, retroflexes manifest "a downward shift of a set of formants or even of all the formants in the spectrum" (Jakobson et al. 1952:31). In the distinctive feature system of Jakobson et al. (1963), retroflexion is subsumed under the cover feature *flat*, together with labialization and pharyngealization (cf. also Bhat 1973:56f; Ohala 1985:223). In the post-*SPE* articulatory-based approach to distinctive features, the phonological status of retroflexes is somewhat controversial. The IPA treats retroflexion as a separate place of articulation (cf. Hamann 2004:54f). In Australian aboriginal linguistics, retroflexes are sometimes regarded as complex segments composed of a retroflex glide [ɻ] and the corresponding nonretroflex apical. Retroflexion is also interpreted as a prosodic feature: the prosodic analysis of retroflexion is supported, e.g., by nonlocal /n/-retroflexion in Sanskrit (cf. Allen 1951), and by the existence in Australian languages of retroflexed syllables in which there are no apical segments capable of serving as the source of retroflexion. Another issue often discussed in connection with retroflexes is their connection with the secondary features of velarization, uvularization, and pharyngealization. Thus, Lin (1989) and Hamann (2003) consider one of these secondary articulations obligatory in retroflexes: considerations adduced in favor or this view include the mechanical retraction of the tongue involved in the retroflexion gesture, the phonological incompatibility of retroflexion with palatalization, and the fact that some of the most important language-internal environments that lead to the development of retroflexes are the position after velars and next to back vowels (Bhat 1973; Evans 1995b). On the other hand, Bhat (1974) considers retroflexion and velarization contrastive in the vowels of a Dravidian

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language Badaga, and Hall (1997:49f) distinguishes between plain and secondarily modified retroflexes, notating the latter set with the usual IPA superscript for velarization.

In view of the above uncertainties surrounding retroflexion, prevocalization may be of service in helping elucidate its phonological status. If all retroflex consonants have a secondary retraction of the tongue, as suggested by Lin (1989) and Hamann (2003), the functioning of prevocalization in the consonant types considered so far leads us to expect that prevowels triggered by retroflexes will always reflect the secondary articulation of the consonant. If, on the other hand, there are retroflexes without a secondary retraction, their prevowels would have to have a different, perhaps retroflex quality. In the latter case, however, the outcome of prevocalization will not help to distinguish between retroflexion as a primary or as a secondary feature: just as, e.g., palatals and palatalized consonants prevocalize with an onglide of the same quality, so will prevocalization of retroflexes yield the same result whether retroflexion represents in them a primary or as a secondary feature.[147]

Prevocalization of retroflexes presents a two-way pattern that effectively supports the distinction between nonretracted and retracted retroflexes, with the former prevocalizing with a retroflex onglide, and the latter with a velar or pharyngeal onglide.

[147] Nevertheless, it would appear useful to make a distinction between retroflexion as a primary or as a secondary feature. The evidence for the functioning of retroflexion as a primary feature is plentiful (cf. Hamann 2003); its functioning as a secondary feature has been discussed much less often, but clearly follows from its functioning as such in vowels (cf. Catford 1977:191), and from its ability to function as prosody. Primary place of articulation features are not known to act in prosodic ways, whereas secondary articulation features do: for example, palatalization and velarization act on a prosodic level in Turkic, with its alternation of palatalized and velarized syllables (cf. Jakobson 1930).

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The former pattern is shown primarily by retroflex stops, flaps, nasals and laterals, i.e., those segments in which the retroflexion gesture of the tongue tip may be executed sequentially with respect to the occlusion gesture. Owing to this, preceding vowels and consonants may, and often do, receive some degree of retroflex coloring.[148] The influence of retroflex consonants on preceding vowels usually takes the form of vowel retroflexion and centralization (Bhat 1973:45f), but in some languages of Australia the preceding vowel is reported to be colored with what Evans (1995b) describes as a more or less pronounced retroflex offglide (739f). The retroflex consonant itself frequently loses retroflexion, thereby turning into the corresponding nonretroflex apicoalveolar. Under these circumstances, the retroflexion of the preceding vowel becomes an important perceptual cue of consonant retroflexion (Dixon 2002:568). This pattern, reported in a number of studies, gave rise to an analysis of retroflex stops, nasals and laterals in Australian languages as underlying sequences of the retroflex glide [ɻ] and the corresponding nonretroflex apicoalveolar (e.g., Glasgow and Glasgow 1967:9; Dixon 2002:572). This view has in turn served as the basis for a theory deriving retroflex consonants in languages of Australia from sequences of rhotics and alveolars. However, Dixon (2002:585) points out that this theory is untenable as the origin of all or most of the retroflex consonants in view of the comparative rarity of rhotic plus alveolar clusters in contemporary languages. It would appear possible, therefore, to

[148] The overwhelmingly anticipatory effect of retroflex stops, flaps, nasals and laterals on the neighboring segments is reported from a vast array of languages; Bhat (1973, 1974) explains the anticipatory rather than perseverative influence of this subset of retroflexes by the fact that, when their production is terminated, the tip of the tongue is released and cannot affect the following segment. Retroflex fricatives and affricates, on the other hand, may affect a following segment since, in their case, the tongue may remain retroflexed after the release of the consonant (Bhat 1973:44ff).

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attribute the existing phonetic clusters of a rhotic approximant [ɻ] and nonretroflex apical stops, nasals and laterals to prevocalization of retroflexes. Such an analysis is suggested, e.g., by the synchronic situation in Guugu Yimidhirr, where oral and nasal retroflex stops may be variably realized as unitary phonemes or as sequences of a retroflex glide and the corresponding alveolar oral or nasal stop (Dixon 2002:573). Such an analysis also derives support from the parallel behavior of palatals, especially oral and nasal palatal stops and palatal lateral. It will be recalled that prevocalization of these categories of palatals frequently involves alveolarization of their C-gesture, and in this respect prevocalization of palatals and that of retroflexes appear to converge.

The second pattern involves secondarily velarized or pharyngealized retroflexes. Allen's (1950) notes on the phonetics of Modern Eastern Armenian include an observation that there is a distinct schwa-like transitional glide between [i] and the following [ɻ] (183), cf. *erbšir* [jeɻbʲʃiˠɻ]. In many dialects of Mandarin Chinese, a schwa onglide develops when the diminutive suffix /r/ [ɻ] is added to words ending in a front vowel. The forms below, from the Beijing dialect, illustrate this pattern (Lin 1989), cf.:

	xua + *r*	>	*xuar*	'flower'
	$t^h u$ + *r*	>	$t^h ur$	'picture'
but	$p^h i$ + *r*	>	$p^h i^ə r$	'skin'
	y + *r*	>	$y^ə r$	'fish'

It will be recalled (4.2.5.1) that a similar transition is inserted between /i/ and the suffix /u/ in the Chengzhou dialect (Lin 1989), cf.:

	ua + *u*	>	*uau*	'socks'
	ia + *u*	>	*iau*	'sprout'
but	*pi* + *u*	>	$pi^ə u$	'nose'

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Finally, Hashimoto's (1973) reconstruction of Ancient Chinese retroflex endings *-ɳ and *-ʈ suggests prevocalization with a rounded velar onglide. The reconstructed endings are mostly reflected in Modern Chinese as -ŋ and -k, respectively, accompanied by rounding of preceding vowels. In some variants of Archaic Chinese and some of the modern Anhwei dialects, however, the posited retroflex finals are reflected as labials. Based on the evidence of a parallel development in the initials, where the labial consonants of some Northwestern Mandarin dialects correspond to Pekinese retroflexes followed by rounded vowels, Hashimoto derives the labial finals from intermediate velars preceded by rounded vowels or the labial glide, which in turn suggests that the original retroflexes prevocalized with a labiovelar onglide, the developmental sequences being *-ɳ > *-^{w}ɳ > *-^{w}ŋ > -*m* and *-ʈ > *-^{w}ʈ > *-^{w}k > -*p*.[149]

4.5 Summary and conclusion

This chapter has provided a representative selection of case studies of prevocalization resulting in nonfront prevowels. It has been shown that central to back onglides are triggered by labials, velars, and velarized consonants, and that rounded onglides are triggered by labialized consonants. It has also been shown that onglides triggered by pharyngeals and uvulars have variable realizations, due in part to overlap with gestures of the preceding vowels, and that retroflex consonants possess a distinctive pattern of

[149] If rounding of preceding vowels may act as a conditioning factor, prevocalization may not be a necessary intermediate stage here: thus, Hamann (2003) lists several cases of a direct rounding of front vowels before retroflex consonants (101).

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prevocalization only when not modified by secondary velarization or pharyngealization. Case studies in this and the preceding chapter bear out the predictions regarding the quality of prevocalization onglides made in Chapter 2, thereby concluding the empirical portion of this study.

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5. Prevocalization and consonant structure

5.1 Introduction

The gestural analysis of prevocalization introduced in Chapter 2 crucially relies on the notion that prevocalization reveals in a direct way the internal composition of consonants. The proposed analysis of the mechanism of prevocalization relies, in its turn, on the syllable-position effects on the mutual timing of intrasegmental gestures. The proposed analysis has led to two related hypotheses. The first of these consists in the notion of both plain and coarticulated consonants having a vocalic and a consonantal component. The second hypothesis consists in the idea that, while the consonantal component corresponds to the consonant's primary stricture, the quality of its vocalic component may be inferred from the prevowel. Both hypotheses stand the test of the empirical data presented in Chapters 3 and 4.

The purpose of the present chapter is twofold. Its first goal is to test briefly how the information on the internal structure of consonants inferred from a study of prevocalization fares by comparison with similar information inferred from a study of other phonological processes capable of revealing the internal structure of consonants. The second goal is to see whether mechanisms analogous to the proposed mechanism of prevocalization may account for at least some cases of compensatory lengthening and vowel intrusion. These particular processes have been chosen for analysis because both sometimes interact with prevocalization in revealing ways by emerging as alternative responses to environments that condition prevocalization.

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The chapter is organized as follows. Sections 5.2 and 5.3 seek evidence for the gestural composition of consonants in postvocalization and the functioning of syllabic consonants, respectively. Section 5.4.1 explores the relationship between prevocalization and vowel intrusion, and Section 5.4.2 looks at interaction between prevocalization of single and geminate consonants, and compensatory lengthening. Section 5.5 is a brief summary of the main points.

5.2 Postvocalization

Though this work has focused primarily on prevocalization, on a few separate occasions it afforded the opportunity to refer to its mirror image, labeled here postvocalization. In postvocalization, which affects consonants in onsets, the consonant's V-gesture is moved in the direction of a nucleus that follows rather than precedes the triggering consonant; mutual ordering of the consonant's V- and C-gestures in onsets is governed by the same tendencies that are responsible for their reverse mutual ordering in syllable codas (e.g., Sproat and Fujimura 1993; Browman and Goldstein 1995). A detailed investigation of postvocalization is outside the scope of this work; however, a few examples will suffice to show that the general functioning of this process leads to the same conclusions as regards the internal structure of consonants as prevocalization. The purpose of the examples below is to demonstrate that the quality of the postvocalizaton offglides is independent of the consonants' C-gestures and derives from their V-gestures instead.

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One example of postvocalization mentioned earlier is the development of a palatal offglide by the palatal consonants in French, both as a historical process (French *chien* [ʃjã] < Vulgar Latin *cane(m)* 'dog') and as a substandard pronunciation in the present-day language ([ʃja] for *chat* 'cat'). Another example of palatal postvocalization that may be mentioned is the historical developments *ʎ* > *lj* and *ɲ* > *nj* in Slovenian (Andersen 1970:39). In addition to palatals, consonants capable of triggering front offglides include palatalized consonants and alveolars. Thus, palatalized labials are realized syllable-initially as sequences of labial plus yod in Lithuanian, Polish and Irish. In Erris Irish, the offglide between palatalized labials and velars /u(:)/ has a [i]-quality, and the one between palatalized velars and /o(:)/ has a [e]-quality (Mhac An Fhailigh 1980:45ff); similar offglides are described for Muskerry Irish (Ó Cuív 1975:52f), Cois Fhairrghe Irish (De Bhaldraithe 1945:43f), Tourmakeady Irish (Búrca 1958:59f), and other dialects. Palatal offglides are sometimes known to consonify, as in the historical development of French, Romanian, or Slavic (cf. Hock 1991:119, 2003b, and the discussion in 3.3.2.4). Postvocalization of allophonically palatalized consonants may be exemplified from Turkana (Nilotic) and Italo-Romance: in Turkana, a palatal offglide optionally occurs after a syllable-initial consonant when there is a [i] in the preceding syllable, cf. *Ni-keno* [Nikjeno] 'fire-places' (Dimmendaal 1983:14), and in south and central Italy the same effect is achieved by a preceding plural masculine definite article, cf. [i kjani] 'the dogs' with Standard Italian *i cani* (Tuttle 1985:26). Postvocalization of

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alveolars may be exemplified with the historical change **s* > *hj* that affected Proto-Polynesian **s* in Moriori and Tokelau (cf. Smith 1988:229).

One feature that pre- and postvocalization have in common is the fact that the C- and V-gestures of the triggering consonant are phased in a way that makes the V-gesture move closer to the nucleus. The ultimate causes behind the two processes are also similar: in the case of secondarily modified consonants, the main drive seems to be synchronic enhancement/diachronic loss of the secondary articulation, and in the case of plain consonants the main reason seems to be lenition. Loss of secondary articulation may be demonstrated, e.g., for Irish, where the palatalization gesture during the occlusion phase of the labial has been completely lost and the offglide is now the only exponent of labial palatalization. Lenition is a natural assumption for the above change **s* > *hj* in Moriori and Tokelau: while onsets are commonly considered a prime site for consonant strengthening (cf. Fougeron 1999:41ff), lenition in that position is neither unusual nor infrequent. Some well-known examples of onset lenition are debuccalization of Proto-Indo-European **s* in Greek, which took place regardless of the syllable position of the sibilant (cf. Pagliuca 1982), and stress- and utterance-initial conditioned onset lenitions in languages of Australia (Blevins 2001).

Noncoronal consonants and consonants with secondary articulations other than palatalization reveal the same internal composition in postvocalization as they do in prevocalization. Labials may trigger the development of a labiovelar offglide, and velars or velarized consonants may trigger the development of a schwa-like, velar, or labiovelar offglide. As postvelarization typically takes place before front vowels, where

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backness rather than roundness of the offglide is expected to be at its most prominent, the rounding of the offglide triggered by velars and velarized consonants may be epiphenomenal, probably the outcome of a routine enhancement of velarity by simultaneous rounding (cf. Stevens et al. 1986). In the case of labials, the reason for offglide rounding may be greater synchronization of the tongue body and lips gesture characteristic of onsets. Conversely, the asynchronous execution of these gestures in codas, where the tongue body gesture precedes the lip gesture, accounts for the lack of rounding in prevocalization onglides (cf. 4.2.3 for details). This analysis is borne out by the pattern of post- and prevocalization in Irish, where offglides following plain labials are always, and those following plain velars sometimes, described as rounded. For example, in Erris Irish there is a [w]-quality offglide between velarized labials /p, b, f, m/ and a following /i(:), e(:)/, and a [o]-quality offglide between velars /k, g, x, ɣ, ŋ/ and a /i(:)/. The offglide between dentals and front vowels, however, has a [ə]-quality (Mhac An Fhailigh 1980:45ff). The absence of rounding in offglides developed by dentals may be accounted for by their less retracted quality. In West Muskerry Irish, only the offglide developed by labials is rounded, while that developed by velars has a [ə]-quality (Ó Cuív 1975:52f). Among other languages with labial postvocalization may be mentioned Moriori and Tokelau, where Proto-Polynesian **f* > *hw* (Smith 1988), and Chichicapan Zapotec, where /p, b, m/ in stressed syllables have developed a labiovelar offglide before the front vowels /i, e/ (Thom Smith Stark, p. c.). Tuttle (1985) notes a parallel development in Ligurian, citing Old Genoese *pucere* 'father', *mucere* 'mother',

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etc., as examples (14 fn. 22). Pope (1934:195f) notes that in Middle French there was a tendency to reduce the [wɛ] product of the diphthong <oi> to [ɛ] in uneducated speech; according to her, this development was checked in part due to the opposite tendency to develop a round offglide between labials and the following [ɛ] (1934:185).

Consonants preceded by rounded vowels may postvocalize with a rounded glide. Thus, in Turkana (Nilotic) a labial glide is optionally inserted after a syllable-initial consonant when there is a *u* in the preceding syllable, cf. *Ni-turkana* [Niturkwana] 'the Turkana people' (Dimmendaal 1983:14). In Birom, a proto-Benue-Congo morpheme of the singular */u-/ is reflected as a glide inside the nominal stem, cf. *rwas* 'man (sg)', *hwa* 'woman (sg)' (basic stems are /-ras/ and /-ha/, respectively) (Tuttle 1991:29). Velars preceded by a round vowel are postlabialized in Trique (Otomanguean), cf. /nuka/ [nuk^{w}a] 'strong' (Silverman 2006:135). Postvocalization triggered by a *u* in the preceding syllable is also common in south and central Italy; the trigger may be a word-internal vowel or the proclitic masculine singular article *u*, cf. *sud^{w}á* 'to sweat', *suk^{w}a* 'to play', *u k^{w}anə* 'the dog', *u n^{w}idə* 'the nest', *u f^{w}iʎʎə* 'son' (cf. Standard Italian *sudare, giocare, il cane, il nido, il figlio*) (Tuttle 1985).

While far from exhaustive, the above picture of consonant postvocalization is nevertheless sufficient to show the existence of a vocalic component in consonants. It also shows that the quality of postvocalization offglides derives from this vocalic component, and that the workings of this process confirm the phonological relevance of

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syllable-based generalizations over the mutual ordering of the consonants' component gestures, applied earlier to the analysis of prevocalization.

5.3 Syllabic consonants

Another area that clearly demonstrates the existence of a V-gesture in consonants is the life-cycle of consonantal syllabics.[150] According to Bell (1978), consonantal syllabics tend to arise through absorption of an adjacent vowel with concomitant shift of syllabicity to the consonant; in time, they tend to develop into consonant-vowel combinations in their turn. Both the origin of consonantal syllabics through absorption of adjacent vowels and their eventual decomposition into consonant-vowel sequences hint at their possessing an internal vocalic component. The observations on the historical development of syllabic nasals in Bell (1978), the most comprehensive cross-language study of syllabic consonants to date, are especially revealing as to the preferred kinds of vowel-nasal consonant combinations that result in their production. According to Bell, "when syllabic nasals are formed, mu, um, ŋu, and uŋ are the first nasal-vowel combinations to be affected" (1978:153); there is also some evidence to show that the most favored diachronic source for a syllabic [n̩] is [ni] (1978:181). These observations indicate that the preferred vowel-nasal consonant combinations, from the

[150] A similar suggestion is made by Reynolds et al. (2000:365): "Though a segmental transcription would record only the consonant quality, the conception of syllabic sonorants in (9d) [i.e., without the accompanying epenthetic vowel, N.O] recognises the simultaneous presence of a vocalic element. . . . We also attribute the syllabicity of the syllabic liquid to the V element in the phonological structure". I am grateful to Brent Vine for bringing this article to my attention.

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viewpoint of formation of nasal syllabics, are those in which the consonant's V-gesture and the gesture of the adjacent vowel closely match, presumably facilitating the merger.

The type of vowels resulting from decomposition of the syllabic consonants apparently reflects their internal structure at the time of decomposition, which in turn may provide clues as to their surface phonetics from a diachronic point of view. Reynolds et al. (2000), e.g., express this as follows: "Even when the vowel and resonant are completely co-extensive, as in (9d), the resulting syllabic contoid has vowel colouring exhibited in the secondary articulation of the resonant" (365). Thus, decomposition of Proto-Indo-European syllabic sonorants *$r̥$, *$l̥$, *$m̥$, *$n̥$ produced clearly differing vowel qualities in different (groups of) languages, presumably reflecting divergences in the articulatory quality of the syllabics. These divergences, interpreted by Meillet (1937) in terms of a dialectal division between Germanic and Balto-Slavic, on the one hand, and Indo-Iranian, Armenian and Greek, on the other, are summarized in Table 12. If taken at face value, the vocalic portions of the reflexes of the syllabic sonorants would indicate the dark (Germanic) or a combination of dark and clear (Balto-Slavic) qualities of the syllabics in the former group of languages, and either a schwa-like or a pharyngealized quality in the latter group.

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Table 12. Development of syllabic sonorants in selected Indo-European languages

Proto-Indo-European	Sanskrit	Avestan	Greek	Armenian	Old Church Slavic	Lithuanian	Gothic
*r̥	r̥	ərə	ra ~ ar	ar	rĭ ~ rŭ	ir̃ ~ ur̃	aur < *ur
*l̥	r̥	ərə	la ~ al	al	lĭ ~ lŭ	il̃ ~ ul̃	ul
*m̥	a	a	a	am	ẽ	im̃ ~ um̃	um
*n̥	a	a	a	an	ẽ	in̄ ~ un̄	un

The synchronic functioning of consonantal syllabics may also be revealing as to their internal structure. Thus, voiceless syllabic stops are always followed by a release, and other syllabic consonants, especially sonorants, are frequently known to alternate with sequences of the corresponding nonsyllabic consonant and a transitional vowel (Bell 1978:185; Hoard 1978:69). According to Hoard (1978:68), the syllable peak of syllabic stops and affricates is contained in their release; this makes the release of these consonants functionally equivalent to the transitional vowel accompanying sequential realizations of other types of consonantal syllabics. When the transitional vowel has an invariable schwa-quality or is fully determined by the surrounding segments, it is apparently uninformative as to the consonant's internal structure; such appears to be the case of transitional vowels in Piro (Arawak) (Matteson and Pike 1958; Matteson 1965). However, in some cases it can be demonstrated that the quality of the transitional vowel or release derives from the consonant's V-gesture. This is easiest to demonstrate for consonants with distinctive secondary articulations: for example, Hoard's (1978) study of syllabic obstruents in languages of the Pacific Northwest includes the observation that the word-final syllable /q^{w}ɬ/ in Quileute may be optionally realized as [q^{w}ʊɬ], as in the words for 'doorknob' and 'shirt' that he cites (60). The releases of word-final stops

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in Russian likewise have the acoustic quality of their secondary articulations (Tomson 1922). Information on the transitional vowels with the quality of nondistinctive secondary articulations of syllabic consonants is also available and has been demonstrated for Tashlhiyt Berber by Coleman (1999, 2001).

Tashlhiyt Berber has figured prominently in recent phonological literature as a language in which any consonant may act as a syllable peak (cf. Dell and Elmedlaoui 1985, 1988). In many cases, syllabic consonants in Tashlhiyt Berber are pronounced with transitional vowels whose quality, according to Dell and Elmedlaoui (1996) arises from a mechanical transition from one consonant to another. Coleman (2001) observes that both syllabic and nonsyllabic consonants in Tashlhiyt Berber have secondary articulations; in the examples below the secondary articulations include palatalization [ʲ], clear quality [ˡ], and central quality [ᵊ]:

[sʲrˡstn̩ᵊt]	'put them down'
[t̩ˡklit]	'you spend the day'
[ʃᵊfᵊtᵊ]	'fishing net'

Coleman also notes that transitional vowels do not always have the same quality in the same consonant sequences, which they would be expected to do had they been purely mechanical transitions between the consonants. He reports the results of his earlier study (Coleman 1999), in which the allophones of transitional vowels were found to fall into three sets: a high front cluster comprising such vowel qualities as [ɪ] and [ə], a high back cluster including [ʊ] and [ɤ], and a low cluster with such members as [ʌ] and [ɐ]. In some cases, the quality of the transitional vowels may be predicted on the basis of

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their segmental environment: for example, the transitional vowels breaking the sequences [gn̩], [ql̩] and [gl̩] in the pairs of words below have a more front quality in the forms on the left and a more retracted quality in the forms on the right. Coleman (2001), whence these examples are taken, interprets the more retracted transitional vowels on the right as the result of backing influence from the segmental /u/ in the same word.

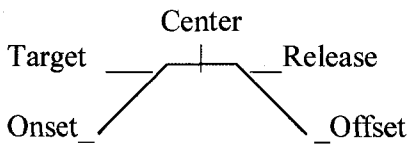
/gn̩/	[gɪnʲ]	'to sleep'	/gn̩nu/	[gən]	'sew'
/iqql̩/	[qə̤l]	'he waited'	/ismuqql̩/	[qʊl]	'he waited'
/tr̩gl̩t/	[gɪlʲ]	'you locked'	/jugl̩/	[gʊl]	'he hung'

However, even after all instances of variation that can be explained by the segmental environment are set aside, "there remains a large residue of unexplained variation" (Coleman 2001:53). Coleman takes this to mean that syllabic consonants in Tashlhiyt Berber are phonologically represented as a combination of an epenthetic vowel and a consonant, "the vocalic component of a consonant" being "its secondary articulation, whether phonologically distinctive (as in /k/ vs. /k^{w}/) or non-distinctive" (2001:54).

Hall (2006) argues that the difference between the schwa and non-schwa realizations of transitional vocoids depends on the precise phasing of the C-gestures of the adjacent consonants. Thus, Gafos (2002) divides each consonantal gesture into a series of landmarks (Figure 18) that refer to the onset of a movement, the target or achievement of maximal constriction, the center of the constriction phase, the release of the constriction, and its offset.

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Figure 18. Landmarks of an articulatory gesture



Based on this suggestion, Hall argues that the percept of a schwa results when the consonantal gestures are phased in the manner shown in Figure 19, i.e., when the center of the first consonant is aligned with the onset of the second.

Figure 19. Gesture phasing producing a schwa-like vowel transition



The gesture phasing as in Figure 19 also allows the transitional vowel to be colored by the gestures of both adjacent consonants, given that both consonantal articulations are active at the time of overlap. However, if the phasing of the adjacent consonant gestures is as shown in Figure 20 below, the transitional vowel can only have the quality of the V-gesture associated with the first consonant. This type of phasing implies that the articulators are no longer actively engaged in producing the first consonant and are on their way to the onset of the second consonant. This allows the intervening space to be filled with the V-gesture of the first consonant, providing a glimpse of its internal composition.

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Figure 20. Gesture phasing producing a non-schwa vowel transition



5.4 Alternatives to prevocalization

5.4.1 Vowel intrusion

In his study of the breaking-related phenomena in Old English, Howell (1991) suggests that liquid plus consonant (LC-) clusters are phonotactically marked in Germanic languages, and views the main purpose of breaking as the elimination of such structures. In the same study, Howell comments on another strategy for eliminating the problematic LC-clusters, which consists in inserting a svarabhakti vowel between the liquid and the following consonant, especially when the two are heterorganic: cf. Old High German *durh* 'through' variously spelled *thuruc, thuruh, dhurah*, etc. (Howell 1991:62). Howell notes that the svarabhakti vowel triggers resyllabification of the liquid which, instead of weakening, is retained in its strong form by moving to a prosodically strong onset position. According to Howell, breaking and the insertion of svarabhakti vowels are complementary in Germanic, even though the two processes may take place in the same dialect or language in a diachronic sequence (1991:65).

Germanic is not the only language area for which the complementarity between prevocalization and vowel intrusion[151] has been noted. For example, in various dialects

[151] I adopt the terms "intrusive vowels" and "vowel intrusion" from Hall (2006); other terms for this phenomenon include "svarabhakti", "excrescent", or "parasitic" vowels.

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of Irish and Scottish Gaelic geminate liquids /r:, l:, n:, m:/ in coda were historically prevocalized with an onglide, whose quality corresponds to the liquids' secondary articulation; the geminates were also degeminated as a result, cf. *dall* > *daul*. Prevocalization, however, takes place only when the sonorant is either in the coda of a monosyllabic word or is followed by a homorganic sonorant, fricative or stop (cf. Ó Dochartaich 1981:225). When the following consonant is nonhomorganic, an intrusive vowel is inserted, which is either a schwa or has the quality of the preceding vowel, cf. *far:g* > *farəg / farag*. The resulting structure is phonetically disyllabic; however, its stress pattern and native speaker intuitions both indicate that it is phonologically a monosyllable (Borgstrøm 1940:153; Ó Dochartaich 1981:231ff; Bosch 1998).

The basic underlying similarity of the Gaelic and Germanic situations above suggests that prevocalization and vowel intrusion interact in a principled way. It was briefly suggested in 4.3 that one possible way of analyzing linking [r] in English is by considering the [ər] ~ [:r] sequences ambisyllabic allophones of /r/. It will also be recalled that the emergence of this allophone was attributed to the tendency to optimize the syllable affiliation of the component gestures of the liquid by phasing its V-gesture with the preceding nucleus and its C-gesture with the following onset. It is possible to explain vowel intrusion by reference to the same tendency, and in this case its complementarity with respect to prevocalization receives a natural explanation in terms of the direction of sliding of the liquid's C-gesture. In prevocalization, the affiliation of the liquid's V-gesture is optimized by being phased with the preceding nucleus; its C-

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gesture, which stays in coda, is subject to weakening. In vowel intrusion, the liquid's C-gesture moves away from the C-gesture of the following consonant, and the acoustic release between them may be interpreted by the listeners as a vowel. If the gestures of the two consonants are phased as in Figure 19, the vowel will be perceived as a schwa, and if they are phased as in Figure 20, the gap may be occupied with the vocalic gesture associated with the preceding or following segmental vowel (cf. Hall 2006:338; Steriade 1990).[152] In time, the intrusive vowel may become lexicalized, which will entail phonologization of the phonetic syllable as well.

5.4.2 Compensatory lengthening

5.4.2.1 Single consonants

In languages with independently motivated quantity distinction in vowels,[153] diphthongal outcomes of prevocalization may be in complementary distribution with lengthened nuclei. The distribution of the two outcomes mostly depends on the articulatory distance between the prevocalization onglide and the preceding vowel: if the two are articulatorily distant, the prevocalization onglide will be perceived as a diphthongal offglide, and if they are close, the onglide will be absorbed by the vowel and reinterpreted as increased duration.

---

[152] Hall (2006) does not mention intrusive vowels with the quality of the secondary articulation of the liquid; the only two choices mentioned by her are a schwa and the vowel gesture of an adjacent vocalic segment. This may be an accidental gap: cf. the optional realization of the syllable /q^{w}ɬ/ in Quileute as [q^{w}ʊɬ], mentioned above (Hoard 1978).

[153] Cf. a discussion of this point in De Chene and Andersen (1979).

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A specific instance of prevocalization that has resulted in a complementary surface distribution of lengthened and diphthongal nuclei is the series of changes in Early Modern English that consisted of pre-/r/ breaking, pre-/r/ lengthening, and /r/-deletion (see 4.3). The combined effect of these changes was the creation of centering diphthongs and triphthongs out of earlier nonlow vowels and diphthongs, respectively, and of long vowels out of earlier /ɑ, ɔ, ə/ (see Table 13).

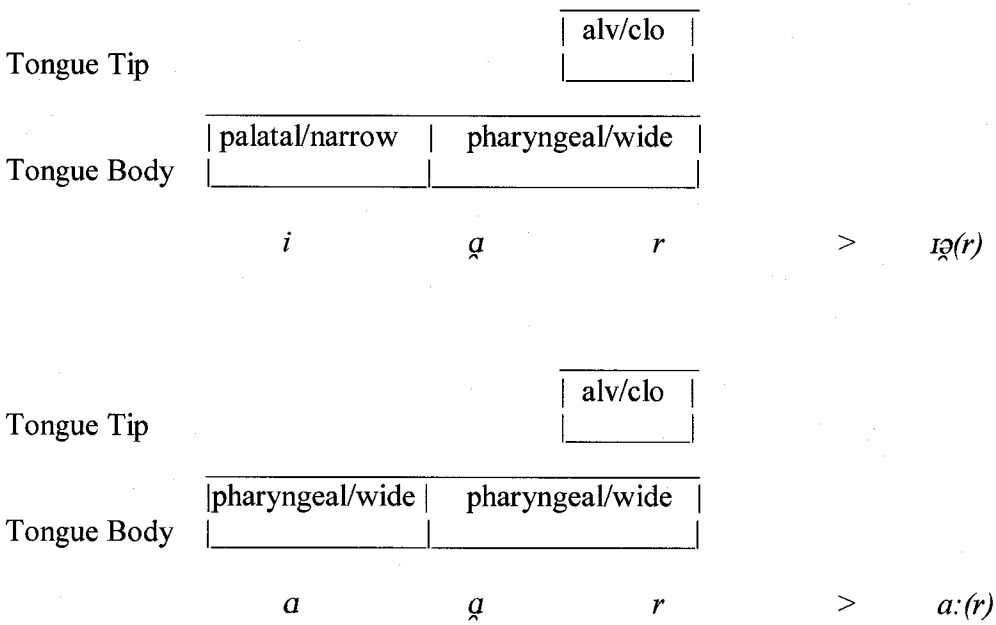
Table 13. Prevocalization of Early Modern English /r/

Affected nucleus	*i:*	*e:*	*o:*	*u:*	*aɪ*	*aʊ*	*ə*	*ɑ*	*ɔ*
Final coda	*beer*	*chair*	*more*	*sure*	*fire*	*tower*	*stir*	*bar*	*for*
Preconsonantal Coda	*beard*	*scarce*	*force*	*gourd*			*bird*	*start*	*north*
Intervocalic	*dreary*	*vary*	*glory*	*jury*					
Surface results	*ɪə(r)*	*ɛə(r)*	*ɔə(r)*	*ʊə(r)*	*aɪə(r)*	*aʊə(r)*	*ə:(r)*	*ɑ:(r)*	*ɔ:(r)*
Surface results more generally	*ə (r)*						*: (r)*		
Underlying result	*ɑ̱ (r)*								

It was suggested in 4.3 that the failure of /ɑ, ɔ, ə/ to produce diphthongs with the prevocalization onglide has to do with its articulatorily proximity to these vowels. Figure 21 expresses these complementary developments in gestural terms.

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Figure 21. Prevocalization of Early Modern English /r/



The above developments clearly show the derived status of the compensatory lengthening outcome of prevocalization. This seems to be generally the case whenever stressed vocalic space is invaded by tautosyllabic liquids. This becomes even more obvious when the historical record clearly indicates that the diphthongal outcome is chronologically the earlier of the two; this is the case, e.g., of Southern Scots, where the contemporary reflexes of Middle Scots /l/-(pre)vocalization consist of diphthongs and long vowels. Since the vowel shape before the /l/ was diphthongal in Middle Scots orthography, present-day long vowel outcomes may be viewed as the result of vowel merger with the prevocalization onglide, cf. Table 14 (based on Taylor 1974:407).

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Table 14. /l/-vocalization in Middle Scots

	Middle Scots spellings	Modern reflexes
a __ l (all, call)	*au, aw*	[ɑ:], [o]
o __ l (boll, knoll)	*ou, ow*	[ou], [ʌu]
u __ l (pull, wool)	*ou, ow*	[u:], [ʌu]

Another type of segments whose prevocalization may result in a complementary distribution between compensatory lengthening and vowel diphthongization, is nasals. For example, /n/-prevocalization in the Visperterminen dialect of Swiss German, discussed in 3.4.3, results in lengthening or diphthongization of the preceding vowel (Wipf 1910:44ff; Anderson 1981:516f). The lengthening outcome is limited to the high vowels /i, u, y/ and to the vowel /a/ before /f/ (only one instance is cited), and diphthongization takes place in the case of the nonhigh vowels /e, a/. The data are repeated below for convenience.

in > *ĩ: / __ ʃ, ç*	*fi:ʃter* 'dark-colored', *trĩ:çu* 'to drink'
un > *ỹ: / __ ʃ, f, s*	*xỹ:ʃt* 'art', *fỹ:f* 'five', *rỹ:s* 'ditch'
yn > *ĩ: / __ ʃ, f*	*ĩ:ʃ* 'us', *frnĩ:ftig* 'prudent'
a > *ã: / __ f*	*sã:ft* 'easily'
an > *ãj / __ s, ʃ, ç*	*Hãjsi* 'Hansi' (diminutive of 'Hans'), *xãjʃt* 'you (sg.) can', *ãjço* 'butter', *bãjç* 'bank'
en > *ẽj / __ ʃ, ç*	*dẽjçu* 'to thank', *pfẽjʃter* 'window'

At first glance, lengthening of /i, u, y/ may be interpreted as compensatory lengthening of vowels conditioned by the loss of the nasal. Nevertheless, at least two considerations force us to conclude that vowel lengthening is secondary to the (pre)vocalization of the nasal. First, lengthening of /a/ is attested only before a /f/: before the coronal and palatal fricatives, this vowel is followed by a palatal offglide. If one assumes that prior to its

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loss /n/ was assimilated to the place of articulation of the fricative, diphthongization of /a/ would be consistent with prepalatalization of the nasal. As no palatalization of the nasal is expected in the case of a following /f/, the onglide to the nasal in that position would not be different enough from the vowel /a/, which would lead to its perception as an extra length. Another consideration is that, while in the case of /i/ extra length resulting from absorption of the prepalatalization onglide would be indistinguishable from "compensatory lengthening" associated with the loss of the nasal, this is not so in the case of the other two high vowels, /u/ and /y/. The expected outcome of their lengthening would be a /ũ:/ and a /ỹ:/, respectively; instead, the results are /ỹ:/ and /ĩ:/. The /ỹ:/ outcome of /u/ is explainable only by assuming a diphthongized stage [ũj], and the prior existence of such a stage is confirmed by developments in the neighboring dialect areas mentioned by Wipf, and also by the single lexical item *xujʃt* 'you (sg.) come' (< *xunʃt*, cf. German *kommst*) in the Visperterminen dialect itself (1910:46). The unrounding of /ỹ:/ to /ĩ:/ also receives a straightforward explanation if one assumes an intermediate stage [ỹj].

5.4.2.2 Geminate consonants

The prevocalization of geminates results in degemination of the consonant and diphthongization or lengthening of the preceding vowel. Such a development is attested in a range of Irish and Scottish Gaelic dialects, where fortis syllable-final liquids and nasals are prevocalized after original short vowels. The sonorant that undergoes

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prevocalization can be in a prepausal coda or preconsonantal coda before homorganic consonants of monosyllabics, or in the preconsonantal coda of disyllabics; when intervocalic, it fails to prevocalize (Ó Dochartaigh 1981:224ff). The quality of the prevocalization onglide corresponds to the secondary articulation of the sonorant. Examples of the long vowel and diphthongal reflexes of velarized geminate sonorants in different dialect areas are given in Table 15 below (based on Ó Dochartaigh 1981:225).

Table 15. Prevocalization of fortis sonorants in Gaelic

	Area 1	Area 2	Area 3
bàrr	ba:r	ba:r	ba:r
dall	dal:	daul	daul
ceann	k^{j}en:	k^{j}eun	k^{j}eun
cam	kam:	kam:	kaum

Ó Baoill (1980:106) derives both the onglides and vowel length in the above forms from the vocalization of the first part of the geminate. Translated into gestural terms, this analysis makes prevocalization of geminates identical to that of single consonants. The difference in the outcomes (deletion in the case of single consonants, degemination in the case of geminates) is explainable by the difference in the length of the triggering consonants. In the case of single consonants, temporal reduction of the C-gesture may be so significant with respect to the overall duration of the consonant that it may be perceived as its complete deletion, while in geminates, temporal reduction of the C-

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gesture is not sufficient to induce consonant deletion, and reduced duration of the C-gesture is perceived as degemination.[154]

A more famous case of a complementary distribution between the compensatory lengthening and diphthongization outcomes of prevocalization concerns the (palatalized) sonorants of Greek discussed in 3.2.2.3; the main descriptive facts are repeated here for convenience. The environment for this development involved intervocalic sonorants **m*, **n*, **r*, **l* followed by a palatal glide; a prevocalization-type development took place after /a, o/ and was shared by all dialects, cf.:

**phan-jō* > *phajnō* 'show'
**g^{w}m̥-jō* > **g^{w}an-jō* > *bajnō* 'go'
**mor-ja* > *mojra* 'lot'

In case the vowels were /e, i, u/, the result was compensatory lengthening of the preceding consonant (in Lesbian and Thessalian) or vowel (other dialects), cf.:

**klin-jō* > *klinnō* (Lesbian/Thessalian), *klīnō* (elsewhere) 'tend'
**awer-jō* > *aerrō* (Lesbian/Thessalian), *aērō* (elsewhere) 'lift'

In 3.2.2.4, the above prevocalization was analyzed as part of the general depalatalization strategy during the relevant historical period of the Greek language (cf. also Hock 2003a). Here attention is only drawn to the fact that the palatalized sonorants that are thought to have resulted from the historical sequences with palatal glide are assumed to have been geminate: geminate outcomes of palatalization are generally assumed, e.g., for the dental, velar and labiovelar stops (cf. Bartoněk 1964; Risch 1981).

[154] For representation of geminates with a single oral gesture cf. Smith (1995).

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Provided the assumption that the original ***lj*, ***rj*, ***nj*/***mj* produced *l^{j}:, *r^{j}:, *n^{j}: is correct, the pattern of sonorant depalatalization would have been as presented schematically in Table 16.

Table 16. Sonorant depalatalization in Proto-Greek

	Simple depalatalization	Vowel lengthening and depalatalization	Vowel diphthongization and depalatalization
*r^{j}:, *n^{j}:, l^{j}:	r:, n:, l:	*V:r^{j}, *V:n^{j}, *V:l^{j} > V:r, V:n, V:l	*ajr^{j}/ojr^{j}, *ajn^{j}/ojn^{j}, *ajl^{j}/ojl^{j} > ajr/ojr, ajn/ojn, ajl/ojl

It can be seen that the initial developments in the second and third columns in Table 14 (i.e., vowel lengthening or diphthongization coupled with degemination of the sonorant) are identical to the developments in Irish and Scottish Gaelic in Table 15: the only difference between the developments in the two language areas lies in the loss of secondary palatalization in Greek.

A discussion of geminate prevocalization will not be complete without a brief mention of partial geminates (= homorganic sonorant-obstruent clusters), which may be exemplified from Maxakalí. As was mentioned earlier (1.5.1.2), all nasal and oral stops in Maxakalí are fully vocalized before homorganic consonants; the relevant data is repeated below for convenience.

/pp/	/-keppa/	[kæʏpa]	'before'
/tt/	/mattɯk/	[mbaɔtɯx]	'toad'
/cc/	/coc cecka/	[ʃoiʃæjʃkaʔ]	'big tooth'
/kk/	/kɯcakkɯk/	[kɯʃaɯkɯx]	'capybara'

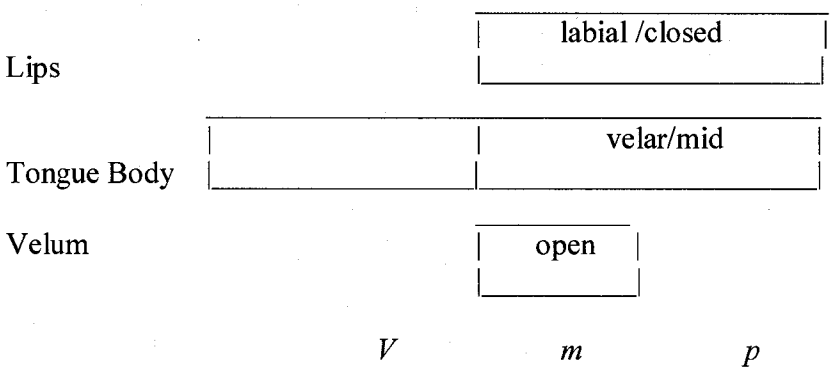
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/mp/	/mĩm pe/	[mĩɤ̃pe]	'bed'
/ɲc/	/mãʔãɲ cecka/	[mãʔãĩʃɛjka]	'large alligator'
/ŋk/	/pɯtɯcnãŋ kɯtɯt/	[pɯtɯɣiçnãɯ̃ k-]	'old bird'

In view of our analysis of geminate prevocalization, it is possible to make the above descriptive statement more precise by analyzing full vocalization of Maxakalí stops as prevocalization of full or partial geminates (cf. also Wetzels and Sluyters 1995). In contrast to geminate prevocalization in Gaelic and Greek, which results in nonsyllabic onglides, prevocalization of geminates in Maxakalí produces vocoids which Gudschinsky et al. (1970) regard as syllabic. As with true geminates, prevocalization of partial geminates may be represented as reduction in the duration of the C-gesture; if the first element of the partial geminate is nasal, the velum lowering gesture is not reduced, causing the (pre)vowel to be nasalized (presented schematically in Figure 22).

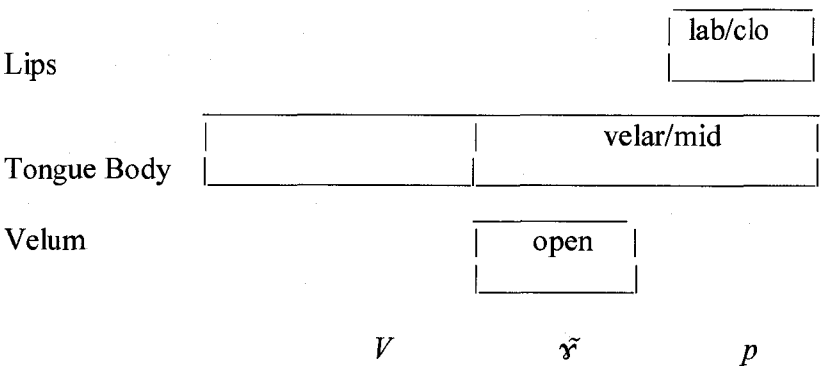
Figure 22. Prevocalization of partial geminates in Maxakalí

(a) Partial geminate /mp/ before prevocalization



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(b) Partial geminate /mp/ after prevocalization



5.5 Summary and conclusion

The preceding discussion has concentrated on additional evidence for the internal structure of consonants and the mechanism of prevocalization advanced in this work. The evidence for the internal structure of consonants was sought in postvocalization, as well as the life-cycle and synchronic functioning of syllabic consonants. The possibility of consonants acting as syllable peaks provides powerful support for the idea that consonants as a group include a vocalic component, as it is hypothesized here that it is precisely the presence of a vocalic component that allows consonants to occupy the nuclear position in a syllable.[155] A syllable peak, however, is a marked position for a consonant to occur in, which is the reason syllabic consonants are diachronically unstable and tend to be replaced by less marked vowel plus consonant sequences. It was suggested that such diachronic developments result from a mechanism similar to that

[155] Incidentally, the failure of vowels to occupy syllable margins may be interpreted as evidence for their lacking a consonantal component: when forced into a syllable margin, vowels are automatically transformed into glides, which may be seen as acquisition of a C-gesture.

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which produces prevocalization, namely, retiming of the constituent gestures of the syllabic consonant. By moving the C-gesture of the consonant to a syllable margin and keeping the V-gesture in the syllable peak, this retiming optimizes the syllable position affiliation of each intrasegmental gesture.

A discussion of compensatory lengthening and vowel intrusion has concentrated on the mechanism of these processes when they interact with prevocalization. It is suggested that when complementary to prevocalization, compensatory lengthening has the same basic mechanism, differing from prevocalization only in the fact that the preceding nucleus is lengthened instead of being diphthongized. It is also suggested that the main difference between prevocalization and vowel intrusion, when the two are complementary, consists in the direction of movement of the C-gesture of the triggering consonant.

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6. Directions for future research

This work has argued that the essence of prevocalization consists in the breaking up of a consonant into a sequence of its internal components that ushers in a vocalic element not present in the original sound chain. It has been suggested that the specification of the vocalic element is present in the underlying gestural score of the consonant, which in turn has led to the conjecture that all consonant-sized units are bigestural in the sense of including an underlying vocalic component. The existence of the vocalic component is uncontroversial in the case of secondarily modified consonants, where it corresponds to the secondary articulation. A study of prevocalization patterns of non-coarticulated consonants has showed that they, too, are bigestural. The nature of their vocalic component has been shown to coincide with the position of the tongue body (and, where applicable, lips) during the consonant's production. The impetus for this analysis was provided by the mismatch between the place specification of the anterior coronals and labials and their prevowels, in Maxakalí and other languages. This hypothesis was tested against a wealth of synchronic and diachronic data from a variety of languages, and turned out to be correct. It was furthermore shown that the vocalic component of plain consonants shows more variation in the anterior coronals, labials and pharyngeals than in tongue body consonants (palatals and velars). In the former consonant group, the vocalic gesture may take on a range of values, from front to central in the case of alveolars, and from central to back in the case of labials, dentals and pharyngeals, with further variations in tongue height.

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The above pairings of plain consonants and prevowels generated by them are strongly reminiscent of the pairings of consonantal and vocalic segments in the preferred CV-type syllables in adult and infant speech, reported in a number of recent studies. Thus, Janson (1986) reports that the favored sequences of consonants and vowels in CV sequences in the languages of the world are dental/alveolar consonant plus front vowel, labial consonant plus back rounded vowel, and velar consonant plus back rounded vowel.[156] MacNeilage and Davis (2000) report that the most frequent patterns of pairing in infant babbling are coronal consonants with front vowels, labial consonants with central vowels, and dorsal consonants with back vowels;[157] the same authors report similar pairings in the global etymologies of Bentson and Ruhlen (1994). These preferential consonant-vowel pairings in CV-syllables are important for the authors' frame-content theory of phonological evolution (cf. MacNeilage and Davis 2005) that views the origin of the protosyllable in the oscillation cycle of the jaw. The theory explains the pairing of labial consonants with central vowels as the mechanical consequence of the tongue being in the position of rest in the center of the mouth when the jaw is in the opening phase; in the case of coronal-front vowel and dorsal-back vowel pairings, the tongue adopts the position common to the consonant and the vowel.

The fact that prevocalization and the research on the origin of speech lead to identical pairings of consonantal and vocalic gestures may be significant. The basis for the connection may at first blush appear mechanical as both instances involve relaxation

[156] Janson's generalizations are based on the data from Finnish, Turkish, Latin, Latvian, and Setswana.

[157] The babies were examined in the English-, Swedish-, French-, Japanese-, Ecuadorian-Quechua- and Brazilian-Portuguese-speaking environments.

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of stricture somewhere in the vocal tract, and the concomitant emergence of a vocoid. However, the similarity is not perfect, for in one case we are dealing with the emergence of a segment-sized, and in another with the emergence of a subsegmental, vocoid. The existence of prevocalization in modern languages and the fact that the component gestures of plain consonants show a striking similarity to the preferred consonant-vowel pairings in CV-type syllables may help conceptualize more clearly the emergence of the earliest units of speech. It may be hypothesized, for example, that originally there was no categorical distinction between the syllable and the phoneme, but rather these speech units differed from each other only in terms of temporal organization of their constituent gestures: the stable pairings of consonantal and vocalic gestures that were sequential in a syllable-sized unit were simultaneous in a phoneme-sized entity. From this point of view, the syllables /pə, ti, kɯ/ and the phonemes /$p^{ə}$, t^{i}, $k^{ɯ}$/ differ from one another only in the timing of their consonantal and vocalic gestures. Further evolution of speech units may have included new types of pairings, whereby the consonantal and vocalic gestures became to be coupled in less optimal ways. This may have resulted in new syllable types in case the pairings were sequential, and in new phonemes in case they were simultaneous. For example, coupling the vocalic gesture normally associated with coronals with the consonantal gesture of labials and velars would have produced both new syllables /pi, ki/ and new phonemes /p^{i}, k^{i}/. It is also likely that the earliest phonemes, when they emerged, would have been closer to present-day consonants than to present-day vowels. One type of evidence for this is the ability of consonants to act as syllable nuclei, which may be viewed as a possible relic

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of their wider functional distribution at an earlier evolutionary stage. When vowels emerged as separate units, at a later stage, they took over the nuclear function. Some evidence for the evolutionary lateness of vocalic segments may lie in the fact that the number of different vowel types is much smaller than that of different consonant types, and in the fact that, unlike consonants, vowels show greater functional specialization in acting exclusively as syllable nuclei.

The above remarks indicate one possible direction for future research by showing how a relatively low-level phonological process may have unexpected and far-reaching consequences for such major areas of inquiry as the phonological evolution and language origins. The notion of the bigestural composition of consonant-sized units introduced in this work, together with the detailed study of the dynamics of prevocalization, open many additional paths of inquiry in such areas as the syllabic consonants, postvelar consonants, consonant gemination, retroflexion, and the synchronic and diachronic mechanism of such processes as umlaut, vocalization, vowel intrusion, metathesis, and compensatory lengthening. Thus, the notion of a bigestural composition of consonants helps put in a new light the functional versatility of consonant-sized units, especially by comparison with the functional specialization of vowels. Accepting the presence of a V-gesture in consonants helps explain not only their ability to act as nuclei or decompose into consonant-vowel or vowel-consonant sequences, but also their ability to bear tone. Interestingly, tone may be borne not only by the syllabic consonants: Clements (1985), for example, shows that tone may appear on the initial portion of geminates, cf. *kùlíɲɲá* 'to climb' or *kùcóp̀pá* 'to become a

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pauper' (60) in Luganda. Since the first part of the geminate is, according to Clements' analysis, in the coda of the preceding syllable, its ability to bear tone receives a natural explanation in terms of a syllable-position optimization of its intrasegmental gestures, with the tone-bearing V-gesture phased with the preceding nucleus, and the tone-free C-gesture with the following one. The data and discussion in 5.4.2 and other parts of this work open up a new way of analyzing the compensatory lengthening patterns of the general type VC > V:. De Chene and Anderson (1979) analyze all such processes as a sequence of consonant gliding (VC > VG) and subsequent merger of the glide with the preceding vowel (VG > V:), thus denying the compensatory lengthening a status as an independent process. More recently, Kavitskaya (2002) proposed to account for such lengthening patterns in terms of phonologization of contextual vowel length triggered by loss of the conditioning consonant (VC > [V:C] > /V:/). The English and Visperterminen data discussed in this work, however, show that these two mechanisms of compensatory lengthening are not mutually exclusive, but can interact in more intricate ways than either of these analyses would suggest. The discussion of retroflexion in 4.4 shows that further study of prevocalization patterns of retroflex consonants promises to throw additional light on this problematic feature as well. Further study of prevocalization patterns of postvelar consonants promises to add to our knowledge of this phonological class; in addition, it may help elucidate various facets of the laryngeal theory, especially as regards the surface phonetics of the laryngeals and the mechanism of their "vocalization" (cf. Reynolds et al. 2000; Vine 2002). In short, the present study opens up a whole new area for both experimental and theoretical

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research, and the suggested model of intrasegmental structure of consonant-level units makes strong predictions relevant to various areas of phonological theory, diachronic linguistics, and articulatory evolution.

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Appendix

Rosapelly's "vocaloid"

Rosapelly (1898a) defines vowels as inherently static articulations produced with an invariable shape of the vocal tract, and consonants as inherently dynamic articulations whose production involves three phases ("temps"): "implosion" (i.e., onset), "le deuxième temps" (i.e., closure), and "explosion" (i.e., release). He observes that the closure phase is the longest of the three as it occupies as much as the onset and the release put together. Using [b] for his illustration of the three phases, Rosapelly draws attention to the fact that the closure phase of this consonant shows remarkable similarity to the articulation of a vowel, both acoustically in the quality of its resonance, and articulatorily in the static position of the articulators. In particular, he observes that the acoustic quality of the closure phase of /b/, although much less clear than that of a vowel due to the closure of the oral passage, nonetheless resembles a vowel in being continuous and in that it can be indefinitely prolonged. In his own words (1898a:73):

> "Pendant la durée de cette occlusion, le tracé des lèvres, comme celui des autres organes, forme une ligne horizontale, de sorte que si, par deux traits pointillés verticaux, on établit des limites des trois temps du B, on obtient pour le deuxième temps une figure qui présente avec le graphique des voyelles une frappante analogie: dans l'un comme dans l'autre on ne trouve que des lignes droites horizontales, indiquant l'immobilité des organes de résonance, et la ligne également horizontale mais ondulée du larynx, indiquant la persistence des vibrations pendant la consonne aussi bien que pendant la voyelle".

Rosapelly then examines other labial articulations including [m, v, f, p, w], and observes than the closure phases of the nasal and the fricatives bear an even closer

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resemblance to a vowel due to a continuous emission of air. He then suggests the term *vocaloid* ("vocaloïde") for the closure phase and notes its affinity to vowel articulations as follows: "je conclus que toute consonne complète est composée de trois temps, don't le deuxième ressemble aux voyelles" (1898a:78). The most salient features of the vocaloid are summarized as follows: (a) it is like a vowel in that the articulators are static throughout its duration; (b) in voiced consonants it has laryngeal vibrations identical to those of vowels in nature and intensity; (c) the duration of the vocaloid is comparable to that of vowels and significantly higher than that of the onset and release phases; in slow speech, both vowels and vocaloids may get lengthened while the duration of the onset and release remains constant; (d) in common with vowels, vocaloids can be pronounced in isolation when stripped of the release and/or onset phases. As an example of a vocaloid stripped of both the onset and release phases Rosapelly cites the syllabic /m̩/: "L'M voyelle c'est le deuxième temps de l'M complet" (1898a:75).

More interesting in the present context is the fact that Rosapelly explicitly connects the resonances of vocaloids with the position of the tongue body during the consonant's closure (1898b). By way of a demonstration he suggests changing the position of the tongue from that of a [a] to that of a [i] during the pronunciation of a [m] or a [v]. When the tongue is in the [a] position, the resonance of the vocaloid approaches that of a back rounded vowel [o] or [u], and when the tongue is in the [i] position, the vocaloid has the quality of a [y]. A similar effect, he suggests, may be obtained by obstructing the mouth

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by hand while pronouncing the corresponding vowels: depending on the degree of closure, a [a] produced in this fashion has a [o], [ɔ] or [u] quality, a [e] has the quality of a [ø] or [œ], and [i] has the quality of a [y].

Of related interest is Rosapelly's view of the speech act as a succession of fixed positions of the vocal tract represented by the vowels and vocaloids connected by means of dynamic transitions in the form of the onset and release phases of the consonants. Well ahead of his time, Rosapelly anticipates much of the current thinking on the complex gestural composition of consonants and the background role of the vocalic articulations (1898a:79), cf.:

> "En dernière analyse, on trouve dans le langage deux catégories de phénomènes; d'une part, les voyelles pures et les vocaloïdes, qui correspondent à des positions fixes des organes pendant une durée relativement longue; et, d'autre part, les plosions qui corresponds à des mouvements dont la durée est beaucoup plus courte. Avec cette manière d'envisager les actes mécaniques de la parole, l'implosion serait le passage de la voyelle au vocaloïde; l'explosion, le passage du vocaloïde à la voyelle, de sorte que, dans une succession alternative de voyelles et de consonnes, comme celles que nous avons étudiées, nous aurions la succession des phénomènes suivants: voyelle, implosion, vocaloïde, explosion, voyelle, etc."

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